

Comply:
ISO 9001:2015
ASME IX

WELDING PROCEDURE REGISTER

Doc No.: DDC-QAQC-VEC26-WPS-001_Rev.0

PROJECT: VIOLA ENERGY CENTER

CLIENT: EVAPCO DRY COOLING, INC. - KIEWIT POWER CONSTRUCTORS CO.

ITEM: CONDENSATE TANK

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Mar 26, 2026	All	Client Review	N.O.Sa	L.V.Nhon
					

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WELDING PROCEDURE REGISTER

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PROJECT: VIOLA ENERGY CENTER
CLIENT: EVAPCO DRY COOLING - KIEWIT POWER CONSTRUCTORS CO.
ITEM: CONDENSATE TANK

No.	WPS No.	Rev.	Standard	Process	Base Metal in PQR	Range Qualified										Supporting PQR No.	Preheat	PWHT	Remarks	
						Product Base Metal		Thickness (mm)		Dia (mm)	Position/ Progression	Consumable/ Filler metal								
								Groove	Fillet			SFA	AWS Class	F.No	Size					Deposited Thick. (mm)
1	WPS-DD-FCAW-251	1	ASME IX:2023	FCAW	ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	08-16	All	All	All, Uphill	5.2	E71T-1C/-9C	6	1.2-1.6	Max.16	PQR-DD-FCAW-251	20°C	N/A	Shell Longitudinal Seam Shell Circumferential Seam Nozzle to Shell (Rolled pipe) Reinforcement Pad to Nozzle; Shell Weldolet Fitting Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate.
2	WPS-DD-FCAW+SAW-252	2	ASME IX:2023	FCAW+ SAW	ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	8-16	All	All	All, uphill for FCAW 1G, 1F for SAW	5.20 5.17	E71T-1C/-9C EM12K-F7A6	6	FCAW Max 1.6 SAW Max 3.2	FCAW: Max. 6 SAW: Max.10	PQR-DD-FCAW+SAW-252	20°C	N/A	Shell Longitudinal Seam Shell Circumferential Seam
3	WPS-DD-FCAW+SAW-378	0	ASME IX:2025	FCAW+ SAW	ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	10-20	All	All	All, uphill for FCAW 1G, 1F for SAW	5.20 5.17	E71T-1C/-9C EM12K-F7A6	6	FCAW Max 1.6 SAW Max 4.0	FCAW: 2 - 8 SAW: 3 - 12	PQR-DD-FCAW+SAW-378	20°C	N/A	Shell Longitudinal Seam Shell Circumferential Seam
4	WPS-DD-FCAW-253	2	ASME IX:2023	FCAW	ASTM A572 Gr.50 to ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	08-40	All	All	All, Uphill	5.20	E71T-1C/-9C	6	1.2-1.6	Max.38	PQR-DD-FCAW-253	10°C	N/A	Shell Longitudinal Seam Shell Circumferential Seam Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate Internal Stiffener.
5	WPS-DD-FCAW-253.1	0	ASME IX:2023	FCAW	ASTM A572 Gr.50 to ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)		All	All	All, Uphill	5.20	E71T-1C/-9C	6	1.2-1.6	Max.38	PQR-DD-FCAW-253	10°C	N/A	Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate
6	WPS-DD-GTAW-254	0	ASME IX: 2023	GTAW	ASME SA53M Gr.B	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	3 - 12	All	All	All,Uphill	5.18	ER70S-6	6	2.4	Max. 12	PQR-DD-GTAW-254	10°C	None	Nozzle to Shell Nozzle Reinforcement Pad to Nozzle Weldolet Fitting to Shell Pipe to fitting and shell
7	WPS-DD-FCAW-134	0	ASME IX: 2019, 2023	FCAW	ASTM A572 Gr.50	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	8 - 16	All	All	All,Uphill	5.20	E71T-1C/-9C	6	1.2-1.6	Max. 16	PQR-DD-FCAW-134	10°C	None	Reinforcement Pad Lifting Lug to Reinforcement Pad Stiffener to Saddle



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						Product Base Metal		Thickness (mm)		Dia (mm)	Position/ Progression	Consumable/ Filler metal								
								Groove	Fillet			SFA	AWS Class	F.No	Size					Deposited Thick. (mm)
8	WPS-DD-FCAW-134/251	0	ASME IX: 2019, 2023	FCAW	ASTM A572 Gr.50 ASME SA516 Gr.70+N	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	8 - 16	All	All	All,Uphill	5.20	E71T-1C/-9C	6	1.2-1.6	Max. 16	PQR-DD-FCAW-134 PQR-DD-FCAW-251	20°C	None	Nozzle to Shell (Small pipe) Reinforcement Pad to Nozzle; Shell Weldolet Fitting Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate.
9	WPS-DD-GTAW+FCAW-196	1	ASME IX: 2021	GTAW+FCAW	EN10025-2 S355J2 to ASTM A572 Gr.50	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	1.5 - 16	All	All	All, Uphill	5.18 5.20	ER70S-6 E71T-1C/-9C	6	1.6 - 2.4 1.2 - 1.6	GTAW: Max. 6 FCAW: Max. 10	PQR-DD-GTAW+FCAW-196	20°C	None	Shell Longitudinal Seam Shell Circumferential Seam Nozzle to Shell (Rolled pipe) Reinforcement Pad to Nozzle; Shell Weldolet Fitting Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate.
10	WPS-DD-GTAW+FCAW-197	1	ASME IX: 2021	GTAW+FCAW	EN10025-2 S355J2 to ASTM A572 Gr.50	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...)	5 - 50	All	All	All, Uphill	5.18 5.20	ER70S-6 E71T-1C/-9C	6	1.6 - 2.4 1.2 - 1.6	GTAW: Max. 10 FCAW: Max. 40	PQR-DD-GTAW+FCAW-197	20°C	None	Shell Longitudinal Seam Shell Circumferential Seam Nozzle to Shell (Rolled pipe) Reinforcement Pad to Nozzle; Shell Weldolet Fitting Lifting Lug Reinforcement Pad, Saddle, Stiffener; Web plate, Web Saddle, Base Plate Fixing support Attachment Plate.
11	WPS-DD-FCAW+SAW-79.1	0	ASMEIX: 2019	FCAW+SAW	JISG3101-SS400 to ASTM A36	P-No.1 Group No.1, 2 (S355, SA516; A105, A36; A572; A53; A106; A500; A1011,...)	JISG3101-SS400 without impact test requirement	5 -50	All	1G	All, uphill for FCAW 1G, 1F for SAW	5.20 5.17	E71T-1C/-9C EM12K-F7A2	6	1.2-1.6 3.2-4.0-4.8	FCAW:Max.24 SAW: Max.26	PQR-DD-FCAW+SAW-79	10°C	N/A	Support Attachment Plate,
12	WPS-DD-FCAW-79.2	0	ASMEIX: 2019	FCAW	JISG3101-SS400 to ASTM A36	P-No.1 Group No.1, 2 (S355, SA516; A105, A36; A572; A53; A106; A500; A1011,...)	JISG3101-SS400 without impact test requirement	5 -24	All	All	All, Uphill	5.20	E71T-1C/-9C	6	1.2-1.6	Max.24	PQR-DD-FCAW+SAW-79	10°C	N/A	Support Attachment Plate,
13	WPS-DD-GTAW-133.1	0	ASME IX: 2019	GTAW	ASTM A240 Gr.304 to ASME SA516 Gr.70+N	P-No.1 Group No.1, 2 (S355, SA516; A105, A36; A572; A53; A106; A500; A1011,...)	P-No.8 Group No. 1, 2 (A240 Gr.304 without impact test requirement)	1.5 - 8	All	All	All, Uphill	5.9	ER309L	6	2.4	Max. 8	PQR-DD-GTAW+FCAW-133	10°C	N/A	Sliding Support Plate
14	WPS-DD-FCAW-133.2	0	ASME IX: 2019	FCAW	ASTM A240 Gr.304 to ASME SA516 Gr.70+N	P-No.1 Group No.1, 2 (S355, SA516; A105, A36; A572; A53; A106; A500; A1011,...)	P-No.8 Group No. 1, 2 (A240 Gr.304 without impact test requirement)	1.5 - 12	All	All	All, Uphill	5.22	E309LT1-1/-4	6	1.2-1.6	Max. 12	PQR-DD-GTAW+FCAW-133	10°C	None	Sliding Support Plate



CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DUNG

123 Bạch Đằng, Phường Tân Sơn Hòa, TP. Hồ Chí Minh, Việt Nam

sale@daidung.vn

(04) 20 3868 1889

www.daidung.com

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						Product Base Metal		Thickness (mm)		Dia (mm)	Position/ Progression	Consumable/ Filler metal								
								Groove	Fillet			SFA	AWS Class	F.No	Size					Deposited Thick. (mm)
15	WPS-DD-GTAW-121	0	ASMEIX: 2019	GTAW	ASTM A312TP304/3 04L	P-No.8 Group No. 1, 2 (A240 Gr.304 without impact test requirement)	P-No.8 Group No. 1, 2 (A240 Gr.304 without impact test requirement)	1.5 -11.4	All	All	All, Uphill	5.9	ER308L	6	2.4	Max.11.4	PQR-DD-GTAW-121	10°C	N/A	Sliding Support Plate
16	WPS-DD-GTAW-240	0	ASME IX: 2023	GTAW	ASTM A572 Gr.50 + EN 10025-2 S355J2	P-No.1 Group No.1 (ASTM A53, A106 GrB, A134, A36, A234 WPB, A572 Gr.50 , ...)	P-No.1 Group No. 2 (ASTM SA516 Gr.70, A105, EN10025-S355J2, ...))	1.5 - 16	All	All	All,Uphill	5.18	ER70S-6	6	2.4	Max. 16	PQR-DD-GTAW-240	10°C	N/A	Nozzle to Shell Nozzle Reinforcement Pad to Nozzle Weldolet Fitting to Shell Pipe to fitting and shell
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION																				
Name: Tran Minh Phung Position QA Welding Signature: 					Name: La Vinh Nhon Position Deputy QA Manager Signature: 															
Date: March 26, 2026					Date: March 26, 2026															

**DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**

● Head Office: Lot No. 38, Zone C, D1 Street, An Ha Industrial Park, Tan Vinh Loc Commune, HCMC, VN

● Rep Office: 123 Bach Dang Street, Tan Son Hoa Ward, HCMC, VN

● sale@daidung.vn

● (84) 28 3868 1689

● www.daidung.com

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ISO 9001:2015
ASME VIII, IX:2023

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-251 Rev.01

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	May 28, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
1	Nov 10, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Add: 38, Zone C, D1 Street, An Ha Industrial Park,
Tan Vinh Loc Commune, Ho Chi Minh City, VN

ĐẶC TÍNH KỸ THUẬT QUÁ TRÌNH HÀN HỒ QUANG

WELDING PROCEDURE SPECIFICATION (WPS)

Số/ WPS N°: WPS-DD-FCAW-251 Rev.1 (10/11/2025)

Quá trình hàn/ *Welding Process(es)* : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / *FCAW*
Phương pháp hàn/ *Type(s)* : Bán tự động, tự động hoặc hàn cơ giới hóa.
Semiautomatic, automatic or machanized welding
Tiêu chuẩn áp dụng/ *Applicable code* : ASME BPVC Section IX, 2023 Edition
Số PQR/ *Supporting PQR No* : KT3-01655BCK4/5 (PQR-DD-FCAW-251)

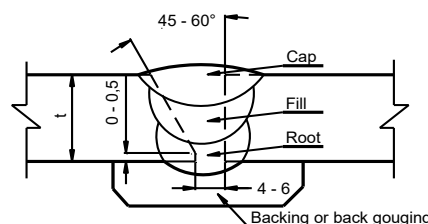
CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ *Joint Design*See sketch or Groove or fillet weld
as per fabrication drawing

Đệm lót/ *Backing* Có / *With*

Loại vật liệu đệm lót/ *Backing Material (Type)*:

- ☒ Kim loại / *Metal* ☒ Phi kim loại/ *Nonmetallic*
☒ Kim loại không nóng chảy/ *Nonfusing Metal*



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Số ký hiệu vật liệu (P-Number)/ *P-Number* 1 với/ *to* 1

Nhóm vật liệu/ *Group* 2 với/ *to* 2

Đặc tính kỹ thuật/ *Specification* ASME SA-516 với/ *to* ASME SA-516

Loại/ *Type/ Grade or UNS N°* Gr. 70 với/ *to* Gr. 70

Khoảng chiều dày (t)/ *Thickness Range (t)*:

Mối hàn rãnh/ *Groove* (8 – 16) mm with CVN; Hàn góc/ *Fillet* Tất cả / *All thickness*

Chiều dày lớp hàn lớn nhất/ *Maximum Pass Thickness* Max. 5 mm

KIM LOẠI HÀN/ FILLER METALS (QW-404)

Đặc tính kỹ thuật/ *Spec. No. (SFA)* SFA-5.20

Loại/ *AWS No. (Class)* E71T-1C/-9C J H8

Số F/ *F-No.* 6

Số A/ *A-No.* 1

Đường kính kim loại hàn/ *Size of Filler Metals* Max. 1,6 mm

Dạng kim loại hàn/ *Filler Metal Product Form* Dây lõi thuốc/ *Flux-cored wire*

Khoảng chiều dày mối hàn/ *Weld Metal Thickness Range*: Mối hàn rãnh/ *Groove* Max. 16 mm

Hàn góc/ *Fillet* Tất cả / *All sizes*

Thuốc hàn/ *Flux*: Loại/ *Type* Không áp dụng/ *Not applicable*

Tên thương mại/ *Trade Name* –

Thông tin khác/ *Other* Filler Metal Trade Name: Supercored 71H or within an SFA Spec.&Class.

THỂ HÀN/ POSITIONS (QW-405)

Thế hàn rãnh/ *Position(s) of Groove* Tất cả các thể/ *All*

Hàn ở vị trí đứng/ *Welding Progression* Hàn leo/ *Uphill*

Thế hàn góc/ *Position(s) of Fillet* Tất cả các thể/ *All*

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ *Preheat Temperature, Minimum* 20°C

Nhiệt độ giữa các lớp/ *Interpass Temperature, Maximum* 205°C

Duy trì/ *Preheat Maintenance* –



**DAI DUNG METALLIC MANUFACTURE CONSTRUCTION
& TRADE CORPORATION**

Add: 38, Zone C, D1 Street, An Ha Industrial Park,
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Số/ WPS N°: WPS-DD-FCAW-251 Rev.1 (10/11/2025)

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Khoảng nhiệt độ/ Temperature Range –

Khoảng thời gian duy trì/ Time Range –

Thông tin khác/ Other –

KHÍ BẢO VỆ/ GAS (QW-408)

	<u>Loại/ Type</u>	<u>Thành phần/ Composition</u>	<u>Lưu lượng/ Flow Rate</u>
Bảo vệ mỗi hàn/ Shielding	CO₂	≥ 99,8 %	(16 – 22) L/min
Bảo vệ chân mỗi hàn/ Backing	---	---	---
Thông tin khác/ Other	---	---	---

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
Root	FCAW	E71T-1C/-9C J	1,2 or 1,6	DCEP	170 – 180	24 – 25	18 – 19	1,29 – 1,50
Fill	FCAW	E71T-1C/-9C J	1,2 or 1,6	DCEP	195 – 205	23 – 24	27 – 28	0,96 – 1,09
Cap	FCAW	E71T-1C/-9C J	1,2 or 1,6	DCEP	195 – 205	23 – 24	26 – 27	1,00 – 1,14

Điện cực Vonfram/ Tungsten Electrode Type and Size..... –

Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer **Cầu / Globular**

Thông tin khác/ Other –

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead **Zic zac/ Weave bead**

Lỗ miệng ống khí/ Gas Cup Size **(14 - 20) mm**

Làm sạch giữa các lớp/ Interpass Cleaning **Cơ học/ Mechanical**

Phương pháp tạo rãnh mặt sau/ Method of Back Gouging **Cơ hoặc nhiệt/ Mechanical or thermal (If required)**

Khoảng cách ống kẹp-vật hàn/ Contact tube to work distance **(14 – 18) mm**

Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) **Nhiều lớp/ Multiple pass**

Hàn một hay nhiều điện cực/ Multiple or Single Electrodes **Một điện cực/ Single**

Khoảng cách điện cực/ Electrode Spacing –

Thông tin khác/ Other **Chiều rộng lớp hàn lớn nhất/ Max. weld bead width: 15 mm**

	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION		
	Prepared by	Checked by	
DATE	10/11/2025	10/11/2025	
SIGN			
NAME	Nguyễn Bảo Tịnh	La Vinh Nhơn	

Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 01/04

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN WELDING PROCEDURE QUALIFICATION RECORD (PQR)

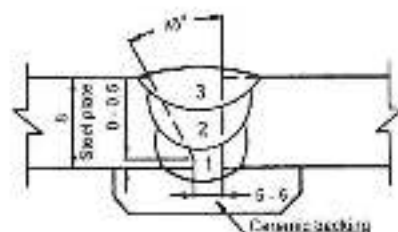
Số PQR/ PQR N° : PQR-DD-FCAW-251 (28/05/2024)
Quá trình hàn/ Welding Process(es) : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp/ Type(s) : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable Code : ASME BPVC Section IX, 2023 Edition
Thợ hàn/ Welder's name (ID) : Nguyễn Trí Hùng (042091005783/ DDC-WD-0716)
Ngày hàn/ Date of welding : 15/05/2024

CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design : Single-Bevel groove weld
Khe hở chân/ Root Spacing : (5 – 6) mm
Đệm lót/ Backing : Có/ Yes

Loại vật liệu đệm lót/ Backing Material (Type):

- ☐ Kim loại hàn/ Weld metal ☒ Phi kim loại/ Nonmetallic
☐ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Đặc tính kỹ thuật/ Specification : ASME SA-516 với/ to ASME SA-516
Loại/ Type/ Grade or UNS N° : Gr. 70 với/ to Gr. 70
Số ký hiệu vật liệu (P-Number)/ P-Number : 1 với/ to 1
Nhóm vật liệu/ Group : 2 với/ to 2
Chiều dày mẫu kiểm tra/ Thickness of Test Coupon : 8 mm
Đường kính mẫu kiểm tra/ Diameter of Test Coupon (OD) : —
Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness : 3,5 mm
Chiều rộng lớp hàn lớn nhất/ Maximum weld bead width : 14 mm
Thông tin khác/ Other : Số nhiệt luyện/ Heat No.: S67929

KIM LOẠI HÀN/ FILLER METALS (QW-404):

Đặc tính kỹ thuật/ SFA Specification : SFA-5.20
Loại/ AWS Classification : E71T-1C
Số F/ F-No. : 6
Số A/ A-No. : 1
Đường kính kim loại hàn/ Size of Filler Metals : 1,2 mm
Dạng kim loại hàn/ Filler Metal Product Form : Dây lõi thuốc/ Flux-cored wire
Chiều dày kim loại mối hàn/ Weld Metal Thickness : 8 mm
Thuốc hàn/ Flux : Loại/ Type : Không áp dụng/ Not applicable
Tên thương mại/ Trade Name :
Thông tin khác/ Other : Filler Metal Trade Name: Hyundai supercored 71H

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove : 1G
Hàn ở vị trí đứng/ Welding Progression : —
Thể hàn góc/ Position(s) of Fillet : —



Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 02/04

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 32 °C (Ambient)
Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 150 °C
Thông tin khác/ Other -

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Nhiệt độ/ Temperature -
Thời gian duy trì/ Time -
Thông tin khác/ Other -

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ môi hàn/ Shielding	CO ₂	99,8 %	(18 – 20) L/min
Bảo vệ chân môi hàn/ Backing	---	---	---
Thông tin khác/ Other	-		

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
1	FCAW	E71T-1C	1,2	DCEP	170 – 180	24 – 25	18 – 19	1,29 – 1,50
2	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	27 – 28	0,96 – 1,09
3	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	26 – 30	1,00 – 1,14

Điện cực Vonfram/ Tungsten Electrode Type and Size -
Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer -
Thông tin khác/ Other -

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Quỹ đạo zic zac/ Weave bead
Lỗ miệng ống khí/ Gas Cup Size Ø14 mm
Làm sạch giữa các lớp/ Interpass Cleaning Mài/ Grinding
Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Không áp dụng/ Not applicable
Khoảng cách ống kẹp-vật hàn/ Contact Tube to Work Distance (16 – 17) mm
Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multile pass
Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single electrode
Khoảng cách điện cực/ Electrode Spacing -
Thông tin khác/ Other -



Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 03/04

KẾT QUẢ KIỂM TRA MỎI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/1)
- Kiểm tra siêu âm/ *Ultrasonic examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/2)
- Thử kéo/ *Tensile test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu <i>Specimen N°</i>	Kích thước mẫu <i>Dimension, mm</i>	Tiết diện <i>Area, mm²</i>	Lực kéo đứt <i>Tensile load, KN</i>	Giới hạn bền <i>Tensile strength, MPa</i>	Vị trí hỏng <i>Broken position</i>	Đánh giá theo/ <i>Assess to ASME, section IX</i>
TS1	19,1 x 7,76	148,2	85,8	579	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>
TS2	18,9 x 7,84	148,2	86,2	582	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>

- Thử uốn/ *Guided bend test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu/ <i>Specimen N°</i>	Kiểu uốn/ <i>Type of bend</i>	Kết quả/ <i>Result</i>	Ghi chú/ <i>Remarks</i>
FB1	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB1	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	
FB2	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB2	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	

- Thử va đập/ *Impact test at -45°C*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference N°*: KT3-01655BCK4/3)

Vị trí rãnh và đập <i>Notch location</i>	Công va đập <i>Absorbed energy (J)</i>			Trung bình <i>Average (J)</i>	Mức chất lượng <i>Criteria(*) (J)</i>	Đánh giá <i>Assessment</i>
	1	2	3			
Kim loại mối hàn/ <i>Weld metal</i>	60	44	48	51	Min. 27	Đạt/ <i>Pass</i>
Vùng ảnh hưởng nhiệt/ <i>Heat affected zone</i>	142	114	120	125		Đạt/ <i>Pass</i>

(*) Theo yêu cầu của khách hàng/ *As customer's request*.



Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 04/04

6. Kiểm tra tổ chức thô đại/ *Macro examination*: Đạt yêu cầu / *Acceptable*.

(Kết quả thử nghiệm tham khảo số/ *Test report reference* N°: KT3-01655BCK4/3)

7. Các thử nghiệm khác/ *Other tests*: Không áp dụng/ *Not applicable*.

Chúng tôi, chứng nhận rằng thông tin trong báo cáo này là đúng, mỗi hàn kiểm tra được chuẩn bị, hàn và thử nghiệm phù hợp với các yêu cầu quy định tại chương IX, Quy phạm nồi hơi và thiết bị áp lực ASME (2023)/ *We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code (2023).*

	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	28/05/2024	28/05/2024
SIGN		
NAME	La Vinh Nhon	Nguyễn Anh Triết

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CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu : MỎI HÀN TRÊN MẪU THÉP TẮM.
Name of sample : *WELDS ON STEEL PLATE COUPON.*
- Mô tả mẫu : Mối hàn trên mẫu thép tấm như hình ảnh ở mục 11.
Sample description : *Welds on steel plate coupon as photograph at item 11.*
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu : 02 mẫu/ Samples.
Testing quantity
- Ngày nhận yêu cầu : 15/05/2024.
Date of receiving
- Ngày thử nghiệm : 15/05/2024.
Date of testing
- Khách hàng : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
Customer : *DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION*
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
Test Location
- Phương pháp thử : ASME BPVC.V-2023 (Article 9).
Test method
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results : *See details on page 02/02.*

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT, GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Điều kiện thử/ Test condition:

☒ Butt weld	☐ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal Thk.: 8 mm
Welding Procedure Spec.: ☐ SMAW	☒ SAW	☒ FCAW	☐ GTAW	
State of Examination : ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation : ☒ As weld	☒ Brushing	☐ Other.....		
Surface Temperature : ☒ (28 + 33) °C	☐ .. °C			
IR Thermometer : FLUKE 561	Serial: 22020241	Next Cal. Due: 06/2024		
Lighting Condition : ☒ Day light > 1000 lux	☐ Artificial > 1000 lux			
Weld Gauge : Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra ngoại quan mối hàn/ Visual examination results:

Tên mẫu Name of test coupon	Chiều dài kiểm tra Tested length, (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	
DD-FCAW+SAW-252 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	

(*) Chú thích/ Legends:

IF : Không ngấu / Incomplete fusion	C : Nứt / crack	CR : Lõm chân / Concave Root
OL : Chồng mép / Overlap	P : Rỗ / Porosity	U : Chảy cạnh / Undercut
SI : Ngậm xỉ / Slag inclusion	R : Quả cầu / External Reinforcement	CS : Lõm mặt / Concave Surface
WP : Biên dạng / Weld bead profile	RO : Ô xi hóa chân / Root Oxidation	PR : Quả thô / Protrusion

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
WELDS ON STEEL PLATE COUPON.
- Mô tả mẫu
Sample description : **Mỗi hàn trên mẫu thép tấm như hình ảnh ở mục 11.**
Welds on steel plate coupon as photograph at item 11.
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu
Testing quantity : **02 mẫu/ Samples.**
- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**
- Ngày thử nghiệm
Date of testing : **17/05/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **ASME BPVC.V-2023 (Article 4).**
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.


Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

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PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

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10. Điều kiện thử nghiệm/ Test condition:

Surface condition ☒ As weld ☐ Brushing ☐ Painting ☐ Grinding	Scanned from ☐ One side ☒ Both sides ☒ One surface ☐ Both surfaces	Preparation ☐ Single V ☐ Double V ☒ Backing ☒ Single Bevel	Couplant ☐ Water ☐ Oil ☒ Cellulose paste ☐	Material ☒ Carbon steel ☐ Stainless steel Diam.: mm Thick.: 8 mm
Ultrasonic instruments ☐ USM 35X DAC (S/N: 9502a) ☐ USM 35X DAC (S/N: 12873a) ☐ USM 35X DAC (S/N: 12875a) ☐ USM 35X DAC (S/N: 14258a) ☒ USM 35X DAC (S/N: 16141a) ☐ POSITECTOR UTG (S/N: 823482)	Probes: Model Angle (°) Freq. (MHz) Serial Ref. (dB) C340151 0 - - - MSEB 0 4 54339 48 MWS 70 4 76970 49	Calibration blocks ☐ V1 - IIW ☐ DAC Ø1,5 mm ☒ V2 - IIW ☐ DAC Ø3,0 mm		
Transfer correction: + 1 dB;		Reference block ☒ T20/Ø2,5 ☐ T35/Ø3,0 (ASME T-434.2.1) ☐		
		Scanning level: Ref. + 12 dB		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra siêu âm/ Ultrasonic testing results:

Tên mẫu Name of test coupon	Chiều dày Thickness (mm)	Chiều dài thử Test length (mm)	Chỉ thị / Indications			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
			Biên độ Actual level (Db)	Loại ¹⁾ Type ¹⁾	Chiều dài Length (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	8	700	-	-	-	✓	
DD-FCAW+SAW-252 SA-516 GR.70	8	700	-	-	-	✓	
¹⁾ Chú thích/ Legends: RI : Sau khi sửa/ After repairing. LOF : Không ngậm / Lack of fusion C : Nứt / crack SI : Ngậm xỉ / Slag inclusion LOP : Không thấm / Lack of penetration P : Rỗ / Porosity U : Cháy cạnh / Undercut							

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
Name of sample **WELDS ON STEEL PLATE COUPON.**
DD-FCAW-251
- Mô tả mẫu : **Mỏi hàn trên thép tấm như hình ảnh bên dưới.**
Sample description **Weld on steel plate as photograph below.**
(700 x 400 x 8) mm.
- Số lượng mẫu : **01 mẫu/ Sample.**
Testing quantity
- Ngày nhận yêu cầu : **15/05/2024.**
Date of receiving
- Ngày thử nghiệm : **22/05/2024.**
Date of testing
- Khách hàng : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DÙNG.**
Customer **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.**
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
Test Location
- Phương pháp thử : **ASME BPVC.IX - 2023.**
Test method **ASME BPVC.VIII.1 - 2023.**
SA-370 (ASME BPVC.III.A-2023)
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết ở trang 02/03 và 03/03.**
Test condition & results **See details on page 02/03 and 03/03.**



TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Kết quả/ *Test results:*

Tên chỉ tiêu Specification		Kết quả thử nghiệm Test result			
A. THỬ KÈO/ TENSILE TEST		Mẫu/Sample 1		Mẫu/Sample 2	
10.1. Kích thước mẫu thử, Dimension of specimen	mm	19,1 x 7,76		18,9 x 7,84	
10.2. Lực kéo đứt, Tensile load	kN	85,8		86,2	
10.3. Giới hạn bền kéo, Tensile strength	MPa	579		582	
10.4. Vị trí đứt Broken position		Kim loại cơ bản Parent metal		Kim loại cơ bản Parent metal	
B. THỬ UỐN/ BEND TEST		FB1	RB1	FB2	RB2
10.5. Kích thước mẫu thử, Dimension of specimen	mm	38,0 x 7,70	38,1 x 7,42	37,9 x 7,70	37,7 x 7,36
10.6. Góc uốn, Bending angle	độ	180	180	180	180
10.7. Đường kính gối uốn, Plunger diameter	mm	32	32	32	32
10.8. Kết quả thử uốn mặt Face bend test result		Đạt Pass	-	Đạt Pass	-
10.9. Kết quả thử uốn chân Root bend test result		-	Đạt Pass	-	Đạt Pass
Tên chỉ tiêu/ Specification		Kết quả thử nghiệm/ Test result			
C. THỬ VA ĐẬP/ IMPACT TEST AT -45°C		Kim loại mối hàn/ Weld metal			
10.10. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.11. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	60	44	48	51
		Vùng ảnh hưởng nhiệt/ Heat affected zone			
10.12. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.13. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	142	114	120	125

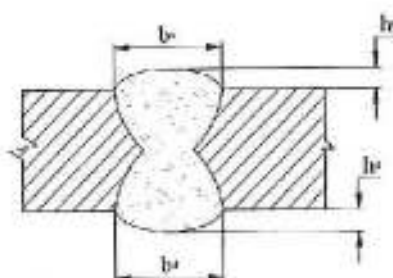
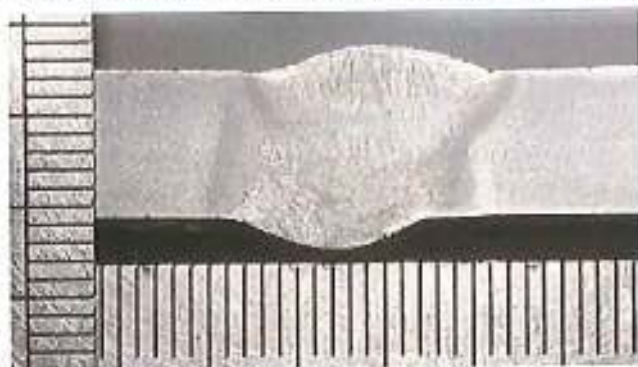


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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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D. TỔ CHỨC THỎ DẠI / MACRO STRUCTURE



QUATEST 3	
☐ Inspected	☒ Reviewed
Ins. No.: 16853	Sign: <i>[Signature]</i>
Date: 15.5.24	<i>[Signature]</i>

검 사 증 명 서

MILL TEST CERTIFICATE

HYUNDAI STEEL

당진3점소: 호남 공진시 호남읍 북부산업로 1480
1489 Buteokan, a-ro, Songok-eup, Dangjin-si, Chungnam, Korea



주문번호
Order No. : F23D400073

품명
Commodity : HOT ROLLED STEEL PLATE

재질규격
Specification : ASME (2021 EDITION) SA516-70-N HT+LTY

고객사
Customer : TUNG WANG STEEL

주요자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20230905-PS-0013-009

발행일자
Date of issue : 2023-09-05

제품사우 Dimension	수량 Quantity	중량 Weight (kg)	세강번호 Heat No.	제품번호 Product No.	Position	연장시험 Tensile Test						충격시험 Impact Test V Notch -40.0 °C			화합성분 Chemical Composition(%)																단위 Unit 2" x 100 3" x 1,000 4" x 10,000 5" x 100,000			
						YP	IS	EL	YP-EL	RA	YR	YP MPa	Energy	SF	LE	Division	C	S	Mn	P	S	Cr	N	B	Cu	Mo	Nb	Ti	V	Si	Al	CSQ		
(N/mm ²)	(%)	(%)	(%)	(%)	(%)	(Joules)	(%)	(%)																										
8.00x2000x8000	1	754	567929	FD10370106	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407			
8.00x2000x8000	1	754	567929	FD10370107	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407			
8.00x2000x8000	1	754	567929	FD10370201	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407			
8.00x2000x8000	1	754	567929	FD10370202	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407			

NOTE

• Position - T:Top, M:Middle, B:Bottom

• Tensile test - Direction:Transverse, Gauge length:200mm (Rectangular), YP Mtd :2 to 2%, off-set) 5 to 0.5% underload, U (Upper yield point), L (Lower yield point)

• Impact test - Direction:Transverse, Spec-size:10mmX5.0mm, SF:Shear fracture, LE:Lateral Expansion

• Division - P:Product analysis, L:Ladle analysis

• The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.

• This M/T Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE SR0175/SR0170 S for hardness test ASME SA516-70 2021 Edition

• The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied + This M/T Test Certificate is issued in accordance with EN 10204 Type 3.1

상기의 제품은 철-사의 합금재질인 규격에 합격하였을 증명합니다.

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Kim in

Chief Of Plate Quality Team



NOTE

- Position - T: Top, M: Middle, B: Bottom
- Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP-Mtd: 1.2 to 2% off-set, S (0.5% underload), U (Upper yield point), L (Lower yield point)
- Impact test - Direction: Transverse, Spec-size: 10mmX5,0mm, SF: 3 shear fracture, LE: Lateral Expansion
- Division - P: Product analysis, L: Lab analysis
- The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.
- This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE MR0175/MR010-3 for hardness test ASME SA516-70 2021 Edition
- The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

입고의 제품은 검사와 검사지침의 규정에 합치하였을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Chief Of Plate Quality Team

[Signature]

- * 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용을 원본에 열거된 URL로부터 확인할 수 있습니다. (<http://www.hyundai-steel.com/csr/login.jsp>)
- * This Mill Test Certificate is a copy that has been printed from original electronic document with digital signature.
- * You are able to check an original electronic document at Hyundai's customer portal: <http://www.hyundai-steel.com/csr/login.jsp> * QR reader scanner App : "QRred"
- * [사용 및 주의사항] 본 검사증명서와 검사의 결과 및 제품사 관련사항 및 안전의 문제가 발생할 수 있습니다.
- * [Caution for Use] Damage on products and safety problems may arise if used other than to the spot stated in this Mill Test Certificate.

CERTIFIED MATERIAL REPORT



CMR No. : HDV - 2024 - 0031

Date of Issue : 2024 - 01 - 05

Customer	-		
Trade Name	Supercored 71H	Size	1.2 mm x 15 kg
Lot No.	MF24340A71	Purchaser's Order No.	-

Specification: EN ISO 17832-A - T42 4 P C1 1 * According to EN 10204-2004 : 3.1
AWS A5.36 E71T1-C1A4-CS1

Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment ("C×hr.)	As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. ("C)	X ₁	X ₂	X ₃	Avg.
	554	619	26	-40	102	95	96	96
EN Spec.	≥420	500~640	≥20	≥47				

Chemical Composition (%)

	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
Weld Metal	0.04	0.40	1.37	0.017	0.005	0.38	0.02	0.01	0.02	0.01	0.01
EN Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05

* Hydrogen Content (nℓ/100g) : 5.76

Others
CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-9J-H8)

QUATEST 3	
☒ Inspected	☒ Reviewed
Ins. No.: 16558	Sign: <i>[Signature]</i>
Date: 15.5.24	Le Tien Han

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

[Signature]

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.

**DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**

Head Office: Lot No. 36, Zone C, D1 Street, An Ha Industrial Park, Tan Vinh Loc Commune, HCMC, VN
Rep Office: 123 Bach Dang Street, Tan Son Hoa Ward, HCMC, VN
sale@daidung.vn | 81 29 3069 1809 | www.daidung.com

Comply:
ISO 9001:2015
ASME VIII, IX:2025

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW+SAW-252 Rev.02

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	May 28, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
1	Nov 10, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
2	Dec 11, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

ĐẶC TÍNH KỸ THUẬT QUÁ TRÌNH HÀN HỒ QUANG

WELDING PROCEDURE SPECIFICATION (WPS)

Số/ WPS N°: WPS-DD-FCAW+SAW-252 Rev.2 (11/12/2025)

Quá trình hàn/ Welding Process(es) : FCAW+SAW (Hàn hồ quang dây lõi thuốc & Hàn hồ quang chìm)

Phương pháp hàn/ Type(s) : Bán tự động, tự động hoặc hàn cơ giới hóa.
Semiautomatic, automatic or mechanized welding

Tiêu chuẩn áp dụng/ Applicable code : ASME BPVC Section IX, 2023 Edition

Số PQR/ Supporting PQR No : KT3-01655BCK4/6 (PQR-DD-FCAW+SAW-252)

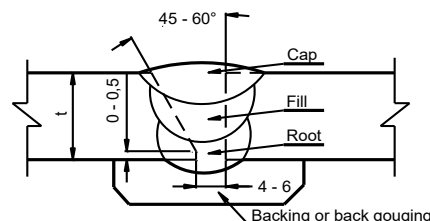
CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design See sketch or Groove or fillet weld
as per fabrication drawing

Đệm lót/ Backing Có / With

Loại vật liệu đệm lót/ Backing Material (Type):

- ☒ Kim loại / Metal ☒ Phi kim loại/ Nonmetallic
☒ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Số ký hiệu vật liệu (P-Number)/ P-Number 1 với/ to 1

Nhóm vật liệu/ Group 2 với/ to 2

Đặc tính kỹ thuật/ Specification ASME SA-516 với/ to ASME SA-516

Loại/ Type/ Grade or UNS N° Gr. 70 với/ to Gr. 70

Khoảng chiều dày (t)/ Thickness Range (t):

Mối hàn rãnh/ Groove (8 – 16) mm with CVN; Hàn góc/ Fillet Tất cả / All thickness

Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness (FCAW&SAW) Max. 5 mm

KIM LOẠI HÀN/ FILLER METALS (QW-404)

	FCAW	SAW
Đặc tính kỹ thuật/ Spec. No. (SFA)	SFA 5.20	SFA 5.17
Loại/ AWS No. (Class)	E71T-1C/-9C J H8	F7A6 +EM12K
Số F/ F-No.	6	6
Số A/ A-No.	1	1
Đường kính kim loại hàn/ Size of Filler Metals	Max. 1,6 mm	Max. 3,2 mm
Dạng kim loại hàn/ Filler Metal Product Form	Dây lõi thuốc/ Flux-cored wire	Dây đặc/ Solid wire
Tên thương mại/ Trade Name	Supercored 71H	M-12K
Thuốc hàn/ Flux - Loại/ Type	-	Agglomerated, Neutral
Tên thương mại/ Trade Name	-	S-717
Khoảng chiều dày mối hàn/ Weld Metal Thickness Range:	≤ 6 mm	≤ 10 mm
Mối hàn rãnh/Groove	Tất cả / All sizes	Tất cả / All sizes
Hàn góc/ Fillet	Tất cả / All sizes	Tất cả / All sizes
Thông tin khác/ Other	Filler Metal Trade Name within SFA Specification & Class.	

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove Tất cả các thể/ All

Hàn ở vị trí đứng/ Welding Progression Hàn leo/ Uphill

Thể hàn góc/ Position(s) of Fillet Tất cả các thể/ All



DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Số/ WPS No: WPS-DD-FCAW+SAW-252 Rev.2 (11/12/2025)

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 20°C

Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 240°C

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Khoảng nhiệt độ/ Temperature Range -

Khoảng thời gian duy trì/ Time Range -

KHÍ BẢO VỆ/ GAS (QW-408)	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ mỗi hàn/ Shielding	CO ₂	≥ 99,8 %	(16 – 22) L/min
Bảo vệ chân mỗi hàn/ Backing	---	---	---
Thông tin khác/ Other	---	---	---

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
Root	FCAW	E71T-1C/-9C J	1,2 or 1,6	DCEP	170 – 180	24 – 25	18 – 19	1,36 – 1,59
Fill/Cap	SAW	F7A6 +EM12K	1,6	DCEP	300 – 310	31 – 32	44 – 46	1,21 – 1,35
Fill/Cap	SAW	F7A6 +EM12K	3,2	DCEP	400 – 450	29 – 31	54 – 56	1,21 – 1,35

Điện cực Vonfram/ Tungsten Electrode Type and Size -

Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer (FCAW) Cầu / Globular

Thông tin khác/ Other -

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Zic zac/ Weave bead (FCAW); Stringer (SAW)

Lỗ miệng ống khí/ Gas Cup Size (14 - 20) mm (FCAW)

Làm sạch giữa các lớp/ Interpass Cleaning Cơ học/ Mechanical

Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Cơ hoặc nhiệt/ Mechanical or thermal (If required)

Khoảng cách ống kẹp-vật hàn/ Contact tube to work distance FCAW:(14-18)mm; SAW:(14-17)mm

Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multiple pass

Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single

Khoảng cách điện cực/ Electrode Spacing -

Thông tin khác/ Other Chiều rộng lớp hàn lớn nhất/ Max. weld bead width (FCAW&SAW): 16 mm

	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION	
	Prepared by	Checked by
DATE	11/12/2025	11/12/2025
SIGN		
NAME	Nguyễn Bảo Tịnh	La Vĩnh Nhơn

Số: KT3-01655B
CK4/6

Ngày: 28/05/2024
Trang số: 01/04

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN WELDING PROCEDURE QUALIFICATION RECORD (PQR)

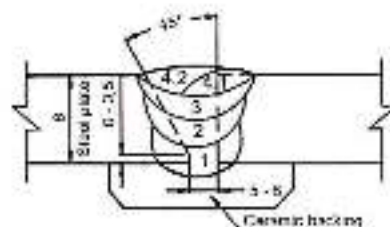
Số PQR/ PQR N° : PQR-DD-FCAW+SAW-252 (28/05/2024)
Quá trình hàn/ Welding Process(es) : FCAW + SAW
Phương pháp/ Type(s) : Bán tự động & Máy / Semiautomatic & Machine welding
Tiêu chuẩn áp dụng/ Applicable Code : ASME BPVC Section IX, 2023 Edition
Thợ hàn/ Welder's name (ID) : NGUYEN TRI HUNG (042091005783/ DDC-WD-0716)
Vận hành máy hàn/ Operator's name (ID) : TRAN VAN TUA (095084008719/ DDC-WD-0715)
Ngày hàn/ Date of welding : 15/05/2024

CHI TIẾT MỐI NỐI JOINTS (QW-402)

Loại mối nối/ Joint Design : Single-Bevel groove weld
Khe hở chân/ Root Spacing : (5 - 6) mm
Đệm lót/ Backing : Có/ Yes

Loại vật liệu đệm lót/ Backing Material (Type):

- ☐ Kim loại hàn/ Weld metal ☒ Phi kim loại/ Nonmetallic
☐ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Đặc tính kỹ thuật/ Specification : ASME SA-516 với/ to ASME SA-516
Loại/ Type or Grade : Gr. 70 với/ to Gr. 70
Số ký hiệu vật liệu (P-Number)/ P-Number : 1 với/ to 1
Nhóm vật liệu/ Group : 2 với/ to 2
Chiều dày mẫu kiểm tra/ Thickness of Test Coupon : 8 mm
Đường kính mẫu kiểm tra/ Diameter of Test Coupon(OD) : -
Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness (FCAW) : 3 mm
Chiều rộng lớp hàn lớn nhất/ Maximum weld bead width : 14 mm
Thông tin khác/ Other : Số nhiệt luyện/ Heat No.: S67929

KIM LOẠI HÀN/ FILLER METALS (QW-404)

Đặc tính kỹ thuật/ SFA Specification:

Loại/ AWS Classification:

Số F/ F-No.:

Số A/ A-No.:

Đường kính kim loại hàn/ Size of Filler Metals:

Dạng kim loại hàn/ Filler Metal Product Form:

Tên thương mại dây hàn/ Filler Metal Trade Name:

Chiều dày kim loại mối hàn/ Weld Metal Thickness:

Thuốc hàn/ Flux(SAW): Loại/ Type : SFA-5.17 F7A6-EM12K (Agglomerated, inactive type flux)

Tên thương mại/ Trade Name : HYUNDAI S-717

Thông tin khác/ Other : -

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove : 1G

Hàn ở vị trí đứng/ Welding Progression : -

Thể hàn góc/ Position(s) of Fillet : -

Số: KT3-01655B
CK4/6

Ngày: 28/05/2024
Trang số: 02/04

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 30 °C (Ambient)
Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 185 °C
Thông tin khác/ Other —

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Nhiệt độ/ Temperature —
Thời gian duy trì/ Time —

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ mối hàn/ Shielding(FCAW)	CO ₂	99,8 %	(18 – 20) L/min
Bảo vệ chân mối hàn/ Backing	—	—	—

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class,	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
1	FCAW	E71T-1C	1,2	DCEP	170 – 180	24 – 25	17 - 18	1,36 – 1,59
2	SAW	EM12K	1,6	DCEP	300 – 310	31 – 32	45 - 46	1,21 – 1,32
3	SAW	EM12K	1,6	DCEP	300 – 310	31 – 32	44 - 45	1,24 – 1,35
4.1	SAW	EM12K	1,6	DCEP	300 – 310	31 – 32	45 - 46	1,21 – 1,32
4.2	SAW	EM12K	1,6	DCEP	300 – 310	31 – 32	45 - 46	1,21 – 1,32

Điện cực Vônfram/ Tungsten Electrode Type and Size (for GTAW) —
Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer (for FCAW) —
Thông tin khác/ Other —

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Zic zac/ Weave bead (FCAW); Stringer (SAW)
Lỗ miệng ống khí/ Gas Cup Size (FCAW) Ø14 mm
Làm sạch giữa các lớp/ Interpass Cleaning Mài/ Grinding
Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Không áp dụng/Not applicable
Khoảng cách ống kẹp-vật hàn/ Contact Tube to Work Distance FCAW: (16 – 17) mm; SAW: (14–15) mm
Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multi pass
Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single electrode
Khoảng cách điện cực/ Electrode Spacing —
Thông tin khác/ Other —

Số: KT3-01655B
CK4/6

Ngày: 28/05/2024
Trang số: 03/04

KẾT QUẢ KIỂM TRA MỎI HÀN/ TEST RESULT

- Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/1)
- Kiểm tra siêu âm/ *Ultrasonic examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/2)
- Thử kéo/ *Tensile test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655CK4/4)

Mẫu/ <i>Specimen N°</i>	Kích thước mẫu/ <i>Dimension, mm</i>	Tiết diện/ <i>Area, mm²</i>	Lực kéo đứt/ <i>Tensile load, KN</i>	Giới hạn bền/ <i>Tensile strength, MPa</i>	Vị trí hỏng/ <i>Broken position</i>	Đánh giá theo/ <i>Assess to ASME, section IX</i>
TS1	19,0 x 8,15	154,8	85,0	549	Kim loại hàn/ <i>Weld metal</i>	Đạt/ <i>Pass</i>
TS2	19,0 x 8,04	152,7	82,7	541	Kim loại hàn/ <i>Weld metal</i>	Đạt/ <i>Pass</i>

- Thử uốn/ *Guided bend test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/4)

Mẫu/ <i>Specimen N°</i>	Kiểu uốn/ <i>Type of bend</i>	Kết quả/ <i>Result</i>	Ghi chú/ <i>Remarks</i>
FB1	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB1	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	
FB2	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB2	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	

- Thử va đập/ *Impact test at -45°C*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference N°*: KT3-01655BCK4/4)

Vị trí rãnh va đập/ <i>Notch location</i>	Công va đập/ <i>Absorbed energy (J)</i>			Trung bình/ <i>Average (J)</i>	Mức chất lượng/ <i>Criteria(*) (J)</i>	Đánh giá/ <i>Assessment</i>
	1	2	3			
Kim loại mối hàn/ <i>Weld metal</i>	132	132	140	135	Min. 27	Đạt/ <i>Pass</i>
Vùng ảnh hưởng nhiệt/ <i>Heat affected zone</i>	34	28	28	30		Đạt/ <i>Pass</i>

(*) Theo yêu cầu của khách hàng/ *As customer's request*.

Số: KT3-01655B
CK4/6

Ngày: 28/05/2024
Trang số: 04/04

6. Kiểm tra tổ chức thô đại/ *Macro examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference No*: KT3-01655BCK4/4)
7. Các thử nghiệm khác/ *Other tests*: Không áp dụng/ *Not applicable*.

Chúng tôi, chứng nhận rằng thông tin trong biên bản này là đúng, mỗi hàn kiểm tra được chuẩn bị, hàn và thử nghiệm phù hợp với các yêu cầu quy định tại chương IX, Quy phạm nồi hơi và thiết bị áp lực ASME (2023)/ *We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code (2023).*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	28/05/2024	28/05/2024
SIGN		
NAME	La Vinh Non	Nguyễn Anh Triết

KT3-01655B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 01/02

- Tên mẫu : MỎI HÀN TRÊN MẪU THÉP TẮM.
Name of sample : **WELDS ON STEEL PLATE COUPON.**
- Mô tả mẫu : Mối hàn trên mẫu thép tấm như hình ảnh ở mục 11.
Sample description : **Welds on steel plate coupon as photograph at item 11.**
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu : 02 mẫu/ Samples.
Testing quantity
- Ngày nhận yêu cầu : 15/05/2024.
Date of receiving
- Ngày thử nghiệm : 15/05/2024.
Date of testing
- Khách hàng : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
Customer : **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
Test Location
- Phương pháp thử : ASME BPVC.V-2023 (Article 9).
Test method
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results : **See details on page 02/02.**

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KI, GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

KT3-01655B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 02/02

10. Điều kiện thử/ Test condition:

☒ Butt weld	☐ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal Thk.: 8 mm
Welding Procedure Spec.: ☐ SMAW	☒ SAW	☒ FCAW	☐ GTAW	
State of Examination : ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation : ☒ As weld	☒ Brushing	☐ Other.....		
Surface Temperature : ☒ (28 + 33) °C	☐ .. °C			
IR Thermometer : FLUKE 561	Serial: 22020241	Next Cal. Due: 06/2024		
Lighting Condition : ☒ Day light > 1000 lux	☐ Artificial > 1000 lux			
Weld Gauge : Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra ngoại quan mối hàn/ Visual examination results:

Tên mẫu Name of test coupon	Chiều dài kiểm tra Tested length, (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	
DD-FCAW+SAW- 252 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	

(*) Chú thích/ Legends:

IF : Không ngấu / Incomplete fusion	C : Nứt / crack	CR : Lõm chân / Concave Root
OL : Chồng mép / Overlap	P : Rỗ / Porosity	U : Cháy cạnh / Undercut
SI : Ngậm xỉ / Slag inclusion	R : Quả cầu / External Reinforcement	CS : Lõm mặt / Concave Surface
WP : Biên dạng / Weld bead profile	RO : Ô xi hóa chân / Root Oxidation	PR : Quả cầu / Protrusion

KT3-01655BC
CK4/2

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 01/02

- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
WELDS ON STEEL PLATE COUPON.
- Mô tả mẫu
Sample description : **Mỗi hàn trên mẫu thép tấm như hình ảnh ở mục 11.**
Welds on steel plate coupon as photograph at item 11.
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu
Testing quantity : **02 mẫu/ Samples.**
- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**
- Ngày thử nghiệm
Date of testing : **17/05/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **ASME BPVC.V-2023 (Article 4).**
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.


Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

KT3-01655BC
CK4/2

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

28/05/2024
Page 02/02

10. Điều kiện thử nghiệm/ Test condition:

Surface condition ☒ As weld ☐ Brushing ☐ Painting ☐ Grinding	Scanned from ☐ One side ☒ Both sides ☒ One surface ☐ Both surfaces	Preparation ☐ Single V ☐ Double V ☒ Backing ☒ Single Bevel	Couplant ☐ Water ☐ Oil ☒ Cellulose paste ☐	Material ☒ Carbon steel ☐ Stainless steel Diam.: mm Thick.: 8 mm
Ultrasonic instruments ☐ USM 35X DAC (S/N: 9502a) ☐ USM 35X DAC (S/N: 12873a) ☐ USM 35X DAC (S/N: 12875a) ☐ USM 35X DAC (S/N: 14258a) ☒ USM 35X DAC (S/N: 16141a) ☐ POSITECTOR UTG (S/N: 823482)	Probes: Model Angle (°) Freq. (MHz) Serial Ref. (dB) C340151 0 - - - MSEB 0 4 54339 48 MWS 70 4 76970 49	Calibration blocks ☐ V1 - IIW ☐ DAC Ø1,5 mm ☒ V2 - IIW ☐ DAC Ø3,0 mm		
		Reference block ☒ T20/Ø2,5 ☐ T35/Ø3,0 (ASME T-434.2.1) ☐		
Transfer correction: +1 dB;		Scanning level: Ref. + 12 dB		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra siêu âm/ Ultrasonic testing results:

Tên mẫu Name of test coupon	Chiều dày Thickness (mm)	Chiều dài thử Test length (mm)	Chỉ thị / Indications			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
			Biên độ Actual level (Db)	Loại ¹⁾ Type ¹⁾	Chiều dài Length (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	8	700	-	-	-	✓	
DD-FCAW+SAW-252 SA-516 GR.70	8	700	-	-	-	✓	
¹⁾ Chú thích/ Legends: RI : Sau khi sửa/ After repairing LOF : Không ngàm / Lack of fusion C : Nứt / crack SI : Ngậm xỉ / Slag inclusion LOP : Không thấm / Lack of penetration P : Rỗ / Porosity U : Cháy cạnh / Undercut							

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TÂM.**
WELDS ON STEEL PLATE COUPON.
DD-FCAW+SAW-252
- Mô tả mẫu
Sample description : **Mối hàn trên thép tấm như hình ảnh bên dưới.**
Weld on steel plate as photograph below.
(700 x 400 x 8) mm.

- Số lượng mẫu
Testing quantity : **01 mẫu/ Sample.**

- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**

- Ngày thử nghiệm
Date of testing : **22/05/2024.**



- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.

- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**

- Phương pháp thử
Test method : **ASME BPVC.IX - 2023.**
ASME BPVC.VIII.1 - 2023.
SA-370 (ASME BPVC.III.A-2023)

- Điều kiện và kết quả thử nghiệm : **Xem chi tiết ở trang 02/03 và 03/03.**
Test condition & results See details on page 02/03 and 03/03.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.


Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

KT3-01655B
CK4/4

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 02/03

10. Kết quả/ Test results:

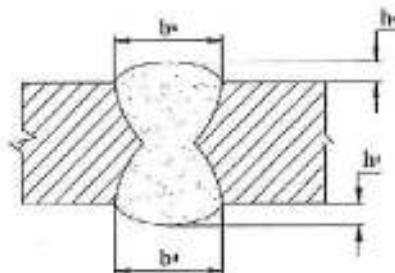
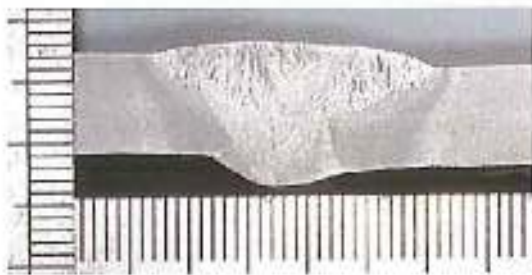
Tên chỉ tiêu Specification		Kết quả thử nghiệm Test result			
A. THỬ KÉO/ TENSILE TEST		Mẫu/Sample 1		Mẫu/Sample 2	
10.1. Kích thước mẫu thử, Dimension of specimen	mm	19,0 x 8,15		19,0 x 8,04	
10.2. Lực kéo đứt, Tensile load	kN	85,0		82,7	
10.3. Giới hạn bền kéo, Tensile strength	MPa	549		541	
10.4. Vị trí đứt Broken position		Kim loại hàn Weld metal		Kim loại hàn Weld metal	
B. THỬ UỐN/ BEND TEST		FB1	RB1	FB2	RB2
10.5. Kích thước mẫu thử, Dimension of specimen	mm	37,6 x 7,68	37,9 x 7,60	37,6 x 7,82	38,0 x 7,77
10.6. Góc uốn, Bending angle	độ	180	180	180	180
10.7. Đường kính gối uốn, Plunger diameter	mm	32	32	32	32
10.8. Kết quả thử uốn mặt Face bend test result		Đạt Pass	-	Đạt Pass	-
10.9. Kết quả thử uốn chân Root bend test result		-	Đạt Pass	-	Đạt Pass
Tên chỉ tiêu/ Specification		Kết quả thử nghiệm/ Test result			
C. THỬ VA ĐẬP/ IMPACT TEST AT -45°C		Kim loại mối hàn/ Weld metal			
10.10. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.11. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	132	132	140	135
		Vùng ảnh hưởng nhiệt/ Heat affected zone			
10.12. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.13. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	34	28	28	30

KT3-01655B
CK4/4

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 03/03

D. TỔ CHỨC THÔ ĐẠI / MACRO STRUCTURE



Kết quả thử nghiệm/ Test result	
Bề mặt/ Surface	Bên trong/ Internal
$b_0 = 24 \text{ mm}$ $h_0 = 1,0 \text{ mm}$ - Đạt/Pass	Mối hàn thấu và ngấu hoàn toàn. Trong mối hàn không có khuyết tật / Full penetration and complete fusion weld. No defect in weld metal.
$b_1 = 11 \text{ mm}$ $h_1 = 1,7 \text{ mm}$ - Đạt/Pass	

QUATEST 3	
☐ Inspected	☒ Reviewed
Ins. No.: 16853	Sign: <i>Lee Tian Heng</i>
Date: 15.5.24	

검 사 증 명 서

MILL TEST CERTIFICATE

HYUNDAI STEEL

당진3점소: 호남 공진시 호남읍 북부산업로 1480
1489 Buteokan, a-ro, Songok-eup, Dangjin-si, Chungnam, Korea



주문번호
Order No. : F23D400073

품명
Commodity : HOT ROLLED STEEL PLATE

재질규격
Specification : ASME (2021 EDITION) SA516-70-N HT+LTY

고객사
Customer : TUNG WANG STEEL

주요자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20230905-PS-0013-009

발행일자
Date of issue : 2023-09-05

제품사우 Dimension	수량 Quantity	중량 Weight (kg)	시각번호 Heat No.	제품번호 Product No.	Position	연장시험 Tensile Test						충격시험 Impact Test V Notch -40.0 °C			화학성분 Chemical Composition(%)																단위 Unit 2" x 100 3" x 1,000 4" x 10,000 5" x 100,000						
						YP	IS	EL	YP-EL	RA	YR	YP MPa	Energy	SF	LE	Division	C	S	Mn	P	S	Cr	N	B	Cu	Mo	Nb	Ti	V	Si	Al	CSQ					
																																	2Ysd (%)	(N)	(%)	(%)	(%)
8.00x2000x8000	1	754	567929	FD10370106	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86				L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407					
8.00x2000x8000	1	754	567929	FD10370107	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						
8.00x2000x8000	1	754	567929	FD10370201	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						
8.00x2000x8000	1	754	567929	FD10370202	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						

NOTE

• Position - T:Top, M:Middle, B:Bottom

• Tensile test - Direction:Transverse, Gauge length:200mm (Rectangular), YP Mtd :2 to 2%, off-set) 5 to 0.5% underload, U (Upper yield point), L (Lower yield point)

• Impact test - Direction:Transverse, Spec-size:10mmX5.0mm, SF:Shear fracture, LE:Lateral Expansion

• Division - P:Product analysis, L:Ladle analysis

• The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.

• This M/T Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE SR0175/SR0170 S for hardness test ASME SA516-70 2021 Edition

• The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied + This M/T Test Certificate is issued in accordance with EN 10204 Type 3.1

상기의 제품은 철-사의 합금재질인 규격에 합격하였음을 증명합니다.

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Kim in

Chief Of Plate Quality Team



NOTE

- Position - T: Top, M: Middle, B: Bottom
- Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mtd: 1.2 to 2% off-set, 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
- Impact test - Direction: Transverse, Spec-size: 10mmX5,0mm, SF: 3 shear fracture, LE: Lateral Expansion
- Division - P: Product analysis, L: Lab analysis
- The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.
- This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE MR0175/MR010-3 for hardness test ASME SA516-70 2021 Edition
- The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

입고의 제품물 질사의 검사시험의 규격에 합치하였을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Chief Of Plate Quality Team

Kim in

* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용을 원본에 열거된 정보로부터 확인할 수 있습니다. (<http://www.hyundai-steel.com/qa/en/login.jsp>)
* This Mill Test Certificate is a copy that has been printed from original electronic document with digital signature.
* You are able to check an original electronic document at Hyundai's customer portal: <http://www.hyundai-steel.com/qa/en/login.jsp> * QRcode scanner App : "QRcode"
* [사용 및 주의사항] 본 검사증명서와 결기의 일치 및 사용시 제품품질 및 안전의 문제가 발생할 수 있습니다.
* [Caution for Use] Damage on products and safety problems may arise if used other than to the spot stated in this Mill Test Certificate.

CERTIFIED MATERIAL REPORT



CMR No. : HDV - 2024 - 0031

Date of Issue : 2024 - 01 - 05

Customer		-									
Trade Name		Supercored 71H				Size		1.2 mm x 15 kg			
Lot No.		MF24340A71				Purchaser's Order No.		-			
Specification		EN ISO 17832-A - T42 4 P C1 1 * According to EN 10204-2004 : 3.1 AWS A5.36 E71T1-C1A4-CS1									
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)				As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)							
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.			
	554	619	26	-40	102	95	96	96			
EN Spec.	≥420	500~640	≥20	≥47							
Chemical Composition (%)											
Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
	0.04	0.40	1.37	0.017	0.005	0.38	0.02	0.01	0.02	0.01	0.01
EN Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05
* Hydrogen Content (nℓ/100g) : 5.76											
Others CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-8J-H8)						QUATEST 3 ☒ Inspected ☒ Reviewed Ins. No.: 16558 Sign: <i>[Signature]</i> Date: 15.5.24 <i>Le Tien Han</i>					
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.											
Approved by QA Manager Pham Dinh Anh Vu											

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.

CERTIFIED MATERIAL REPORT

Y.D.KIM / Representative Director

CMR No. : HWK-2023-036-3

Date of Issue : 2023-03-23

Customer										
Trade Name	S-717 X M-12K			Size	10 x 48 MESH 1.5mm x 20kg					
Lot No.	232003121X 230611113			Purchaser's Order No.						
Applicable Specification	EN ISO 14171-A S 42 4 AB S2Si									
Welding Current	AC	Shielding Gas	*		Post Weld Heat Treatment (°C x hr.)			As Welded		
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)						
	YS (Mpa)	TS (Mpa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Ave.		
	505	583	29	-40	81	84	86	84		
Spec.	≥420	500-640	≥20	≥47						
Chemical Composition (%) (Zirc)	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	Al
	0.10	0.23	1.02	0.014	0.007	0.03	0.04	0.01	0.09	0.002
Spec.	0.07 ~0.15	0.15 ~0.40	0.08 ~1.30	≤0.025	≤0.025	≤0.13	≤0.15	≤0.15	≤0.30	≤0.03
Others: QUATEST 3 ☐ Inspected / ☒ Reviewed Ins. No.: 1655 Sign: <i>[Signature]</i> Date: 15.5.24 <i>[Signature]</i>										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements. Approved by:  QA Manager Liu Yi										

Comply:
ISO 9001:2015
ASME IX:2025

WELDING PROCEDURE

Doc. No.: WPS&PQR -DD-FCAW+SAW-378

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Jan 20, 2026	All	Client Review	N.Quynh Sa	La Vinh Nhon
					

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DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Add: 123 Bach Dang Street, Tan Son Hoa Ward,
Ho Chi Minh City, Vietnam



ĐẶC TÍNH KỸ THUẬT QUÁ TRÌNH HÀN HỒ QUANG WELDING PROCEDURE SPECIFICATION (WPS)

Số/ WPS No: WPS-DD-FCAW+ SAW-378 Rev.0 (20/01/2026)

Quá trình hàn/ Welding Process(es) : FCAW + SAW
Phương pháp hàn/ Type(s) : Bán tự động, tự động hoặc hàn máy
Semiautomatic, automatic or machine welding
Tiêu chuẩn áp dụng/ Applicable code : ASME BPVC Section IX, 2025 Edition
Số PQR/ Supporting PQR No : CKIN26000066.04 (PQR-DD-FCAW+SAW-378 For manufacturer)

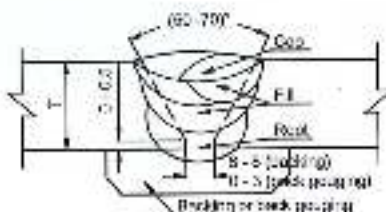
CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design See sketch or Groove or fillet weld
as per fabrication drawing

Đệm lót/ Backing Backing or back gouging

Loại vật liệu đệm lót/ Backing Material (Type):

- ☒ Kim loại/ Metallic ☒ Phi kim loại/ Nonmetallic
☒ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Số ký hiệu vật liệu (P-Number)/ P-Number 1 với/ to 1

Nhóm vật liệu/ Group 2 với/ to 2

Đặc tính kỹ thuật/ Specification ASME SA-516 với/ to ASME SA-516

Loại/ Type or Grade Gr. 70 với/ to Gr. 70

Khoảng chiều dày/ Thickness Range (T):

Mỗi hàn rãnh/ Groove (10 – 20) mm; Hàn góc/ Fillet Tất cả/ All thickness

Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness (FCAW & SAW) ≤ 6,0 mm

KIM LOẠI HÀN/ FILLER METALS (QW-404)

Đặc tính kỹ thuật/ Spec. No. (SFA):

Loại/ AWS No. (Class):

Số F/ F-No.:

Số A/ A-No.:

Đường kính kim loại hàn/ Size of Filler Metals:

Dạng kim loại hàn/ Filler Metal Product Form:

Tên thương mại dây hàn/ Filler Metal Trade Name

Khoảng chiều dày mối hàn/ Weld Metal Thickness Range:

- Mỗi hàn rãnh/ Groove

- Hàn góc/ Fillet

FCAW	SAW
5.20	5.17
E71T-1C/-9C-J (*)	EM12K
6	6
1	1
Max. 1,6 mm	Max. 4,0 mm
Dây lõi thuốc Flux-cored wire	Dây đặc/ Solid wire
Bất kỳ/ Any (within *)	Bất kỳ/ Any (within **)
≤ 8 mm	≤ 12 mm
Tất cả/ All sizes	Tất cả/ All sizes

Thuốc hàn/ Flux: Loại/ Type SFA-5.17, F7A6 (Agglomerated, inactive type flux)

Tên thương mại/ Trade Name Bất kỳ/ Any (within**)

Khác/ Other: Flux-wire class. : SFA-5.17, F7A6-EM12K(**)

THẾ HÀN/ POSITIONS (QW-405)

Thế hàn rãnh/ Position(s) of Groove Tất cả các thế/ All (for FCAW); 1G (for SAW)

Hàn ở vị trí đứng/ Welding Progression Hàn lên/ Uphill (for FCAW)

Thế hàn góc/ Position(s) of Fillet Tất cả các thế/ All (for FCAW); 1F, 2F (for SAW)

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 20°C

Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 228°C

Duy trì/ Preheat Maintenance



**DAI DUNG METALLIC MANUFACTURE CONSTRUCTION
& TRADE CORPORATION**

Add: 123 Bach Dang Street, Tan Son Hoa Ward,
Ho Chi Minh City, Vietnam

Số/ WPS N^o: WPS-DD-FCAW+ SAW-378 Rev.0 (20/01/2026)

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Khoảng nhiệt độ/ Temperature Range

Khoảng thời gian duy trì/ Time Range

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ mối hàn/ Shielding(FCAW)	CO ₂	≥ 99,8 %	(16 – 30) L/min
Bảo vệ chân mối hàn/ Backing	—	—	—
Thông tin khác/ Other	—		

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class	Dường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
Root	FCAW	E71T-1C/-9C-J	Max. 1,6	DCEP	140 - 173	25 - 30	10 - 15	1,40 – 3,11
Fill	SAW	EM12K	Max. 4.0	DCEP	400 - 550	25 - 30	45 - 60	1,00 – 2,20
Cap	SAW	EM12K	Max. 4.0	DCEP	460- 580	25 - 30	45 - 60	1,15 – 2,32

Điện cực Vonfram/ Tungsten Electrode Type and Size (GTAW)

Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer

Thông tin khác/ Other

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Zíc zắc/ Weave bead (FCAW); Stringer (SAW)

Lỗ miệng ống khí/ Gas Cup Size (14 – 20) mm-FCAW

Làm sạch giữa các lớp/ Interpass Cleaning Cơ học/ Mechanical

Phương pháp tạo lõi nhiệt sau/ Method of Back Gouging Cơ hoặc nhiệt/ Mechanical or thermal (If required)

Khoảng cách ống kẹp-vật hàn/ Contact tube to work distance FCAW: (14 – 20) mm; SAW: (20–28) mm

Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multiple pass

Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single

Khoảng cách điện cực/ Electrode Spacing

Thông tin khác/ Other Split layers

if w > 16 mm with no bead width exceeding 25 mm

DAI DUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION		EXAMINING BODY PP. DIRECTOR OF QUATEST 3	
	Prepared by	Checked by	
DATE	20/01/2026	20/01/2026	
SIGN			
NAME	Nguyễn Quỳnh Sa	La Vinh Nhon	Nguyễn Tấn Tùng



Số: Q326B000849
(CKTN26000066.04)

Ngày: 20/01/2026
Trang số: 01/04

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN WELDING PROCEDURE QUALIFICATION RECORD (PQR)

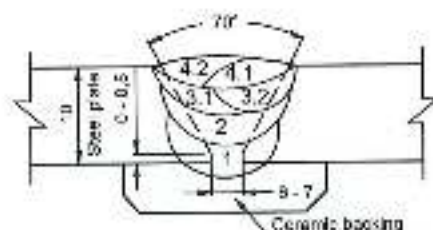
Số PQR/ PQR № : PQR-DD-FCAW+SAW-378 (20/01/2026)
Quá trình hàn/ Welding Process(es) : FCAW + SAW
Phương pháp/ Type(s) : Bán tự động & Máy / Semiautomatic & Machine welding
Tiêu chuẩn áp dụng/ Applicable Code : ASME BPVC Section IX, 2025 Edition
Thợ hàn/ Welder's name (ID) : NGUYEN VAN HIEU (083085007092/ CT-WD-0581)
Vận hành máy hàn/ Operator's name (ID) : NGUYEN THANH DUOC (091099007567/ DDC-WD-0694)
Ngày hàn/ Date of welding : 05/01/2026

CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design : Single-V, groove weld
Khe hở chân/ Root Spacing : (6 - 7) mm
Đệm lót/ Backing : Có/ Yes

Loại vật liệu đệm lót/ Backing Material (Type):

- ☐ Kim loại hàn/ Weld metal ☒ Phi kim loại/ Nonmetallic
☐ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Đặc tính kỹ thuật/ Specification : ASME SA-516 với/ to ASME SA-516
Loại/ Type or Grade : Gr. 70 với/ to Gr. 70
Số ký hiệu vật liệu (P-Number)/ P-Number : 1 với/ to 1
Nhóm vật liệu/ Group : 2 với/ to 2
Chiều dày mẫu kiểm tra/ Thickness of Test Coupon : 10 mm
Đường kính mẫu kiểm tra/ Diameter of Test Coupon(OD) : -
Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness (FCAW) : 4 mm
Chiều rộng lớp hàn lớn nhất/ Maximum weld bead width : 16 mm
Thông tin khác/ Other : Số nhiệt luyện/ Cast No.: KA5908

KIM LOẠI HÀN/ FILLER METALS (QW-404)

Đặc tính kỹ thuật/ SFA Specification:

Loại/ AWS Classification:

Số F/ F-No.:

Số A/ A-No.:

Đường kính kim loại hàn/ Size of Filler Metals:

Dạng kim loại hàn/ Filler Metal Product Form:

Tên thương mại dây hàn/ Filler Metal Trade Name

Chiều dày kim loại mối hàn/ Weld Metal Thickness:

Thuốc hàn/ Flux(SAW): Loại/ Type : SFA-5.17, F7A6 (Agglomerated, inactive type flux)

Tên thương mại/ Trade Name : HYUNDAI S-717

Thông tin khác/ Other : Flux-wire class: SFA-5.17, F7A6-EM12K

FCAW	SAW
SFA-5.20	SFA-5.17
E71T-1C/-9C-J	EM12K
6	6
1	1
1,2 mm	3,2 mm
Dây lõi thuốc/Flux cored	Dây đặc/ Solid
Hyundai superoxored 71H	Hyundai M-12K
Approx. 4 mm	Approx. 6 mm

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove : 1G

Hàn ở vị trí đứng/ Welding Progression : -

Thể hàn góc/ Position(s) of Fillet : -

Số: Q326B000849
(CKTN26000066.04)

Ngày: 20/01/2026
Trang số: 02/04

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 32 °C (Ambient)
Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 178 °C
Thông tin khác/ Other

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Nhiệt độ/ Temperature
Thời gian duy trì/ Time

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ mối hàn/ Shielding (FCAW)	CO ₂	99,8 %	(23 – 24) L/min
Bảo vệ chân mối hàn/ Backing	–	–	–

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
1	FCAW	E71T-1C/-9C-J	1,2	DCEP	143 – 157	27 – 28	10,5	2,21 – 2,51
2	SAW	EM12K	3,2	DCEP	445 – 465	25 – 26	54,4	1,23 – 1,33
3.1	SAW	EM12K	3,2	DCEP	490 – 500	27 – 28	51	1,56 – 1,65
3.2	SAW	EM12K	3,2	DCEP	480 – 490	27 – 28	54,5	1,43 – 1,51
4.1	SAW	EM12K	3,2	DCEP	530 – 535	25 – 26	55	1,45 – 1,52
4.2	SAW	EM12K	3,2	DCEP	520 – 530	26 – 27	51	1,59 – 1,68

Điện cực Vonfram/ Tungsten Electrode Type and Size (for GTAW)
Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer (for FCAW) Phun/ Spray
Thông tin khác/ Other

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Zíc zắc/ Weave bead (FCAW); Stringer (SAW)
Lỗ miệng ống khí/ Gas Cup Size (FCAW) Ø15 mm
Làm sạch giữa các lớp/ Interpass Cleaning Mài/ Grinding
Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Không áp dụng/ Not applicable
Khoảng cách ống kẹp-vật hàn/ Contact Tube to Work Distance FCAW: (18 – 20) mm; SAW: (24 – 25) mm
Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multi pass
Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single electrode
Khoảng cách điện cực/ Electrode Spacing
Thông tin khác/ Other

Số: Q326B000849
 (CKTN26000066.04)

 Ngày: 20/01/2026
 Trang số: 03/04

KẾT QUẢ KIỂM TRA MỐI HÀN/ TEST RESULT

- Kiểm tra ngoại quan/ Visual examination:** Đạt yêu cầu / Acceptable.
 (Kết quả thử nghiệm tham khảo số/ Test report N°: CKTN26000066.01)
- Kiểm tra chụp ảnh bức xạ/ Radiographic examination:** Đạt yêu cầu / Acceptable.
 (Kết quả thử nghiệm tham khảo số/ Test report N°: CKTN26000066.02)
- Thử kéo/ Tensile test:** Đạt yêu cầu / Acceptable.
 (Kết quả thử nghiệm tham khảo số/ Test report N°: CKTN26000066.03)

Mẫu Specimen N°	Kích thước mẫu Dimension, mm	Tiết diện Area, mm ²	Lực kéo đứt Tensile load, KN	Giới hạn bền Tensile strength, MPa	Vị trí hỏng Broken position	Đánh giá theo/ Assess to ASME, section IX
TS1	19,5 x 10,0	195	106,6	547	Kim loại hàn Weld metal	Đạt/ Pass
TS2	19,5 x 10,0	195	106,5	546	Kim loại hàn Weld metal	Đạt/ Pass

- Thử uốn/ Guided bend test:** Đạt yêu cầu / Acceptable.
 (Kết quả thử nghiệm tham khảo số/ Test report N°: CKTN26000066.03)

Mẫu/ Specimen N°	Kiểu uốn/ Type of bend	Kết quả/ Result	Ghi chú/ Remarks
FB1	Uốn mặt/ Face bend	Đạt/ Pass	
RB1	Uốn chân/ Root bend	Đạt/ Pass	
FB2	Uốn mặt/ Face bend	Đạt/ Pass	
RB2	Uốn chân/ Root bend	Đạt/ Pass	

- Thử va đập/ Impact test at -45°C:** Đạt yêu cầu / Acceptable.
 (Kết quả thử nghiệm tham khảo số/ Test report reference N°: CKTN26000066.03)

Vị trí rãnh va đập Notch location	Công va đập Absorbed energy (J)			Trung bình Average (J)	Mức chất lượng Criteria(*) (J)	Đánh giá Assessment
	1	2	3			
Kim loại mối hàn/ Weld metal	44	72	74	63,3	Min. 27	Đạt/ Pass
Vùng ảnh hưởng nhiệt/ Heat affected zone	68	34	66	56		Đạt/ Pass

(*) Theo yêu cầu của khách hàng/ As customer's request.

Số: Q326B000849
(CKTN26000066.04)

Ngày: 20/01/2026
Trang số: 04/04

6. Kiểm tra tổ chức thô đại/ *Macro examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference No*: CKTN26000066.03)

7. Các thử nghiệm khác/ *Other tests*: Không áp dụng/ *Not applicable*.

Chúng tôi, chứng nhận rằng thông tin trong biên bản này là đúng, mỗi hàn kiểm tra được chuẩn bị, hàn và thử nghiệm phù hợp với các yêu cầu quy định tại chương IX, Quy phạm nồi hơi và thiết bị áp lực ASME (2025)/ *We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code (2025).*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY PP. DIRECTOR OF QUATEST 3
DATE	20/01/2026	20/01/2026
SIGN	 DDC QA/QC DEPT	
NAME	Nguyễn Quỳnh Sa	Nguyễn Tấn Tùng

Q326B000849
(CKTN26000066.01)

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

20/01/2026
Page 01/02

1. Tên mẫu : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
Name of sample **WELDS ON STEEL PLATE COUPON.**
2. Mô tả mẫu : **Mỏi hàn trên mẫu thép tấm như hình ảnh ở mục 11.**
Sample description **Welds on steel plate coupon as photograph at item 11.**
Kích thước/ Dimension: (800 x 400 x 10) mm.
3. Số lượng mẫu : **01 mẫu/ Sample.**
Testing quantity
4. Ngày nhận yêu cầu : **05/01/2026.**
Date of receiving
5. Ngày thử nghiệm : **05/01/2026.**
Date of testing
6. Khách hàng : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
Customer **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**
Lô số 38, Khu C, Đường D1, Khu công nghiệp An Hạ, Xã Tân Vĩnh Lộc, Thành phố Hồ Chí Minh.
7. Nơi thử nghiệm : **Trung Tâm Kỹ Thuật 3, KCN Biên Hòa 1, phường Trăn Biên, Tỉnh Đồng Nai.**
Test Location
8. Phương pháp thử : **ASME BPVC.V-2025 (Article 9).**
Test method
9. Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results **See details on page 02/02.**

TRƯỞNG PTN CƠ KHÍ & NDT
HEAD OF MECHANICAL & NDT
TESTING LAB.


Phạm Văn Út

TL. GIÁM ĐỐC/ PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.



Nguyễn Tấn Tùng



Q326B000849
 (CKTN26000066.01)

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

20/01/2026
 Page 02/02

10. Điều kiện thử/ Test condition:

☒ Butt weld	☐ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal Thk.: 10 mm
Welding Procedure Spec.: ☐ SMAW	☒ SAW	☒ FCAW	☐ GTAW	
State of Examination: ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation: ☒ As weld	☒ Brushing	☐ Other:		
Surface Temperature: ☒ (28 ± 3) °C	☐ ... °C			
IR Thermometer: FLUKE 561	Serial: 61560053WS	Next Cal. Due: 10/11/2026		
Lighting Condition: ☒ Day light > 1076 lux	☐ Artificial > 1076 lux			
Weld Gauge: Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of tested coupon:



12. Kết quả kiểm tra ngoại quan mối hàn/ Visual examination results:

Tên mẫu Name of test coupon	Chiều dài kiểm tra Tested length (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo/ Assess to ASMR BPVC.IX-2025	
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable
FCAW+SAW-1G-PL10-A516Gr.70-BK	640 x 2 Surfaces	-	-	-	✓	

(*) Chú thích/ Legends:

IF : Không ngấu / Incomplete fusion
 OL : Chấu nối / Overlap
 SI : Ngậm xỉ / Slag inclusion
 WI : Biên dạng / Weld bead profile

C : Nứt / crack
 P : Rỗ / Porosity
 R : Quấn xỉ External slag/encasement
 RO : Ô xi hóa chân/ Root Oxidation

CR : Lõm chân / Concave Root
 U : Chảy cạnh / Undercut
 CS : Lõm mặt / Concave Surface
 PR : Qôn thẩu / Protrusion

Q326B000849
(CKTN26000066.02)

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

20/01/2026
Page 01/02

1. Tên mẫu : **MỎI HÀN TRÊN MẪU THÉP TẦM.**
Name of sample **WELDS ON STEEL PLATE TEST COUPON.**
2. Mô tả mẫu : **Mỏi hàn trên thép tấm như hình ảnh ở mục 11.**
Sample description **Weld on steel plate as photograph at item 11.**
Kích thước/ Dimension: (800 x 400 x 10) mm.
3. Số lượng mẫu : **01 mẫu/ Sample.**
Testing quantity
4. Ngày nhận yêu cầu : **05/01/2026.**
Date of receiving
5. Ngày thử nghiệm : **13/01/2026.**
Date of testing
6. Khách hàng : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
Customer **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.**
Lô số 38, Khu C, Đường D1, Khu công nghiệp An Hạ, Xã Tân Vĩnh Lộc, Thành phố Hồ Chí Minh.
7. Nơi thử nghiệm : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Phường Trăn Biên, Tỉnh Đồng Nai.**
Test location
8. Phương pháp thử : **ASME BPVC.V-2025 (Article 2).**
Test method
9. Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results **See details from page 02/02.**

TRƯỞNG PTN CƠ KHÍ & NDT
HEAD OF MECHANICAL & NDT
TESTING LAB.


Phạm Văn Út

TL. GIÁM ĐỐC/ PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.


Nguyễn Tấn Tùng

Q326B000849
(CKTN26000066.02)

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

20/01/2026
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10. Điều kiện thử/ Test condition:

☒ X-RAYS		☐ GAMMA RAYS		FILM	
Unit : CME PDUc 250		Source Type : ☐ Ir 192		☐ AGFA GEVAERT D4	
Focus size : 2,0 x 2,0 mm		☐ Co 60		☐ AGFA GEVAERT D5	
Voltage : 180 kV		Activity : Ci		☐ AGFA GEVAERT D7	
Intensity : 5 mA		Dimensions : mm		☐ FUJI IX 50	
SFD : 600 mm		SFD : mm		☐ FUJI IX 80	
Exposure time : 1,0 min		Exposure time : min		☒ FUJI IX 100	
☒ Single film	☒ Single viewing	IQI (wire) : ☐ DIN10 FE	☐ DIN10 FE	Material : ☒ C. Steel	☐ S. Steel
☐ Double film	☐ Double viewing			Thickness : 10 mm	
☒ Single wall	☐ Panoramic	Processing : ☒ Manual	☐ Automatic	Welds : Reinforcement: Max.3 mm	
☐ Double wall	☒ One by one			☒ SAW	☐ GTAW ☒ PCAW
Lead screen thick. : Front 0,125 mm	Back 0,125 mm	Filters : ☐ Lead	☐ ...	☐ Spiral	☐ Circular ☒ Longitudinal
				☐ Heat treatment	

11. Hình ảnh mẫu thử/ Photograph of tested coupon:



FCAW+SAW-1G-PL10-A516 Gr.70-BK, N-V-HIEU/ N-T-DUOC

12. Kết quả chụp ảnh phóng xạ/ Radiographic testing results:

Tên mẫu Name of samples or Film's ID	IQI (wire)	Độ đen Density	Chỉ thị / Indications			Đánh giá theo/ Assess to ASME BPVC.IX-2025		Ghi chú Remarks
			Loại / Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable	
FCAW+SAW-1G-PL10-A516Gr.70-BK	0-1	0,16	2,4 - 2,6	-	-	✓		
	1-2	0,16	2,4 - 2,6	-	-	✓		

Chú thích/ Legends:

LOF : Không ngấu/ Lack of fusion	P : Rỗ/ Porosity	SD : Khuyết tật bề mặt/ Surface defect
LOP : Không thấu/ Lack of penetration	S : Ngậm xỉ/ Slag inclusion	RS : Chụp lại/ Reshoot
C : Nứt/ Crack	U : Chảy rãnh/ Undercut	RC : Lỗ hổng rễ/ Root concavity

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu và khách hàng ghi tên và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the samples identified/ specified only, and this is not a certificate of product.

2. Tên mẫu, tên khách hàng được ghi trên đây chỉ có giá trị khi được ký duyệt và chấp thuận theo thẩm quyền của khách hàng.
The name of the sample and the customer's name are valid only when they are signed and approved by the customer's request.

3. Độ không đảm bảo độ tin cậy được tính từ độ không đảm bảo đo của chuẩn mẫu với hệ số phủ k=2, phần bù chuẩn tương ứng với 95% độ tin cậy.
The reported accuracy of measurement is based on the standard uncertainty of measurement multiplied by the coverage factor k=2, at 95% confidence level.

4. Không được trích ra từ phần phía bên cuối thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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5. Mọi thắc mắc về kết quả, cách tiếp nhận và theo dõi chỉ định kỹ thuật, hãy liên hệ để biết thêm thông tin.
Please contact Quatest 3 at the email address or telephone number for further information about this report.

Q326B000849
(CKTN26000066.03)

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

20/01/2026
Page 01/03

- Tên mẫu
Name of sample : MỎI HÀN TRÊN MẪU THÉP TẦM.
WELDS ON STEEL PLATE COUPON.
DD-FCAW+SAW-378
- Mô tả mẫu
Sample description : Mỏ hàn trên thép tấm như hình ảnh bên dưới.
Weld on steel plate as photograph below.
(800 x 400 x 10) mm.
- Số lượng mẫu
Testing quantity : 01 mẫu/ Sample.
- Ngày nhận yêu cầu
Date of receiving : 05/01/2025.
- Ngày thử nghiệm
Date of testing : 19/01/2025.
- Khách hàng
Customer : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, Khu C, Đường D1, Khu công nghiệp An Hạ,
Xã Tân Vĩnh Lộc, Thành phố Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Phường Trảng Biên,
Tỉnh Đồng Nai.
- Phương pháp thử
Test method : ASME BPVC.IX - 2025.
ASME BPVC.VIII.1 - 2025.
SA-370 (ASME BPVC.III.A-2025)
- Điều kiện và kết quả thử nghiệm
Test condition & results : Xem chi tiết ở trang 02/03 và 03/03.
See details on page 02/03 and 03/03.



TRƯỞNG PTN CƠ KHÍ & NDT
HEAD OF MECHANICAL & NDT
TESTING LAB.


Phạm Văn Út

TL. GIÁM ĐỐC/ PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.



Nguyễn Văn Tuấn

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

20/01/2026
Page 02/03

10. Kết quả/ Test results:

Tên chỉ tiêu Specification		Kết quả thử nghiệm Test result			
A. THỬ KÉO/ TENSILE TEST		Mẫu/Sample 1		Mẫu/Sample 2	
10.1. Kích thước mẫu thử, Dimension of specimen	mm	19,5 x 10,0		19,5 x 10,0	
10.2. Lực kéo đứt, Tensile load	kN	106,6		106,5	
10.3. Giới hạn bền kéo, Tensile strength	MPa	547		546	
10.4. Vị trí đứt Broken position		Kim loại hàn Weld metal		Kim loại hàn Weld metal	
B. THỬ UỐN/ BEND TEST		FB1	RB1	FB2	RB2
10.5. Kích thước mẫu thử, Dimension of specimen	mm	38,2 x 10,2	38,3 x 10,1	38,4 x 10,0	38,7 x 10,0
10.6. Góc uốn, Bending angle	độ	180	180	180	180
10.7. Đường kính gối uốn, Plunger diameter	mm	40	40	40	40
10.8. Kết quả thử uốn mặt Face bend test result		Đạt Pass	-	Đạt Pass	-
10.9. Kết quả thử uốn chân Root bend test result		-	Đạt Pass	-	Đạt Pass
Tên chỉ tiêu/ Specification		Kết quả thử nghiệm/ Test result			
C. THỬ VA ĐẬP/ IMPACT TEST AT -45°C		Kim loại mối hàn/ Weld metal			
10.10. Kích thước mẫu thử, Dimension of specimen	mm	10 x 7,5	10 x 7,5	10 x 7,5	Trung bình Average
10.11. Năng lượng hấp thụ (KV _s), Absorbed energy	J	44	72	74	63,3
		Vùng ảnh hưởng nhiệt/ Heat affected zone			
10.12. Kích thước mẫu thử, Dimension of specimen	mm	10 x 7,5	10 x 7,5	10 x 7,5	Trung bình Average
10.13. Năng lượng hấp thụ (KV _s), Absorbed energy	J	68	34	66	56

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu thử được kiểm tra và không phải là giấy chứng nhận sản phẩm.

Test results are valid for the sample submitted (sample) only and they are not a certificate of product.

2. Các mẫu thử khách hàng được ghi theo phần của các tài liệu. / Sample of inspection and customer are written in customer's request.

3. Độ không đảm bảo đo mà chúng được tính ra từ độ không đảm bảo đo chuẩn nhân với hệ số phủ k = 2, phần bù chuẩn tương ứng với 95 % độ tin cậy.

The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

4. Không được trích ra các phần phân bổ qua thử nghiệm này cho những cơ sở dùng để làm cơ sở của Trung tâm Kỹ thuật 3.

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5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ được ghi trên phiếu này và để biết thêm thông tin.

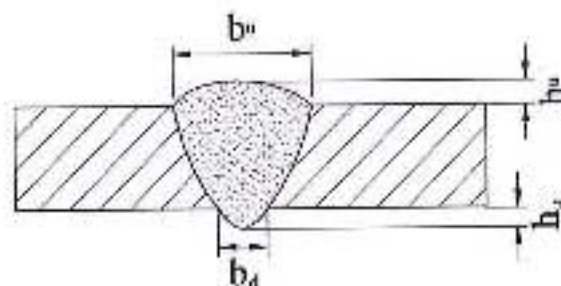
Please contact QUATEST 3 at the email addresses given on report for further information about our report.

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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D. TỔ CHỨC THỎ ĐAI / MACRO STRUCTURE



Kết quả thử nghiệm/ Test result	
Bề mặt/ Surface	Bên trong/ Internal
$b_0 = 24,8 \text{ mm}$ $b_1 = 1,9 \text{ mm}$ - Đạt/Pass $b_2 = 12,2 \text{ mm}$ $b_3 = 2,5 \text{ mm}$ - Đạt/Pass	Mỗi hàn thấu và ngấu hoàn toàn. Trong mỗi hàn không có khuyết tật / Full penetration and complete fusion weld. No defect in weld metal.

[Handwritten signature]

COMMODITY

1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 2680, 26

SPECIFICATION + ACCORDING TO EN 10204:2004 TYPE 3.1

DATE 2023-08-28 P#

S= SURFACE

VR=時伏比 field Ratio, RA=絞り Reduction Area, UE=一様伸び Uniform Elongation, LE=異伸び Lateral Elongation, SF=終極せん断 Shear Fracture, CF=脆性破壊 Cleavage Fracture, SI=モバレーションインデックス Separation Index

CERTIFIED MATERIAL REPORT



CMR No. : HDV – 2025 – 2593

Date of Issue : 2025 – 12 – 04

Customer		G-TECH CORPORATION			
Trade Name		Supercored 71H	Size	1.2 mm x 15 kg	
Lot No.		MF25340K71	Purchaser's Order No.	91-2025/GT/HW/PO	
Specification		AWS A5.20 E71T-9C-J * According to EN 10204-2004 : 3.1			
Welding Current	DCEP	Shielding Gas	C1	Post Weld Heat Treatment (°C×hr.)	As Welded

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	531	585	28.1	-40	90	81	83	85
AWS Spec.	≥390	490~670	≥22	≥27				

Chemical Composition (%)

Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
	0.04	0.49	1.28	0.017	0.008	0.36	0.02	0.02	0.01	0.01
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.03	≤0.03	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35

* Hydrogen Content (mℓ/100g) (AWS A4.3) : 2.52

* Radiographic Test : ACCEPT

* Fillet Weld Test : ACCEPT

Others

* Q'ty: 6,300 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Nhon Trach Ward, Dong Nai Province, Vietnam.

CERTIFIED MATERIAL REPORT



CMR No. : HDV – 2025 – 2361 (Rev.1)

Date of Issue : 2025 – 11 – 07

Customer	G-TECH CORPORATION		
Trade Name	M-12K	Size	3.2 mm x 25 kg
Lot No.	252002101	Purchaser's Order No.	67-2025/GT/HW/PO
Specification	AWS A5.17 EM12K * According to EN 10204-2004 : 3.1 JIS Z 3351 YS-S3 Certified by CWB to CSA W4 EM12K		
Welding Current	-	Shielding Gas	-
		Post Weld Heat Treatment (°C×hr.)	-

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	-	-	-	-	-	-	-	-
AWS Spec.	-	-	-	-				

Chemical Composition (%)

Wire	C	Si	Mn	P	S	Cu				
	0.09	0.23	1.04	0.010	0.006	0.10				
AWS Spec.	0.05 ~0.15	0.10 ~0.35	0.80 ~1.25	≤0.030	≤0.030	≤0.35				

Others

* Q'ty: 1,200 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Nhon Trach Ward, Dong Nai Province, Vietnam.

CERTIFIED MATERIAL REPORT



CMR No. : HDV – 2025 – 2367/1

Date of Issue : 2025 – 11 – 07

Customer	G-TECH CORPORATION		
Trade Name	S-717 x M-12K	Size	10 x 48 Mesh
Lot No.	25051321	Purchaser's Order No.	74-2025/GT/HW/PO
Specification	AWS A5.17 F7A6 x EM12K * According to EN 10204-2004 : 3.1 AS/NZS EN ISO 14171-A S 42 4 AB S2Si Certified by CWB to CSA W48 F49A3-EM12K-H8		
Welding Current	AC	Shielding Gas	-
		Post Weld Heat Treatment (°C×hr.)	As Welded

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	498	575	28.3	-51	64	60	71	65
AWS Spec.	≥400	480~660	≥22	≥27				

Chemical Composition (%)

Wire	C	Si	Mn	P	S	Cu				
	0.10	0.22	1.02	0.009	0.007	0.08				
AWS Spec.	0.05 ~0.15	0.10 ~0.35	0.80 ~1.25	≤0.030	≤0.030	≤0.35				

* Hydrogen Content (mℓ/100g) (AWS A4.3) : 4.66

* Radiographic Test : ACCEPT

Others

* Identical with JIS Z 3183 S502-H (S3-AB1)

* Q'ty: 2,400 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.



0035-CPR-C771
0035 24 / HWV-501
EN13479

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Nhon Trach Ward, Dong Nai Province, Vietnam.

Comply:
ISO 9001:2015
ASME VIII, IX:2025

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-253

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Jun 12, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
1	Nov 20, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
2	Mar 23, 2026	All	Client Review	L.V.Nhon	T.H.P.Vu
				 DDC QA/QC DEPT	

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME VIII, IX - Boiler and Pressure Vessel Code-2025)

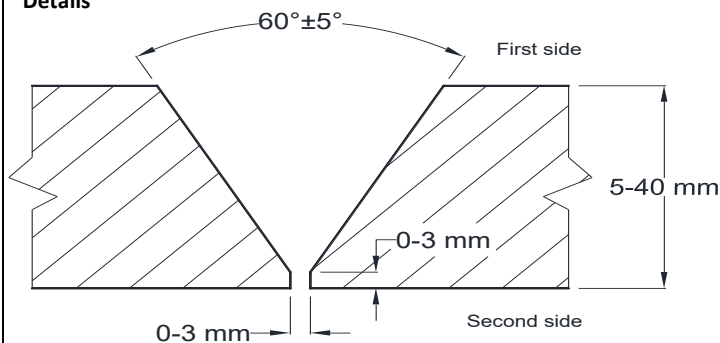
Page: 1/2

Applicable Code: **ASME Section VIII, IX - 2025 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-253** Revision No. **2** Date **Mar 23, 2026**
Supporting PQR No.(s): **PQR-DD-FCAW-253 & PQR-DD-FCAW-251**
Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Semi-Automatic, Mechanized welding**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**
Root Spacing **As per fabrication drawing**
Backing **With backing or back gouging**
Retainer **None**
Backing Material **Base metal or weld metal or Nonmetallic**

Details



Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **1** Group No. **1, 2** to P-No. **1** Group No. **1, 2**
OR
Specification and type/grade or UNS Number **ASTM A572 Gr.50 or ASME SA516 Gr.70 & any same P-No, Group No**
to Specification and type/grade or UNS Number **ASTM A572 Gr.50 or ASME SA516 Gr.70 & any same P-No, Group No**
OR
Chem. Analysis and Mech. Prop. **None**
to Chem. Analysis and Mech. Prop. **None**
Thickness Range: **5 - 40 (mm) - without CVN requirement**
Base Metal: **8 - 40 (mm) - with CVN requirement** **All for fillet**
Maximum Pass Thickness ≤ 13 mm: **Yes** ☒
Other **None**

FILLER METALS (QW-404)

Process **FCAW**
Spec. No. (SFA) **SFA/A-5.20 or 5.36**
AWS No. (Class) **E71T-1C/-9C-J or E71T1-C1A4-CS1**
F-No. **6**
A-No. **1**
Size of Filler Metals **Any (1.2 to 1.6 mm)**
Supplemental Filler Metal **None**
Filler Metal Product Form **Flux cored wire**
Filler Metal Brand Name **Supercored 71H or Any within SFA specification & classification.**
Weld Metal
Deposited Thickness:
Groove **Max. 40 mm**
Fillet **All**
Electrode-Flux (Class) **None**
Flux Type **None**
Flux Trade Name **None**
Consumable Insert **None**
Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME VIII, IX - Boiler and Pressure Vessel Code-2025)

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Applicable Code: **ASME Section VIII, IX - 2025 Edition**Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**By: **La Vinh Nhon**Welding Procedure Specification No.: **WPS-DD-FCAW-253**Revision No. **2**Date **Mar 23, 2026****POSITIONS (QW-405)**

Position(s) of Groove **All**

Position(s) of Fillet **All**

Welding Progression

Vertical up **Yes**

Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**

Time Range **None**

Other **None**

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	99.8%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: only for FCAW Process

PREHEAT (QW-406)

Preheat Temperature, Minimum **20 °C**

Interpass Temperature, Maximum **182 °C**

Preheat Maintenance **Yes**

Other

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	180 - 250	None	0.96 - 1.50	23 - 25	250 - 260	None
Fill	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	195 - 265	None	0.93 - 1.50	23 - 26	275 - 290	None
Cap	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	195 - 265	None	1.00 - 1.50	23 - 25	265 - 270	None

Pulsing Current **None** Heat Input (max.) **1.50** (kJ/mm)Tungsten Electrode Size and Type **None**Mode of Metal Transfer for GMAW (FCAW) **Globular**Other **None****TECHNIQUE (QW-410)**Process **FCAW**String or Weave Bead **Weave bead**Orifice, Nozzle, or Gas Cup Size **No. 6**Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**Method of Back Gouging **Grinding or Air carbon arc**Oscillation **None**Contact Tube to Work Distance **15 - 25 (mm)**Multiple or Single Pass (Per Side) **Multiple**Multiple or Single Electrode **Single**Electrode Spacing **None**Peening **None**Other **None**

Organization

DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon

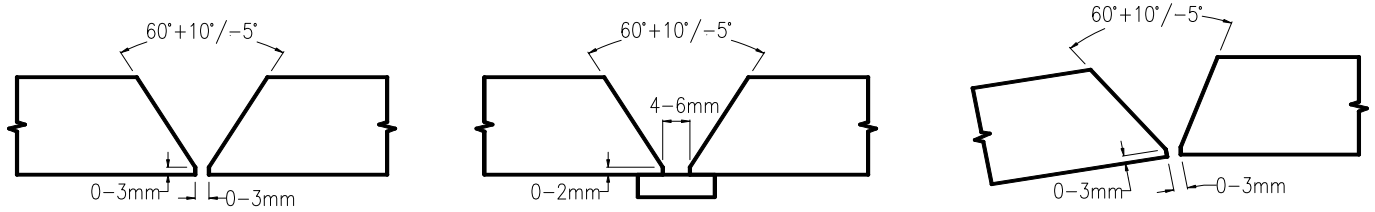
Date: **Mar 23, 2026**Date: **Mar 23, 2026**

Welding Procedure Specification No.: **WPS-DD-FCAW-253**

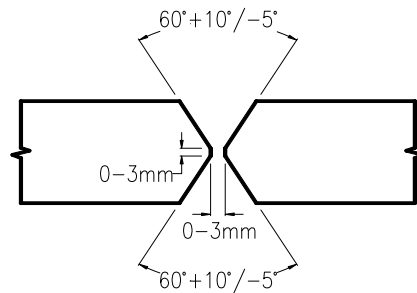
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

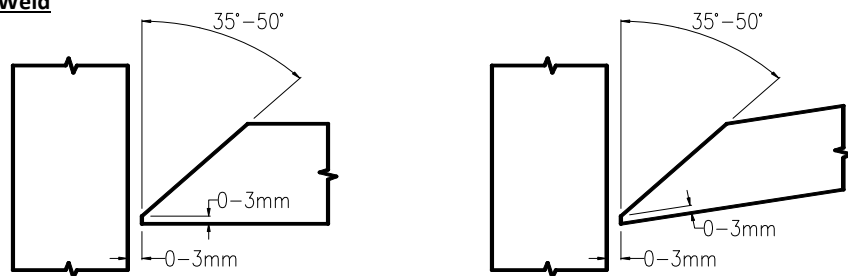
1. Single V Groove Butt Joint



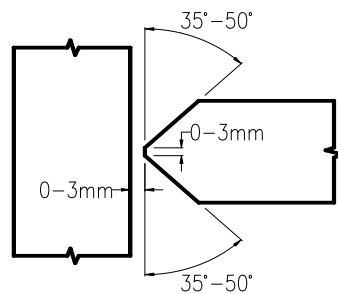
2. Double V Groove Butt Joint



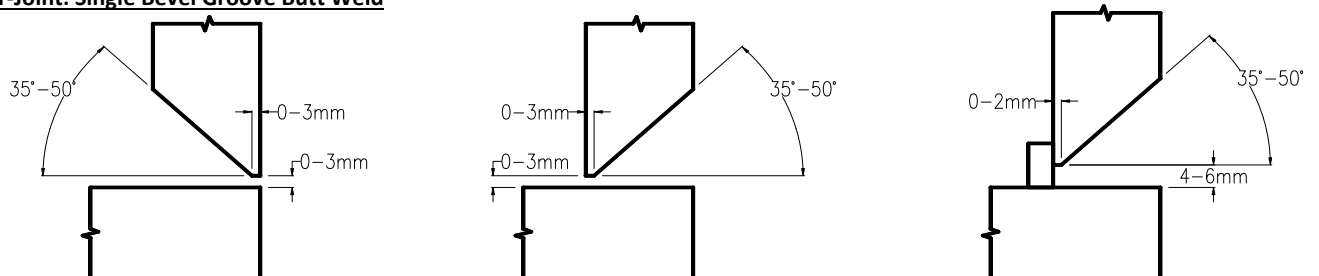
3. T-Joint: Single Bevel Groove Butt Weld



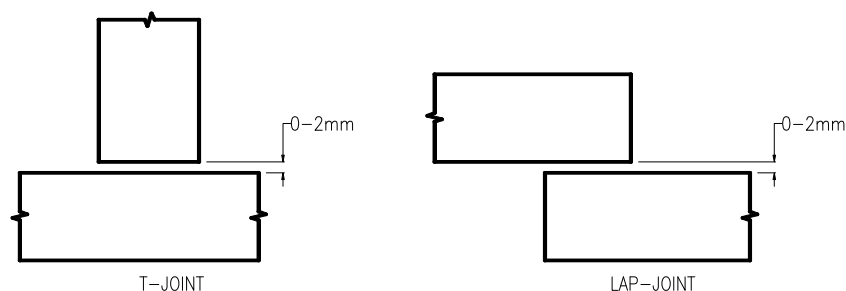
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld





WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2023)

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Applicable Code: **ASME Section IX - 2023 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-FCAW-253.1** Revision No. **0** Date **Jun 16, 2025**

Supporting PQR No.(s): **PQR-DD-FCAW-253** Date **Jun 12, 2024**

Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Mechanized welding**

JOINTS (QW-402)

Joint Design **Fillet and T joints as per fabrication drawing**

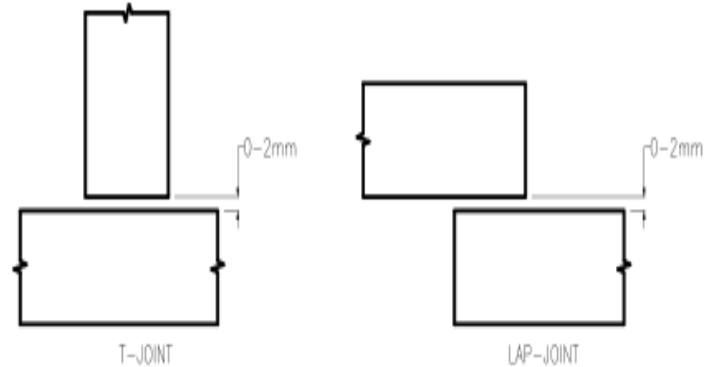
Root Spacing **As per fabrication drawing**

Backing **N/a**

Retainer **None**

Backing Material **-**

Details



BASE METALS (QW-403)

P-No. **1** Group No. **1, 2** to P-No. **1** Group No. **1, 2**

OR

Specification and type/grade or UNS Number **ASTM A572 Gr.50; SA516 Gr.70 (see Table QW/QB-422) to**

Specification and type/grade or UNS Number **ASTM A572 Gr.50; SA516 Gr.70 (see Table QW/QB-422)**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other **None**

FILLER METALS (QW-404)

Process **FCAW**

Spec. No. (SFA) **SFA/A-5.20 or 5.36**

AWS No. (Class) **E71T-1C/-9C-J or E71T1-C1A4-CS1**

F-No. **6**

A-No. **1**

Size of Filler Metals **$\Phi 1.2$ (mm)**

Supplemental Filler Metal **None**

Filler Metal Product Form **Flux cored wire**

Filler Metal Brand Name **Hyundai Supercored 71H or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 40 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2023)

Page: 2/2

Applicable Code: **ASME Section IX - 2023 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-253.1** Revision No. **0** Date **Jun 16, 2025**

POSITIONS (QW-405)

Position(s) of Groove **-**
Position(s) of Fillet **2F, 3F**
Welding Progression
Vertical up **Yes**
Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C if 5 ≤ T ≤ 32 (mm) or
50 °C if 32 < T ≤ 38 mm (T: Thickness of Base Metal)**
Interpass Temperature, Maximum **182 °C**
Preheat Maintenance **Yes**
Other

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	99.8%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: only for FCAW Process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Position	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter							
2F	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 mm	DC EP	252	None	2.1	22 - 33	250 - 260	None
3F	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 mm	DC EP	252	None	2.1	22 - 33	250 - 260	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **None**
Mode of Metal Transfer for GMAW (FCAW) **Globular**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or Weave Bead **Weave bead**
Orifice, Nozzle, or Gas Cup Size **No. 6**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding or Air carbon arc**
Oscillation **Transverse Length 11 mm; Weaving speed 29 mm/s; Swing angle 2; dwell Time 0.2s.**
Contact Tube to Work Distance **15 - 25 (mm)**
Multiple or Single Pass (Per Side) **Multiple or single pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Certified by Welding Coordinator:
La Vinh NhonDate: **Jun 16, 2025**Approved by: **Truong Hoai Phong Vu (QA Manager)**Date: **Jun 16, 2025**



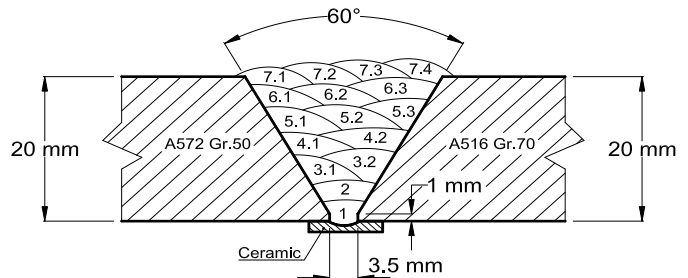
PROCEDURE QUALIFICATION RECORD (PQR) (ASME VIII, IX - Boiler and Pressure Vessel Code-2023)

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Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
Procedure Qualification Record No. **PQR-DD-FCAW-253** Date: **Jun 12, 2024**
Welding Process(es) **FCAW**
Type(s)(Manual, Automatic, Semi-Automatic) **Semi-Automatic**

JOINTS (QW-402)

Joint Type **Butt joint, single V groove**
Backing **Ceramic**
Retainer **None**



BASE METALS (QW-403)

Material Spec. **ASTM A572 to ASME SA516**
Type/Grade, or UNS Number **Gr.50 to Gr.70+N**
P-No. **1** Group No. **1** to P-No. **1** Group No. **2**
Thickness of Test Coupon **20**
Diameter of Test Coupon **Plate**
Maximum Pass Thickness **< 13 mm**
Other **None**

FILLER METALS (QW-404)

Process **FCAW**
SFA Specification **SFA 5.20, AWS A5.20/A5.36**
AWS Classification **E71T-9C-J / E71T1-C1A4-CS1**
Filler Metal F-No. **6**
Weld Metal Analysis A-No. **1**
Size of Filler Metal **1.2 mm**
Filler Metal Product Form **Flux cored wire**
Supplemental Filler Metal **None**
Filler Metal Trade Name **SUPERCORED 71H**
Electrode Flux Classification **-**
Flux Type **-**
Flux Trade Name **-**
Weld Metal Thickness **20 mm**
Other **None**

POSITION (QW-405)

Process **FCAW**
Position(s) **1G**
Weld Progression (Uphill, Downhill) **None**
Other **None**

PREHEAT (QW-406)

Preheat Temperature **31 °C**
Interpass Temperature **Max. 127 °C**
Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**
Time **None**
Other **None**

GAS (QW-408)

	Gas(es)	Percent Composition (Mixture)	Flow Rate
Shielding	CO₂	99.8 %	20 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process **FCAW**
Current and Polarity **DC EP**
Amps. **See table below**
Volts **See table below**
Mode of Metal Transfer for GMAW (FCAW) **Globular**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or weave Bead **Weave (Max. width 12 mm)**
Orifice, Nozzle, or Gas Cup Size **14 mm**
Method cleaning **Grinding, brushing**
Method back gouge **None**
Contact tube to work distance **16 - 17 (mm)**
Multipass or single pass (per side) **Multipass**
Multipass or single Electrodes **Single**
Closed to out chamber **None**
Peening **None**
Use of thermal processes **None**
Other **None**

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	240 - 245	29 - 30	231	31	3	20	1.81 - 1.91
2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31	224	57	3	20	2.21 - 2.37
3.1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31	221	67	4	20	2.24 - 2.40
3.2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31	242	84		20	2.05 - 2.19



PROCEDURE QUALIFICATION RECORD (PQR) (ASME VIII, IX - Boiler and Pressure Vessel Code-2023)

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Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Procedure Qualification Record No. **PQR-DD-FCAW-253** Date: **Jun 12, 2024**Welding Process(es) **FCAW**Type(s)(Manual, Automatic, Semi-Automatic) **Semi-Automatic****WELDING PARAMETER**

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
4.1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31	250	93	4	20	1.98 - 2.12
4.2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31	254	64		20	1.95 - 2.09
5.1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	273	91	3	20	1.88 - 1.98
5.2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	261	112		20	1.97 - 2.07
5.3	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	252	127		20	2.04 - 2.14
6.1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	240	47	3	20	2.14 - 2.25
6.2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	268	56		20	1.91 - 2.01
6.3	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31	273	64		20	1.88 - 1.98
7.1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29	300	34	3	20	1.20 - 1.31
7.2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29	341	64		20	1.06 - 1.15
7.3	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29	333	76		20	1.08 - 1.18
7.4	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29	316	91		20	1.14 - 1.24



PROCEDURE QUALIFICATION RECORD (PQR) (ASME VIII, IX - Boiler and Pressure Vessel Code-2023)

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Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Procedure Qualification Record No. **PQR-DD-FCAW-253** Date: **Jun 12, 2024**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	20.5	19.0	389.5	211	542	Ductile and base metal
Sample 2	20.1	19.0	381.9	212	554	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1(Side bend)	Pass
Specimen 2(Side bend)	Pass
Specimen 3(Side bend)	Pass
Specimen 4(Side bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM1	Weld metal	10x10	-45°C	60 (J)			N
WM2	Weld metal	10x10	-45°C	140 (J)			N
WM3	Weld metal	10x10	-45°C	84 (J)			N
HAZ1	HAZ (A572)	10x10	-45°C	164 (J)			N
HAZ2	HAZ (A572)	10x10	-45°C	172 (J)			N
HAZ3	HAZ (A572)	10x10	-45°C	126 (J)			N
BM1	Base metal (A572)	10x10	-45°C	136 (J)			N
BM2	Base metal (A572)	10x10	-45°C	168 (J)			N
BM3	Base metal (A572)	10x10	-45°C	168 (J)			N
HAZ4	HAZ (A516)	10x10	-45°C	60 (J)			N
HAZ5	HAZ (A516)	10x10	-45°C	50 (J)			N
HAZ6	HAZ (A516)	10x10	-45°C	78 (J)			N
BM4	Base metal (SA516)	10x10	-45°C	152 (J)			N
BM5	Base metal (SA516)	10x10	-45°C	160 (J)			N
BM6	Base metal (SA516)	10x10	-45°C	156 (J)			N

Comments **Acceptable (> 27J)**

Fillet-Weld Test (QW-180)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: **Acceptable**

Other Tests (Not Apply)

Type of Test _____
Deposit Analysis _____
Other _____Welder's Name **Truong Hoang Thien** Clock No. **None** Stamp (ID) No. **DDC-WD-0378**
Welder's Name **-** Clock No. **-** Stamp (ID) No. **-**
Tests Conducted by **Quatest 3** Laboratory Test No. **KT3-02134BCK4**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX - 2023 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon

Date: **Jun 12, 2024**Date: **Jun 12, 2024**

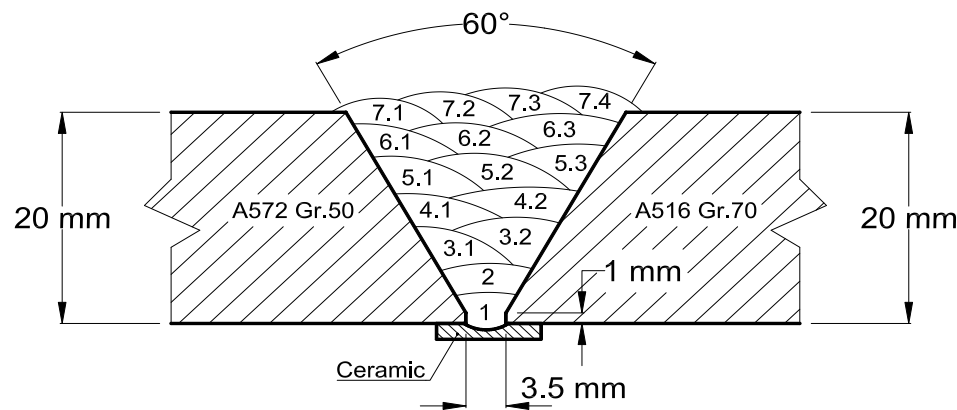
WELDING PARAMETERS RECORD

(Doc No.: WPR-FCAW-253)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-FCAW-253
 Test Coupon: DD-FCAW-253
 Test position: 1G
 Base metal:
 - Specification: ASTM A572 Gr.50 ASME SA-516 Gr.70+N
 - Dimension: PL20x200x600mm PL20x200x600mm
 - Heat/Product No.: SB98414 S43043
 - Certificate No.: 240229-FP01PS-0051A1-0002 20230601-PS-0017-010
 Filler metal:
 - Specification: AWS A-5.36
 - Classification: E71T1-C1A4-CS1
 - Trade name: Hyundai Supercored 71H
 - Batch/Lot No.: MF24340A71
 - Certificate No.: HDV-2024-0031
 Gas: CO2 (99.8%)
 Welder Name: Truong Hoang Thien
 I.D.: DDC-WD-0378

Joint Sketch:

Date: Jun 7, 2024



Root face: 1 mm

Root gap: 2 mm

Groove angle: 60°

Backing: Ceramic

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.						Width	Depth	
1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	240 - 245	29 - 30		130	500	231	31	20	1.81 - 1.91	6	3	
2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31		134	500	224	57	20	2.21 - 2.37	8	3	
3.1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31		136	500	221	67	20	2.24 - 2.4	9	4	
3.2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31		124	500	242	84	20	2.05 - 2.19	9		
4.1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31		120	500	250	93	20	1.98 - 2.12	10	4	
4.2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	275 - 285	30 - 31		118	500	254	64	20	1.95 - 2.09	10		

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-253)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
5.1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		110	500	273	91	20	1.88 - 1.98	9	3	
5.2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		115	500	261	112	20	1.97 - 2.07	10		
5.3	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		119	500	252	127	20	2.04 - 2.14	10		
6.1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		125	500	240	47	20	2.14 - 2.25	10	3	
6.2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		112	500	268	56	20	1.91 - 2.01	10		
6.3	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	285 - 290	30 - 31		110	500	273	64	20	1.88 - 1.98	12		
7.1	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29		100	500	300	34	20	1.2 - 1.31	9	3	
7.2	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29		88	500	341	64	20	1.06 - 1.15	10		
7.3	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29		90	500	333	76	20	1.08 - 1.18	10		
7.4	FCAW	AWS A-5.36 E71T1-C1A4-CS1	1.2	DC EP	215 - 225	28 - 29		95	500	316	91	20	1.14 - 1.24	11		
		Prepared by				Reviewed by										
Name		Nguyen Bao Tinh				La Vinh Nhon										
Signature																
Date		Jun 7, 2024				Jun 7, 2024										

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- | | |
|--|---|
| 1. Tên mẫu :
<i>Name of sample</i> | Mẫu thử nghiệm quy trình hàn - Welding Procedure Qualification
Sample DD-FCAW-253 |
| 2. Số lượng mẫu :
<i>Quantity</i> | 01 |
| 3. Mô tả mẫu :
<i>Sample description</i> | Thép tấm có mối hàn (400 x 600 x 20) mm
Welded steel plate |
| 4. Ngày nhận mẫu :
<i>Date of receiving</i> | 10/06/2024 |
| 5. Ngày thử nghiệm :
<i>Date of testing</i> | 11/06/2024 |
| 6. Nơi gửi mẫu :
<i>Customer</i> | DAI DUNG METALLIC MANUFACTURE CONSTRUCTION
AND TRADE CORPORATION
Lot 38, Zone C, D1 Street, An Ha Industrial Park,
Pham Van Hai Commune, Binh Chanh District, Ho Chi Minh City,
Vietnam |
| 7. Kết quả thử nghiệm :
<i>Test results</i> | Xem trang 02/04...04/04
See pages |

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL TESTING LAB.


Phạm Văn Út

TL.GIÁM ĐỐC / PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.



Nguyễn Văn Tùng

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TEST REPORT

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification		Phương pháp thử Test method	Kết quả thử nghiệm Test result			
THỬ KÉO MỐI HÀN TENSILE TEST OF WELD		ASME IX - 2023	TS1		TS2	
7.1. Kích thước mẫu thử, Dimension of specimen	mm		19,0 x 20,5		19,0 x 20,1	
7.2. Lực kéo đứt, Tensile load	kN		211		212	
7.3. Giới hạn bền kéo, Tensile strength	MPa		542		554	
7.4. Vị trí đứt Broken position			Kim loại cơ bản Parent metal (A572)		Kim loại cơ bản Parent metal (A572)	
THỬ UỐN MỐI HÀN BEND TEST OF WELD		ASME IX - 2023	SB1	SB2	SB3	SB4
7.5. Kích thước mẫu thử, Dimension of specimen	mm		10,1 x 20,1	9,75 x 20,3	10,0 x 20,3	9,83 x 20,3
7.6. Góc uốn, Bending angle	độ		180	180	180	180
7.7. Đường kính gối uốn, Plunger diameter	mm		40	40	40	40
7.8. Kết quả thử uốn cạnh, Side bend test result			Đạt Pass	Đạt Pass	Đạt Pass	Đạt Pass



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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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7. Kết quả thử nghiệm : *Test results*

Tên chỉ tiêu Specification		Phương pháp thử Test method	Kết quả thử nghiệm Test result		
THỬ VA ĐẬP Ở - 45 °C IMPACT TEST AT - 45 °C		ASTM E 23 - 18	Kim loại mối hàn/ Weld metal		
7.9. Kích thước mẫu thử, Dimension of specimen	mm		10 x 10	10 x 10	10 x 10
7.10. Năng lượng hấp thụ (KV _s), Absorbed energy	J		60	140	84
			Vùng ảnh hưởng nhiệt/ Heat affected zone (A572)		
7.11. Kích thước mẫu thử, Dimension of specimen	mm		10 x 10	10 x 10	10 x 10
7.12. Năng lượng hấp thụ (KV _s), Absorbed energy	J		164	172	126
			Kim loại cơ bản/ Parent metal (A572)		
7.13. Kích thước mẫu thử, Dimension of specimen	mm		10 x 10	10 x 10	10 x 10
7.14. Năng lượng hấp thụ (KV _s), Absorbed energy	J		136	168	168
			Vùng ảnh hưởng nhiệt/ Heat affected zone (SA516)		
7.15. Kích thước mẫu thử, Dimension of specimen	mm		10 x 10	10 x 10	10 x 10
7.16. Năng lượng hấp thụ (KV _s), Absorbed energy	J		60	50	78
		Kim loại cơ bản/ Parent metal (SA516)			
7.17. Kích thước mẫu thử, Dimension of specimen	mm	10 x 10	10 x 10	10 x 10	
7.18. Năng lượng hấp thụ (KV _s), Absorbed energy	J	152	160	156	

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the sample submitted complete only, and they is not a certificate of analysis.

2. Tên nhà, tên khách hàng được ghi theo yêu cầu của nười gửi nhà. / Name of sender(s) and customer are written as customer's request.

3). Độ không đảm bảo đo mô men được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ $k = 2$, phần bỏ chuẩn tương ứng với 95 % độ tin cậy.

The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor $k = 2$, at 95 % confidence level.

4. Không được trích sao một phần hoặc toàn bộ kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.

This Test Report shall not be reproduced, except in full, without the written permission by Qvantel S.

5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ cs@quatest3.com, vụ để biết thêm thông tin.

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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7. Kết quả thử nghiệm :
Test results



Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result
TỔ CHỨC THỂ ĐẠI MACRO STRUCTURE	ASME IX - 2023	Mối hàn thấu và ngấu hoàn toàn, trong mối hàn không có khuyết tật. Full penetration and complete fusion weld, no defect in weld metal



1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the specially submitted sample(s) only, and this is not a certificate of product.
2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của đơn gửi mẫu. / Name of sample(s) and customer are written as customer's request.
The reported expanded uncertainty of measurement is stated at the specified uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
3. Không được trích sau một phần phiếu kết quả thử nghiệm này mà không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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4. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ email: info@quatest3.com.vn để biết thêm thông tin.
Please contact QUATEST 3 at the email addresses at info@quatest3.com.vn for further information about test report.

Order No./계약번호 : 01S6629972

PO No./주문번호 : 01S6629972

Supplier
/주문자 : POSCO INTERNATIONAL CORPORATIONCommodity : STEEL PLTAE
/품명Customer : DDC
/고객사Spec & Type : ASTM A572-50-TY1
/규격

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Country of Melt & Pour	Position	Tensile Test /인장시험			Division	Chemical Composition/화합성분											
							YP (MPa)	TS (MPa)	EL (%)		C (%)	Si (%)	Mn (%)	P (%)	S (%)	Cr (%)	Ni (%)	Cu (%)	Mo (%)	Nb (%)	V (%)	
16x2001x12000	PB92803501-3505	5	15,080	SB98437	KOR	T	421	550	26	L	0.1523	0.345	1.390	0.0097	0.0026	0.026	0.013	0.025	0.004	0.024	0.003	
16x2001x12000	PB92851501-1502	2	8,032	SB98914	KOR	T	398	537	26	P	0.1566	0.344	1.392	0.0102	0.0027	0.024	0.014	0.024	0.005	0.023	0.002	
16x2001x12000	PB92884603-4605	3	9,048	SB98914	KOR	T	398	537	26	L	0.1482	0.342	1.406	0.0102	0.0024	0.029	0.009	0.010	0.002	0.024	0.002	
										P	0.1453	0.337	1.386	0.0103	0.0023	0.027	0.010	0.010	0.004	0.022	0.002	
*** Sub Total (040) ***		79	238,264 (kg)							L	0.1482	0.342	1.406	0.0102	0.0024	0.029	0.009	0.010	0.002	0.024	0.002	
										P	0.1453	0.337	1.386	0.0103	0.0023	0.027	0.010	0.010	0.004	0.022	0.002	
20x2001x12000	PB92493901-3904	4	15,080	SB98414	KOR	T	398	529	30	L	0.1503	0.330	1.390	0.0133	0.0022	0.037	0.011	0.012	0.004	0.026	0.003	
20x2001x12000	PB92495201-5204	4	15,080	SB98414	KOR	T	398	529	30	P	0.1497	0.333	1.387	0.0134	0.0022	0.036	0.012	0.012	0.005	0.024	0.002	
20x2001x12000	PB92495601-5604	4	15,080	SB98414	KOR	T	404	526	28	L	0.1503	0.330	1.390	0.0133	0.0022	0.037	0.011	0.012	0.004	0.026	0.003	
20x2001x12000	PB92524501-4504	4	15,080	SB98414	KOR	T	404	526	28	P	0.1497	0.333	1.387	0.0134	0.0022	0.036	0.012	0.012	0.005	0.024	0.002	
20x2001x12000	PB92524601-4604	4	15,080	SB98414	KOR	T	404	526	28	L	0.1503	0.330	1.390	0.0133	0.0022	0.037	0.011	0.012	0.004	0.026	0.003	
20x2001x12000	PB92530601-0604	4	15,080	SB98415	KOR	T	400	530	27	P	0.1497	0.333	1.387	0.0134	0.0022	0.036	0.012	0.012	0.005	0.024	0.002	
20x2001x12000	PB92530901-0904	4	15,080	SB98415	KOR	T	400	530	27	L	0.1503	0.330	1.390	0.0133	0.0022	0.037	0.011	0.012	0.004	0.026	0.003	
20x2001x12000	PB92529501-9504	4	15,080	SB98416	KOR	T	397	531	29	P	0.1497	0.333	1.387	0.0134	0.0022	0.036	0.012	0.012	0.005	0.024	0.002	
20x2001x12000	PB92529601-9604	4	15,080	SB98416	KOR	T	397	531	29	L	0.1519	0.347	1.383	0.0076	0.0019	0.029	0.014	0.012	0.003	0.024	0.002	
20x2001x12000	PB92650701-0704	4	15,080	SB98914	KOR	T	393	529	28	P	0.1517	0.345	1.342	0.0070	0.0017	0.027	0.014	0.012	0.011	0.023	0.002	
										L	0.1466	0.342	1.383	0.0095	0.0022	0.032	0.012	0.015	0.002	0.024	0.002	
										P	0.1470	0.343	1.377	0.0102	0.0020	0.031	0.013	0.015	0.005	0.024	0.002	
										L	0.1486	0.342	1.383	0.0095	0.0022	0.032	0.012	0.015	0.002	0.024	0.002	
										P	0.1470	0.343	1.377	0.0102	0.0020	0.031	0.013	0.015	0.005	0.024	0.002	
										L	0.1482	0.342	1.406	0.0102	0.0024	0.029	0.009	0.010	0.002	0.024	0.002	
										P	0.1453	0.337	1.386	0.0103	0.0023	0.027	0.010	0.010	0.004	0.022	0.002	

* Position - T : Top, M : Middle, B : Bottom

* Tensile Test, Direction : Transversal, Gauge Length : 200 mm(Rectangular),

YP Method : 0.2 % off-set

* Division - L : Ladle Analysis, P : Products Analysis

* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.

This material has been made by vacuum degassing process.

Test Certificate is issued according to ISO 10474/EN 10204 3.1.

We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)

We hereby certify that the material herein has been made in accordance with the order and above specification.

Legal sanction can be imposed on forging. Improper use of product can cause safety issues.

Surveyor To :

B.S. Kim



검 사 증 명 서

MILL TEST CERTIFICATE



당진공장: 충남 당진시 송악읍 복곡신로 1488
1488 Bulgusan-ro, Songak-eup, Dangjin-si, Chungnam, Korea



주문번호
Order No. : F230400073

품 명
Commodity : HOT ROLLED STEEL PLATE

제품규격
Specification : ASME (2021) E01-5A516-70+N MT+LTV

고객사
Customer : TUNG WANG STEEL

주요지
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20230601-PS-0017-010

발행일자
Date of Issue : 2023-06-01

제품사수 Dimension	수량 Quantity	중량 Weight (kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V Notch -40.0℃			화합성분 Chemical Composition(%)														단위 Unit 2 : x10C 3 : x1,000 4 : x10,000 5 : x100,000		
						YP	TS	EL	YP-EL	RA	YR	YP Mtd	Energy	SF	LE	Division	C	Si	Mn	P	S	Cl	Ni	B	Cu	Mo	Nb	Ti	V	Sn	As
						(N/mm ²)	(%)	(%)	(%)	(%)	(Joules)		(%)	(mm)	4		3	3	4	4	2	2	4	2	2	3	3	3	3	3	
20.00x2000x12000	1	3,708	S43043	FC59855201	T	375.3	561.8	29				2	1:152 2:162 3:157 A:157			L	1.613	349	1,108	100	13	17	17	4	141	0	17	2	2	31	431
20.00x2000x12000	1	3,708	S43043	FC59855202	T	375.3	561.8	29				2	1:152 2:162 3:157 A:157			L	1.613	349	1,108	100	13	17	17	4	141	0	17	2	2	31	431
20.00x2000x12000	1	3,766	S59844	FC59951801	T	364.4	542.0	27				2	1:151 2:86 3:177 A:138			L	1.681	333	1,146	103	11	18	17	2	136	0	16	2	2	35	414
20.00x2000x12000	1	3,766	S59844	FC59951802	T	364.4	542.0	27				2	1:151 2:86 3:177 A:138			L	1.681	333	1,146	103	11	18	17	2	136	0	16	2	2	35	414
Sub Total(kg)						4 15,072kg																									
NOTE	Position - T:Top, M:Middle, B:Bottom																														
	Tensile test - Direction: Transverse, Gauge length 200mm (Rectangular), YP Mtd : 2 ID, 2% off-weld, 5 ID, 5% underload, U (Upper yield point), L (Lower yield point)																														
	Impact test - Direction: Transverse, Size: 10mmX10mm, SF: Shear fracture, LE: Lateral Expansion																														
	Division - P: Product analysis, L: Ladle analysis																														
* This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE MR0175/MR010 3 for hardness test ASME SA516-70 2021 Edition																															
* The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. * This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1																															
성기와 제품명 검사의 결과 지정된 규격에 합격한 것을 증명합니다.																															
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS																															
Signature																															
Chief Of Plate Quality Team																															



* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 현대제철 고객포털에서 확인하실 수 있습니다. (<http://am.hyundai-steel.com/cs/cw/login.jsp>)
* This Mill Test Certificate is a copy that has been printed from original electronic document(with digital signing).
* You are able to check an original electronic document at hyundai-steel's customer portal (<http://am.hyundai-steel.com/cs/cw/login.jsp>). * QRcode scanner App : "QRReal"
* [사용상 주의사항] 본 검사증명서에 명시된 규격 외 사용시 제품손상 및 안전상 문제가 발생할 수 있습니다.
* [Caution for Use] Damage on products and safety problems may arise if used other than to the spec stated in this Mill Test Certificate.

검 사 증 명 서

MILL TEST CERTIFICATE

(Attached Sheet)



당첨업체명: 충남 공진시 동약읍 북부신영로 1480
1480 Buhusan-ro, Gongjin-si, Chungnam, Korea



주문번호
Order No.

F230400073

품 명
Commodity

HOT ROLLED STEEL PLATE

고객사
Customer

TUNG WANG STEEL

증명서 번호
Certificate No.

20230601-PS-0017-019

제품규격
Specification

ASME (2021 ED) SA516-70+H MT+LTV

주요자
Contractor

HYUNDAI CORPORATION

발행일자
Date of Issue

2023-06-01

제품번호 Product No.	Position	경도시험 Hardness Test			
		HRC			
		1	2	3	Avg
✓ FC9885201	T	159.0	159.0	161.0	159.7
FC9885202	T	159.0	159.0	161.0	159.7
FC98851801	T	152.0	153.0	152.0	152.3
✓ FC98851802	T	152.0	153.0	152.0	152.3
*** Sub Total(D00) ***		4	15.072Mg		

TUNG WANG STEEL CO., LTD.
ACCEPTED
TRUE CERTIFIED COPY
Quality Assurance Dept

NOT E	* This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. NACE MR0175/MR010 3 for hardness test ASME SA516-70 2021 Edition.	
	* The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. * This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1	
	상기의 제품은 검사의 결과 지정된 규격에 합격한 것을 증명합니다. WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS	

Signature
Kim in
Chief Of Sales Quality Team



* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 현대제철 고객포털에서 확인하실 수 있습니다.(<http://sm.hyundai-steel.com/customer/login.jsp>)
 * This Mill Test Certificate is a copy that has been printed from original electronic document with digital signing.
 * You are able to check an original electronic document at hyundai-steel's customer portal: <http://sm.hyundai-steel.com/customer/login.jsp> * QRcode scanner App: 'QReal'
 * (사용상 주의사항) 본 검사증명서에 명기된 규격 외 사용시 제품손상 및 안전상 문제가 발생할 수 있습니다.
 * [Caution for Use] Damage on products and safety problems may arise if used other than to the spec stated in this Mill Test Certification.

CERTIFIED MATERIAL REPORT



CMR No. : HDV - 2024 - 0031

Date of Issue : 2024 - 01 - 05

Customer	-		
Trade Name	Supercored 71H	Size	1.2 mm x 15 kg
Lot No.	MF24340A71	Purchaser's Order No.	-

Specification: EN ISO 17832-A - T42 4 P C1 1 * According to EN 10204-2004 : 3.1
AWS A5.36 E71T1-C1A4-CS1

Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment ("C×hr.)	As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. ("C)	X ₁	X ₂	X ₃	Avg.
	554	619	26	-40	102	95	96	96
EN Spec.	≥420	500~640	≥20	≥47				

Chemical Composition (%)

	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
Weld Metal	0.04	0.40	1.37	0.017	0.005	0.38	0.02	0.01	0.02	0.01	0.01
EN Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05

* Hydrogen Content (nℓ/100g) : 5.76

Others
CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-9J-H8)

QUATEST 3	
☒ Inspected	☒ Reviewed
Ins. No.: 16558	Sign: <i>[Signature]</i>
Date: 15.5.24	Le Tien Han

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

[Signature]

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.

Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 01/04

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN WELDING PROCEDURE QUALIFICATION RECORD (PQR)

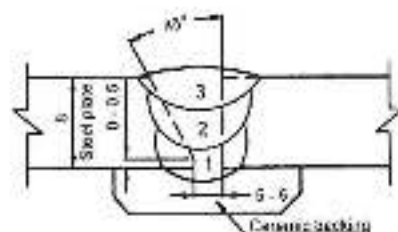
Số PQR/ PQR N° : PQR-DD-FCAW-251 (28/05/2024)
Quá trình hàn/ Welding Process(es) : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp/ Type(s) : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable Code : ASME BPVC Section IX, 2023 Edition
Thợ hàn/ Welder's name (ID) : Nguyễn Trí Hùng (042091005783/ DDC-WD-0716)
Ngày hàn/ Date of welding : 15/05/2024

CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design : Single-Bevel groove weld
Khe hở chân/ Root Spacing : (5 – 6) mm
Đệm lót/ Backing : Có/ Yes

Loại vật liệu đệm lót/ Backing Material (Type):

- ☐ Kim loại hàn/ Weld metal ☒ Phi kim loại/ Nonmetallic
☐ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Đặc tính kỹ thuật/ Specification : ASME SA-516 với/ to ASME SA-516
Loại/ Type/ Grade or UNS N° : Gr. 70 với/ to Gr. 70
Số ký hiệu vật liệu (P-Number)/ P-Number : 1 với/ to 1
Nhóm vật liệu/ Group : 2 với/ to 2
Chiều dày mẫu kiểm tra/ Thickness of Test Coupon : 8 mm
Đường kính mẫu kiểm tra/ Diameter of Test Coupon (OD) : —
Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness : 3,5 mm
Chiều rộng lớp hàn lớn nhất/ Maximum weld bead width : 14 mm
Thông tin khác/ Other : Số nhiệt luyện/ Heat No.: S67929

KIM LOẠI HÀN/ FILLER METALS (QW-404):

Đặc tính kỹ thuật/ SFA Specification : SFA-5.20
Loại/ AWS Classification : E71T-1C
Số F/ F-No. : 6
Số A/ A-No. : 1
Đường kính kim loại hàn/ Size of Filler Metals : 1,2 mm
Dạng kim loại hàn/ Filler Metal Product Form : Dây lõi thuốc/ Flux-cored wire
Chiều dày kim loại mối hàn/ Weld Metal Thickness : 8 mm
Thuốc hàn/ Flux : Loại/ Type : Không áp dụng/ Not applicable
Tên thương mại/ Trade Name :
Thông tin khác/ Other : Filler Metal Trade Name: Hyundai supercored 71H

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove : 1G
Hàn ở vị trí đứng/ Welding Progression : —
Thể hàn góc/ Position(s) of Fillet : —

Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 02/04

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 32 °C (Ambient)
Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 150 °C
Thông tin khác/ Other -

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Nhiệt độ/ Temperature -
Thời gian duy trì/ Time -
Thông tin khác/ Other -

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ môi hàn/ Shielding	CO ₂	99,8 %	(18 – 20) L/min
Bảo vệ chân môi hàn/ Backing	---	---	---
Thông tin khác/ Other	---		

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
1	FCAW	E71T-1C	1,2	DCEP	170 – 180	24 – 25	18 – 19	1,29 – 1,50
2	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	27 – 28	0,96 – 1,09
3	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	26 – 30	1,00 – 1,14

Điện cực Vonfram/ Tungsten Electrode Type and Size -
Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer -
Thông tin khác/ Other -

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Quỹ đạo zíc zắc/ Weave bead
Lỗ miệng ống khí/ Gas Cup Size Ø14 mm
Làm sạch giữa các lớp/ Interpass Cleaning Mài/ Grinding
Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Không áp dụng/ Not applicable
Khoảng cách ống kẹp-vật hàn/ Contact Tube to Work Distance (16 – 17) mm
Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multile pass
Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single electrode
Khoảng cách điện cực/ Electrode Spacing -
Thông tin khác/ Other -

Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 03/04

KẾT QUẢ KIỂM TRA MỎI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/1)
- Kiểm tra siêu âm/ *Ultrasonic examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/2)
- Thử kéo/ *Tensile test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu <i>Specimen N°</i>	Kích thước mẫu <i>Dimension, mm</i>	Tiết diện <i>Area, mm²</i>	Lực kéo đứt <i>Tensile load, KN</i>	Giới hạn bền <i>Tensile strength, MPa</i>	Vị trí hỏng <i>Broken position</i>	Đánh giá theo/ <i>Assess to ASME, section IX</i>
TS1	19,1 x 7,76	148,2	85,8	579	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>
TS2	18,9 x 7,84	148,2	86,2	582	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>

- Thử uốn/ *Guided bend test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu/ <i>Specimen N°</i>	Kiểu uốn/ <i>Type of bend</i>	Kết quả/ <i>Result</i>	Ghi chú/ <i>Remarks</i>
FB1	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB1	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	
FB2	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB2	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	

- Thử va đập/ *Impact test at -45°C*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference N°*: KT3-01655BCK4/3)

Vị trí rãnh và đập <i>Notch location</i>	Công va đập <i>Absorbed energy (J)</i>			Trung bình <i>Average (J)</i>	Mức chất lượng <i>Criteria(*) (J)</i>	Đánh giá <i>Assessment</i>
	1	2	3			
Kim loại mối hàn/ <i>Weld metal</i>	60	44	48	51	Min. 27	Đạt/ <i>Pass</i>
Vùng ảnh hưởng nhiệt/ <i>Heat affected zone</i>	142	114	120	125		Đạt/ <i>Pass</i>

(*) Theo yêu cầu của khách hàng/ *As customer's request*.

Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 04/04

6. Kiểm tra tổ chức thô đại/ *Macro examination*: Đạt yêu cầu / *Acceptable*.

(Kết quả thử nghiệm tham khảo số/ *Test report reference* N°: KT3-01655BCK4/3)

7. Các thử nghiệm khác/ *Other tests*: Không áp dụng/ *Not applicable*.

Chúng tôi, chứng nhận rằng thông tin trong báo cáo này là đúng, mỗi hàn kiểm tra được chuẩn bị, hàn và thử nghiệm phù hợp với các yêu cầu quy định tại chương IX, Quy phạm nồi hơi và thiết bị áp lực ASME (2023)/ *We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code (2023).*

	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	28/05/2024	28/05/2024
SIGN	 DDC QA/QC DEPT	 TRUNG TÂM KỸ THUẬT TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG 3 TỔNG CỤC TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG
NAME	La Vinh Nhon	Nguyễn Anh Triết

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CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu : MỎI HÀN TRÊN MẪU THÉP TẮM.
Name of sample : **WELDS ON STEEL PLATE COUPON.**
- Mô tả mẫu : Mối hàn trên mẫu thép tấm như hình ảnh ở mục 11.
Sample description : **Welds on steel plate coupon as photograph at item 11.**
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu : 02 mẫu/ Samples.
Testing quantity
- Ngày nhận yêu cầu : 15/05/2024.
Date of receiving
- Ngày thử nghiệm : 15/05/2024.
Date of testing
- Khách hàng : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
Customer : **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
Test Location
- Phương pháp thử : ASME BPVC.V-2023 (Article 9).
Test method
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results : **See details on page 02/02.**

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KI, GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

K13-01655B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Điều kiện thử/ Test condition:

☒ Butt weld	☐ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal Thk.: 8 mm
Welding Procedure Spec.: ☐ SMAW	☒ SAW	☒ FCAW	☐ GTAW	
State of Examination : ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation : ☒ As weld	☒ Brushing	☐ Other.....		
Surface Temperature : ☒ (28 + 33) °C	☐ .. °C			
IR Thermometer : FLUKE 561	Serial: 22020241	Next Cal. Due: 06/2024		
Lighting Condition : ☒ Day light > 1000 lux	☐ Artificial > 1000 lux			
Weld Gauge : Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra ngoại quan mối hàn/ Visual examination results:

Tên mẫu Name of test coupon	Chiều dài kiểm tra Tested length, (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	
DD-FCAW+SAW- 252 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	

(*) Chú thích/ Legends:

IF : Không ngấu / Incomplete fusion	C : Nứt / crack	CR : Lõm chân / Concave Root
OL : Chồng mép / Overlap	P : Rỗ / Porosity	U : Cháy cạnh / Undercut
SI : Ngậm xỉ / Slag inclusion	R : Quả cầu / External Reinforcement	CS : Lõm mặt / Concave Surface
WP : Biên dạng / Weld bead profile	RO : Ô xi hóa chân / Root Oxidation	PR : Quả thô / Protrusion

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CK4/2

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
WELDS ON STEEL PLATE COUPON.
- Mô tả mẫu
Sample description : **Mỗi hàn trên mẫu thép tấm như hình ảnh ở mục 11.**
Welds on steel plate coupon as photograph at item 11.
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu
Testing quantity : **02 mẫu/ Samples.**
- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**
- Ngày thử nghiệm
Date of testing : **17/05/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **ASME BPVC.V-2023 (Article 4).**
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.


Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

KT3-01655BC
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Điều kiện thử nghiệm/ Test condition:

<u>Surface condition</u> ☒ As weld ☐ Brushing ☐ Painting ☐ Grinding	<u>Scanned from</u> ☐ One side ☒ Both sides ☒ One surface ☐ Both surfaces	<u>Preparation</u> ☐ Single V ☐ Double V ☒ Backing ☒ Single Bevel	<u>Couplant</u> ☐ Water ☐ Oil ☒ Cellulose paste ☐	<u>Material</u> ☒ Carbon steel ☐ Stainless steel Diam.: mm Thick.: 8 mm
<u>Ultrasonic instruments</u> ☐ USM 35X DAC (S/N: 9502a) ☐ USM 35X DAC (S/N: 12873a) ☐ USM 35X DAC (S/N: 12875a) ☐ USM 35X DAC (S/N: 14258a) ☒ USM 35X DAC (S/N: 16141a) ☐ POSITECTOR UTG (S/N: 823482)	<u>Probes:</u> Model C340451 MSEB MWS Angle (°) 0 0 70 Freq. (MHz) - 4 4 Serial - 54339 76970 Ref. (dB) - 48 49	<u>Calibration blocks</u> ☐ V1 - IIW ☒ V2 - IIW ☐ DAC Ø1,5 mm ☐ DAC Ø3,0 mm	<u>Reference block</u> ☒ T20/Ø2,5 ☐ T35/Ø3,0 (ASME T-434.2.1) ☐	
Transfer correction: + 1 dB;			Scanning level: Ref. + 12 dB	

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra siêu âm/ Ultrasonic testing results:

Tên mẫu Name of test coupon	Chiều dày Thickness (mm)	Chiều dài thử Test length (mm)	Chỉ thị / Indications			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
			Biên độ Actual level (Db)	Loại ¹⁾ Type ¹⁾	Chiều dài Length (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	8	700	-	-	-	✓	
DD-FCAW+SAW-252 SA-516 GR.70	8	700	-	-	-	✓	
¹⁾ Chú thích/ Legends: RI : Sau khi sửa/ After repairing. LOF : Không ngậm / Lack of fusion C : Nứt / crack SI : Ngậm xỉ / Slag inclusion LOP : Không thấm / Lack of penetration P : Rỗ / Porosity U : Cháy cạnh / Undercut							

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
WELDS ON STEEL PLATE COUPON.
DD-FCAW-251
- Mô tả mẫu
Sample description : **Mỏi hàn trên thép tấm như hình ảnh bên dưới.**
Weld on steel plate as photograph below.
(700 x 400 x 8) mm.
- Số lượng mẫu
Testing quantity : **01 mẫu/ Sample.**
- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**
- Ngày thử nghiệm
Date of testing : **22/05/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DÙNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **ASME BPVC.IX - 2023.**
ASME BPVC.VIII.1 - 2023.
SA-370 (ASME BPVC.III.A-2023)
- Điều kiện và kết quả thử nghiệm
Test condition & results : **Xem chi tiết ở trang 02/03 và 03/03.**
See details on page 02/03 and 03/03.



TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Kết quả/ *Test results:*

Tên chỉ tiêu Specification		Kết quả thử nghiệm Test result			
A. THỬ KÈO/ TENSILE TEST		Mẫu/Sample 1		Mẫu/Sample 2	
10.1. Kích thước mẫu thử, Dimension of specimen	mm	19,1 x 7,76		18,9 x 7,84	
10.2. Lực kéo đứt, Tensile load	kN	85,8		86,2	
10.3. Giới hạn bền kéo, Tensile strength	MPa	579		582	
10.4. Vị trí đứt Broken position		Kim loại cơ bản Parent metal		Kim loại cơ bản Parent metal	
B. THỬ UỐN/ BEND TEST		FB1	RB1	FB2	RB2
10.5. Kích thước mẫu thử, Dimension of specimen	mm	38,0 x 7,70	38,1 x 7,42	37,9 x 7,70	37,7 x 7,36
10.6. Góc uốn, Bending angle	độ	180	180	180	180
10.7. Đường kính gồ uốn, Plunger diameter	mm	32	32	32	32
10.8. Kết quả thử uốn mặt Face bend test result		Đạt Pass	-	Đạt Pass	-
10.9. Kết quả thử uốn chân Root bend test result		-	Đạt Pass	-	Đạt Pass
Tên chỉ tiêu/ Specification		Kết quả thử nghiệm/ Test result			
C. THỬ VA ĐẬP/ IMPACT TEST AT -45°C		Kim loại mối hàn/ Weld metal			
10.10. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.11. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	60	44	48	51
		Vùng ảnh hưởng nhiệt/ Heat affected zone			
10.12. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.13. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	142	114	120	125

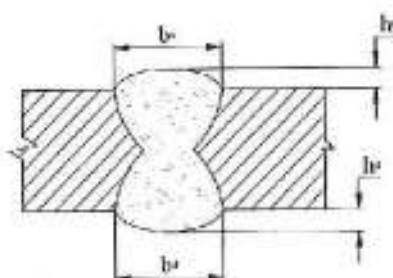
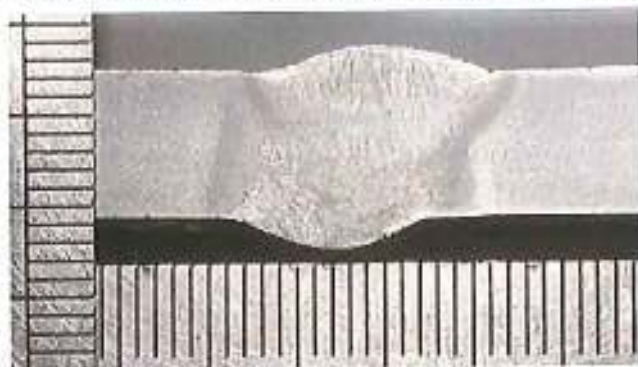


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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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D. TỔ CHỨC THỎ DẠI / MACRO STRUCTURE



Kết quả thử nghiệm/ Test result	
Bề mặt/ Surface	Bên trong/ Internal
$b_0 = 14 \text{ mm}$	Mối hàn thấu và ngấu hoàn toàn. Trong mối hàn không có khuyết tật / Full penetration and complete fusion weld. No defect in weld metal.
$h_u = 1,5 \text{ mm}$ - Đạt/Pass	
$b_d = 10 \text{ mm}$	
$h_d = 1,6 \text{ mm}$ - Đạt/Pass	

QUATEST 3®

- Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the exactly submitted samples only, and this is not a certificate of product.
- Tên mẫu, tên khách hàng được ghi theo yêu cầu của nơi gửi mẫu. / Name of sample(s) and customer are written as customer's request.
- Hệ không chắc chắn do mở rộng được tính từ độ không đảm bảo đo chuẩn hóa với hệ số phủ $k = 2$, phần bù chưa tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated at the standard uncertainty of measurement multiplied by the coverage factor $k = 2$, at 95 % confidence level.
- Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ Thuật 3.
This Test Report shall not be reproduced, except in full, without the written permission by Quatest 3.
- Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ info@quatest3.com.vn để biết thêm thông tin.
Please contact QUATEST 3 at the email addresses info@quatest3.com.vn for further information about test report.

QUATEST 3	
☐ Inspected	☒ Reviewed
Ins. No.: 16853	Sign: <i>[Signature]</i>
Date: 15.5.24	<i>[Signature]</i>

검 사 증 명 서

MILL TEST CERTIFICATE

HYUNDAI STEEL

당진3점소: 호남 공진시 호남읍 북부산업로 1480
1489 Buteokan, P.O. Songok-6up, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : F23D400073

품명
Commodity : HOT ROLLED STEEL PLATE

재질규격
Specification : ASME (2021 EDITION) SA516-70-N HT+LTY

고객사
Customer : TUNG WANG STEEL

주요자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20230905-PS-0013-009

발행일자
Date of issue : 2023-09-05

제품사우 Dimension	수량 Quantity	중량 Weight (kg)	세강번호 Heat No.	제품번호 Product No.	Position	연장시험 Tensile Test						충격시험 Impact Test V Notch -40.0 °C			화합성분 Chemical Composition(%)																단위 Unit 2" x 100 3" x 1,000 4" x 10,000 5" x 100,000						
						YP	IS	EL	YP-EL	RA	YR	YP MPa	Energy	SF	LE	Division	C	S	Mn	P	S	Cr	N	B	Cu	Mo	Nb	Ti	V	Si	Al	CSQ					
																																	2%o.d (%)	%	%	%	%
8.00x2000x8000	1	754	567929	FD10370106	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86				L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407					
8.00x2000x8000	1	754	567929	FD10370107	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						
8.00x2000x8000	1	754	567929	FD10370201	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						
8.00x2000x8000	1	754	567929	FD10370202	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407						
NOTE	• Position - T:Top, M:Middle, B:Bottom • Tensile test - Direction:Transverse, Gauge length:200mm (Rectangular), YP Mtd :2 to 2%, off-set) 5 to 0.5% underload, U (Upper yield point), L (Lower yield point) • Impact test - Direction:Transverse, Spec-size:10mmX5.0mm, SF:Shear fracture, LE:Lateral Expansion • Division - P:Product analysis, L:Ladle analysis • The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test. • This M/T Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE SR0175/SR0170 S for hardness test ASME SA516-70 2021 Edition • The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied • This M/T Test Certificate is issued in accordance with EN 10204 Type 3.1																																				
	상기의 제품은 질소의 합과 저철의 규격에 합여한것을 증명합니다. WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS																																				
	Signature																																				
	Kim in																																				
	Chief Of Plate Quality Team																																				



NOTE

- Position - T: Top, M: Middle, B: Bottom
- Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mtd: 1.2 to 2% off-set, 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
- Impact test - Direction: Transverse, Spec-size: 10mmX5,0mm, SF: 3 shear fracture, LE: Lateral Expansion
- Division - P: Product analysis, L: Lab analysis
- The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.
- This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE MR0175/MR010-3 for hardness test ASME SA516-70 2021 Edition
- The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

입고의 제품물 검사에 따라 시험한 규격에 합격하였을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Chief Of Plate Quality Team

[Signature]

- * 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용을 원본에 열거된 정보로부터 확인할 수 있습니다. (<http://www.hyundai-steel.com/qa/en/login.jsp>)
- * This Mill Test Certificate is a copy that has been printed from original electronic document with digital signature.
- * You are able to check an original electronic document at Hyundai's customer portal: <http://www.hyundai-steel.com/qa/en/login.jsp> * QRcode scanner App : "QRcode"
- * [사용 및 주의사항] 본 검사증명서와 검사된 금속 재 사용시 제품품질 및 안전의 문제가 발생할 수 있습니다.
- * [Caution for Use] Damage on products and safety problems may arise if used other than to the spot stated in this Mill Test Certification.

CERTIFIED MATERIAL REPORT



CMR No. : HDV - 2024 - 0031

Date of Issue : 2024 - 01 - 05

Customer	-		
Trade Name	Supercored 71H	Size	1.2 mm x 15 kg
Lot No.	MF24340A71	Purchaser's Order No.	-

Specification: EN ISO 17832-A - T42 4 P C1 1 * According to EN 10204-2004 : 3.1
AWS A5.36 E71T1-C1A4-CS1

Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment ("C×hr.)	As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. ("C)	X ₁	X ₂	X ₃	Avg.
	554	619	26	-40	102	95	96	96
EN Spec.	≥420	500~640	≥20	≥47				

Chemical Composition (%)

	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
Weld Metal	0.04	0.40	1.37	0.017	0.005	0.38	0.02	0.01	0.02	0.01	0.01
EN Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05

* Hydrogen Content (nℓ/100g) : 5.76

Others
CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-9J-H8)

QUATEST 3	
☒ Inspected	☒ Reviewed
Ins. No.: 16558	Sign: <i>[Signature]</i>
Date: 15.5.24	Le Tin Han

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

[Signature]

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
ASME VIII, IX:2023

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW-254

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Jul 29, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 1/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-GTAW-254** Revision No. **0** Date **Jul 29, 2024**

Supporting PQR No.(s): **PQR-DD-GTAW-254** Date **Jul 29, 2024**

Welding Process(es) **GTAW** Type(s)(Manual, Automatic, Semi-Automatic): **Manual**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**

Root Spacing **As per fabrication drawing**

Backing **With or without**

Retainer **None**

Backing Material **Base metal or weld metal**

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**

OR

Specification and type/grade or UNS Number **None**

to Specification and type/grade or UNS Number **None**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **3 - 12 (mm)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other

FILLER METALS (QW-404)

Process **GTAW**

Spec. No. (SFA) **SFA-5.18**

AWS No. (Class) **ER70S-6 with Mn $\leq 1.6\%$ and Si $\leq 1.0\%$**

F-No. **6**

A-No. **1**

Size of Filler Metals **$\varnothing 1.6 - 2.4$ (mm)**

Supplemental Filler Metal **None**

Filler Metal Product Form **Solid rod**

Filler Metal Brand Name **Hyundai ST-50.6 or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 12 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 2/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-GTAW-254** Revision No. **0** Date **Jul 29, 2024**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**
Interpass Temperature, Maximum **255 °C**
Preheat Maintenance **None**
Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Ar	99.99%	12 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: Minimum purity 99.99%Ar for GTAW process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	GTAW	ER70S-6	1.6 - 2.4	DC EN	93 - 105	None	0.9 - 2.0	13 - 17	51 - 83	None
Hot	GTAW	ER70S-6	1.6 - 2.4	DC EN	115 - 130	None	1.1 - 1.5	14 - 18	91 - 94	None
Fill	GTAW	ER70S-6	1.6 - 2.4	DC EN	110 - 150	None	1.1 - 1.7	15 - 18	79 - 107	None
Cap	GTAW	ER70S-6	1.6 - 2.4	DC EN	135 - 155	None	1.4 - 2.0	15 - 19	82 - 91	None

Pulsing Current **None** Heat Input (max.) **None**

Tungsten Electrode Size and Type **Ø2.4 mm, 2% Thoriated Tungsten (EWTh-2)**

Mode of Metal Transfer for GMAW (FCAW) **None**

Other **None**

TECHNIQUE (QW-410)

Process **GTAW**
String or Weave Bead **Weave Bead**
Orifice, Nozzle, or Gas Cup Size **All size**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding or Air carbon arc**
Oscillation **None**
Contact Tube to Work Distance **None**
Multiple or Single Pass (Per Side) **Multiple pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:
La Vinh Nhon

Approved by: **Truong Hoai Phong Vu (QA Manager)**



Date: **Jul 29, 2024**

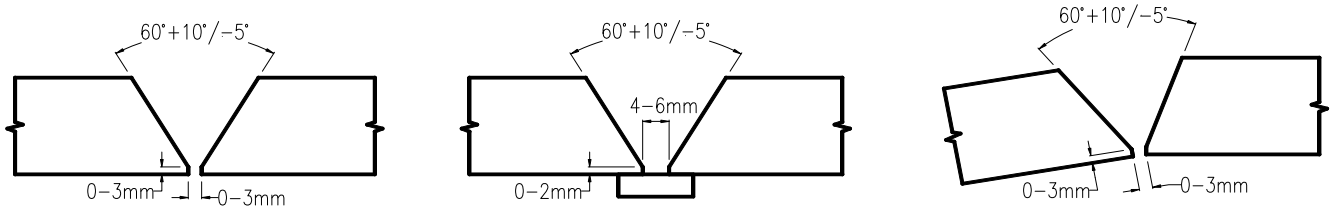
Date: **Jul 29, 2024**

Welding Procedure Specification No.: **WPS-DD-GTAW-254**

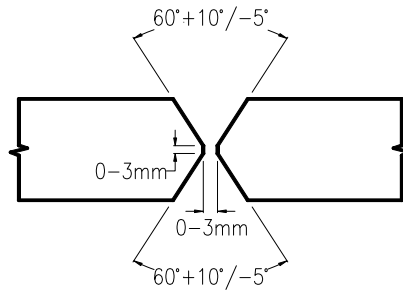
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

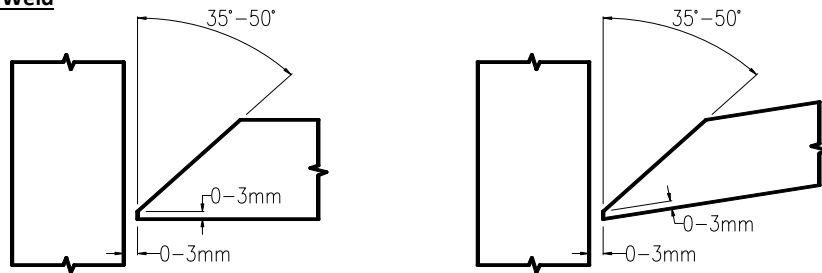
1. Single V Groove Butt Joint



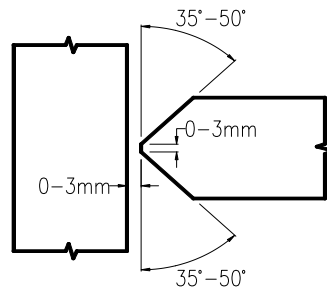
2. Double V Groove Butt Joint



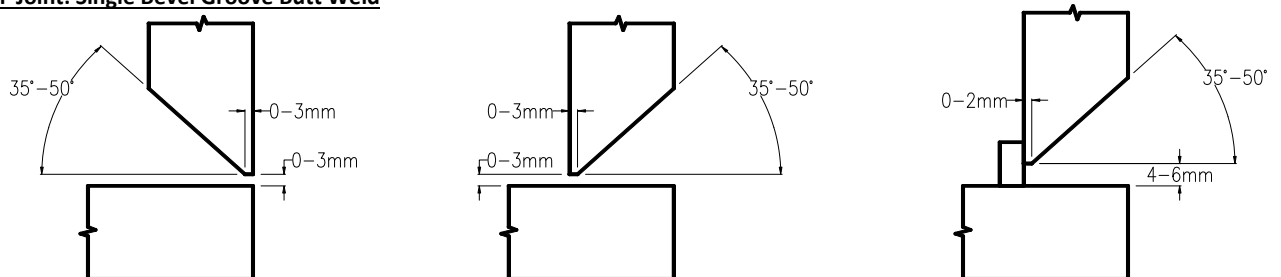
3. T-Joint: Single Bevel Groove Butt Weld



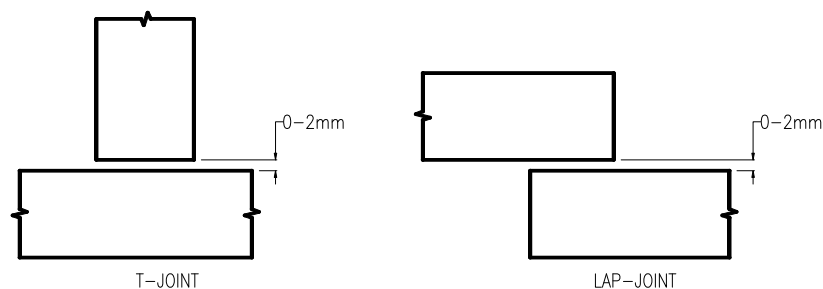
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld

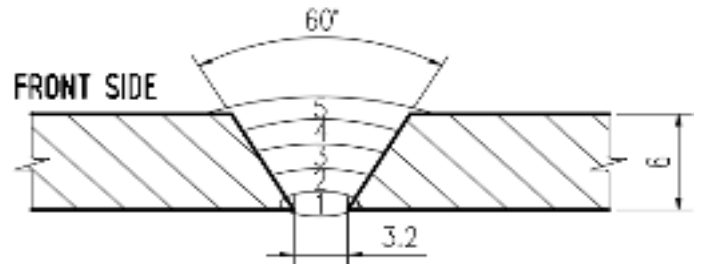


Signature



PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2023)

Page: 1/2

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Procedure Qualification Record No. **PQR-DD-GTAW-254** Date: **Jul 29, 2024**Welding Process(es) **GTAW**Type(s)(Manual, Automatic, Semi-Automatic) **Manual****JOINTS (QW-402)**Joint Type **Butt joint, single V groove**Backing **None**Retainer **None****BASE METALS (QW-403)**Material Spec. **ASME SA-53M**Type/Grade, or UNS Number **Gr.B**P-No. **1** Group No. **1** to P-No. **1** Group No. **1**Thickness of Test Coupon **6 mm**Diameter of Test Coupon **114.3**Maximum Pass Thickness **< 6 mm**Other **None****POSTWELD HEAT TREATMENT (QW-407)**Temperature **None**Time **None**Other **None****GAS (QW-408)**

	Percent Composition		Flow Rate
	Gas(es)	(Mixture)	
Shielding	Ar	99.99%	12 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)Process **GTAW**SFA Specification **ASME SFA-5.18**AWS Classification **ER70S-6**Filler Metal F-No. **6**Weld Metal Analysis A-No. **1**Size of Filler Metal **Ø2.4 mm**Filler Metal Product Form **Solid rod**Supplemental Filler Metal **None**Filler Metal Trade Name **Hyundai ST-50.6**Electrode Flux Classification **-**Flux Type **-**Flux Trade Name **-**Weld Metal Thickness **6 mm**Other **None****ELECTRICAL CHARACTERISTICS (QW-409)**Process **GTAW**Current and Polarity **DC EN**Amps. **See table below**Volts **See table below**

Mode of Metal Transfer

for GMAW (FCAW) **-**Other **-****TECHNIQUE (QW-410)**Process **GTAW**String or weave Bead **Weave**Oscillation **None**Method cleaning **Grinding, brushing**Method back gouge **None**Multipass or single pass (per side) **Multipass**Multiple or single Electrodes **Single**Peening **None**Use of thermal processes **None**Other **None****POSITION (QW-405)**Process **GTAW**Position(s) **1G**Weld Progression (Uphill, Downhill) **-**Other **None****PREHEAT (QW-406)**Preheat Temperature **34 °C (Ambient)**Interpass Temperature **Max. 200 °C**Other **None****WELDING PARAMETER**

Weld Layer	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
Root	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	93 - 105	13 - 17	51 - 83	34 (Ambient)	1.2	12	0.9 - 2.0
Hot	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	115 - 130	14 - 18	91 - 94	110	1.2	12	1.1 - 1.5
Fill	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	110 - 150	15 - 18	79 - 107	180	2.8	12	1.1 - 1.7
Cap	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	135 - 155	15 - 19	82 - 91	200	1.8	12	1.4 - 2.0



PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2023)

Page: 2/2

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Procedure Qualification Record No. **PQR-DD-GTAW-254** Date: **Jul 29, 2024**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
TS1	19.15	5.24	100.3	50.1	500	Ductile and in base metal
TS2	19.15	5.22	99.96	49	490	Ductile and in base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
FB1 (Face bend)	Pass
RB1 (Root bend)	Pass
FB2 (Face bend)	Pass
RB2 (Root bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.01 x 3.99	-52.3 °C	43 (J)			
WM 2	Weld metal	10.03 x 4.00	-52.3 °C	48 (J)			
WM 3	Weld metal	10.01 x 4.00	-52.3 °C	54 (J)			
HAZ 1	HAZ (A572)	10.01 x 4.00	-52.3 °C	25 (J)			
HAZ 2	HAZ (A572)	10.01 x 3.99	-52.3 °C	24 (J)			
HAZ 3	HAZ (A572)	10.00 x 4.00	-52.3 °C	21 (J)			

Comments **Acceptable**

Fillet-Weld Test (QW-180)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: **Acceptable**

Other Tests (Not Apply)

Type of Test _____
Deposit Analysis _____
Other _____Welder's Name **Nguyen Van Dem** Clock No. **None** Stamp (ID) No. **DDC-WD-0214**
Welder's Name **-** Clock No. **-** Stamp (ID) No. **-**
Tests Conducted by **SGS** Laboratory Test No. **INL24/512130-525**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX - 2023 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Certified by Welding Coordinator:
La Vinh Nhon

Approved by: Truong Hoai Phong Vu (QA Manager)

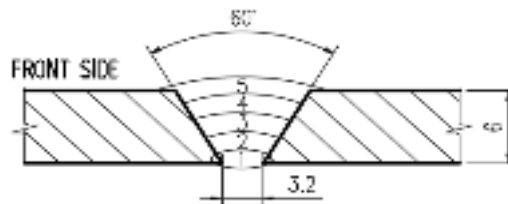
Date: **Jul 29, 2024**Date: **Jul 29, 2024**

WELDING PARAMETERS RECORD (Doc No.: WPR-GTAW-254)

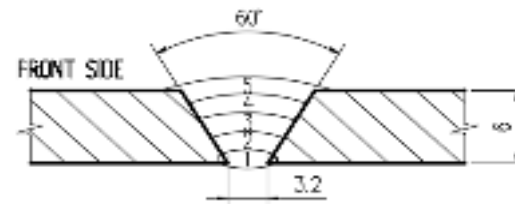
Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GTAW-254
 Test Coupon: DD-GTAW-254.1 and DD-GTAW-254.2
 Test position: 1G and 6G
 Base metal:
 - Specification: ASME SA-53M Gr.B
 - Dimension: O.D 114.3 x W.T 6.02 x L180
 - Heat/Product No.: J1LC900
 - Certificate No.: BYYF2497
 Filler metal:
 - Specification: AWS A5.18
 - Classification: ER70S-6 (Ø2.4)
 - Trade name: Hyundai ST-50.6
 - Batch/Lot No.: 23696310
 - Certificates No.: 2310303300003
 Gas: Argon (99.99%)
 Welder Name: Nguyen Van Dem I.D: DDC-WD-0214

Joint Sketch:

Date: Jul 16, 2024



For Test Position 1G



For Test Position 6G

Root face: 0 mm
 Root gap: 3.2 mm
 Groove angle: 60°



Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
DD-GTAW-254.1 (1G)																
1	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	93-103	14-17	1	58	160	81	34 (Ambient)	12	1-1.3	5	1.2	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	93-103	14-17	1	56	160	83	50	12	0.9-1.3	5		
2	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	120-130	15-18	1	43	160	93	90	12	1.2-1.5	7	1.3	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	120-130	15-18	1	42	160	94	110	12	1.1-1.5	7		
3	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	130-140	15-18	1	37	160	99	120	12	1.2-1.5	8	1	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	130-140	15-18	1	40	160	96	130	12	1.2-1.6	8		
4	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	140-150	15-18	1	42	160	94	150	12	1.3-1.7	10	1.8	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	140-150	15-18	1	40	160	96	180	12	1.3-1.7	10		

WELDING PARAMETERS RECORD (Doc No.: WPR-GTAW-254)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
5	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	145-155	15-18	1	48	160	89	190	12	1.5-1.9	10	1.7	Reinforcement: 1 mm
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	145-155	15-18	1	46	160	91	200	12	1.4-1.8	10		
DD-GTAW-254.2 (6G)																
1	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	93-103	12-15	3	10	160	51	34 (Ambient)	12	1.3-1.8	5	1.2	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	95-105	13-16	3	7	160	51	58	12	1.5-2	5		
2	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	115-125	14-17	1	45	160	92	90	12	1.1-1.4	6.5	1.1	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	115-125	14-17	1	46	160	91	110	12	1.1-1.4	6.5		
3	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	125-135	15-18	1	50	160	87	120	12	1.3-1.7	7.5	1	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	125-135	15-18	1	30	160	107	130	12	1.1-1.4	7.5		
4	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	130-110	15-18	2	2	160	79	150	12	1.5-1.5	9	1.7	
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	130-140	15-18	1	40	160	96	170	12	1.2-1.6	9		
5	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	135-145	16-19	1	57	160	82	190	12	1.6-2	11	2	Reinforcement: 1 mm
	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	135-145	16-19	1	54	160	84	200	12	1.5-2	11		
		Conducted by Welding Coordinator														
Name		La Vinh Nhon														
Signature																
Date		Jul 16, 2024														

VILAS 237

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MÔI HÀN
DD-GTAW-254.1 & DD-GTAW-254.2
Joint type: Single V Butt Joint
Welding process: GTAW-1G & 6G
Base metal: ASME SA-53M Gr. B
Dimension: OD 114.3mm x T 6.02mm x L 360mm
Filler metal: AWS A5.18 ER70S-6
Trade name: Hyundai ST-50.6 (Ø2.4mm)
Welder: Nguyen Van Dem DDC-WD-0214

Project / Dự án : Evapco – Aspen Power Station Project
DaiDung Metallic Manufacture Construction & Trade Corporation

Item / Hàng mục : Procedure Qualification Record (PQR) Test for Ducting

Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)

Date of receiving / Ngày nhận mẫu : 19-Jul-2024

Date of testing / Ngày thử nghiệm : From / từ 19-Jul-2024 to / đến 23-Jul-2024

Test method / Phương pháp thử : ASME BPVC.IX-2023 Welding, Brazing, and Fusing Qualifications
ASME BPVC.VIII.1-2023 Rules for Construction of Pressure Vessels
(SA-370 (ASME BPVC.II.A-2023 Materials))

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of

SGS Vietnam LTD

Thay mặt công ty

SGS

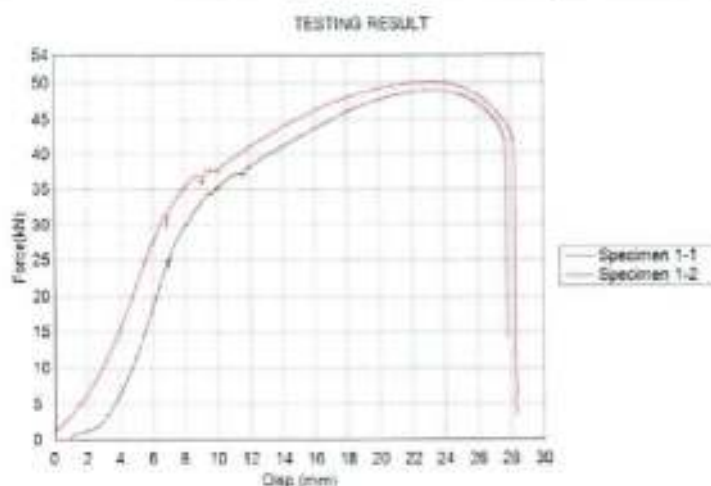
SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test ⁽¹⁾ / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	5.24	5.22
• Width of specimen / Chiều rộng mẫu mm	19.15	19.15
• Cross section area / Tiết diện mm ²	100.3	99.96
• Tensile load / Lực kéo đứt kN	50.1	49.0
• Tensile strength / Giới hạn bền kéo MPa	500	490
• Broken position / Vị trí mẫu đứt	In base metal	In base metal



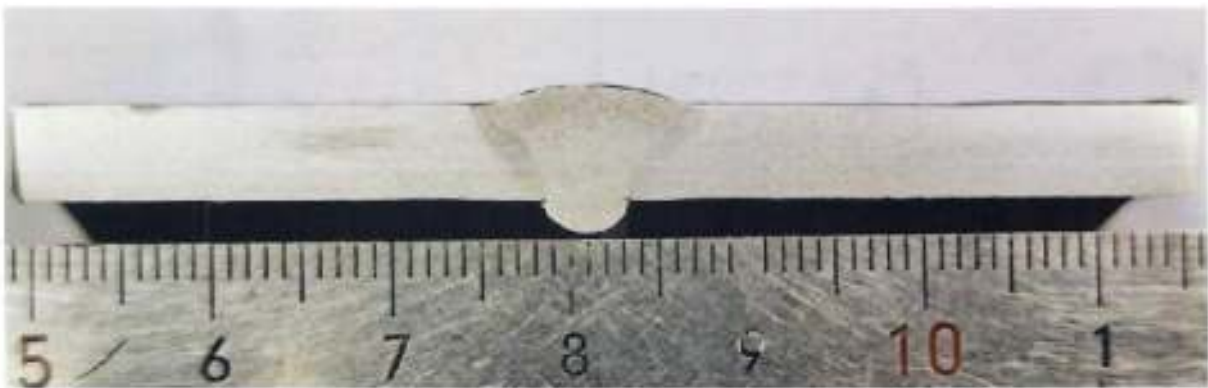
The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test ⁽¹⁾ / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	5.24	5.24	5.22	5.22
• Width of specimen / Chiều rộng mm	38.0	38.0	38.0	38.0
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	21	21	21	21
• Test result / Kết quả	No Crack/ Không nứt	Crack/ Nứt: 0.5+0.6mm	No Crack/ Không nứt	No Crack/ Không nứt

*** TO BE CONTINUED ***

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
C. Impact test at -52.3°C / Thử va đập ở -52,3°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens / Kích thước mẫu thử	mm	10.01 x 3.99	10.03 x 4.00	10.01 x 4.00
• Impact energy / Công va đập	J	43	48	54
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens / Kích thước mẫu thử	mm	10.01 x 4.00	10.01 x 3.99	10.00 x 4.00
• Impact energy / Công va đập	J	25	24	21
D. Characteristics / Tên chỉ tiêu: Macro structure (*) / Tổ chức thô đại				
• Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn. • No defect in weld metal / Trong mối hàn không có khuyết tật.				
				

*** TO BE CONTINUED ***

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Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

Note / Ghi chú :

(*) The test was accredited ISO 17025:2017 by VILAS / Chỉ tiêu được công nhận ISO 17025:2017 bởi VILAS.

ILAC: International Laboratory Accreditation / Tổ chức hợp tác công nhận Phòng thử nghiệm quốc tế.

MRA: Mutual Recognition Arrangement / Thỏa ước thừa nhận lẫn nhau.

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*** END OF THE REPORT ***

Please extract your time to give us your feedback as this will help our service better. Thank you!



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NIPPON STEEL

NIPPON STEEL CORPORATION
KANSAI WORKS WAKAYAMA AREA (KATNAN)
260-100, FUNDO, KATNAN, JAPAN

INSPECTION CERTIFICATE

CERTIFICATE NO. :BYYP2497

PAGE : 1/2 DATE :2022-02-03

CUSTOMER :NIPPON STEEL PTE. LTD.
ORDER NO. :P/O NO.17000182
SHIPPER :NIPPON STEEL TRADING CORPORATION 111 5246-AMS1665D 2-842-N3-5-3-V442-08
COMMODITY :SEAMLESS HOT FINISHED CARBON STEEL LINE PIPE
STANDARD :API 5L GR.L290 PSL 1(46TH ED., APR. 2018)
API 5L GR.L245 PSL 1(46TH ED., APR. 2018)
ASTM A106M-19A / ASME 2019 EDITION SA-106M GR.B
ASTM A53M-20 / ASME 2019 EDITION SA-53M GR.B
SPECIFICATION :EL7-L-0198RM
MILL WORK NO. :BYYP2497 O.D.:114.3mm W.T.:6.02mm LENGTH:6000mm QUANTITY:367pcs. MASS:35379kg

HEAT NO. :J1LC900
PRODUCTS PCS. :367

HEAT TREATMENT :AS ROLLED

CHEMICAL COMPOSITION(%)

		*1	C	Si	Mn	P	S	Cu	Cr	Ni	Mo	Ti	V	Nb	B	*D4	*VN	*V1	*CEQ
						*2	*2					*2		*2	*2				
SPEC. MIN.		R	-	10	29	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MAX.		R	23	-	106	30	15	40	40	40	15	-	8	-	1	100	8	16	42
HEAT NO.																			
J1LC900		L	19	29	95	14	3	2	4	2	1	10	0	10	0	9	1	2	36
		P	18	30	94	12	2	2	4	2	1	10	0	10	0	9	1	2	35
		P	18	30	94	13	2	2	4	2	1	10	0	10	0	9	1	2	35
*1 R:LADLE & PRODUCT ANALYSIS L:LADLE ANALYSIS P:PRODUCT ANALYSIS *2:X1000 OTHER:X100 *D4:Cu+Ni+Cr+Mo+V *VN:Nb+V *V1:Nb+V+Ti *CEQ:C+Mn/6+(Cr+Mo+V)/5+(Cu+Ni)/15																			

TENSILE TEST

		*1	*2	YS	TS	EL		
				*3	*3	%		
SPEC. MIN.		L	B	M	290	M	415	24
MAX.		L	B	M	-	M	620	-
HEAT NO.								
J1LC900		L	B	M	411	M	536	36
TYPE OF SPECIMEN:STRIP 25.4mm WIDTH *1 SAMPLING DIRECTION L:LONGITUDINAL *2 SAMPLING POSITION B:BASE METAL *3 UNIT M:MPa GAUGE LENGTH:50.0mm KIND OF YS:0.5% EXTENSION UNDER LOAD								

HARDNESS TEST(HV 10)

		*1	*2	*3
SPEC. MIN.		B C	-	-
MAX.		B C	248	-
HEAT NO.			(1)	(2) (3)
J1LC900		B C	148	149 150
*1 SAMPLING POSITION B:BASE METAL *2 LOCATION C:CROSS SECTION *3 EACH (1)~(3): IMPRESSION NO.				

FLATTENING TEST: ACCEPTABLE

WE HEREBY CERTIFY THAT THE MATERIAL DESCRIBED HEREIN HAS BEEN MADE IN ACCORDANCE WITH THE RULES OF THE CONTRACT.

TAKAFUMI SATSUKI

HEAD OF DEPARTMENT
QUALITY ASSURANCE DEPARTMENT

(Notice)
Modification to or unauthorized use of this certificate is strictly prohibited. If you have any questions on the authenticity of this certificate, you can contact us by E-mail, pipe-ipp@jp.nipponsteel.com. NO.314347322



This certificate is a printed copy of the original electric document as of 2022-03-10



CERTIFIED MATERIAL REPORT

CMR No. : 2310303300003

Date of Issue : 2023-10-30

Customer	GLOBAL TECHNOLOGY COMMERCIAL CORPORA		
Trade Name	ST-50.6	Size	2.4mm*1000mm*5kg
Lot No.	23696310	Purchaser's Order No.	3823_HM_GT
Specification	AWS A5.18 ER70S-6 EN ISO 636-A - W 3Si1		

Welding Current	OCEN	Shielding Gas	Ar	Post Weld Heat Treatment (°C × hr.)	AS WELDED
-----------------	------	---------------	----	-------------------------------------	-----------

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS(MPa)	TS(MPa)	EL(%)	Temp.(°C)	X1	X2	X3	Avg.
	460	570	28.2	-30	110	120	120	117
AWS Spec.	≥400	≥490	≥22	≥27				

Chemical Composition (%)

* According to EN 10204-2004 : 3.1

WIRE	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V	
	0.08	0.86	1.47	0.012	0.019	0.14	0.01	0.04	0.01	0.01	
AWS Spec.	0.05 ~0.15	0.80 ~1.15	1.40 ~1.85	≤0.025	≤0.035	≤0.50	≤0.15	≤0.15	≤0.15	≤0.03	

*Radiographic Test : ACCEPT

Others

* Q'ty : 1,100kg

ORIGINAL

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

K.W. Park

QA Manager Park K. W.



HYUNDAI
WELDING

HYUNDAI WELDING CO., LTD.

Head Office

507,Teheran-ro,Gangnam-gu,Seoul,Republic of Korea

Pohang Plant

100,Daesong-ro,Nam-gu,Pohang,Republic of Korea



Comply:
ISO 9001:2015
ASME VIII, IX:2023

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-134

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Aug 5, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 1/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-FCAW-134** Revision No. **0** Date **Aug 5, 2024**

Supporting PQR No.(s): **PQR-DD-FCAW-134** Date **Sep 27, 2021**

Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Semi-Automatic, Mechanized** Date **Aug 5, 2024**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**

Root Spacing **As per fabrication drawing**

Backing **With backing or back gouging**

Retainer **None**

Backing Material **Base metal or weld metal or Nonmetallic**

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**

OR

Specification and type/grade or UNS Number **None**

to Specification and type/grade or UNS Number **None**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **8 - 16 (mm)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other

FILLER METALS (QW-404)

Process **FCAW**

Spec. No. (SFA) **SFA/A-5.20 or 5.36**

AWS No. (Class) **E71T-1C/-9C-J or E71T1-C1A4-CS1**

F-No. **6**

A-No. **1**

Size of Filler Metals **Ø1.2 - Ø1.6 (mm)**

Supplemental Filler Metal **None**

Filler Metal Product Form **Flux cored wire**

Filler Metal Brand Name **Hyundai Supercored 71H or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 16 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 2/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-134** Revision No. **0** Date **Aug 5, 2024**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**
Interpass Temperature, Maximum **197 °C**
Preheat Maintenance **None**
Other **None**

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	99.8%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: Minimum purity 99.8% for FCAW process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	135 - 165	None	0.7 - 0.9	24 - 30	264 - 337	None
Fill	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	153 - 198	None	0.9 - 1.1	23 - 30	227 - 343	None
Cap	FCAW	E71T-1C/-9C-J or E71T1-C1A4-CS1	1.2 - 1.6	DC EP	153 - 187	None	1.2 - 1.4	23 - 30	181 - 232	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **None**
Mode of Metal Transfer for GMAW (FCAW) **Globular, Spray or Pulsed**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or Weave Bead **Weave bead (Max. weld bead width: 15 mm)**
Orifice, Nozzle, or Gas Cup Size **No. 6**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding or Air carbon arc**
Oscillation **None**
Contact Tube to Work Distance **15 - 25 (mm)**
Multiple or Single Pass (Per Side) **Multiple**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:
La Vinh Nhon

Date: **Aug 5, 2024**

Approved by: **Truong Hoai Phong Vu (QA Manager)**



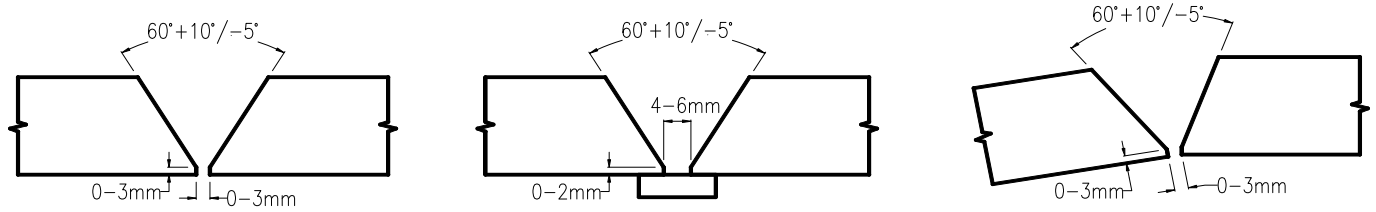
Date: **Aug 5, 2024**

Welding Procedure Specification No.: **WPS-DD-FCAW-134**

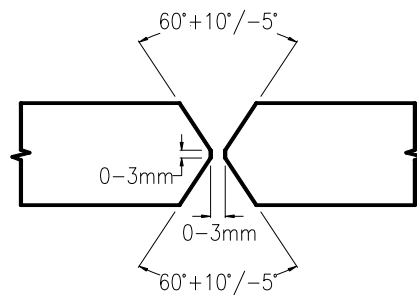
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

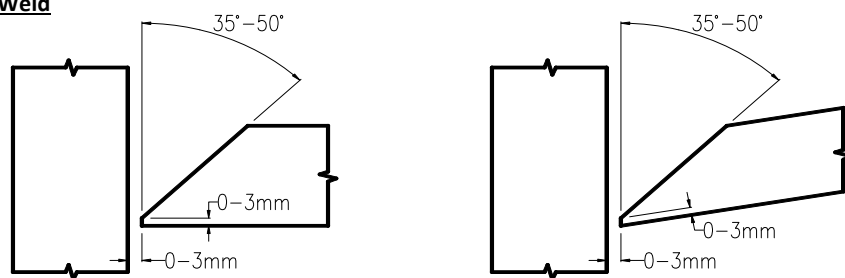
1. Single V Groove Butt Joint



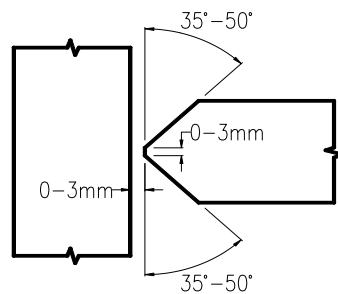
2. Double V Groove Butt Joint



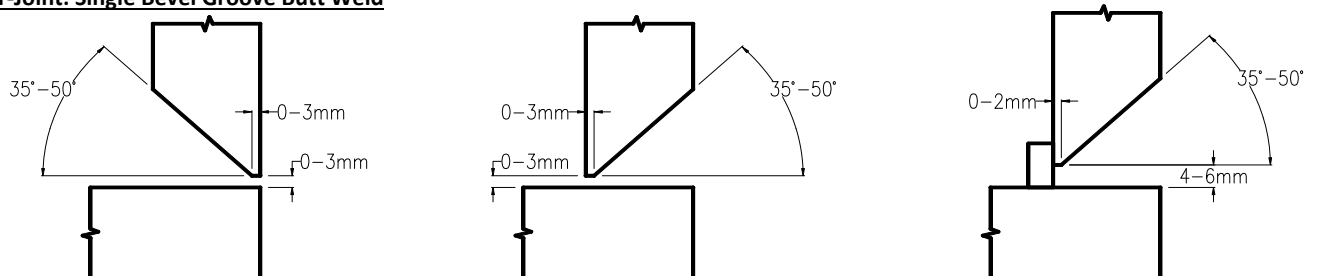
3. T-Joint: Single Bevel Groove Butt Weld



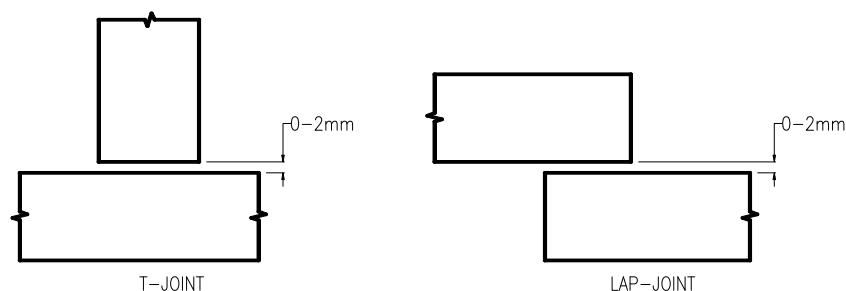
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld



Signature



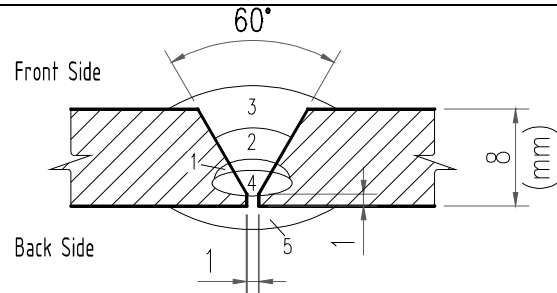
PROCEDURE QUALIFICATION RECORD (PQR) (ASME VIII, IX - Boiler and Pressure Vessel Code-2019)

Page: 1/2

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
Procedure Qualification Record No. **PQR-DD-FCAW-134** Date: **Sep 27, 2021**
Welding Process(es) **FCAW**
Type(s)(Manual, Automatic, Semi-Automatic) **Semi-Automatic**

JOINTS (QW-402)

Joint Type **Butt joint, single V groove**
Backing **None**
Retainer **None**

**BASE METALS (QW-403)**

Material Spec. **ASTM A572**
Type/Grade, or UNS Number **Gr.50**
P-No. **1** Group No. **1** to P-No. **1** Group No. **1**
Thickness of Test Coupon **8**
Diameter of Test Coupon **Plate**
Maximum Pass Thickness **< 13 mm**
Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**
Time **None**
Other **None**

GAS (QW-408)

	Percent Composition		Flow Rate
	Gas(es)	(Mixture)	
Shielding	CO₂	99,99%	20 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process **FCAW**
SFA Specification **AWS A5.20 / 5.36**
AWS Classification **E71T-9C-J / E71T1-C1A4-CS1**
Filler Metal F-No. **6**
Weld Metal Analysis A-No. **1**
Size of Filler Metal **1.2 mm**
Filler Metal Product Form **Flux cored wire**
Supplemental Filler Metal **None**
Filler Metal Trade Name **SUPERCORED 71H**
Electrode Flux Classification **-**
Flux Type **-**
Flux Trade Name **-**
Weld Metal Thickness **8 mm**
Other **None**

ELECTRICAL CHARACTERISTICS (QW-409)

Process **FCAW**
Current and Polarity **DC EP**
Amps. **See table below**
Volts **See table below**
Mode of Metal Transfer for GMAW (FCAW) **None**
Other **None**

POSITION (QW-405)

Process **FCAW**
Position(s) **1G**
Weld Progression (Uphill, Downhill) **None**
Other **None**

PREHEAT (QW-406)

Preheat Temperature **30 °C (Ambient)**
Interpass Temperature **Max. 142 °C**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or weave Bead **Weave**
Oscillation **None**
Method cleaning **Grinding, brushing**
Method back gouge **Air carbon arc, grinding**
Multipass or single pass (per side) **Multipass**
Multipass or single Electrodes **Single**
Peening **None**
Use of thermal processes **None**
Other **None**

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	150	27	293	30 (Ambient)	4	20	0.8
2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	27	252	105	3	20	1.1
3	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	27	202	137	3	20	1.4
4	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	180	26	298	116	2	20	0.9
5	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	26	201	142	4	20	1.3

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW-134** Date: **Sep 27, 2021**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	19.0	8.18	155.4	86.4	556	Ductile and base metal
Sample 2	19.0	8.16	155.0	86.4	557	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1 (face bend)	Pass
Specimen 2 (face bend)	Pass
Specimen 3 (root bend)	Pass
Specimen 4 (root bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.0 x 6.69	-40 °C	114.8 (J)			N
WM 2	Weld metal	10.0 x 6.71	-40 °C	85.3 (J)			N
WM 3	Weld metal	10.0 x 6.70	-40 °C	93.2 (J)			N
HAZ 1	HAZ	10.0 x 6.70	-40 °C	152.1 (J)			N
HAZ 2	HAZ	10.0 x 6.71	-40 °C	149.1(J)			N
HAZ 3	HAZ	10.0 x 6.71	-40 °C	101.0 (J)			N

Comments **Acceptable**

Fillet-Weld Test (QW-180)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: **Acceptable**

Other Tests (Not Apply)

Type of Test _____

Deposit Analysis _____

Other _____

Welder's Name **Le Minh Loi** Clock No. **None** Stamp (ID) No. **DDC-WD-0340**

Welder's Name **-** Clock No. **-** Stamp (ID) No. **-**

Tests Conducted by **SGS** Laboratory Test No. **INL21-506868-477**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX:2019 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon

Date: **Sep 27, 2021**

Date: **Sep 27, 2021**




PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2023)

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Additional

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW-134**

Date: **Aug 5, 2024**

Additional Toughness Tests (QW-170)

Test Report No.: **INL24/512130-541-1**

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.01 x 6.70	-45 °C	95 (J)			
WM 2	Weld metal	10.01 x 6.71	-45 °C	82 (J)			
WM 3	Weld metal	10.00 x 6.70	-45 °C	95 (J)			
HAZ 1	HAZ	10.01 x 6.71	-45 °C	28 (J)			
HAZ 2	HAZ	10.01 x 6.71	-45 °C	34 (J)			
HAZ 3	HAZ	10.01 x 6.71	-45 °C	39 (J)			

Test Report No.: **INL24/512130-541-2**

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.01 x 10.00	-45 °C	88 (J)			
WM 2	Weld metal	10.01 x 10.01	-45 °C	69 (J)			
WM 3	Weld metal	10.01 x 10.01	-45 °C	112 (J)			
HAZ 1	HAZ	10.01 x 10.01	-45 °C	99 (J)			
HAZ 2	HAZ	10.00 x 10.01	-45 °C	66 (J)			
HAZ 3	HAZ	10.01 x 10.01	-45 °C	55 (J)			

Comments **Acceptable**

Welder's Name **Le Minh Loi**

Clock No. **None**

Stamp (ID) No. **DDC-WD-0340**

Welder's Name **-**

Clock No. **-**

Stamp (ID) No. **-**

Tests Conducted by **SGS**

Laboratory Test No. **INL24/512130-541-1; INL24/512130-541-2**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX:2023 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon



Date: **Aug 5, 2024**

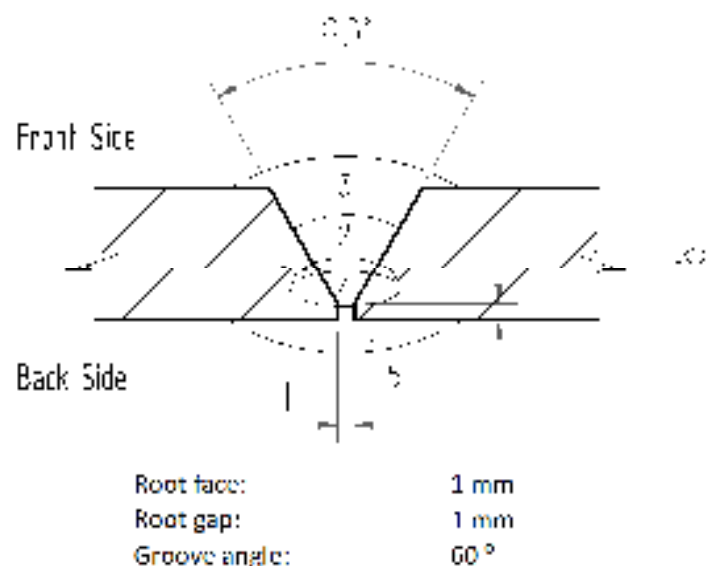
Date: **Aug 5, 2024**

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-134)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR DD FCAW 134
 Test Coupon: DD-FCAW-134
 Test position: 1G
 Base metal:
 - Specification: ASTM A572 Gr.50
 - Dimension: PL8 x 180 x 660 (mm) x 2 pcs
 Heat/Product No.: R58087/CU562620
 Certification No.: 20201002 IIS 001 004
 Filler metal:
 - Specification: A5.20 / A5.36
 Classification: E71T 9C J / E71T1 C1A4 C51
 Trade name: Hyundai Supercore 7111
 Batch/Lot No.: MF21340E71
 Certificates No.: HV 2021 0464
 Gas: CO₂ (100%)
 Welder Name: Le Minh Loi
 I.D.: DDC WD 0340

Joint Sketch:

Date: Sep 22, 2021



Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (l/min.)	Heat Input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Front Side (First Side):																
1	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	150	27	2	15	660	293	30 (Ambient)	20	0.8	8	4	
2	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	170	27	2	37	660	252	105	20	1.1	10	3	
3	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	170	27	3	16	660	202	137	20	1.4	15	3	Reinforcement: 2 mm



WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-134)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (l/min.)	Heat Input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Back Side (Second Side) with back gouging:																
4	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	180	26	2	13	660	298	116	20	0.9	10	2	
5	FCAW	AWS A5.20 / E71T-9C-I	1.2	DC EP	170	26	3	17	660	201	142	20	1.3	15	4	Reinforcement: 2 mm
Conducted by Welding Coordinator																
Name		La Vinh Nhon														
Signature																
Date		Sep 22, 2021														

VILAS 237

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Client's address / Địa chỉ : B23/474C TRẦN ĐẠI NGHĨA, ẤP 2, XÃ TÂN NHỰT, HUYỆN BÌNH CHÁNH, TP.HCM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING STEEL PLATE / MỎI HÀN THÉP TẤM
DD-FCAW-134; FCAW - 1G, Single V
Size / Kích thước : 660 x 360 x 8 (mm)
Material / Vật liệu : ASTM A572 Gr.50
Welder / Mã thợ hàn : DDC-WD-0340

Project / Dự án : Great Plains Power Station (Moose Jaw, Canada)
Evapco Dry Cooling, Inc.

Item / Hạng mục : Procedure Qualification Record (PQR) Test For Duct

Quantity / Số lượng mẫu : 01 sample / mẫu

Date of receiving / Ngày nhận mẫu : 23-Sep-2021

Date of testing / Ngày thử nghiệm : From / từ 23-Sep-2021 to / đến 27-Sep-2021

Test method / Phương pháp thử : ASME BPVC.IX-2019 Welding, Brazing, and Fusing Qualifications
ASME BPVC.VIII.1-2019 Rules for Construction of Pressure Vessels
ASTM A370 - 19 Standard Test Methods and Definitions for Mechanical Testing of Steel Products

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Note :

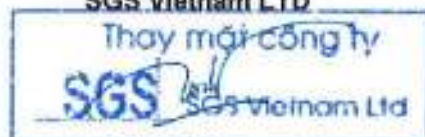
(*) The test was accredited ISO 17025:2017 by VILAS / Chỉ tiêu được công nhận ISO 17025:2017 bởi VILAS.

ILAC : International Laboratory Accreditation / Tổ chức hợp tác công nhận Phòng thử nghiệm quốc tế.

MRA : Mutual Recognition Arrangement / Thỏa ước thừa nhận lẫn nhau.

ILAC-MRA -VILAS : VILAS is the full member and signatory of Mutual Recognition Arrangement with International Laboratory Accreditation / VILAS là thành viên đầy đủ và ký thỏa ước thừa nhận lẫn nhau với tổ chức công nhận phòng thử nghiệm quốc tế.

Signed for and on behalf of
SGS Vietnam LTD

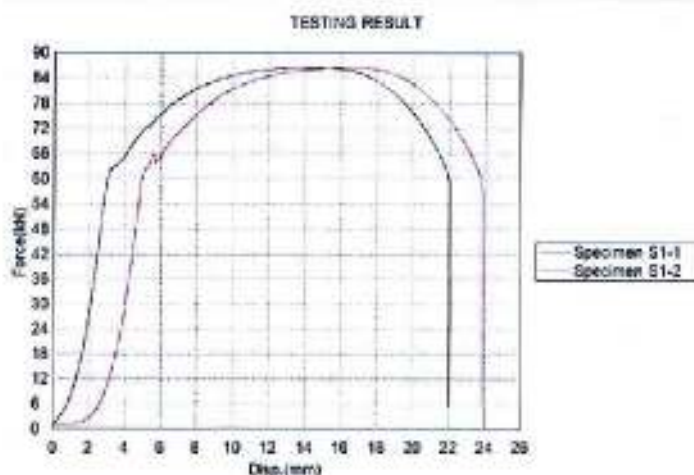


Lê Tiếp Du
Lab Assistant Manager
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test ⁽¹⁾ / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	8.18	8.16
• Width of specimen / Chiều rộng mẫu mm	19.0	19.0
• Cross section area / Tiết diện thực đo mm ²	155.4	155.0
• Tensile load / Lực kéo đứt kN	86.4	86.4
• Tensile strength / Giới hạn bền kéo MPa	556	557
• Broken position / Vị trí mẫu đứt	In base metal	In base metal



The relationship between tensile load and deformation
Biểu đồ tương quan giữa lực thử kéo và biến dạng

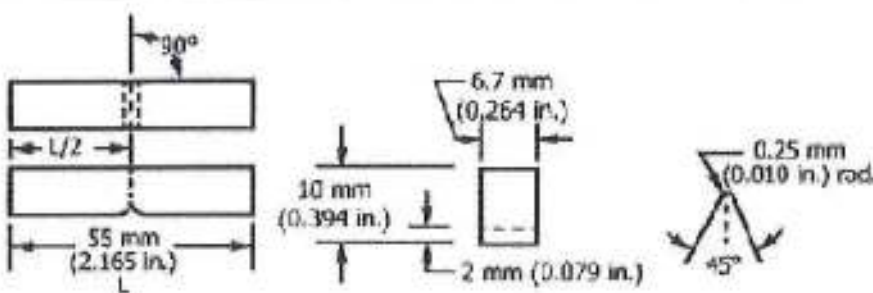
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test ⁽¹⁾ / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	8.16	8.18	8.16	8.16
• Width of specimen / Chiều rộng mm	38	38	38	38
• Bending angle / Góc uốn ° độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	32	32	32	32
• Test result / Kết quả	No Crack	No Crack	No Crack	No Crack
• Remark / Nhận xét	Pass	Pass	Pass	Pass

*** TO BE CONTINUED ***

063692

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
C. Impact test at -40°C ⁽¹⁾ / Thử va đập ở -40°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.0 x 6.69	10.0 x 6.71	10.0 x 6.70
• Impact energy / Công va đập	J	114.8	85.3	93.2
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.0 x 6.70	10.0 x 6.71	10.0 x 6.71
• Impact energy / Công va đập	J	152.1	149.1	101.0
				
Figure / Hình ảnh minh họa				

*** TO BE CONTINUED ***

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D. Characteristic / Tên chỉ tiêu : Macro etch (*) / Tổ chức thô đại

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn
- No defect in weld metal / Trong mối hàn không có khuyết tật



Sample Photo(s) / Hình ảnh mẫu nhân:



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*** END OF THE REPORT ***

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검 사 증 명 서

MILL TEST CERTIFICATE



당진제철소 중남 당진시 송악읍 북부산업로 1480
1480 Bukbusanup-Ro, Songak-Eup, Dangin-Si, Chungnam, Korea



주문번호
Order No. : F200600214

품 명
Commodity : Hot Rolled Coil

재품규격
Specification : ASTM A572 GRADE 50

고객사
Customer : DAIDUNG CORP

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20201009-HS-001-004

발행일자
Date of Issue : 2020-10-09

지름치수 Dimension (mm)	수량 Quantity	중량 Weight (kg)	재고번호 Hot No.	제품번호 Product No.	단위 Unit (kg)	인장시험 Tensile Test							화학성분 Chemical Composition(%)							
						YP	TS	EL	Y+EL	RA	YR	YP	C		Si		Mn		P	
						(N/㎡)	(%)	(%)	(%)	(%)	(%)	(N/㎡)	4	3	3	4	4	3	3	3
8.00x1940xC	1	23,410	R58087	CL562520	B	447.2	545.5	31				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	22,300	R58087	CL578900	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	23,290	R58086	CL578910	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949
8.00x1940xC	1	23,420	R58086	CL578940	B	447.4	550.9	29				2	L	1.660	15	1.013	155	61	18	1.949
8.00x1940xC	1	23,530	R58086	CL578930	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949
8.00x1940xC	1	23,310	R58086	CL578940	B	442.1	543.4	30				2	L	1.560	15	1.013	155	61	18	1.949
8.00x1940xC	1	23,400	R58087	CL578950	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	21,360	R58087	CL578960	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	21,360	R58087	CL578970	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	21,370	R58087	CL578980	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	24,060	R58087	CL578990	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	24,160	R58087	CL579000	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	24,220	R58087	CL579010	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	23,260	R58086	CL579020	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949
8.00x1940xC	1	23,540	R58087	CL579030	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017
8.00x1940xC	1	24,600	R47006	CVS12570	B	386.4	531.0	34				2	L	1.964	17	1.024	140	41	17	1.962

*** Sub Total(000) *** 15 373,000Kgt

N O T E	- Position - T:Top, M:Middle, B:Bottom - Tensile test - Direction: Transverse. Gauge length: 50mm (Rectangular). YF Min. 2 (0.2% off-load), 5 (0.5% underload), U (Upper yield point), L (Lower yield point) - Division - P: Product analysis, L: Lab analysis

본사의 제품과 검사에 관한 자세한 규격에 합당한 것을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS.

Signature

S. B. Park

Chief Of Quality Assurance Team

- * 본 검사증명서는 원본의 전자문서(검사서)와 정본, 도본(도판)을 통해만 사용 가능합니다. 전자문서의 내용을 원본과 동일하게 출력하여 사용하실 수 있습니다. (http://smhyundai-steel.com/online/sign.asp)
- * The Mill Test Certificate is a copy that has been printed from original electronic document (with digital signing).
- * You are able to check an original electronic document at Hyundai's customer portal (http://smhyundai-steel.com/online/sign.asp) * QR Code scanner App : QR&I
- * [사용상 주의사항] 이 검사증명서에 기재된 규격 및 사용처 사용과 관련하여 문의할 수 있습니다.
- * [Caution for Use] Caution on products and safety problems may arise if used other than the ones stated in the Mill Test Certificate.

ORIGINAL

CERTIFIED MATERIAL REPORT

CMR No.: HV - 2021 - 0464

Date of Issue: 2021 - 05 - 1



Customer		G-TECH CORPORATION								
Trade Name		Supercored 71H		Size		1.2 mm x 18 kg				
Lot No.		MF21340E71		Purchaser's Order No.		25-2021/GT/HW/PO				
Specification		AWS A5.20 E71T-9C-J * According to EN 10204-2004 : 3.1								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)						
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	502	560	28	-40	65	81	72	73		
AWS Spec.	≥350	490~670	≥22	≥27						
Chemical Composition (%)										
	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
(Weld Metal)	0.04	0.47	1.21	0.017	0.009	0.36	0.02	0.01	0.01	0.03
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35
Others * Identical with AWS A5.36 E71T1-C1A4-CS1 * Q'ty : 900 kg DDC - QA/QC DEPT TRUE COPY Date: 26/05/2021										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements. Approved by QA Manager Pham Dinh Anh Vu										

HYUNDAI WELDING VINA CO., LTD.

Road No. 2, Nhon trach 5 IZ, Nhon Trach District, Dong Nai Province, Vietnam

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MỐI HÀN
DD-FCAW-134A
WPS-DD-FCAW-134
Base material : PL8, A572 Gr. 50
Size: PLATE 300x200x8 mm to 300x200x8 mm
Filler material : AWS A5.20 E71T-1C/-9C-J
Trade name : Hyundai Supercored 71H (Ø1.2mm)
Lot No: MF24340E71

Project / Dự án : Evapco – Aspen Power Station Project
Item / Hàng mục : Procedure Qualification Record (PQR) Test for Ducting
Quantity / Số lượng mẫu : 01 sample / mẫu
Description / Mô tả mẫu : 500 x 340 x 8 (mm)
Date of receiving / Ngày nhận mẫu : 01-Aug-2024
Date of testing / Ngày thử nghiệm : From / từ 01-Aug-2024 to / đến 02-Aug-2024
Test method / Phương pháp thử : ASME BPVC.VIII.1-2023 Rules for Construction of Pressure Vessels
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

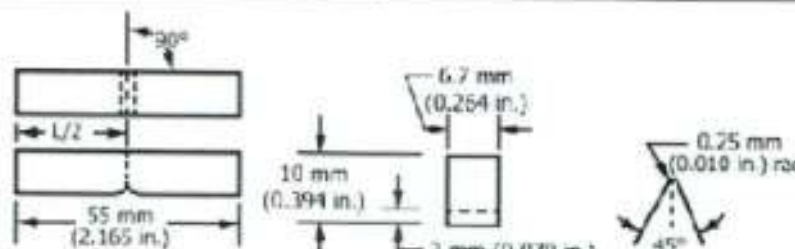
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033103

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
Impact test at -45°C / Thử va đập ở -45°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 6.70	10.01 x 6.71	10.00 x 6.70
• Impact energy Công va đập	J	95	82	95
• Average / Trung bình	J	91		
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 6.71	10.01 x 6.71	10.01 x 6.71
• Impact energy Công va đập	J	28	34	39
• Average / Trung bình	J	34		
				
Figure / Hình ảnh minh họa				

Original sample photo(s) / Hình ảnh mẫu nhận:


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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MÔI HÁN
DD-FCAW-134B
WPS-DD-FCAW-134
Base material : PL12, A572 Gr. 50
Size: PLATE 300x200x12 mm to 300x200x12 mm
Filler material : AWS A5.20 E71T-1C/-9C-J
Trade name : Hyundai Supercored 71H (Ø1.2mm)
Lot No: MF24340E71

Project / Dự án : Evapco – Aspen Power Station Project

Item / Hàng mục : Procedure Qualification Record (PQR) Test for Ducting

Quantity / Số lượng mẫu : 01 sample / mẫu

Description / Mô tả mẫu : 500 x 340 x 12 (mm)

Date of receiving / Ngày nhận mẫu : 01-Aug-2024

Date of testing / Ngày thử nghiệm : From / từ 01-Aug-2024 to / đến 02-Aug-2024

Test method / Phương pháp thử : ASME BPVC.VIII.1-2023 Rules for Construction of Pressure Vessels

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of

SGS Vietnam LTD
Thay mặt công ty

SGS Vietnam Ltd

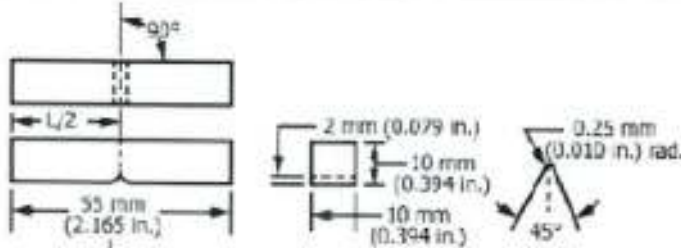
Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
Impact test at -45°C / Thử va đập ở -45°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 10.00	10.01 x 10.01	10.01 x 10.01
• Impact energy Công va đập	J	88	69	112
• Average / Trung bình	J	90		
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 10.01	10.00 x 10.01	10.01 x 10.01
• Impact energy Công va đập	J	99	66	55
• Average / Trung bình	J	73		
				
Figure / Hình ảnh minh họa				

Original sample photo(s) / *Hình ảnh mẫu nhận:*



SGS authenticate the photos on original report only / *SGS chỉ xác thực các hình ảnh trong báo cáo chính thức*

*** END OF THE REPORT ***

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검 사 증 명 서

MILL TEST CERTIFICATE



당진개칭소 : 충남 당진시 송악읍 북부산업로 1480
1480 Bukbusanup-Ro, Songak-Eup, Dengjin-Si, Chungnam, Korea



주문번호
Order No. : F240100222

품 명
Commodity : Hot Rolled Coil

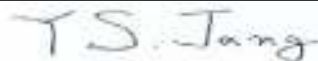
제품규격
Specification : ASTM A572 GRADE 50

고객사
Customer : DAIDUNG CORP

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20240319-HS-0012-004

발행일자
Date of Issue : 2024-03-19

제품치수 Dimension (mm)	수량 Quantity	중량 Weight (Kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test							화학성분 Chemical Composition(%)																단위 Unit		
						YP	TS	EL	YP-EL	RA	YR	YP Mtd	Division	C	Si	Mn	P	S	Cr	Ni	B	Cu	Mo	Nb	V	DR	2 : x100	3 : x1,000	4 : x10,000	5 : x100,000	
						(N/mm ²) ()	(%)	(%)	(%)	(%)	4			3	3	4	4	2	2	4	3	2	3	3	3						
8.00x1999xC	1	23,830	S97136	CPA46250	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987					
8.00x1999xC	1	24,050	S97136	CPA46260	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987					
8.00x1999xC	1	23,990	S97136	CPA46280	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987					
8.00x1999xC	1	24,060	S97136	CPA46290	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987					
8.00x1999xC	1	24,190	S97136	CQA03650	B	397.4	537.1	25				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987					
*** Sub Total(004) *** 5 120,120(Kg)																															
*** Grade Total *** 22 521,580(Kg) DR = 6XC+Mn																															
*** Last Item ***																															
NOTE	* Position – T:Top, M:Middle, B:Bottom * Tensile test – Direction:Transverse, Gauge length:50mm (Rectangular), YP Mtd : 2 (0.2% off-set),5 (0.5% underload), U (Upper yield point), L (Lower yield point) * Division – P:Product analysis, L:Ladle analysis																														
	* Mill Test Certificate according to EN 10204 Type 3.1.																														
상기의 제품은 검사의 결과 지정된 규격에 합격한 것을 증명합니다. WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS																								Signature  Chief Of Hot Rolled Quality Team							

* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 현대제철 고객포탈에서 확인하실 수 있습니다.(http://sm.hyundai-steel.com/cs/cm/login.jsp)
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You are able to check an original electronic document at hyundaisteel's customer portal.(http://sm.hyundai-steel.com/cs/cm/login.jsp) * QRcode scanner App : 'QReal'
* [사용상 주의사항] 본 검사증명서에 명기된 규격 외 사용시 제품손상 및 안전상 문제가 발생할 수 있습니다.
* [Caution for Use] Damage on products and safety problems may arise if used other than to the spec stated in this Mill Test Certification.

Order No./계약번호 : 01S6648703

PO No./주문번호 : 01S6648703

Supplier /주문자 : POSCO INTERNATIONAL CORPORATION

Commodity /품명 : STEEL PLTAE

Customer /고객사 : DDC

Spec & Type /규격 : ASTM A572-50-TY1

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Country of Melt & Pour	Position	Tensile Test /인장시험			Division	Chemical Composition/화합성분										
							YP	TS	EL		C	Si	Mn	P	S	Cr	Ni	Cu	Mo	Nb	V
							(MPa)		(%)		(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)
*** Sub Total (0+0) ***		62	116,870 (kg)																		
12x2001x12000	PB93389601-9605	5	11,310	SB99466	KOR	T	432	553	23	L	0.1532	0.376	1.435	0.0110	0.0018	0.028	0.012	0.015	0.002	0.026	0.002
										P	0.1560	0.353	1.404	0.0113	0.0022	0.022	0.013	0.013	0.012	0.024	0.002
12x2001x12000	PB93491901-1905	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93537701-7705	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93642901-2905	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93537601-7605	5	11,310	SB99606	KOR	T	425	557	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93570701-0706	6	13,572	SB99685	KOR	T	431	552	22	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662501-2506	6	13,572	SB99685	KOR	T	431	552	22	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93593601-3605	5	11,310	SB99685	KOR	T	420	554	25	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662401-2406	6	13,572	SB99685	KOR	T	420	554	25	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662201-2206	6	13,572	SB99689	KOR	T	441	553	24	L	0.1464	0.338	1.395	0.0096	0.0020	0.027	0.015	0.014	0.003	0.024	0.002
										P	0.1492	0.329	1.375	0.0099	0.0021	0.026	0.015	0.012	0.009	0.022	0.002
12x2001x12000	PB93662001-2006	6	13,572	SB99747	KOR	T	440	557	23	L	0.1497	0.366	1.396	0.0114	0.0028	0.026	0.011	0.009	0.003	0.025	0.002
										P	0.1526	0.302	1.276	0.0115	0.0035	0.030	0.013	0.013	0.005	0.022	0.007
12x2001x12000	PB93662301-2305	5	11,310	SB99747	KOR	T	440	557	23	L	0.1497	0.366	1.396	0.0114	0.0028	0.026	0.011	0.009	0.003	0.025	0.002
										P	0.1526	0.302	1.276	0.0115	0.0035	0.030	0.013	0.013	0.005	0.022	0.007
12x2001x12000	PB93643001-3005	5	11,310	SB78995	KOR	T	412	554	24	L	0.1521	0.366	1.392	0.0101	0.0027	0.029	0.011	0.014	0.004	0.021	0.002
										P	0.1500	0.368	1.406	0.0098	0.0024	0.027	0.011	0.015	0.006	0.021	0.002

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test. Direction : Transversal, Gauge Length : 200 mm(Rectangular),
YP Method : 0.2 % off-set
* Division - L : Ladle Analysis, P : Products Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in
South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

Legal sanction can be imposed on forging. Improper use of product can cause safety issues.

Surveyor To :

B.S. Kim



CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2024 – 0744/2

Date of Issue : 2024 – 05 – 24



Customer		G-TECH CORPORATION						
Trade Name		Supercored 71H		Size		1.2 mm x 15 kg		
Lot No.		MF24340E71		Purchaser's Order No.		38-2024/GT/HW/PO		
Specification		AWS A5.20 E71T-9C-J * According to EN 10204-2004 : 3.1						
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded	
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	534	589	27	-40	72	70	78	73
AWS Spec.	≥390	490~670	≥22	≥27				

Chemical Composition (%)

Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
	0.04	0.49	1.29	0.017	0.008	0.37	0.02	0.01	0.01	0.02
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35

* Hydrogen Content (mℓ/100g) (AWS A4.3) : 2.86
* Radiographic Test : ACCEPT
* Fillet Weld Test : GOOD

Others
CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-9J-H8)

* Q'ty: 3,600 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

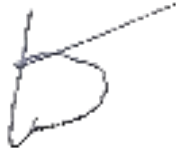
Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
ASME VIII, IX:2023

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-134/251

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Aug 17, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 1/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-134/251** Revision No. **0** Date **Aug 17, 2024**
Supporting PQR No.(s): **PQR-DD-FCAW-134 (Sep 27, 2021 & Aug 5, 2024) + PQR-DD-FCAW-251 (May 28, 2024)**
Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Semi-Automatic, Mechanized**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**
Root Spacing **As per fabrication drawing**
Backing **With baking or back gouging**
Retainer **None**
Backing Material **Ceramic or base metal or weld metal**

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **1** Group No. **1, 2** to P-No. **1** Group No. **1, 2**
OR
Specification and type/grade or UNS Number **None**
to Specification and type/grade or UNS Number **None**
OR
Chem. Analysis and Mech. Prop. **None**
to Chem. Analysis and Mech. Prop. **None**
Thickness Range:
Base Metal: **8 - 16 (mm)** **All for fillet welds**
Maximum Pass Thickness ≤ 13 mm: **Yes** ☒
Other

FILLER METALS (QW-404)

Process **FCAW**
Spec. No. (SFA) **SFA-5.20**
AWS No. (Class) **E71T-1C/-9C-J or E71T1-C1A4-CS1**
F-No. **6**
A-No. **1**
Size of Filler Metals **$\varnothing 1.2 - \varnothing 1.6$ (mm)**
Supplemental Filler Metal **None**
Filler Metal Product Form **Flux cored wire**
Filler Metal Brand Name **Hyundai Supercored 71H or the same spec & class**
Weld Metal
Deposited Thickness:
Groove **Max. 16 mm**
Fillet **All**
Electrode-Flux (Class) **None**
Flux Type **None**
Flux Trade Name **None**
Consumable Insert **None**
Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2023)

Page: 2/2

Applicable Code: **ASME Section II.A (SA-370), VIII, IX - 2023 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-134/251** Revision No. **0** Date **Aug 17, 2024**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **20 °C**
Interpass Temperature, Maximum **197 °C**
Preheat Maintenance **None**
Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	99.8%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: Minimum purity 99.8% CO₂ for FCAW process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
2 or more	FCAW	E71T-1C/-9C-J	1.2 - 1.6	DC EP	170 - 200	None	0.96 - 1.4	23 - 28	180 - 300	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **None**
Mode of Metal Transfer for GMAW (FCAW) **Globular, Spray or Pulsed**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or Weave Bead **String & Weave (Max. width 14 mm)**
Orifice, Nozzle, or Gas Cup Size **No. 6**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding or Air carbon arc**
Oscillation **None**
Contact Tube to Work Distance **15 - 25 (mm)**
Multiple or Single Pass (Per Side) **Multiple pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:
La Vinh Nhon

Approved by: **Truong Hoai Phong Vu (QA Manager)**



Date: **Aug 17, 2024**

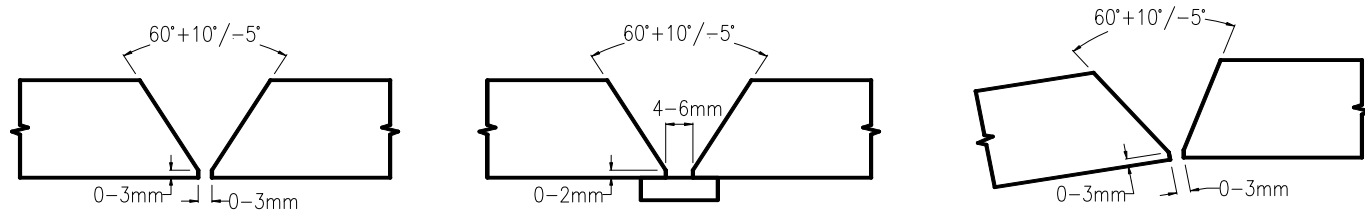
Date: **Aug 17, 2024**

Welding Procedure Specification No.: **WPS-DD-FCAW-134/251**

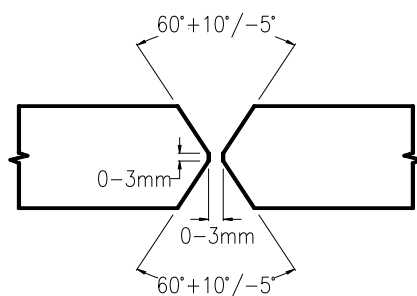
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

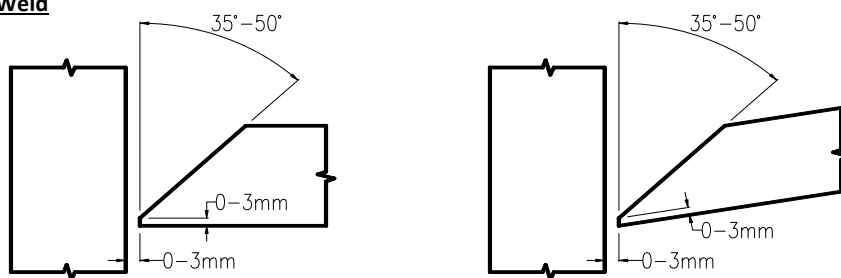
1. Single V Groove Butt Joint



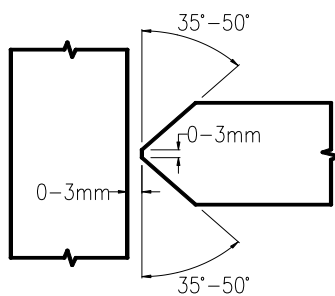
2. Double V Groove Butt Joint



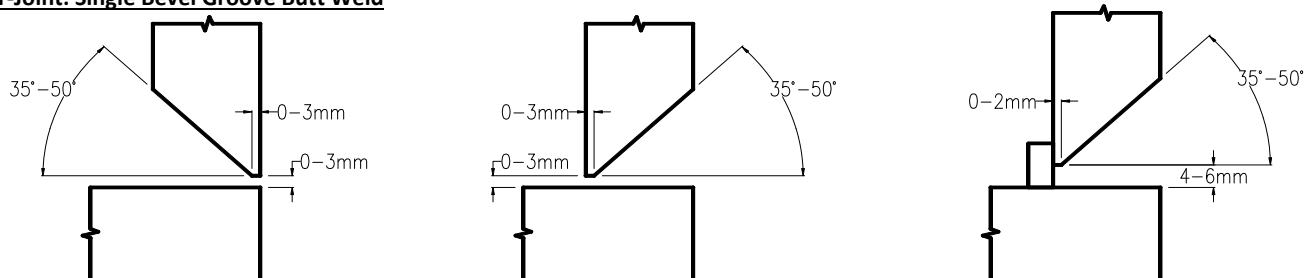
3. T-Joint: Single Bevel Groove Butt Weld



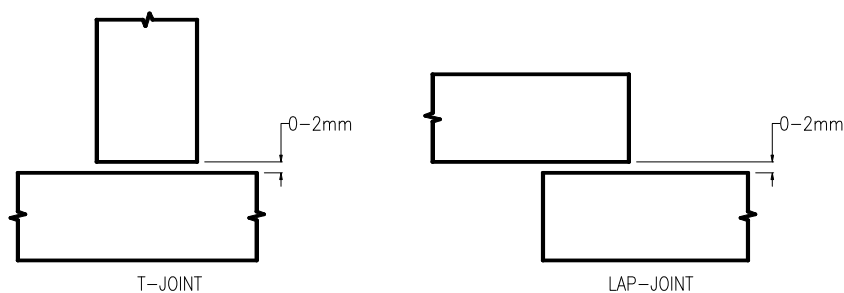
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld





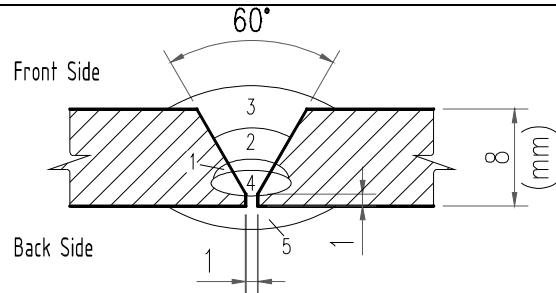
PROCEDURE QUALIFICATION RECORD (PQR) (ASME VIII, IX - Boiler and Pressure Vessel Code-2019)

Page: 1/2

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
 Procedure Qualification Record No. **PQR-DD-FCAW-134** Date: **Sep 27, 2021**
 Welding Process(es) **FCAW**
 Type(s)(Manual, Automatic, Semi-Automatic) **Semi-Automatic**

JOINTS (QW-402)

Joint Type **Butt joint, single V groove**
 Backing **None**
 Retainer **None**



BASE METALS (QW-403)

Material Spec. **ASTM A572**
 Type/Grade, or UNS Number **Gr.50**
 P-No. **1** Group No. **1** to P-No. **1** Group No. **1**
 Thickness of Test Coupon **8**
 Diameter of Test Coupon **Plate**
 Maximum Pass Thickness **< 13 mm**
 Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**
 Time **None**
 Other **None**

GAS (QW-408)

	Percent Composition		Flow Rate
	Gas(es)	(Mixture)	
Shielding	CO₂	99,99%	20 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process **FCAW**
 SFA Specification **AWS A5.20 / 5.36**
 AWS Classification **E71T-9C-J / E71T1-C1A4-CS1**
 Filler Metal F-No. **6**
 Weld Metal Analysis A-No. **1**
 Size of Filler Metal **1.2 mm**
 Filler Metal Product Form **Flux cored wire**
 Supplemental Filler Metal **None**
 Filler Metal Trade Name **SUPERCORED 71H**
 Electrode Flux Classification **-**
 Flux Type **-**
 Flux Trade Name **-**
 Weld Metal Thickness **8 mm**
 Other **None**

ELECTRICAL CHARACTERISTICS (QW-409)

Process **FCAW**
 Current and Polarity **DC EP**
 Amps. **See table below**
 Volts **See table below**
 Mode of Metal Transfer for GMAW (FCAW) **None**
 Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
 String or weave Bead **Weave**
 Oscillation **None**
 Method cleaning **Grinding, brushing**
 Method back gouge **Air carbon arc, grinding**
 Multipass or single pass (per side) **Multipass**
 Multipass or single Electrodes **Single**
 Peening **None**
 Use of thermal processes **None**
 Other **None**

POSITION (QW-405)

Process **FCAW**
 Position(s) **1G**
 Weld Progression (Uphill, Downhill) **None**
 Other **None**

PREHEAT (QW-406)

Preheat Temperature **30 °C (Ambient)**
 Interpass Temperature **Max. 142 °C**
 Other **None**

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	150	27	293	30 (Ambient)	4	20	0.8
2	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	27	252	105	3	20	1.1
3	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	27	202	137	3	20	1.4
4	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	180	26	298	116	2	20	0.9
5	FCAW	E71T-9C-J / E71T1-C1A4-CS1	1.2	DC EP	170	26	201	142	4	20	1.3

Handwritten signature

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW-134** Date: **Sep 27, 2021**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	19.0	8.18	155.4	86.4	556	Ductile and base metal
Sample 2	19.0	8.16	155.0	86.4	557	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1 (face bend)	Pass
Specimen 2 (face bend)	Pass
Specimen 3 (root bend)	Pass
Specimen 4 (root bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.0 x 6.69	-40 °C	114.8 (J)			N
WM 2	Weld metal	10.0 x 6.71	-40 °C	85.3 (J)			N
WM 3	Weld metal	10.0 x 6.70	-40 °C	93.2 (J)			N
HAZ 1	HAZ	10.0 x 6.70	-40 °C	152.1 (J)			N
HAZ 2	HAZ	10.0 x 6.71	-40 °C	149.1(J)			N
HAZ 3	HAZ	10.0 x 6.71	-40 °C	101.0 (J)			N

Comments **Acceptable**

Fillet-Weld Test (QW-180)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: **Acceptable**

Other Tests (Not Apply)

Type of Test _____

Deposit Analysis _____

Other _____

Welder's Name **Le Minh Loi** Clock No. **None** Stamp (ID) No. **DDC-WD-0340**

Welder's Name **-** Clock No. **-** Stamp (ID) No. **-**

Tests Conducted by **SGS** Laboratory Test No. **INL21-506868-477**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX:2019 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon

Date: **Sep 27, 2021**

Date: **Sep 27, 2021**




PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2023)

Page: 2
Additional

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW-134**

Date: **Aug 5, 2024**

Additional Toughness Tests (QW-170)

Test Report No.: **INL24/512130-541-1**

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.01 x 6.70	-45 °C	95 (J)			
WM 2	Weld metal	10.01 x 6.71	-45 °C	82 (J)			
WM 3	Weld metal	10.00 x 6.70	-45 °C	95 (J)			
HAZ 1	HAZ	10.01 x 6.71	-45 °C	28 (J)			
HAZ 2	HAZ	10.01 x 6.71	-45 °C	34 (J)			
HAZ 3	HAZ	10.01 x 6.71	-45 °C	39 (J)			

Test Report No.: **INL24/512130-541-2**

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10.01 x 10.00	-45 °C	88 (J)			
WM 2	Weld metal	10.01 x 10.01	-45 °C	69 (J)			
WM 3	Weld metal	10.01 x 10.01	-45 °C	112 (J)			
HAZ 1	HAZ	10.01 x 10.01	-45 °C	99 (J)			
HAZ 2	HAZ	10.00 x 10.01	-45 °C	66 (J)			
HAZ 3	HAZ	10.01 x 10.01	-45 °C	55 (J)			

Comments **Acceptable**

Welder's Name **Le Minh Loi**

Clock No. **None**

Stamp (ID) No. **DDC-WD-0340**

Welder's Name **-**

Clock No. **-**

Stamp (ID) No. **-**

Tests Conducted by **SGS**

Laboratory Test No. **INL24/512130-541-1; INL24/512130-541-2**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX:2023 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

Approved by: **Truong Hoai Phong Vu (QA Manager)**

La Vinh Nhon



Date: **Aug 5, 2024**

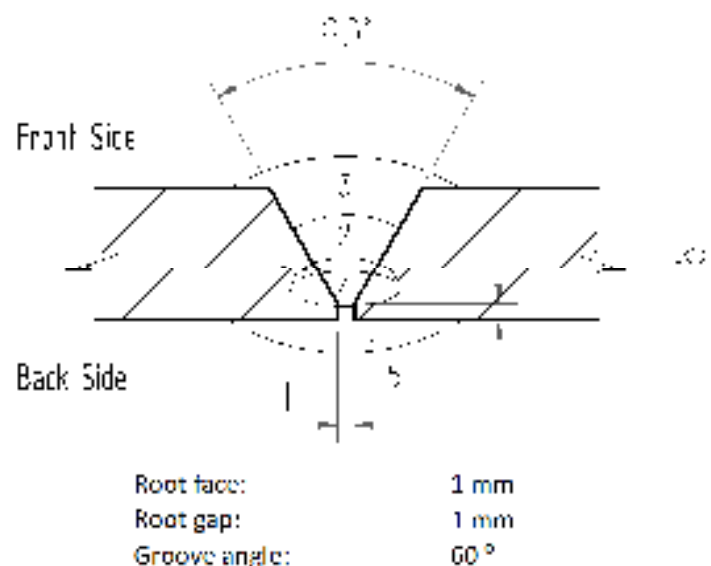
Date: **Aug 5, 2024**

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-134)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR DD FCAW 134
 Test Coupon: DD-FCAW-134
 Test position: 1G
 Base metal:
 - Specification: ASTM A572 Gr.50
 - Dimension: PL8 x 180 x 660 (mm) x 2 pcs
 Heat/Product No.: R58087/CU562620
 Certification No.: 20201002 IIS 001 004
 Filler metal:
 - Specification: A5.20 / A5.36
 Classification: E71T 9C J / E71T1 C1A4 C51
 Trade name: Hyundai Supercore 7111
 Batch/Lot No.: MF21340E71
 Certificates No.: HV 2021 0464
 Gas: CO₂ (100%)
 Welder Name: Le Minh Loi
 I.D.: DDC WD 0340

Joint Sketch:

Date: Sep 22, 2021



Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (l/min.)	Heat Input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Front Side (First Side):																
1	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	150	27	2	15	660	293	30 (Ambient)	20	0.8	8	4	
2	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	170	27	2	37	660	252	105	20	1.1	10	3	
3	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	170	27	3	16	660	202	137	20	1.4	15	3	Reinforcement: 2 mm



WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-134)

Layer/ pass	Process	Filler metal		Electric Current		Volts [V]	Times		Length [mm]	Travel Speed [mm/min.]	Preheat/ Interpass Temp. (°C)	Shielding Gas Flow Rate [l/min.]	Heat Input [kJ/mm]	Weld bead		Remark
		Specification/ Classification	Dia. [mm]	Polarity	Current [A]		min.	sec.						Width	Depth	
Back Side (Second Side) with back gouging:																
4	FCAW	AWS A5.20 / E71T-9C J	1.2	DC EP	180	26	2	13	660	298	116	20	0.9	10	2	
5	FCAW	AWS A5.20 / E71T-9C-I	1.2	DC EP	170	26	3	17	660	201	142	20	1.3	15	4	Reinforcement: 2 mm
Conducted by Welding Coordinator																
Name		La Vinh Nhon														
Signature																
Date		Sep 22, 2021														

VILAS 237

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Client's address / Địa chỉ : B23/474C TRẦN ĐẠI NGHĨA, ẤP 2, XÃ TÂN NHỰT, HUYỆN BÌNH CHÁNH, TP.HCM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING STEEL PLATE / MỎI HÀN THÉP TẤM
DD-FCAW-134; FCAW - 1G, Single V
Size / Kích thước : 660 x 360 x 8 (mm)
Material / Vật liệu : ASTM A572 Gr.50
Welder / Mã thợ hàn : DDC-WD-0340

Project / Dự án : Great Plains Power Station (Moose Jaw, Canada)
Evapco Dry Cooling, Inc.

Item / Hạng mục : Procedure Qualification Record (PQR) Test For Duct

Quantity / Số lượng mẫu : 01 sample / mẫu

Date of receiving / Ngày nhận mẫu : 23-Sep-2021

Date of testing / Ngày thử nghiệm : From / từ 23-Sep-2021 to / đến 27-Sep-2021

Test method / Phương pháp thử : ASME BPVC.IX-2019 Welding, Brazing, and Fusing Qualifications
ASME BPVC.VIII.1-2019 Rules for Construction of Pressure Vessels
ASTM A370 - 19 Standard Test Methods and Definitions for Mechanical Testing of Steel Products

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Note :

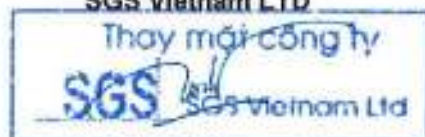
(*) The test was accredited ISO 17025:2017 by VILAS / Chỉ tiêu được công nhận ISO 17025:2017 bởi VILAS.

ILAC : International Laboratory Accreditation / Tổ chức hợp tác công nhận Phòng thử nghiệm quốc tế.

MRA : Mutual Recognition Arrangement / Thỏa ước thừa nhận lẫn nhau.

ILAC-MRA -VILAS : VILAS is the full member and signatory of Mutual Recognition Arrangement with International Laboratory Accreditation / VILAS là thành viên đầy đủ và ký thỏa ước thừa nhận lẫn nhau với tổ chức công nhận phòng thử nghiệm quốc tế.

Signed for and on behalf of
SGS Vietnam LTD

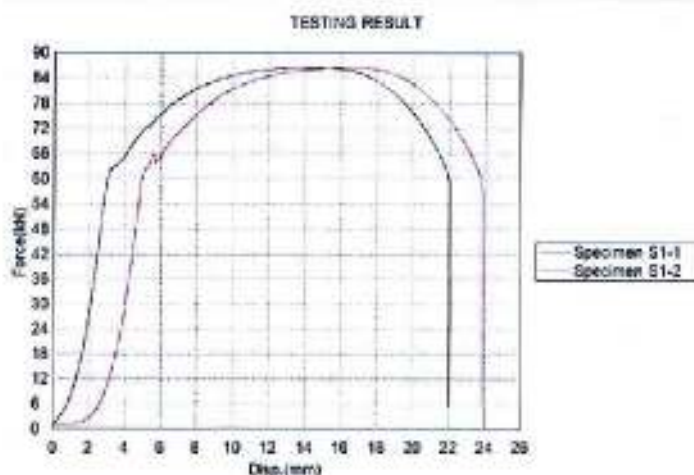


Lê Tiếp Du
Lab Assistant Manager
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test ⁽¹⁾ / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	8.18	8.16
• Width of specimen / Chiều rộng mẫu mm	19.0	19.0
• Cross section area / Tiết diện thực đo mm ²	155.4	155.0
• Tensile load / Lực kéo đứt kN	86.4	86.4
• Tensile strength / Giới hạn bền kéo MPa	556	557
• Broken position / Vị trí mẫu đứt	In base metal	In base metal



The relationship between tensile load and deformation
Biểu đồ tương quan giữa lực thử kéo và biến dạng

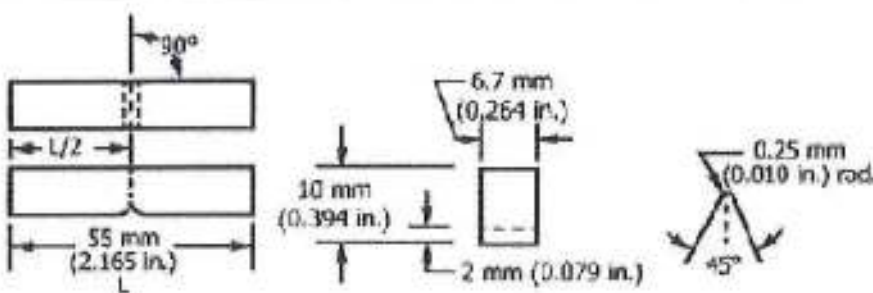
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test ⁽¹⁾ / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	8.16	8.18	8.16	8.16
• Width of specimen / Chiều rộng mm	38	38	38	38
• Bending angle / Góc uốn ° độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	32	32	32	32
• Test result / Kết quả	No Crack	No Crack	No Crack	No Crack
• Remark / Nhận xét	Pass	Pass	Pass	Pass

*** TO BE CONTINUED ***

063692

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
C. Impact test at -40°C ⁽¹⁾ / Thử va đập ở -40°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.0 x 6.69	10.0 x 6.71	10.0 x 6.70
• Impact energy / Công va đập	J	114.8	85.3	93.2
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.0 x 6.70	10.0 x 6.71	10.0 x 6.71
• Impact energy / Công va đập	J	152.1	149.1	101.0
				
Figure / Hình ảnh minh họa				

*** TO BE CONTINUED ***

063694

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D. Characteristic / Tên chỉ tiêu : Macro etch (*) / Tổ chức thô đại

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn
- No defect in weld metal / Trong mối hàn không có khuyết tật



Sample Photo(s) / Hình ảnh mẫu nhân:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

*** END OF THE REPORT ***

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검 사 증 명 서

MILL TEST CERTIFICATE



당진제철소 중남 당진시 송악읍 북부산업로 1480
1480 Bukbusanup-Ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : F200600214

품 명
Commodity : Hot Rolled Coil

제품규격
Specification : ASTM A572 GRADE 50

고객사
Customer : DAIDUNG CORP

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20201009-HS-001-004

발행일자
Date of Issue : 2020-10-09

제품치수 Dimension (mm)	수량 Quantity	중량 Weight (kg)	재질번호 Mat.No.	제품번호 Product No.	단위 Unit (kg)	인장시험 Tensile Test						화학성분 Chemical Composition(%)									
						Y _P (N/mm ²)	T _S (%)	E _L (%)	Y _P +E _L (%)	RA (%)	Y _R (%)	Y _P Mn	500 Unit 2 : x 100 3 : x 1,000 4 : x 10,000 5 : x 100,000								
													C	Si	Mn	P	S	Al	DR	4	3
8,00x1940xC	1	23,410	R58087	CL562520	B	447.2	545.5	31				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	22,300	R58087	CL578900	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	23,290	R58086	CL578910	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949	
8,00x1940xC	1	23,420	R58086	CL578940	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949	
8,00x1940xC	1	23,530	R58086	CL578930	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949	
8,00x1940xC	1	23,310	R58086	CL578940	B	442.1	543.4	30				2	L	1.560	15	1.013	155	61	18	1.949	
8,00x1940xC	1	23,400	R58087	CL578950	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	21,360	R58087	CL578960	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	21,360	R58087	CL578970	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	21,370	R58087	CL578980	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	24,060	R58087	CL578990	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	24,160	R58087	CL579000	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	24,220	R58087	CL579010	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	23,260	R58086	CL579020	B	447.4	550.9	29				2	L	1.560	15	1.013	155	61	18	1.949	
8,00x1940xC	1	23,540	R58087	CL579030	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017	
8,00x1940xC	1	24,600	R47006	CVS12570	B	386.4	531.0	34				2	L	1.564	17	1.024	140	41	17	1.962	

*** Sub Total(000) *** 15 373,000Kgt

NOTES	* Position - T:Top, M:Middle, B:Bottom * Tensile test - Direction: Transverse, Gauge length: 50mm (Rectangular), YF Min. : 2 (0.2% offset), 5 (0.5% underload), U (Upper yield point), L (Lower yield point) * Division - P: Product analysis, L: Lab. analysis

본사의 제품과 검사 및 결과 지정된 규격에 합당한 것을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS.

Signature

S. B. Park

Chief Of Quality Assurance Team

- * 본 검사증명서는 원본의 전자문서(검사서와 검사 결과)로만 유효하며, 사본은 무효입니다. 전자문서의 내용을 원본과 동일하게 인쇄하여도 효력이 없습니다. (http://smhyundai-steel.com/onlinehelp.asp)
- * The Mill Test Certificate is a copy that has been printed from original electronic document (with digital signing).
- * You are able to check an original electronic document in Hyundai's customer portal (http://smhyundai-steel.com/onlinehelp.asp) * QR Code scanner App : QR&I
- * [제품상 주의사항] 이 검사증명서에 기재된 규격 외 사용시 제품과 관련된 안전사고 발생될 수 있습니다.
- * [Caution for Use] Caution on products and safety problems may arise if used other than the spec. stated in the Mill Test Certificate.

ORIGINAL

CERTIFIED MATERIAL REPORT

CMR No.: HV - 2021 - 0464

Date of Issue: 2021 - 05 - 14



Customer		G-TECH CORPORATION								
Trade Name		Supercored 71H			Size		1.2 mm x 18 kg			
Lot No.		MF21340E71			Purchaser's Order No.		25-2021/GT/HW/PO			
Specification		AWS A5.20 E71T-9C-J * According to EN 10204-2004 : 3.1								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)						
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	502	560	28	-40	65	81	72	73		
AWS Spec.	≥350	490~670	≥22	≥27						
Chemical Composition (%)										
	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
(Weld Metal)	0.04	0.47	1.21	0.017	0.009	0.36	0.02	0.01	0.01	0.03
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35
Others * Identical with AWS A5.36 E71T1-C1A4-CS1 * Q'ty : 900 kg DDC - QA/QC DEPT TRUE COPY Date: 26/05/2021										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements. Approved by QA Manager Phạm Đình Anh Vũ										

HYUNDAI WELDING VINA CO., LTD.

Road No. 2, Nhon trach 5 IZ, Nhon Trach District, Dong Nai Province, Vietnam

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MỐI HÀN
DD-FCAW-134A
WPS-DD-FCAW-134
Base material : PL8, A572 Gr. 50
Size: PLATE 300x200x8 mm to 300x200x8 mm
Filler material : AWS A5.20 E71T-1C/-9C-J
Trade name : Hyundai Supercored 71H (Ø1.2mm)
Lot No: MF24340E71

Project / Dự án : Evapco – Aspen Power Station Project
Item / Hàng mục : Procedure Qualification Record (PQR) Test for Ducting
Quantity / Số lượng mẫu : 01 sample / mẫu
Description / Mô tả mẫu : 500 x 340 x 8 (mm)
Date of receiving / Ngày nhận mẫu : 01-Aug-2024
Date of testing / Ngày thử nghiệm : From / từ 01-Aug-2024 to / đến 02-Aug-2024
Test method / Phương pháp thử : ASME BPVC.VIII.1-2023 Rules for Construction of Pressure Vessels
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

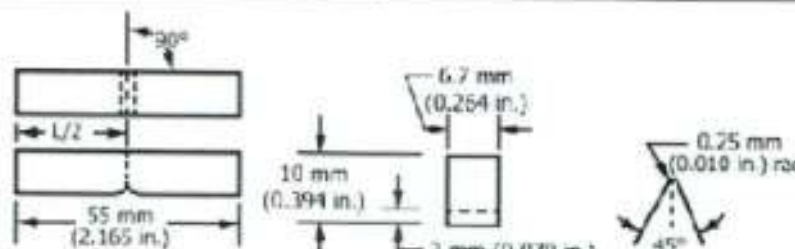
Please extract your time to give us your feedback as this will help our service better. Thank you!



033103

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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
Impact test at -45°C / Thử va đập ở -45°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 6.70	10.01 x 6.71	10.00 x 6.70
• Impact energy Công va đập	J	95	82	95
• Average / Trung bình	J	91		
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 6.71	10.01 x 6.71	10.01 x 6.71
• Impact energy Công va đập	J	28	34	39
• Average / Trung bình	J	34		
				
Figure / Hình ảnh minh họa				

Original sample photo(s) / Hình ảnh mẫu nhận:



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*** END OF THE REPORT ***

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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MÔI HÁN
DD-FCAW-134B
WPS-DD-FCAW-134
Base material : PL12, A572 Gr. 50
Size: PLATE 300x200x12 mm to 300x200x12 mm
Filler material : AWS A5.20 E71T-1C/-9C-J
Trade name : Hyundai Supercored 71H (Ø1.2mm)
Lot No: MF24340E71

Project / Dự án : Evapco – Aspen Power Station Project

Item / Hàng mục : Procedure Qualification Record (PQR) Test for Ducting

Quantity / Số lượng mẫu : 01 sample / mẫu

Description / Mô tả mẫu : 500 x 340 x 12 (mm)

Date of receiving / Ngày nhận mẫu : 01-Aug-2024

Date of testing / Ngày thử nghiệm : From / từ 01-Aug-2024 to / đến 02-Aug-2024

Test method / Phương pháp thử : ASME BPVC.VIII.1-2023 Rules for Construction of Pressure Vessels

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of

SGS Vietnam LTD
Thay mặt công ty

SGS Vietnam Ltd

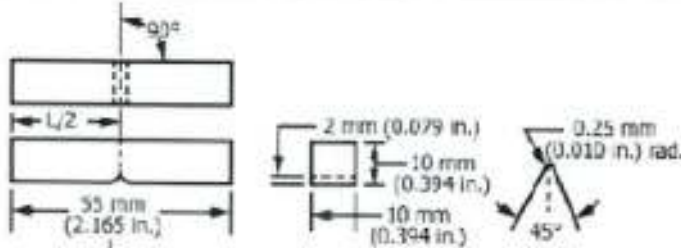
Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

Please extract your time to give us your feedback as this will help our service better. Thank you!



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Characteristics / Tên chỉ tiêu		Test result / Kết quả thử nghiệm		
Impact test at -45°C / Thử va đập ở -45°C		1	2	3
Weld metal / Kim loại mối hàn				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 10.00	10.01 x 10.01	10.01 x 10.01
• Impact energy Công va đập	J	88	69	112
• Average / Trung bình	J	90		
Heat affected zone / Vùng ảnh hưởng nhiệt				
• Dimension of specimens Kích thước mẫu thử	mm	10.01 x 10.01	10.00 x 10.01	10.01 x 10.01
• Impact energy Công va đập	J	99	66	55
• Average / Trung bình	J	73		
				
Figure / Hình ảnh minh họa				

Original sample photo(s) / *Hình ảnh mẫu nhận:*



SGS authenticate the photos on original report only / *SGS chỉ xác thực các hình ảnh trong báo cáo chính thức*

*** END OF THE REPORT ***

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검 사 증 명 서

MILL TEST CERTIFICATE



당진개청소 : 충남 당진시 송악읍 북부산로 1480
1480 Bukbusan-ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : F240100222

품 명
Commodity : Hot Rolled Coil

제품규격
Specification : ASTM A572 GRADE 50

고객사
Customer : DAIDUNG CORP

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20240319-HS-0012-004

발행일자
Date of Issue : 2024-03-19

제품치수 Dimension (mm)	수량 Quantity	중량 Weight (Kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test							화학성분 Chemical Composition(%)																	단위 Unit 2 : x100 3 : x1,000 4 : x10,000 5 : x100,000		
						YP	TS	EL	YP-EL	RA	YR	YP Mtd	Division	C	Si	Mn	P	S	Cr	Ni	B	Cu	Mo	Nb	V	DR						
						(N/mm ²) ()	(%)	(%)	(%)	(%)	4			3	3	4	4	2	2	4	3	2	3	3	3							
8.00x1999xC	1	23,830	S97136	CPA46250	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987						
8.00x1999xC	1	24,050	S97136	CPA46260	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987						
8.00x1999xC	1	23,990	S97136	CPA46280	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987						
8.00x1999xC	1	24,060	S97136	CPA46290	B	424.3	546.1	21				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987						
8.00x1999xC	1	24,190	S97136	CQA03650	B	397.4	537.1	25				2	L	1,590	3	1,033	158	48	3	1	1	13	0	17	2	1,987						
*** Sub Total(004) *** 5 120,120(Kg)																																
*** Grade Total *** 22 521,580(Kg) DR = 6XC+Mn																																
*** Last Item ***																																
NOTE	* Position – T:Top, M:Middle, B:Bottom * Tensile test – Direction:Transverse, Gauge length:50mm (Rectangular), YP Mtd : 2 (0.2% off-set),5 (0.5% underload), U (Upper yield point), L (Lower yield point) * Division – P:Product analysis, L:Ladle analysis																															
	* Mill Test Certificate according to EN 10204 Type 3.1.																															
상기의 제품은 검사의 결과 지정된 규격에 합격한 것을 증명합니다. WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS																							Signature YS. Jang Chief Of Hot Rolled Quality Team									

* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 현대제철 고객포탈에서 확인하실 수 있습니다.(<http://sm.hyundai-steel.com/cs/cm/login.jsp>)
* This Mill Test Certificate is a copy that has been printed from original electronic document(with digital signing).
You are able to check an original electronic document at hyundaisteel's customer portal.(<http://sm.hyundai-steel.com/cs/cm/login.jsp>) * QRcode scanner App : 'QReal'
* [사용상 주의사항] 본 검사증명서에 명기된 규격 외 사용시 제품손상 및 안전상 문제가 발생할 수 있습니다.
* [Caution for Use] Damage on products and safety problems may arise if used other than to the spec stated in this Mill Test Certification.

Order No./계약번호 : 01S6648703

PO No./주문번호 : 01S6648703

Supplier /주문자 : POSCO INTERNATIONAL CORPORATION

Commodity /품명 : STEEL PLTAE

Customer /고객사 : DDC

Spec & Type /규격 : ASTM A572-50-TY1

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Country of Melt & Pour	Position	Tensile Test /인장시험			Division	Chemical Composition/화합성분										
							YP	TS	EL		C	Si	Mn	P	S	Cr	Ni	Cu	Mo	Nb	V
							(MPa)		(%)		(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)
*** Sub Total (0+0) ***		62	116,870 (kg)																		
12x2001x12000	PB93389601-9605	5	11,310	SB99466	KOR	T	432	553	23	L	0.1532	0.376	1.435	0.0110	0.0018	0.028	0.012	0.015	0.002	0.026	0.002
										P	0.1560	0.353	1.404	0.0113	0.0022	0.022	0.013	0.013	0.012	0.024	0.002
12x2001x12000	PB93491901-1905	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93537701-7705	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93642901-2905	5	11,310	SB99606	KOR	T	412	554	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93537601-7605	5	11,310	SB99606	KOR	T	425	557	24	L	0.1566	0.337	1.423	0.0113	0.0028	0.023	0.024	0.010	0.003	0.024	0.002
										P	0.1511	0.340	1.423	0.0107	0.0024	0.022	0.023	0.012	0.005	0.023	0.002
12x2001x12000	PB93570701-0706	6	13,572	SB99685	KOR	T	431	552	22	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662501-2506	6	13,572	SB99685	KOR	T	431	552	22	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93593601-3605	5	11,310	SB99685	KOR	T	420	554	25	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662401-2406	6	13,572	SB99685	KOR	T	420	554	25	L	0.1487	0.357	1.395	0.0110	0.0021	0.027	0.018	0.032	0.004	0.025	0.002
										P	0.1484	0.346	1.379	0.0109	0.0020	0.026	0.019	0.030	0.006	0.023	0.002
12x2001x12000	PB93662201-2206	6	13,572	SB99689	KOR	T	441	553	24	L	0.1464	0.338	1.395	0.0096	0.0020	0.027	0.015	0.014	0.003	0.024	0.002
										P	0.1492	0.329	1.375	0.0099	0.0021	0.026	0.015	0.012	0.009	0.022	0.002
12x2001x12000	PB93662001-2006	6	13,572	SB99747	KOR	T	440	557	23	L	0.1497	0.366	1.396	0.0114	0.0028	0.026	0.011	0.009	0.003	0.025	0.002
										P	0.1526	0.302	1.276	0.0115	0.0035	0.030	0.013	0.013	0.005	0.022	0.007
12x2001x12000	PB93662301-2305	5	11,310	SB99747	KOR	T	440	557	23	L	0.1497	0.366	1.396	0.0114	0.0028	0.026	0.011	0.009	0.003	0.025	0.002
										P	0.1526	0.302	1.276	0.0115	0.0035	0.030	0.013	0.013	0.005	0.022	0.007
12x2001x12000	PB93643001-3005	5	11,310	SB78995	KOR	T	412	554	24	L	0.1521	0.366	1.392	0.0101	0.0027	0.029	0.011	0.014	0.004	0.021	0.002
										P	0.1500	0.368	1.406	0.0098	0.0024	0.027	0.011	0.015	0.006	0.021	0.002

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test. Direction : Transversal, Gauge Length : 200 mm(Rectangular),
YP Method : 0.2 % off-set
* Division - L : Ladle Analysis, P : Products Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in
South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

Legal sanction can be imposed on forging. Improper use of product can cause safety issues.

Surveyor To :

B.S. Kim



CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2024 – 0744/2

Date of Issue : 2024 – 05 – 24



Customer		G-TECH CORPORATION						
Trade Name		Supercored 71H		Size		1.2 mm x 15 kg		
Lot No.		MF24340E71		Purchaser's Order No.		38-2024/GT/HW/PO		
Specification		AWS A5.20 E71T-9C-J * According to EN 10204-2004 : 3.1						
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded	
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	534	589	27	-40	72	70	78	73
AWS Spec.	≥390	490~670	≥22	≥27				

Chemical Composition (%)

Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
	0.04	0.49	1.29	0.017	0.008	0.37	0.02	0.01	0.01	0.02
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35

- * Hydrogen Content (mℓ/100g) (AWS A4.3) : 2.86
- * Radiographic Test : ACCEPT
- * Fillet Weld Test : GOOD

Others
CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-9J-H8)

* Q'ty: 3,600 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.

Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 01/04

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN

WELDING PROCEDURE QUALIFICATION RECORD (PQR)

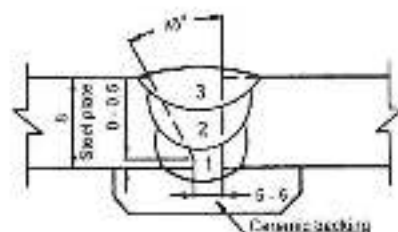
Số PQR/ PQR N^o : PQR-DD-FCAW-251 (28/05/2024)
Quá trình hàn/ Welding Process(es) : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp/ Type(s) : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable Code : ASME BPVC Section IX, 2023 Edition
Thợ hàn/ Welder's name (ID) : Nguyễn Trí Hùng (042091005783/ DDC-WD-0716)
Ngày hàn/ Date of welding : 15/05/2024

CHI TIẾT MỐI NỐI/ JOINTS (QW-402)

Loại mối nối/ Joint Design : Single-Bevel groove weld
Khe hở chân/ Root Spacing : (5 – 6) mm
Đệm lót/ Backing : Có/ Yes

Loại vật liệu đệm lót/ Backing Material (Type):

- ☐ Kim loại hàn/ Weld metal ☒ Phi kim loại/ Nonmetallic
☐ Kim loại không nóng chảy/ Nonfusing Metal



KIM LOẠI CƠ BẢN/ BASE METALS (QW-403)

Đặc tính kỹ thuật/ Specification : ASME SA-516 với/ to ASME SA-516
Loại/ Type/ Grade or UNS N^o : Gr. 70 với/ to Gr. 70
Số ký hiệu vật liệu (P-Number)/ P-Number : 1 với/ to 1
Nhóm vật liệu/ Group : 2 với/ to 2
Chiều dày mẫu kiểm tra/ Thickness of Test Coupon : 8 mm
Đường kính mẫu kiểm tra/ Diameter of Test Coupon (OD) : —
Chiều dày lớp hàn lớn nhất/ Maximum Pass Thickness : 3,5 mm
Chiều rộng lớp hàn lớn nhất/ Maximum weld bead width : 14 mm
Thông tin khác/ Other : Số nhiệt luyện/ Heat No.: S67929

KIM LOẠI HÀN/ FILLER METALS (QW-404):

Đặc tính kỹ thuật/ SFA Specification : SFA-5.20
Loại/ AWS Classification : E71T-1C
Số F/ F-No. : 6
Số A/ A-No. : 1
Đường kính kim loại hàn/ Size of Filler Metals : 1,2 mm
Dạng kim loại hàn/ Filler Metal Product Form : Dây lõi thuốc/ Flux-cored wire
Chiều dày kim loại mối hàn/ Weld Metal Thickness : 8 mm
Thuốc hàn/ Flux : Loại/ Type : Không áp dụng/ Not applicable
Tên thương mại/ Trade Name :
Thông tin khác/ Other : Filler Metal Trade Name: Hyundai supercored 71H

THỂ HÀN/ POSITIONS (QW-405)

Thể hàn rãnh/ Position(s) of Groove : 1G
Hàn ở vị trí đứng/ Welding Progression : —
Thể hàn góc/ Position(s) of Fillet : —



Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 02/04

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT (QW-406)

Nhiệt độ nung nóng/ Preheat Temperature, Minimum 32 °C (Ambient)
Nhiệt độ giữa các lớp/ Interpass Temperature, Maximum 150 °C
Thông tin khác/ Other -

XỬ LÝ NHIỆT SAU KHI HÀN/ POSWELD HEAT TREATMENT (QW-407): N/A

Nhiệt độ/ Temperature -
Thời gian duy trì/ Time -
Thông tin khác/ Other -

KHÍ BẢO VỆ/ GAS (QW-408)

	Loại/ Type	Thành phần/ Composition	Lưu lượng/ Flow Rate
Bảo vệ môi hàn/ Shielding	CO ₂	99,8 %	(18 – 20) L/min
Bảo vệ chân môi hàn/ Backing	---	---	---
Thông tin khác/ Other	---		

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS (QW-409)

Lớp hàn Pass	Quá trình hàn Process	Kim loại hàn/ Filler metal		Dòng điện/ Current		Điện thế/ Volt Range, (V)	Tốc độ hàn Travel speed range, (cm/min)	Nhiệt cấp vào Heat input (kJ/mm)
		Loại Class.	Đường kính/ Diam. (mm)	Loại, phân cực/ Type, polarity	Cường độ Amps range, (A)			
1	FCAW	E71T-1C	1,2	DCEP	170 – 180	24 – 25	18 – 19	1,29 – 1,50
2	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	27 – 28	0,96 – 1,09
3	FCAW	E71T-1C	1,2	DCEP	195 – 205	23 – 24	26 – 30	1,00 – 1,14

Điện cực Vonfram/ Tungsten Electrode Type and Size -
Kiểu chuyển dịch kim loại hàn/ Mode of Metal Transfer -
Thông tin khác/ Other -

KỸ THUẬT HÀN/ TECHNIQUE (QW-410)

Dạng quỹ đạo điện cực/ String or Weave Bead Quỹ đạo zic zac/ Weave bead
Lỗ miệng ống khí/ Gas Cup Size Ø14 mm
Làm sạch giữa các lớp/ Interpass Cleaning Mài/ Grinding
Phương pháp tạo rãnh mặt sau/ Method of Back Gouging Không áp dụng/ Not applicable
Khoảng cách ống kẹp-vật hàn/ Contact Tube to Work Distance (16 – 17) mm
Hàn một hay nhiều lớp/ Multiple or Single Pass (Per Side) Nhiều lớp/ Multile pass
Hàn một hay nhiều điện cực/ Multiple or Single Electrodes Một điện cực/ Single electrode
Khoảng cách điện cực/ Electrode Spacing -
Thông tin khác/ Other -

Số: KT3-01655BCK4/5

Ngày: 28/05/2024
Trang số: 03/04

KẾT QUẢ KIỂM TRA MỎI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/1)
- Kiểm tra siêu âm/ *Ultrasonic examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/2)
- Thử kéo/ *Tensile test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu <i>Specimen N°</i>	Kích thước mẫu <i>Dimension, mm</i>	Tiết diện <i>Area, mm²</i>	Lực kéo đứt <i>Tensile load, KN</i>	Giới hạn bền <i>Tensile strength, MPa</i>	Vị trí hỏng <i>Broken position</i>	Đánh giá theo/ <i>Assess to ASME, section IX</i>
TS1	19,1 x 7,76	148,2	85,8	579	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>
TS2	18,9 x 7,84	148,2	86,2	582	Kim loại cơ bản <i>Base metal</i>	Đạt/ <i>Pass</i>

- Thử uốn/ *Guided bend test*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-01655BCK4/3)

Mẫu/ <i>Specimen N°</i>	Kiểu uốn/ <i>Type of bend</i>	Kết quả/ <i>Result</i>	Ghi chú/ <i>Remarks</i>
FB1	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB1	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	
FB2	Uốn mặt/ <i>Face bend</i>	Đạt/ <i>Pass</i>	
RB2	Uốn chân/ <i>Root bend</i>	Đạt/ <i>Pass</i>	

- Thử va đập/ *Impact test at -45°C*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report reference N°*: KT3-01655BCK4/3)

Vị trí rãnh và đập <i>Notch location</i>	Công va đập <i>Absorbed energy (J)</i>			Trung bình <i>Average (J)</i>	Mức chất lượng <i>Criteria(*) (J)</i>	Đánh giá <i>Assessment</i>
	1	2	3			
Kim loại mối hàn/ <i>Weld metal</i>	60	44	48	51	Min. 27	Đạt/ <i>Pass</i>
Vùng ảnh hưởng nhiệt/ <i>Heat affected zone</i>	142	114	120	125		Đạt/ <i>Pass</i>

(*) Theo yêu cầu của khách hàng/ *As customer's request*.

Số: KT3-01655BCK4/5

Ngày: 28/05/2024

Trang số: 04/04

6. Kiểm tra tổ chức thô đại/ *Macro examination*: Đạt yêu cầu / *Acceptable*.

(Kết quả thử nghiệm tham khảo số/ *Test report reference No*: KT3-01655BCK4/3)

7. Các thử nghiệm khác/ *Other tests*: Không áp dụng/ *Not applicable*.

Chúng tôi, chứng nhận rằng thông tin trong báo cáo này là đúng, mỗi hàn kiểm tra được chuẩn bị, hàn và thử nghiệm phù hợp với các yêu cầu quy định tại chương IX, Quy phạm nồi hơi và thiết bị áp lực ASME (2023)/ *We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code (2023).*

	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	28/05/2024	28/05/2024
SIGN		
NAME	La Vinh Nhon	Nguyễn Anh Triết

KT3-01655B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 01/02

- Tên mẫu : MỎI HÀN TRÊN MẪU THÉP TẮM.
Name of sample : **WELDS ON STEEL PLATE COUPON.**
- Mô tả mẫu : Mối hàn trên mẫu thép tấm như hình ảnh ở mục 11.
Sample description : **Welds on steel plate coupon as photograph at item 11.**
Kích thước/ Dimension: (700 x 400 x 8) mm.
- Số lượng mẫu : 02 mẫu/ Samples.
Testing quantity
- Ngày nhận yêu cầu : 15/05/2024.
Date of receiving
- Ngày thử nghiệm : 15/05/2024.
Date of testing
- Khách hàng : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
Customer : **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
Test Location
- Phương pháp thử : ASME BPVC.V-2023 (Article 9).
Test method
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results : See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KI, GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

KT3-01655B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 02/02

10. Điều kiện thử/ Test condition:

☒ Butt weld	☐ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal Thk.: 8 mm
Welding Procedure Spec.: ☐ SMAW	☒ SAW	☒ FCAW	☐ GTAW	
State of Examination : ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation : ☒ As weld	☒ Brushing	☐ Other.....		
Surface Temperature : ☒ (28 + 33) °C	☐ .. °C			
IR Thermometer : FLUKE 561	Serial: 22020241	Next Cal. Due: 06/2024		
Lighting Condition : ☒ Day light > 1000 lux	☐ Artificial > 1000 lux			
Weld Gauge : Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra ngoại quan mối hàn/ Visual examination results:

Tên mẫu Name of test coupon	Chiều dài kiểm tra Tested length, (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	
DD-FCAW+SAW- 252 SA-516 GR.70	700 x 2 Surfaces	-	-	-	✓	

(*) Chú thích/ Legends:

IF : Không ngấu / Incomplete fusion	C : Nứt / crack	CR : Lõm chân / Concave Root
OL : Chồng mép / Overlap	P : Rỗ / Porosity	U : Cháy cạnh / Undercut
SI : Ngậm xỉ / Slag inclusion	R : Quả lồi / External Reinforcement	CS : Lõm mặt / Concave Surface
WP : Biên dạng / Weld bead profile	RO : Ô xi hóa chân / Root Oxidation	PR : Quả thô / Protrusion

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 01/02

1. Tên mẫu : MỐI HÀN TRÊN MẪU THÉP TẮM.
Name of sample : *WELDS ON STEEL PLATE COUPON.*
2. Mô tả mẫu : Mối hàn trên mẫu thép tấm như hình ảnh ở mục 11.
Sample description : *Welds on steel plate coupon as photograph at item 11.*
Kích thước/ Dimension: (700 x 400 x 8) mm.
3. Số lượng mẫu : 02 mẫu/ Samples.
Testing quantity
4. Ngày nhận yêu cầu : 15/05/2024.
Date of receiving
5. Ngày thử nghiệm : 17/05/2024.
Date of testing
6. Khách hàng : CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.
Customer : *DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.*
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
7. Nơi thử nghiệm : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
Test Location
8. Phương pháp thử : ASME BPVC.V-2023 (Article 4).
Test method
9. Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results : *See details on page 02/02.*

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

KT3-01655BC
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
Page 02/02

10. Điều kiện thử nghiệm/ Test condition:

Surface condition ☒ As weld ☐ Brushing ☐ Painting ☐ Grinding	Scanned from ☐ One side ☒ Both sides ☒ One surface ☐ Both surfaces	Preparation ☐ Single V ☐ Double V ☒ Backing ☒ Single Bevel	Couplant ☐ Water ☐ Oil ☒ Cellulose paste ☐	Material ☒ Carbon steel ☐ Stainless steel Diam.: mm Thick.: 8 mm	
Ultrasonic instruments ☐ USM 35X DAC (S/N: 9502a) ☐ USM 35X DAC (S/N: 12873a) ☐ USM 35X DAC (S/N: 12875a) ☐ USM 35X DAC (S/N: 14258a) ☒ USM 35X DAC (S/N: 16141a) ☐ POSITECTOR UTG (S/N: 823482)	Probes: Model Angle (°) C340151 0 MSEB 0 MWS 70	Calibration blocks ☐ V1 – IIW ☐ DAC Ø1,5 mm ☒ V2 – IIW ☐ DAC Ø3,0 mm	Freq. (MHz) - 4 4	Serial - 54339 76970	Ref. (dB) - 48 49
Transfer correction: + 1 dB;			Reference block ☒ T20/Ø2,5 ☐ T35/Ø3,0 (ASME T-434.2.1) ☐		
			Scanning level: Ref. + 12 dB		

11. Hình ảnh của mẫu/ Photograph of tested coupons:



12. Kết quả kiểm tra siêu âm/ Ultrasonic testing results:

Tên mẫu Name of test coupon	Chiều dày Thickness (mm)	Chiều dài thử Test length (mm)	Chỉ thị / Indications			Đánh giá theo/ Assess to ASME BPVC.IX-2023	
			Biên độ Actual level (Db)	Loại ¹⁾ Type ¹⁾	Chiều dài Length (mm)	Đạt Acceptable	Không đạt Unacceptable
DD-FCAW-251 SA-516 GR.70	8	700	-	-	-	✓	
DD-FCAW+SAW-252 SA-516 GR.70	8	700	-	-	-	✓	
¹⁾ Chú thích/ Legends: RI : Sau khi sửa/ After repairing. LOF : Không ngậm / Lack of fusion C : Nứt / crack SI : Ngậm xỉ / Slag inclusion LOP : Không thấm / Lack of penetration P : Rỗ / Porosity U : Cháy cạnh / Undercut							

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM.**
WELDS ON STEEL PLATE COUPON.
DD-FCAW-251
- Mô tả mẫu
Sample description : **Mỏi hàn trên thép tấm như hình ảnh bên dưới.**
Weld on steel plate as photograph below.
(700 x 400 x 8) mm.
- Số lượng mẫu
Testing quantity : **01 mẫu/ Sample.**
- Ngày nhận yêu cầu
Date of receiving : **15/05/2024.**
- Ngày thử nghiệm
Date of testing : **22/05/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DÙNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **ASME BPVC.IX - 2023.**
ASME BPVC.VIII.1 - 2023.
SA-370 (ASME BPVC.III.A-2023)
- Điều kiện và kết quả thử nghiệm
Test condition & results : **Xem chi tiết ở trang 02/03 và 03/03.**
See details on page 02/03 and 03/03.



TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

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PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

28/05/2024
Page 02/03

10. Kết quả/ Test results:

Tên chỉ tiêu Specification		Kết quả thử nghiệm Test result			
A. THỬ KÉO/ TENSILE TEST		Mẫu/Sample 1		Mẫu/Sample 2	
10.1. Kích thước mẫu thử, Dimension of specimen	mm	19,1 x 7,76		18,9 x 7,84	
10.2. Lực kéo đứt, Tensile load	kN	85,8		86,2	
10.3. Giới hạn bền kéo, Tensile strength	MPa	579		582	
10.4. Vị trí đứt Broken position		Kim loại cơ bản Parent metal		Kim loại cơ bản Parent metal	
B. THỬ UỐN/ BEND TEST		FB1	RB1	FB2	RB2
10.5. Kích thước mẫu thử, Dimension of specimen	mm	38,0 x 7,70	38,1 x 7,42	37,9 x 7,70	37,7 x 7,36
10.6. Góc uốn, Bending angle	độ	180	180	180	180
10.7. Đường kính gối uốn, Plunger diameter	mm	32	32	32	32
10.8. Kết quả thử uốn mặt Face bend test result		Đạt Pass	-	Đạt Pass	-
10.9. Kết quả thử uốn chân Root bend test result		-	Đạt Pass	-	Đạt Pass
Tên chỉ tiêu/ Specification		Kết quả thử nghiệm/ Test result			
C. THỬ VA ĐẬP/ IMPACT TEST AT -45°C		Kim loại mối hàn/ Weld metal			
10.10. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.11. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	60	44	48	51
		Vùng ảnh hưởng nhiệt/ Heat affected zone			
10.12. Kích thước mẫu thử, Dimension of specimen	mm	10 x 6,7	10 x 6,7	10 x 6,7	Trung bình Average
10.13. Năng lượng hấp thụ (KV ₂), Absorbed energy	J	142	114	120	125

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
 Test results are valid for the samples submitted only, and they are not a certificate of product.
 2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của mỗi gia mẫu. / Name of sample(s) and customer are written at customer's request.
 3. Độ không đảm bảo đo mở rộng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ k = 2, phù hợp chuẩn tương ứng với 95 % độ tin cậy.
 The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
 4. Không được trích sao một phần phiếu kết quả thử nghiệm này của khách có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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 5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ email info@quatest3.com.vn để biết thêm thông tin.
 Please contact QUATEST 3 at the email info@quatest3.com.vn for further information about test report.

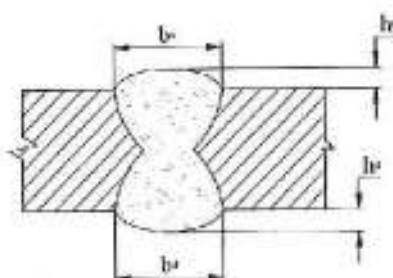
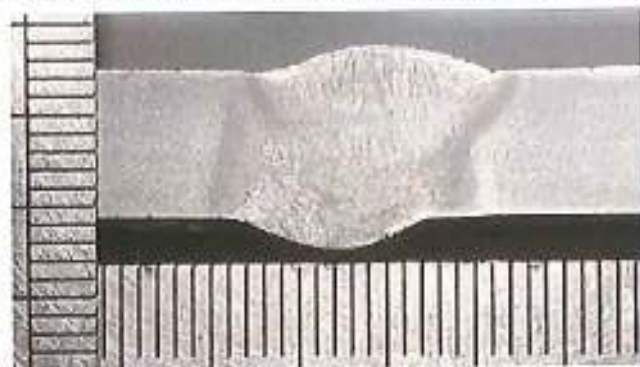


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 CK4/3

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

28/05/2024
 Page 03/03

D. TỔ CHỨC THỎ DẠI / MACRO STRUCTURE



Kết quả thử nghiệm/ Test result	
Bề mặt/ Surface	Bên trong/ Internal
$b_0 = 14 \text{ mm}$ $h_u = 1,5 \text{ mm}$ - Đạt/Pass $b_a = 10 \text{ mm}$ $h_d = 1,6 \text{ mm}$ - Đạt/Pass	Mỗi hàn thấu và ngấu hoàn toàn. Trong mỗi hàn không có khuyết tật / Full penetration and complete fusion weld. No defect in weld metal.

QUATEST 3®

- Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the exactly submitted samples only, and this is not a certificate of product.
- Tên mẫu, tên khách hàng được ghi theo yêu cầu của nơi gửi mẫu. / Name of sample(s) and customer are written as customer's request.
- Hệ không chắc chắn do mở rộng được tính từ độ không đảm bảo đo chuẩn hóa với hệ số phủ k = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated at the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
- Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ Thuật 3.
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- Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ dxn@quatest3.com.vn để biết thêm thông tin.
Please contact QUATEST 3 at the email addresses dxn@quatest3.com.vn for further information about test report.

QUATEST 3	
☐ Inspected	☒ Reviewed
Ins. No.: 16853	Sign: <i>[Signature]</i>
Date: 15.5.24	<i>[Signature]</i>

검 사 증 명 서

MILL TEST CERTIFICATE

HYUNDAI STEEL

당진3점소: 호남 공진시 호남읍 북부산업로 1480
14898 Buteokan, P.O. Songok-6up, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : F23D400073

품명
Commodity : HOT ROLLED STEEL PLATE

재질규격
Specification : ASME (2021 EDITION) SA516-70-N HT+LTY

고객사
Customer : TUNG WANG STEEL

주요자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20230905-PS-0013-009

발행일자
Date of issue : 2023-09-05

제품사우 Dimension	수량 Quantity	중량 Weight (kg)	세강번호 Heat No.	제품번호 Product No.	Position	연장시험 Tensile Test						충격시험 Impact Test V Notch -40.0 °C			화합성분 Chemical Composition(%)																단위 Unit 2" x 100 3" x 1,000 4" x 10,000 5" x 100,000			
						YP	IS	EL	YP-EL	RA	YR	YP MPa	Energy	SF	LE	Oxide	C	S	Mn	P	S	Cr	N	B	Cu	Mo	Nb	Ti	V	Si	Al	CSQ		
(N/mm ²)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)								

8.00x2000x8000	1	754	567929	FD10370106	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407
8.00x2000x8000	1	754	567929	FD10370107	T	397.6	540.6	24				2	1-83 2-92 3-94 A-86			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407
8.00x2000x8000	1	754	567929	FD10370201	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407
8.00x2000x8000	1	754	567929	FD10370202	T	394.1	539.0	26				2	1-77 2-86 3-76 A-83			L	1.804	358	1.154	123	22	17	16	4	138	0	17	2	3	34	407

NOTE

• Position - T:Top, M:Middle, B:Bottom

• Tensile test - Direction:Transverse, Gauge length:200mm (Rectangular), YP Mtd :2 to 2%, off-set) 5 to 0.5% underload, U (Upper yield point), L (Lower yield point)

• Impact test - Direction:Transverse, Spec-size:10mmX5.0mm, SF:Shear fracture, LE:Lateral Expansion

• Division - P:Product analysis, L:Laboratory analysis

• The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.

• This M/T Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE SR0175/SR0170 S for hardness test ASME SA516-70 2021 Edition

• The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied • This M/T Test Certificate is issued in accordance with EN 10204 Type 3.1

합격의 제품인 동시에 합격자점의 규정에 합치한 것을 증명합니다.

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Kim in

Chief Of Plate Quality Team



NOTE

- Position - T: Top, M: Middle, B: Bottom
- Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mtd: 1.2 to 2% off-set, 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
- Impact test - Direction: Transverse, Spec-size: 10mmX5,0mm, SF: Shear fracture, LE: Lateral Expansion
- Division - P: Product analysis, L: Lab analysis
- The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.
- This Mill Test Certificate is issued in accordance with EN 10204 (2004) TYPE 3.1, fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied, NACE MR0175/MR010-3 for hardness test ASME SA516-70 2021 Edition
- The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

입고의 제품은 검사의 결과 시험의 규격에 합격하였을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Chief Of Plate Quality Team

[Signature]

- * 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 원본에 열거된 정보로부터 확인하실 수 있습니다. (<http://www.hyundai-steel.com/qa/en/login.jsp>)
- * This Mill Test Certificate is a copy that has been printed from original electronic document with digital signature.
- * You are able to check an original electronic document at Hyundai's customer portal: <http://www.hyundai-steel.com/qa/en/login.jsp> * QRcode scanner App : "QRcode"
- * [사용 및 주의사항] 본 검사증명서와 검사의 결과 및 제품사 관련사항 및 안전의 문제가 발생할 수 있습니다.
- * [Caution for Use] Damage on products and safety problems may arise if used other than to the spot stated in this Mill Test Certificate.

CERTIFIED MATERIAL REPORT



CMR No. : HDV - 2024 - 0031

Date of Issue : 2024 - 01 - 05

Customer		-									
Trade Name		Supercored 71H				Size		1.2 mm x 15 kg			
Lot No.		MF24340A71				Purchaser's Order No.		-			
Specification		EN ISO 17832-A - T42 4 P C1 1 * According to EN 10204-2004 : 3.1 AWS A5.36 E71T1-C1A4-CS1									
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)				As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)							
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.			
	554	619	26	-40	102	95	96	96			
EN Spec.	≥420	500~640	≥20	≥47							
Chemical Composition (%)											
Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
	0.04	0.40	1.37	0.017	0.005	0.38	0.02	0.01	0.02	0.01	0.01
EN Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05
* Hydrogen Content (nℓ/100g) : 5.76											
Others CSA W48-18 E491T1-C1A4-CS1-H8 (E491T-8J-H8)						QUATEST 3 ☒ Inspected ☒ Reviewed Ins. No.: 16558 Sign: <i>[Signature]</i> Date: 15.5.24 <i>Le Tien Han</i>					
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.											
Approved by QA Manager Pham Dinh Anh Vu											

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
ASME IX:2021

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW+FCAW-196

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Mar 10, 2023	All	Client Review	L.V.Nhon	T.H.P.Vu
1	Nov 20, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2021)

Page: 1/2

Applicable Code: **ASME Section IX - 2021 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-GTAW+FCAW-196** Revision No. **1** Date **20 Nov 2025**

Supporting PQR No.(s): **PQR-DD-GTAW+FCAW-196** Date **10 Mar 2023**

Welding Process(es) **GTAW+FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Manual, Semi-Automatic, Mechanized**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**

Root Spacing **As per fabrication drawing**

Backing **With or without**

Retainer **None**

Backing Material **Base metal or weld metal**

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **1** Group No. **1, 2** to P-No. **1** Group No. **1, 2**

OR

Specification and type/grade or UNS Number **None**

to Specification and type/grade or UNS Number **None**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **1.5 - 16 (mm) for groove welds without toughness requirement** **All for fillet welds**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other

FILLER METALS (QW-404)

Process	GTAW	FCAW
Spec. No. (SFA)	SFA-5.18	SFA-5.20
AWS No. (Class)	ER70S-6 with Mn ≤ 1.6% and Si ≤ 1.0%	E71T-1C
F-No.	6	6
A-No.	1	1
Size of Filler Metals	Ø1.6 - 2.4 (mm)	Ø1.2 - 1.6 (mm)
Supplemental Filler Metal	None	None
Filler Metal Product Form	Solid rod	Flux cored wire
Filler Metal Brand Name	TG-S51T or Any within SFA specification & classification.	SC-71LH or Any within SFA specification & classification.
Weld Metal		
Deposited Thickness:		
Groove	Max. 6 mm	Max. 10 mm
Fillet	All	All
Electrode-Flux (Class)	None	None
Flux Type	None	None
Flux Trade Name	None	None
Consumable Insert	None	None
Other	None	None



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2021)

Page: 2/2

Applicable Code: **ASME Section IX - 2021 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-GTAW+FCAW-196** Revision No. **1** Date **20 Nov 2025**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **GTAW FCAW**
Yes Yes
Vertical down **None None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**
Interpass Temperature, Maximum **161 °C**
Preheat Maintenance **None**
Other **None**

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Ar	100%	10 - 20 L/min.
Shielding	CO₂	100%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: Minimum purity 99.99%Ar for GTAW process and 99.8% CO₂ for FCAW process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	GTAW	ER70S-6	1.6 - 2.4	DC EN	104 - 127	None	0.7 - 0.9	10 - 12	75 - 91	None
Hot	GTAW	ER70S-6	1.6 - 2.4	DC EN	167 - 204	None	0.7 - 0.9	11 - 13	167 - 205	None
Fill	FCAW	E71T-1C	1.2 - 1.6	DC EP	180 - 220	None	0.6 - 1.2	26 - 32	493 - 603	None
Cap	FCAW	E71T-1C	1.2 - 1.6	DC EP	180 - 220	None	0.6 - 1.2	26 - 32	256 - 312	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **Ø2.4 mm, 2% Thoriated Tungsten (EWTh-2)**
Mode of Metal Transfer for GMAW (FCAW) **Globular**
Other **None**

TECHNIQUE (QW-410)

Process	GTAW	FCAW
String or Weave Bead	Weave Bead	String & Weave (Max. width 15 mm)
Orifice, Nozzle, or Gas Cup Size	All size	No. 6
Initial and Interpass Cleaning (Brushing, Grinding, etc.)	Brushing or Grinding	Brushing or Grinding
Method of Back Gouging	Grinding or Air carbon arc	Grinding or Air carbon arc
Oscillation	None	None
Contact Tube to Work Distance	None	15 - 25 (mm)
Multiple or Single Pass (Per Side)	Multiple pass	Multiple pass
Multiple or Single Electrode	Single	Single
Electrode Spacing	None	None
Peening	None	None
Other	None	None

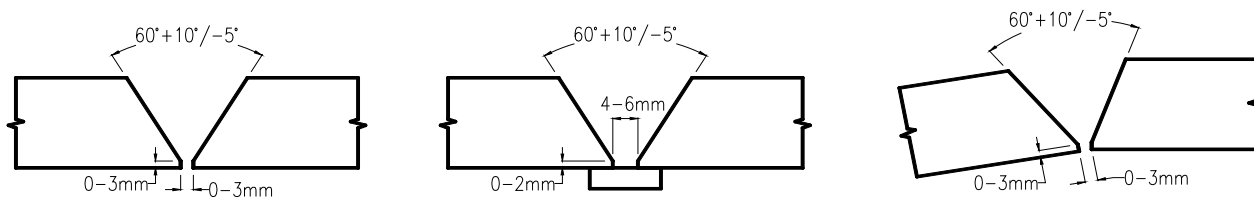
Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Certified by Welding Coordinator:
La Vinh NhonApproved by: **Truong Hoai Phong Vu (QA Manager)**Date: **20 Nov 2025**Date: **20/Nov/2025**

Welding Procedure Specification No.: WPS-DD-GTAW+FCAW-196

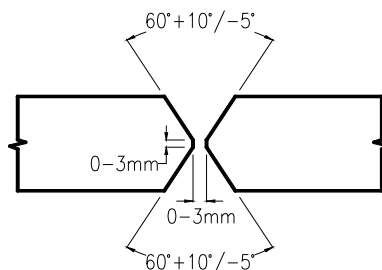
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

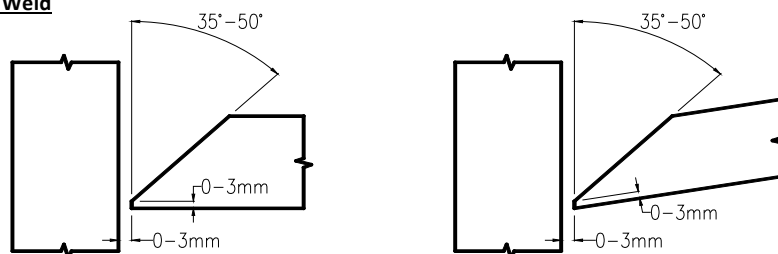
1. Single V Groove Butt Joint



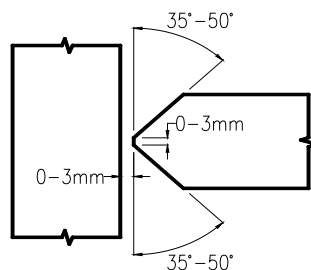
2. Double V Groove Butt Joint



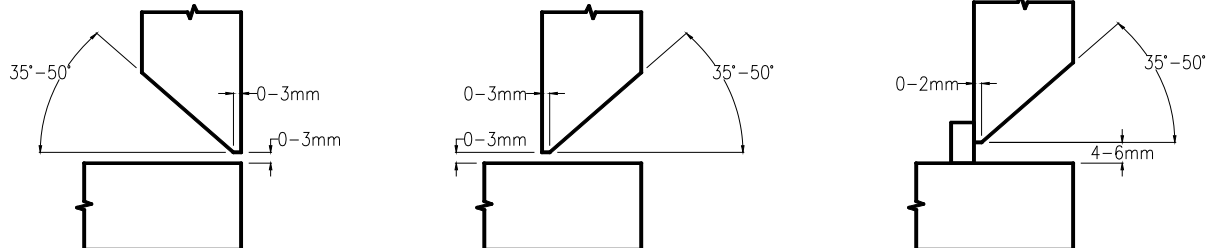
3. T-Joint: Single Bevel Groove Butt Weld



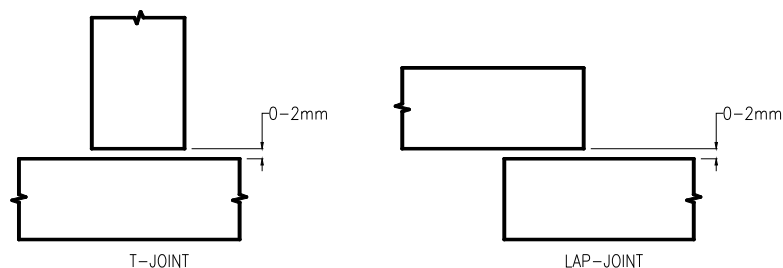
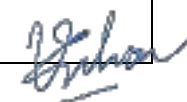
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-GTAW+FCAW-196** Date **Mar 10, 2023**

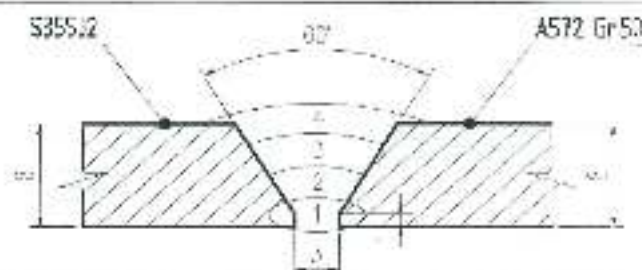
Welding Process(es) **GTAW+FCAW**

Type(s) (Manual, Automatic, Semi-Automatic) **Manual + Semi-Automatic**

JOINTS (QW-402)

Joint Type **Butt joint, single V groove**

Backing **None**

Retainer **None**


BASE METALS (QW-403)

Material Spec. **EN 10025-2 to ASTM A572**

Type/Grade, or UNS Number **S355J2 to A572 Gr.50**

P-No. **1** Group No. **2** to P-No. **1** Group No. **1**

Thickness of Test Coupon **8 mm**

Diameter of Test Coupon **Plate**

Maximum Pass Thickness **< 13 mm**

Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**

Time **None**

Other **None**

GAS (QW-408)

	Gas(es)	Percent Composition (Mixture)	Flow Rate
Shielding	Ar	99.99%	12 L/min
Shielding	CO ₂	99.8%	23 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process	GTAW	FCAW
SFA Specification	ASME SFA-5.18	ASME SFA-5.20
AWS Classification	ER70S-6	E71T-1C
Filler Metal F No.	6	6
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	Ø2.4 mm	Ø1.2 mm
Filler Metal Product Form	Solid rod	Flux cored wire
Supplemental Filler Metal	None	None
Filler Metal Trade Name	TG-551T	SC-71LH
Electrode Flux Classification	-	-
Flux Type	-	-
Flux Trade Name	-	-
Weld Metal Thickness	3 mm	5 mm
Other	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process	GTAW	FCAW
Current and Polarity	DC EN	DC EP
Amps.	See table below	
Volts	See table below	
Mode of Metal Transfer for GMAW (FCAW)	None	Globular
Other	None	None

POSITION (QW-405)

Process	GTAW	FCAW
Position(s)	1G	1G
Weld Progression (Up/Down)	-	-
Other	None	None

PREHEAT (QW-406)

Preheat Temperature	35 °C (Ambient)
Interpass Temperature	Max. 106 °C
Other	None

TECHNIQUE (QW-410)

Process	GTAW	FCAW
String or weave Bead	Weave	Weave
Oscillation	None	None
Method cleaning	Grinding, brushing	
Method back gouge	None	None
Multipass or single pass (per side)	Multipass	Multipass
Multiple or single Electrodes	Single	Single
Peening	None	None
Use of thermal processes	None	None
Other	None	None

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler	Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/Interpass Temp. (°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
1	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	115	11	83	85 (Ambient)	1.5	Ar, 12 L/min, 0.5
2	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	185	12	186	55	1.5	Ar, 12 L/min, 0.7
3	FCAW	AWS A5.20 / E71T-1C	1.2	DC EP	200	29	548	31	3.5	CO ₂ , 22 L/min, 0.6
4	FCAW	AWS A5.20 / E71T-1C	1.2	DC EP	200	29	548	31	3.5	CO ₂ , 22 L/min, 1.2



Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No.: **PQR-DD-GTAW-FCAW-196**

Date: **Mar 10, 2023**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
T51	19.0	8.0	152.0	87.8	578	Ductile and base metal (S355J2)
T52	19.1	7.9	150.9	86.7	574	Ductile and base metal (S355J2)

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
FB1 (Face bend)	Pass
RB1 (Root bend)	Pass
FB2 (Face bend)	Pass
RB2 (Root bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break [Y/N]
				ft lb or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	6.7 x 10.0	-20 °C	80 (J)			N
WM 2	Weld metal	6.7 x 10.0	-20 °C	66 (J)			N
WM 3	Weld metal	6.7 x 10.0	-20 °C	70 (J)			N
HAZ 1	HAZ (A572)	6.7 x 10.0	-20 °C	136 (J)			N
HAZ 2	HAZ (A572)	6.7 x 10.0	-20 °C	142 (J)			N
HAZ 3	HAZ (A572)	6.7 x 10.0	-20 °C	148 (J)			N
HAZ 7	HAZ (S355J2)	6.7 x 10.0	-20 °C	58 (J)			N
HAZ 8	HAZ (S355J2)	6.7 x 10.0	-20 °C	62 (J)			N
HAZ 9	HAZ (S355J2)	6.7 x 10.0	-20 °C	58 (J)			N

Comments: **Acceptable**

Fillet-Weld Test (QW-180)

Result - Satisfactory: **Yes**
No
Yes
No

Macro - Results: **Acceptable**

Hardness HV10 (Other Tests)

Line	Base metal (Q355D)			HAZ (Q355D)			Weld metal			HAZ (S355J2)			Base metal (S355J2)			Results
A	167	163	171	193	193	204	209	202	189	226	226	210	174	186	180	Acceptable
B	183	170	171	179	182	188	181	187	184	212	197	184	175	178	180	Acceptable

Welder's Name: **Phan Hong Thanh**

Clock No.: **None**

Stamp (ID) No.: **CT-WD-0541**

Welder's Name: **Tran Ngoc Em**

Clock No.: **None**

Stamp (ID) No.: **DDC-WD-0446**

Tests Conducted by: **Quatest 3**

Laboratory Test No.: **KT3-00375ACK3/3**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX - 2021 Edition of the ASME Boiler and Pressure Vessel Code.

Organization: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:
La Vinh Nhon

Approved by: **Truong Hoai Phong VU (QA Manager)**

Date: **Mar 10, 2023**

Date: **Mar 10, 2023**

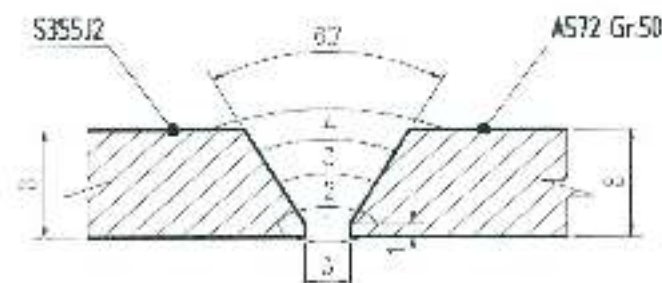
MAI HOANG KHANH

WELDING PARAMETERS RECORD

(Doc No.: WPR-GTAW+FCAW-196)

Company name: DAIDUNG CORPORATION
PQR No: PQR-DD-GTAW-FCAW-196
Test Coupon: DD-GTAW+FCAW-196
Test position: 1G for both of GTAW and FCAW
Base metal:
- Specification: EN 10025-2 S355J2 / ASTM A572 Gr 50
- Dimension: PL8 x 200 x 650 / PL8 x 180 x 650
- Heat/Product No.: 22201711A / R58087
- Certificate No.: 220331Z00358 / 20201009-IIS-001-004
Filler metal:
- Specification: AWS A5.18 / A5.20
- Classification: ER70S-6 (Ø2.4) / E71T-1C (Ø1.2)
- Trade name: Kobelco TG-851T / Hyundai SC-711H
- Batch/Lot No.: FK5008R1255 / M722120K71
- Certificate No.: KG20 D02 07-20 / HV-2022-1143
Gas: Argon (99.99%) / CO₂ (99.8%)
Welder Name: Phan Hong Thanh; Tran Ngoc Em
I.D: CT-WD-0541; DDC-WD-0446

Joint Sketch: Date: Feb 17, 2023



Root face: 1 mm
Root gap: 3 mm
Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
1	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	115	11	7	38	635	85	55 (Ambient)	Ar, 12 L/min.	0.9	5.5	1.5	
2	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	185	12	3	25	635	185	55	Ar, 12 L/min.	0.7	7	1.5	
3	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	200	29	1	9	630	548	31	CO ₂ , 22 L/min.	0.6	9	3.5	
4	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	200	29	7	13	630	264	105	CO ₂ , 22 L/min.	1.2	15	3.5	
		Conducted by Welding Coordinator														
Name:		La Vinh Nhon														
Signature:																
Date:		Feb 17, 2023														
																



KT3-00375ACK3/3

PHIẾU KẾT QUẢ THỬ NGHIỆM

TEST REPORT

09/03/2023
Page 01/05

1. Tên mẫu : **QUY TRÌNH HÀN DD - GTAW + FCAW - 196**
Name of sample **GTAW+FCAW - 1G - Single V**
Material: S355J2 + A572 Gr.50
Filler: Kobeleo TG-S5IT (Ø2.4)
Hyundai SC-71LH (Ø1.2)
Welder: CT-WD-0541;
DDC-WD-0446
2. Số lượng mẫu : **01**
Quantity
3. Mô tả mẫu : **Mẫu là thép tấm có mối hàn, kích thước (8 x 380 x 650) mm**
Sample description **The as-received sample is welded steel plate**
4. Ngày nhận mẫu : **20/02/2023**
Date of receiving
5. Ngày thử nghiệm : **27/02/2023 & 08/03/2023**
Date of testing
6. Nơi gửi mẫu: **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG**
Customer **Lô số 38, Khu C, Đường D1, Khu công nghiệp An Hạ,**
Xã Phạm Văn Hai, Huyện Bình Chánh, Thành phố Hồ Chí Minh
7. Kết quả thử nghiệm : **Xem trang 02/05...05/05**
Test results **See pages**

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL TESTING LAB.


Phạm Văn Út

TL.GIÁM ĐỐC / PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.


Nguyễn Văn Tùng

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là bằng chứng chất lượng sản phẩm.
Test results are valid for the sample submitted/complex only, and this is not a certificate of product.

2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của nơi gửi mẫu. / Name of sample and customer are written as customer's request.

3. Độ không đảm bảo đo mô phỏng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ hồi = 2, phân bố chuẩn trong ứng với 95 % độ tin cậy.
The reported expanded uncertainty is based on the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

4. Không được trích sao một phần hoặc bản gốc thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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5. Mọi thắc mắc về kết quả khách hàng liên hệ bộ theo dõi của Quatest 3 tại địa chỉ: cs@quatest3.com.vn và ng@quatest3.com.vn để biết thêm thông tin.
Please contact Quatest 3 at the email addresses địa chỉ: cs@quatest3.com.vn and ng@quatest3.com.vn for further information about test report.

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu <i>Specification</i>	Phương pháp thử <i>Test method</i>	Kết quả thử nghiệm <i>Test result</i>			
		Sample TS1		Sample TS2	
THỬ KÉO/ TENSILE TEST	ASME IX - 2021	19,0 x 8,00		19,1 x 7,90	
7.1. Kích thước mẫu thử, mm <i>Dimension of specimen</i>		87,8		86,7	
7.2. Lực kéo đứt, kN <i>Tensile load</i>		578		574	
7.3. Giới hạn bền kéo, MPa <i>Tensile strength</i>		Kim loại cơ bản <i>Parent metal</i> (S355J2)		Kim loại cơ bản <i>Parent metal</i> (S355J2)	
7.4. Vị trí đứt <i>Broken position</i>		FB1	RB1	FB2	RB2
THỬ UỐN/ BEND TEST	ASME IX - 2021	37,8 x 7,72	37,6 x 7,60	37,8 x 7,66	38,1 x 7,80
7.5. Kích thước mẫu thử, mm <i>Dimension of specimen</i>		180	180	180	180
7.6. Góc uốn, độ <i>Bending angle</i>		32	32	32	32
7.7. Đường kính gồi uốn, mm <i>Puncher diameter</i>		Đạt <i>Pass</i>	-	Đạt <i>Pass</i>	-
7.8. Kết quả thử uốn mặt <i>Face bend test result</i>		-	Đạt <i>Pass</i>	-	Đạt <i>Pass</i>
7.9. Kết quả thử uốn chân <i>Root bend test result</i>					



1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu đã được kiểm tra và được phân bổ này không phải là sản phẩm.
Test results are valid for the sample submitted completely only, and this is not a certificate of product.

2. Tên mẫu, số kích thước được ghi theo yêu cầu của bên gửi mẫu. 3. Nếu có khuyết tật và/hoặc các sai lệch do khách hàng gây ra.
The name, size and dimensions are given according to the requirements of the customer's request.

4. Hệ thống đánh giá độ tin cậy được tính từ dữ liệu không đánh giá độ chính xác của kết quả thử nghiệm k=1, phân bố chuẩn tương ứng với 95% độ tin cậy.
The system of evaluation of the reliability of the results is calculated on the basis of the results of the tests without taking into account the accuracy of the results. k=1, at 95% confidence level.

5. Không được trích sao mà phải phân phối lại: kết quả thử nghiệm này nên không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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6. Mọi thắc mắc về kết quả khách hàng liên hệ theo địa chỉ info@quatest3.com.vn và www.quatest3.com.vn để biết thêm thông tin.
Please contact Quatest 3 at the email address info@quatest3.com.vn and by www.quatest3.com.vn for further information about our report.

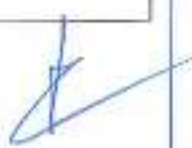
KT3-00375ACK3/3

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TEST REPORT

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result		
THỬ VA ĐẬP Ở - 20 °C IMPACT TEST AT - 20 °C	ASME IX - 2021	Kim loại mối hàn Weld metal		
7.10. Kích thước mẫu thử, Dimension of specimen	mm	6,7 x 10	6,7 x 10	6,7 x 10
7.11. Năng lượng hấp thụ (KV _s), Absorbed energy	J	80	66	70
		Vùng ảnh hưởng nhiệt – A572 Heat affected zone		
7.16. Kích thước mẫu thử, Dimension of specimen	mm	6,7 x 10	6,7 x 10	6,7 x 10
7.17. Năng lượng hấp thụ (KV _s), Absorbed energy	J	136	142	148
		Vùng ảnh hưởng nhiệt - S355J2 Heat affected zone		
7.18. Kích thước mẫu thử, Dimension of specimen	mm	6,7 x 10	6,7 x 10	6,7 x 10
7.19. Năng lượng hấp thụ (KV _s), Absorbed energy	J	58	62	58



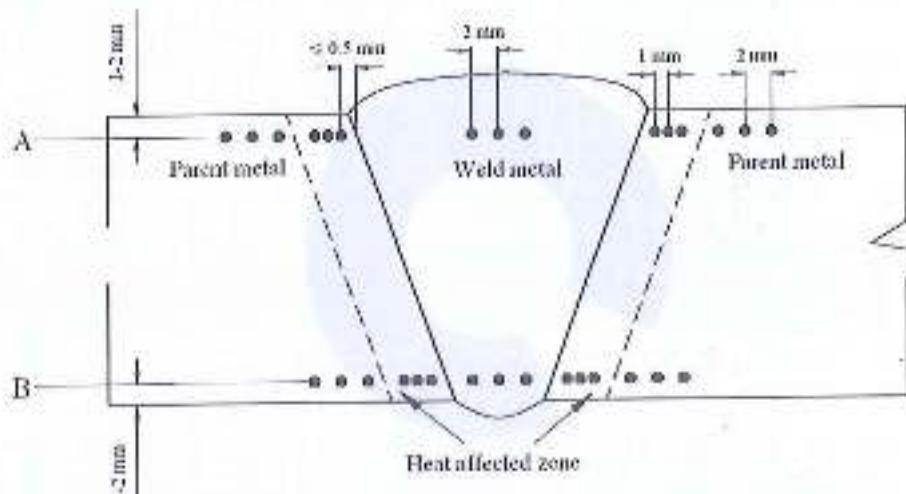
1. Các kết quả thử nghiệm ghi trong phần này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the sample submitted/accepted only, and they is not a certificate of product.
2. Tần suất, tần kích ứng được ghi theo yêu cầu của khách hàng. / Number of frequency and exposure are written as customer's request.
3. Độ không đảm bảo do mô phỏng được tính từ độ không đảm bảo đo chuẩn hóa với hệ số phủ k = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.
The uncertainty of simulation is stated as the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
4. Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung Tâm Kỹ Thuật 3.
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5. Mọi thắc mắc về kết quả kích ứng liên hệ theo địa chỉ email info@quatest3.com.vn và info@quatest3.com.vn và www.quatest3.com.vn và www.quatest3.com.vn để biết thêm thông tin.
Please contact Quatest 3 at the email addresses info@quatest3.com.vn and info@quatest3.com.vn and www.quatest3.com.vn and www.quatest3.com.vn for further information about test report.

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification			Phương pháp thử Test method ASME IX - 2021			Kết quả thử nghiệm Test result									
ĐO CỨNG/ <i>HARDNESS</i> HV 10															
															
	Kim loại cơ bản <i>Parent metal</i> (A572 Gr.50)			Vùng ảnh hưởng nhiệt <i>Heat affected zone</i> (A572 Gr.50)			Kim loại mối hàn <i>Weld metal</i>			Vùng ảnh hưởng nhiệt <i>Heat affected zone</i> (S355J2)			Kim loại cơ bản <i>Parent metal</i> (S355J2)		
A	167	163	171	193	199	204	209	202	189	226	226	210	174	186	180
B	183	170	171	179	182	183	181	187	184	212	197	184	175	178	180

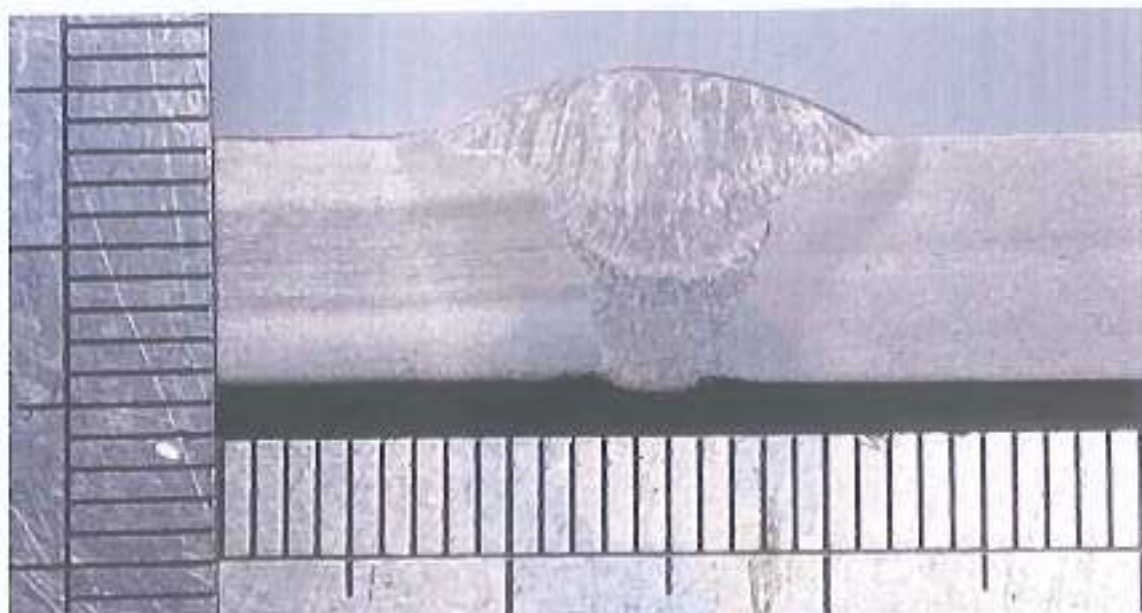


1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu đo kích thước ghi trên và không phải là bằng chứng nhận sản phẩm.
Test results are valid for the exactly submitted sample/s only, and this is not a certificate of product.
2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của tài gửi thử. / Sample of sample/s and customer are written as customer's request.
3. Độ chính xác của kết quả thử nghiệm được tính theo độ lệch chuẩn của các kết quả thử nghiệm. / The accuracy of test results is calculated according to the standard deviation of test results.
4. Không được trích sao một phần hoặc toàn bộ kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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5. Mọi thắc mắc về kết quả khách hàng liên hệ theo địa chỉ info@quatest3.com.vn và số điện thoại [0938294274](tel:0938294274) để được hỗ trợ thêm thông tin.
Please contact Quatest 3 at the email addresses info@quatest3.com.vn and by tel: [0938294274](tel:0938294274) for further information about test report.

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result
TỔ CHỨC THỐ ĐẠI MACRO STRUCTURE	ASME IX - 2021	Mỗi hàn thấu và ngấu hoàn toàn, trong mỗi hàn không có khuyết tật. Full penetration and complete fusion weld, no defect in weld metal

8. Chứng kiến và đã ký trong bản lưu hồ sơ thử nghiệm / Witnessed and signed in the test records :

BUREAU VERITAS VIETNAM
 Mai Hoang Khanh



1. Các kết quả thử nghiệm chỉ trong phạm vi có giá trị đối với vấn đề khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
 Test results are valid for the narrowest submitted sample(s) only, and this is not a certificate of product.
 2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của tài gửi mẫu. / Name of sample(s) and customer are written as customer's request.
 3. Độ không đảm bảo đo mà phòng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.
 The uncertainty of measurement is stated on the standard deviation of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
 4. Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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 5. Mọi thắc mắc về kết quả thử nghiệm liên hệ theo địa chỉ (info@quatest.vn) và (info@quatest.vn) để biết thêm thông tin.
 Please contact Quatest 3 at the email address info@quatest.vn and (info@quatest.vn) for further information about test report.

CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

0 Tungsten Carbide Coating 2006 During 2006, KSCS built 76,000g of Tungsten Carbide Coating. Tungsten Carbide Coating 2006
 0 DMS Coating 2006 During 2006, KSCS built 76,000g of DMS Coating. DMS Coating 2006
 0 Tungsten Carbide Coating 2006 During 2006, KSCS built 76,000g of Tungsten Carbide Coating. Tungsten Carbide Coating 2006



SS KOT W/Report No.	1802-J1-Q15-QERICKRISER 0000
Request No.	NOT PCR-Q15 06 23
SS in they is Newton;	33
Ngng/Cat:	10/Feb/2023
Trans s&A/Place No:	

KẾT QUẢ KIỂM TRA SIÊU ÂM/
ULTRASONIC EXAMINATION REPORT

Đơn vị yêu cầu/Client:	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
Công trình/Project:	DERRICK & RISER
Hạng mục/item:	DERRICK STRUCTURE, LADDERS & PLATFORM, RISER AND PIPING
Địa điểm location:	DASHIGULAZERMAN
Kho kiểm tra/inspection place:	DAI DUNG FACTORY
Ngày kiểm tra/inspection date:	18-Feb-2023

Bản vẽ số/ Drawing No.	N/A	Số lượng/ Quantity	1 Pcs	Kiểm tra đúng về Checklist	N/A
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THÔNG SỐ VẬT LIỆU/MATERIAL INFORMATION

Chứng loại vật liệu/Material class: EN-10025-2 GR S355J2 + ASTM A572 Gr50		Loại sản xuất/Product type: DERRICK STRUCTURE, LADDERS & PLATFORM, RISER AND PIPING			Tiêu chuẩn/Standard: ASME SEC V - DWI	
NDT mở rộng/NDT Extent: N/A		Quy trình hàn/Welding process: FCAW & GTAW			Cấp/Capex: Lemo	
Loại thiết bị/Equipment type: EPOCH650		Số seri/Seriai No : 10255206			Chất lấp âm/Compound: Cellulose paste Mẫu chuẩn và khối chuẩn/ Calibration and Reference blocks	
Góc mở góc/Probe angle beam:	0°	40°	60°	70°	Mẫu chuẩn/ Calibration block:	IM-VI
Nhãn hiệu/Số seri/ Brand/Serial	PK 1307310	N/A	N/A	N/A	Số seri/Seriai No	0023
Tần số/Frequency (MHz)	5	N/A	N/A	4	Mẫu đối chứng/ Reference block	DA0
Kích thước/Size (mm x mm)	810	N/A	N/A	8 x 9	Số seri/Seriai No	0023
Đường chuẩn (trường)/ Reference curve (dB)	+0dB	N/A	N/A	-6dB	Số seri/Seriai No	N/A
Hệ số chuyển đổi/ Transfer correction (dB)	+2dB	N/A	N/A	+2dB	Số seri/Seriai No	N/A
Thêm độ quét/ Extra when scanning (dB)	+6/-14dB	N/A	N/A	+14dB	Phạm vi/Range	(100 + 200)mm

DANH GIẢI (VÀ TÌM)

Khai định mối hàn Weld Identification		Chiều dài & dày Length & Thickness (mm)	Đỉnh gờ > 50% DAC Amplitude > 20% DAC	Bản đồ kiểm tra DyePen/Grind			Đánh giá Evaluation		Ngày kiểm tra Inspected date (dd-mm-yy)
Biểu ID joint	ID thợ hàn Welder ID			Loại Type	Vị trí Location (mm)	Dài Length (mm)	Sâu Depth (mm)	Loại bề Bleed	

1. Test coupon: DD-GTAW+FC&W-19B

[illegible]

<u>Viết tắt/Abbreviation:</u>	CD-RS là: long/Gusta Family	SD chuyển tải bề mặt/Surface Defect	CF-A là CF-ak	UT-Grey map/Undercut
<u>ACC-Chức năng/Aspect:</u>	CA-Lam/Corrosion	On-Chấm mấp (chảy tràn)/Corrosion	SI-Nguyên tử/Sig-Inclusion	IF-RS chỉ: Acid/acid-corrosion
<u>Thuật ngữ và/hoặc thuật ngữ/Definition:</u>	COE-Chuyển dòng chất/Lack of Insulation			

IQTC		INSPECTOR	3RD PARTY	CLIENT
Evaluated: HOANG CHI THANH IIT LEVEL 2 Signature:  Date: 19-Feb-2023	Name: Signature: Date:	Name: Signature: Date:	Name: Signature: Date:	



产品质量证明书

INSPECTION CERTIFICATE

[illegible]

	本产品已按照标准要求进行制造和检验，其结果符合要求，特此证明。 WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HAS MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENTS OF THE ABOVE MATERIAL SPECIFICATION	冶金技术处处长 DIRECTOR OF METALLURGICAL DEPARTMENT <i>许继勇</i>
--	---	---

검 사 증 명 서

MILL TEST CERTIFICATE



당진제철소 중남 당진시 송악읍 북부산업로 1480
1480 Bukbusanup-Ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : F200600214

품 명
Commodity : Hot Rolled Coil

제품규격
Specification : ASTM A572 GRADE 50

고객사
Customer : DAIDUNG CORP

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20201009-HS-001-004

발행일자
Date of Issue : 2020-10-09

제품치수 Dimension (mm)	수량 Quantity	중량 Weight (kg)	재질번호 Mat.No.	제품번호 Product No.	단위 Unit	인장시험 Tensile Test						화학성분 Chemical Composition(%)													
						500 Unit						2 : x 100 3 : x 1,000 4 : x 10,000 5 : x 100,000													
						YP	TS	EL	Y+EL	RA	YR	YP	C	S	Mn	P	S	Mo	DR						
(N/mm ²)	(%)	(%)	(%)	(%)	(%)	(%)	4	5	3	4	5	3	3												
8.00x1990xC	1	23,410	R58087	CL578520	B	447.2	545.5	31				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	22,300	R58087	CL578500	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	23,290	R58086	CL578510	B	447.4	550.9	29				2	L	1.590	15	1.013	155	61	18	1.949					
8.00x1990xC	1	23,420	R58086	CL578540	B	447.4	550.9	29				2	L	1.600	15	1.013	155	61	18	1.949					
8.00x1990xC	1	23,530	R58086	CL578530	B	447.4	550.9	29				2	L	1.590	15	1.013	155	61	18	1.949					
8.00x1990xC	1	23,310	R58086	CL578540	B	442.1	543.4	30				2	L	1.590	15	1.013	155	61	18	1.949					
8.00x1990xC	1	23,400	R58087	CL578550	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	21,360	R58087	CL578560	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	21,360	R58087	CL578570	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	21,370	R58087	CL578580	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	24,060	R58087	CL578590	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	24,160	R58087	CL579000	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	24,220	R58087	CL579010	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	23,260	R58086	CL579020	B	447.4	550.9	29				2	L	1.590	15	1.013	155	61	18	1.949					
8.00x1990xC	1	23,540	R58087	CL579030	B	438.9	531.2	30				2	L	1.620	19	1.045	152	74	17	2.017					
8.00x1990xC	1	24,600	R47006	CVS12570	B	386.4	531.0	34				2	L	1.564	17	1.024	140	41	17	1.962					

*** Sub Total(000) *** 15 373,000Kgt

N O T E	- Position - T:Top, M:Middle, B:Bottom - Tensile test - Direction:Transverse. Gauge length:50mm (Rectangular), YF Min. ± 10.2% (1-sec), 5 (0.5% underload), U (Upper yield point), L (Lower yield point) - Division - P:Product analysis, L:Ladle analysis

본사의 제품과 검사에 결과 지정한 규격에 합당한 것을 증명합니다.
WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

S. B. Park

Chief Of Quality Assurance Team

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- * The Mill Test Certificate is a copy that has been printed from original electronic document (with digital signing).
- * You are able to check an original electronic document in Hyundai's customer portal (http://smhyundai-steel.com/onlinehelp.asp) * QR Code scanner App : QR&I
- * [제품상 주의사항] 이 검사증명서에 기재된 규격 및 사용처 사용과량 및 인허가 유무가 발생될 수 있습니다.
- * [Caution for Use] Caution on products and safety problems may arise if used other than the ones stated in the Mill Test Certificate.

ORIGINAL

PURCHASER

HAN VINH JOINT STOCK COMPANY

INSPECTION CERTIFICATE

Date of Issue 05/Oct/2026

Certificate No. KG20-002-07-20

TRADE DESIGNATION	SIZE	MFG. No.		APPLICABLE SPECIFICATION & CLASSIFICATION							
TG-S51T	2.4	FKS008B1255		AWS A5.18 ER70S-6							
CHEMICAL COMPOSITION (%) ACCORDING TO EN10204 TYPE 3.1											
Elements	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V	
ROD OR WIRE	0.10	0.86	1.54	0.012	0.013	0.16	0.02	0.04	< 0.01	0.01	
MECHANICAL PROPERTY ACCORDING TO EN10204 TYPE 3.1											
Tensile Test of Weld Metal				Impact Test of Weld Metal							
Yield Strength at 0.2% offset	Tensile Strength		Elongation	Test Temp	Absorbed Energy						
531 MPa	635 MPa		33 %	-30 °C	Avg.	102 J					
Welding Conditions								REMARKS			
Type of Current	DCEN		Shielding Gas		Ar						
Welding Current	180 A										
Welding Voltage	12 V										

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

KOBELCO MIG WIRE (THAILAND) CO., LTD.

CHIEF INSPECTOR



CERTIFIED MATERIAL REPORT

CMR No. : HV - 2022 - 1133

Date of Issue : 2022

 CITY ENERHYUNDAI WELDING VINA
 CÔNG TY TNHH HỖ TRỢ CÔNG NGHỆ
 11-09 3600921745

Customer		G-TECH CORPORATION								
Trade Name		SG-71LH		Size		1.2 mm x 16 kg				
Lot No.		MF22120K71		Purchase's Order No.		B1-2022/GT:HW/PO				
Specification		AWS A5.20 E71T-1C EN ISO 17832-A-T 42 2 P C1 1 H5								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C/hr.)			As Welded			
Mechanical Properties	Tensile Test				Impact Test (CVN - Joule)					
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	527	591	28	-20	87	90	81	85		
AWS Spec.	≥390	450~670	≥22	≥27						
Chemical Composition (%)										
	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
(Weld Metal)	0.04	0.57	1.35	0.016	0.008	0.02	0.02	0.01	0.01	0.02
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.50	≤0.06	≤0.35
Others * AS/NZS ISO 17832-A-T 42 2 P C1 1 H5 * Qty: 6,300 kg										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.										
Approved by  QA Manager: Phạm Đình Anh Vũ										

HYUNDAI WELDING VINA CO., LTD.

Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
ASME IX:2021

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW+FCAW-197

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Mar 10, 2023	All	Client Review	L.V.Nhon	T.H.P.Vu
1	Nov 20, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2021)

Page: 1/2

Applicable Code: ASME Section IX - 2021 Edition
Organization Name: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION By: La Vinh Nhon
Welding Procedure Specification No.: WPS-DD-GTAW+FCAW-197 Revision No. 1 Date 20/Nov/2025
Supporting PQR No.(s): PQR-DD-GTAW+FCAW-197 Date 10/Mar/2023
Welding Process(es) GTAW+FCAW Type(s)(Manual, Automatic, Semi-Automatic): Manual, Semi-Automatic, Mechanized

JOINTS (QW-402)

Joint Design Groove or fillet as per fabrication drawing
Root Spacing As per fabrication drawing
Backing With or without
Retainer None
Backing Material Base metal or weld metal

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. 1 Group No. 1, 2 to P-No. 1 Group No. 1, 2
OR
Specification and type/grade or UNS Number None
to Specification and type/grade or UNS Number None
OR
Chem. Analysis and Mech. Prop. None
to Chem. Analysis and Mech. Prop. None
Thickness Range:
Base Metal: 5 - 50 (mm) for groove welds without toughness requirement All for fillet welds
Maximum Pass Thickness ≤ 13 mm: Yes ☒
Other Refer ASME VIII_Table UCS-56-1_General Notes: (b)_ (2)

FILLER METALS (QW-404)

Process	GTAW	FCAW
Spec. No. (SFA)	SFA-5.18	SFA-5.20
AWS No. (Class)	ER70S-6 with Mn $\leq 1.6\%$ and Si $\leq 1.0\%$	E71T-1C
F-No.	6	6
A-No.	1	1
Size of Filler Metals	$\varnothing 1.6 - 2.4$ (mm)	$\varnothing 1.2 - 1.6$ (mm)
Supplemental Filler Metal	None	None
Filler Metal Product Form	Solid rod	Flux cored wire
Filler Metal Brand Name	TG-S51T or Any within SFA specification & classification.	SC-71LH or Any within SFA specification & classification.
Weld Metal		
Deposited Thickness:		
Groove	Max. 10 mm	Max. 40 mm
Fillet	All	All
Electrode-Flux (Class)	None	None
Flux Type	None	None
Flux Trade Name	None	None
Consumable Insert	None	None
Other	None	None



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2021)

Page: 2/2

Applicable Code: ASME Section IX - 2021 Edition

Organization Name: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

By: La Vinh Nhon

Welding Procedure Specification No.: WPS-DD-GTAW+FCAW-197

Revision No. 1

Date 20/Nov/2025

POSITIONS (QW-405)

Position(s) of Groove	All		
Position(s) of Fillet	All		
Welding Progression		GTAW	FCAW
Vertical up		Yes	Yes
Vertical down		None	None

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range	None
Time Range	None
Other	None

PREHEAT (QW-406)

Preheat Temperature, Minimum	10 °C
Interpass Temperature, Maximum	222 °C
Preheat Maintenance	None
Other	None

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Ar	100%	10 - 20 L/min.
Shielding	CO ₂	100%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: Minimum purity 99.99%Ar for GTAW process and 99.8% CO₂ for FCAW process

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	GTAW	ER70S-6	1.6 - 2.4	DC EN	126 - 154	None	1.8 - 2.5	12 - 14	55 - 67	None
Hot	GTAW	ER70S-6	1.6 - 2.4	DC EN	187 - 229	None	1.8 - 2.5	12 - 14	89 - 109	None
Fill	FCAW	E71T-1C	1.2 - 1.6	DC EP	189 - 253	None	1.4 - 1.9	25 - 31	188 - 484	None
Cap	FCAW	E71T-1C	1.2 - 1.6	DC EP	198 - 242	None	1.4 - 1.9	25 - 31	366 - 518	None

Pulsing Current None Heat Input (max.) None

Tungsten Electrode Size and Type Ø2.4 mm, 2% Thoriated Tungsten (EWTh-2)

Mode of Metal Transfer for GMAW (FCAW) Globular

Other None

TECHNIQUE (QW-410)

Process	GTAW	FCAW
String or Weave Bead	Weave Bead	String & Weave (Max. width 15 mm)
Orifice, Nozzle, or Gas Cup Size	All size	All size
Initial and Interpass Cleaning (Brushing, Grinding, etc.)	Brushing or Grinding	Brushing or Grinding
Method of Back Gouging	Grinding or Air carbon arc	Grinding or Air carbon arc
Oscillation	None	None
Contact Tube to Work Distance	None	15 - 25 (mm)
Multiple or Single Pass (Per Side)	Multiple pass	Multiple pass
Multiple or Single Electrode	Single	Single
Electrode Spacing	None	None
Peening	None	None
Other	None	None

Organization

DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Certified by Welding Coordinator:

Approved by: Truong Hoai Phong Vu (QA Manager)

La Vinh Nhon



Date: 20/Nov/2025

Date: 20/Nov/2025



WELDING PROCEDURE SPECIFICATION (WPS) (ASME Boiler and Pressure Vessel Code:2021)

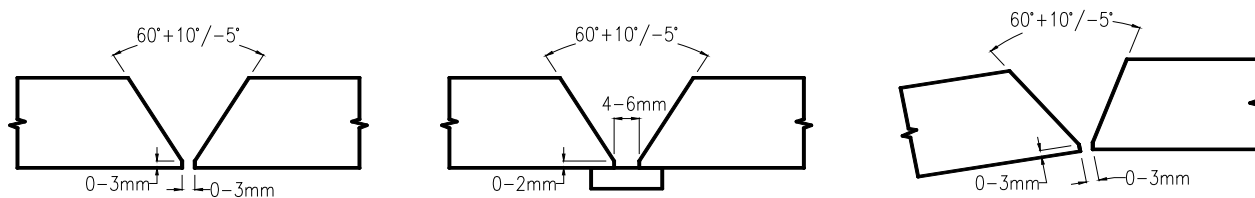
Page for
Sketches

Welding Procedure Specification No.: WPS-DD-GTAW+FCAW-197

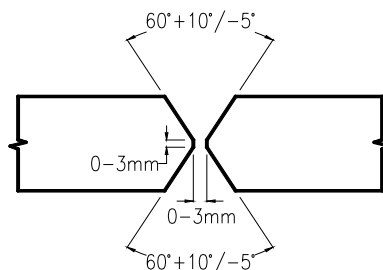
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

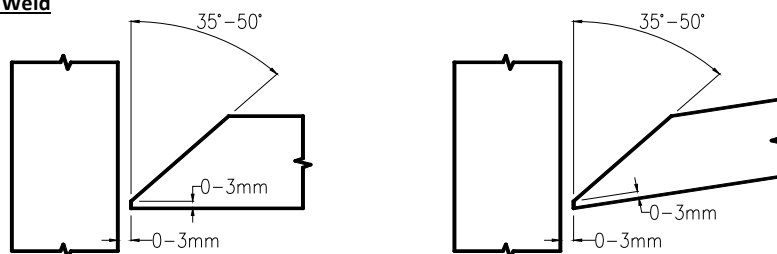
1. Single V Groove Butt Joint



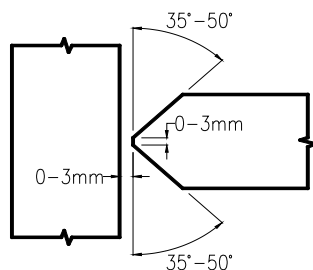
2. Double V Groove Butt Joint



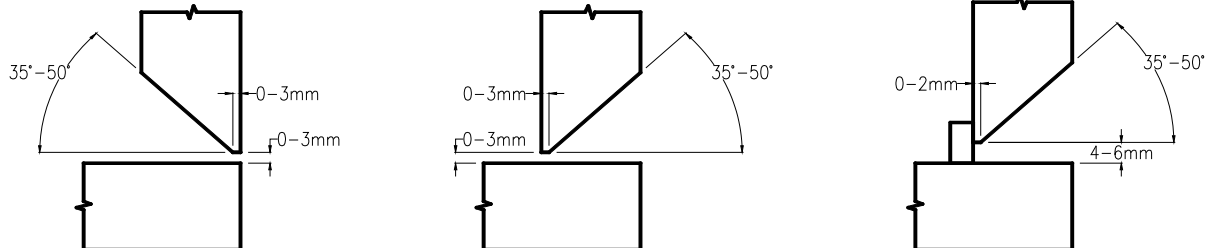
3. T-Joint: Single Bevel Groove Butt Weld



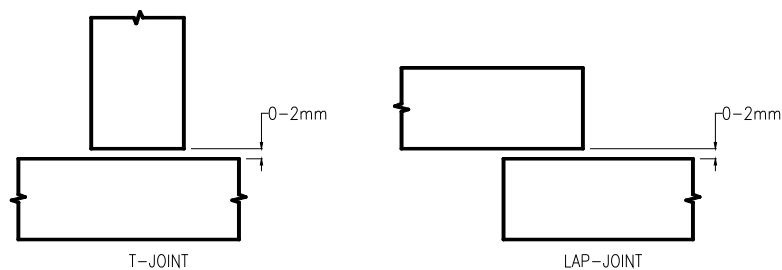
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld



Handwritten signature

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No.: **PQR-DD-GTAW+FCAW-197** Date: **Mar 10, 2023**

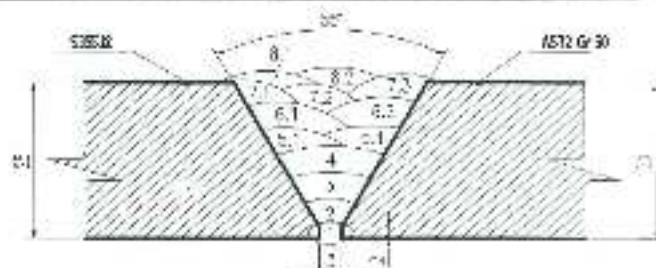
Welding Process(es): **GTAW+FCAW**

Type(s) (Manual, Automatic, Semi-Automatic): **Manual + Semi-Automatic**

JOINTS (QW-402)

Joint Type: **Butt joint, single V groove**

Backing: **None**

Retainer: **None**


BASE METALS (QW-403)

Material Spec.: **EN 10025-2 to ASTM A572**

Type/Grade, or UNS Number: **S355J2 to A572 Gr.50**

P-No.: **1** Group No.: **2** to P No.: **1** Group No.: **1**

Thickness of Test Coupon: **25 mm**

Diameter of Test Coupon: **Plate**

Maximum Pass Thickness: **< 13 mm**

Other: **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature: **None**

Time: **None**

Other: **None**

GAS (QW-408)

	Gas(es)	Percent Composition (Mixture)	Flow Rate
Shielding	Ar	99.99%	12 L/min
Shielding	CO ₂	99.8%	22 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process	GTAW	FCAW
SFA Specification	ASME SFA-5.18	ASME SFA-5.20
AWS Classification	ER70S-6	E71T-1C
Filler Metal P-No.	6	6
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	Ø2.4 mm	Ø1.2 mm
Filler Metal Product Form	Solid rod	Flux cored wire
Supplemental Filler Metal	None	None
Filler Metal Trade Name	TG-551T	5C-71LH
Electrode Flux Classification	-	-
Flux Type	-	-
Flux Trade Name	-	-
Weld Metal Thickness	5 mm	20 mm
Other	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process	GTAW	FCAW
Current and Polarity	DC EN	DC EP
Amps.	See table below	
Volts	See table below	
Mode of Metal Transfer for GMAW (FCAW)	None	Globular
Other	None	None

TECHNIQUE (QW-410)

Process	GTAW	FCAW
String or weave Bead	Weave	Weave
Oscillation	None	None
Method cleaning	Grinding, brushing	
Method back gouge	None	None
Multipass or single pass (per side)	Multipass	Multipass
Multiple or single Electrodes	Single	Single
Peening	None	None
Use of thermal processes	None	None
Other	None	None

POSITION (QW-405)

Process	GTAW	FCAW
Position(s)	1G	1G
Weld Progression (Uphill, Downhill)	-	-
Other	None	None

PREHEAT (QW-406)

Preheat Temperature	35 °C (Ambient)
Interpass Temperature	Max. 167 °C
Other	None

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler	Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/Interpass Temp. (°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
1	GTAW	AWS A5.18 / ER70S-6	DC EN	140	13	61	35 (Ambient)	2.5	Ar, 12 L/min.	1.8
2	GTAW	AWS A5.18 / ER70S-6	DC EN	208	13	99	55	2.5	Ar, 12 L/min.	1.6
3	FCAW	AWS A5.20 / E71T-1C	DC EP	230	28	270	35	3.5	CO ₂ , 22 L/min.	1.4
4	FCAW	AWS A5.20 / E71T-1C	DC EP	220	29	209	35	3.5	CO ₂ , 22 L/min.	1.8



Signature



PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2021)

Page: 2/3

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Procedure Qualification Record No. **PQR-DD GTAW+FCAW 197** Date: **Mar 10, 2023**Welding Process(es) **GTAW+FCAW**Type(s) (Manual, Automatic, Semi-Automatic) **Manual + Semi-Automatic****WELDING PARAMETER**

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp. (°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
5.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	300	90	3	CO ₂ , 22 L/min.	1.2
5.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	287	120	3	CO ₂ , 22 L/min.	1.3
6.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	210	28	277	130	5	CO ₂ , 22 L/min.	1.3
6.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	300	167	5	CO ₂ , 22 L/min.	1.4
7.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	230	28	317	42	5	CO ₂ , 22 L/min.	1.2
7.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	379	78	4	CO ₂ , 22 L/min.	1.0
7.3	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	440	115	4	CO ₂ , 22 L/min.	0.8
8.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	407	106	3.5	CO ₂ , 22 L/min.	0.9
8.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	471	117	3.5	CO ₂ , 22 L/min.	0.8

*Signature*



PROCEDURE QUALIFICATION RECORD (PQR) (ASME Boiler and Pressure Vessel Code:2021)

Page: 3/3

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No.: **PQR-DD-GTAW-FCAW-297**

Date: **Mar 10, 2023**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
T51	19.0	24.9	473.1	255	505	Ductile end base metal (S355J2)
T52	19.1	25.0	477.5	275	575	Ductile end base metal (S355J2)

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
SB1 (Side bend)	Pass
SB2 (Side bend)	Pass
SB3 (Side bend)	Pass
SB4 (Side bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (J/N)
				1/4-HB or J	% Shear	Mils (in.) or mm	
WM 1	Weld metal	10 x 10	-20 °C	100 (J)			N
WM 2	Weld metal	10 x 10	-20 °C	65 (J)			N
WM 3	Weld metal	10 x 10	-20 °C	94 (J)			N
WM 4	Weld metal	10 x 10	-20 °C	82 (J)			N
WM 5	Weld metal	10 x 10	-20 °C	45 (J)			N
WM 6	Weld metal	10 x 10	-20 °C	80 (J)			N
HAZ 1	HAZ (S355J2)	10 x 10	-20 °C	158 (J)			N
HAZ 2	HAZ (S355J2)	10 x 10	-20 °C	132 (J)			N
HAZ 3	HAZ (S355J2)	10 x 10	-20 °C	172 (J)			N
HAZ 4	HAZ (S355J2)	10 x 10	-20 °C	182 (J)			N
HAZ 5	HAZ (S355J2)	10 x 10	-20 °C	154 (J)			N
HAZ 6	HAZ (S355J2)	10 x 10	-20 °C	152 (J)			N
HAZ 7	HAZ (A572 Gr.50)	10 x 10	-20 °C	90 (J)			N
HAZ 8	HAZ (A572 Gr.50)	10 x 10	-20 °C	96 (J)			N
HAZ 9	HAZ (A572 Gr.50)	10 x 10	-20 °C	38 (J)			N
HAZ 10	HAZ (A572 Gr.50)	10 x 10	-20 °C	122 (J)			N
HAZ 11	HAZ (A572 Gr.50)	10 x 10	-20 °C	94 (J)			N
HAZ 12	HAZ (A572 Gr.50)	10 x 10	-20 °C	102 (J)			N

Comments: **Acceptable**

Fillet-Weld Test (QW-180)

Result - Satisfactory: ☒ Yes ☐ No
Macro - Results: **Acceptable**

Hardness HV10 (Other Tests)

Line	Base metal (S355J2)			HAZ (S355J2)			Weld metal			HAZ (A572)			Base metal (A572)			Results
A	191	191	196	253	266	260	205	210	201	245	270	247	191	185	186	Acceptable
B	168	168	159	170	175	183	160	185	173	181	181	177	176	177	178	Acceptable

Welder's Name: **Phan Hong Thanh**

Clock No.: **None** Stamp (I.D.) No.: **CT-WD-0541**

Welder's Name: **Tran Ngoc Em**

Clock No.: **None** Stamp (I.D.) No.: **DDC-WD-0446**

Tests Conducted by: **Quatest 3**

Laboratory Test No.: **KT3-00375ACK3/4**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX - 2021 Edition of the ASME Boiler and Pressure Vessel Code.

Organization: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
Certified by Welding Coordinator:

Approved by: **Truong Hoi Phap (QC Manager)**

La Vinh Khon

Date: **Mar 10, 2023**

Date: **Mar 10, 2023**



MAI HOANG KHANH

WELDING PARAMETERS RECORD

(Doc No.: WPR-GTAW+FCAW-197)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GTAW+FCAW-197
 Test Coupon: DD-GTAW+FCAW-197
 Test position: 1G for both of GTAW and FCAW
 Base metal:

- Specification	EN 10025-2 S355J2	AS 1M A5/2 Gr.50
- Dimension	PL25 x 200 x 650	PL25 x 200 x 550
- Heat/Product No.	22102673D	21005
- Certification No.	220216200009	201028-ASHEX 4186-002

Filler metal:

- Specification	AWS A5.18	A5.20
- Classification	ER70S-6 (Ø2.4)	E711-1C (Ø1.2)
- Trade name	Kobelco TG-S51T	Hyundai SC-711H
- Batch/Lot No.	FK500831255	MF22120K71
- Certificates No.	KG20-002 07-20	HV 2022-1133

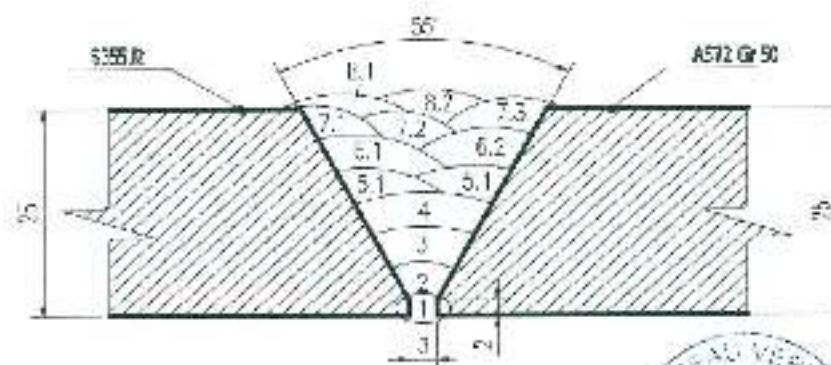
Gas: Argon (99.99%) CO₂ (99.8%)

Welder Name: Phan Hong Thanh; Tran Ngoc Em

I.D.: CT-WD-0541; DDC-WD-0446

Joint Sketch:

Date: Feb 17, 2023



Root face: 3 mm
 Root gap: 2 mm
 Groove angle: 55°



Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (l/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.						Width	Depth	
1	GTAW	AWS A5.18 / ER70S-6	2.4	DC EN	140	13	8	40	525	61	35 (Ambient)	Ar, 12 l/min.	1.8	5.5	2.5	
2	GTAW	AWS A5.18 / ER70S-6	2.4	DC FN	208	13	5	17	525	99	55	Ar, 12 l/min.	1.6	8	2.5	
3	FCAW	AWS A5.20/ E711-1C	1.2	DC EP	230	28	2	2	550	270	35	CO ₂ , 22 l/min.	1.4	9.5	3.5	
4	FCAW	AWS A5.20/ E711-1C	1.2	DC EP	220	28	2	38	550	209	65	CO ₂ , 22 l/min.	1.8	12	3.5	

WELDING PARAMETERS RECORD [Doc.No.: WPR-GTAW+FCAW-197]

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.						Width	Depth	
5.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	50	550	300	90	CO ₂ , 22 L/min.	1.2	11	3	
5.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	55	550	287	120	CO ₂ , 22 L/min.	1.3	12	3	
6.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	210	28	1	59	550	277	130	CO ₂ , 22 L/min.	1.3	13	5	
6.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	50	550	300	167	CO ₂ , 22 L/min.	1.2	14	5	
7.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	230	28	1	44	550	317	42	CO ₂ , 22 L/min.	1.2	15	5	
7.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	27	550	379	78	CO ₂ , 22 L/min.	1	13	4	
7.3	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	15	550	440	119	CO ₂ , 22 L/min.	0.8	12.5	4	
8.1	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	21	550	407	106	CO ₂ , 22 L/min.	0.9	12.5	3.5	
8.2	FCAW	AWS A5.20/ E71T-1C	1.2	DC EP	220	28	1	10	550	471	117	CO ₂ , 22 L/min.	0.8	12.5	3.5	
Conducted by Welding Coordinator																
Name		La Vinh Nhon														
Signature		 														
Date		Feb 17, 2023														

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

09/03/2023
 Page 01/06

- Tên mẫu :**
Name of sample

QUY TRÌNH HÀN DD - GTAW + FCAW - 197
GTAW+FCAW - 1G - Single V
Material: S355J2 + A572 Gr.50
Filler: Kobelco TG-S51T (Ø2.4);
Hyundai SC-71LH (Ø1.2)
Welder: CT-WD-0541;
DDC-WD-0446
- Số lượng mẫu :**
Quantity

01
- Mô tả mẫu :**
Sample description

Mẫu là thép tấm có mối hàn, kích thước (25 x 380 x 550) mm
The as-received sample is welded steel plate
- Ngày nhận mẫu :**
Date of receiving

20/02/2023
- Ngày thử nghiệm :**
Date of testing

27/02/2023 & 08/03/2023
- Nơi gửi mẫu:**
Customer

CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Lô số 38, Khu C, Đường D1, Khu công nghiệp An Hạ,
Xã Phạm Văn Hai, Huyện Bình Chánh, Thành phố Hồ Chí Minh
- Kết quả thử nghiệm :**
Test results

Xem trang 02/06...06/06
See pages

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL TESTING LAB.


Phạm Văn Út

TL.GIÁM ĐỐC / PP. DIRECTOR
TRƯỞNG PHÒNG THỬ NGHIỆM
HEAD OF TESTING LAB.


Nguyễn Tấn Tùng

1. Các kết quả thử nghiệm gửi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
 This report only valid for the sample submitted completely only, and this is not a certificate of product.
 2. Tên của, và kích thước được ghi theo yêu cầu của nơi gửi mẫu. / Name of sample and dimension are written as customer's request.
 3. Độ không đều của độ dày được tính từ độ không đều của độ chuẩn nhận với hệ số phân bố = 2, phần bổ chuẩn tương ứng với 95% độ tin cậy.
 The reported standard deviation of measurement is based on the standard deviation of measurement calculated by the average factor k = 2, at 95% confidence level.
 4. Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ Thuật 3.
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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7. Kết quả thử nghiệm :
 Test results

Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result			
THỬ KÉO/ TENSILE TEST	ASME IX - 2021	Sample TS1	Sample TS2		
7.1. Kích thước mẫu thử, mm Dimension of specimen		19,0 x 24,9	19,1 x 25,0		
7.2. Lực kéo đứt, kN Tensile load		269	275		
7.3. Giới hạn bền kéo, MPa Tensile strength		569	575		
7.4. Vị trí đứt Broken position		Kim loại cơ bản Parent metal (S355J2)	Kim loại cơ bản Parent metal (S355J2)		
THỬ UỐN/ BEND TEST	ASME IX - 2021	SB1	SB2	SB3	SB4
7.5. Kích thước mẫu thử, mm Dimension of specimen		9,90 x 25,0	10,0 x 25,0	10,1 x 25,0	10,0 x 25,0
7.6. Góc uốn, độ Bending angle		180	180	180	180
7.7. Đường kính gồi uốn, mm Plunger diameter		40	40	40	40
7.8. Kết quả thử uốn cạnh Side bend test result		Đạt Pass	Đạt Pass	Đạt Pass	Đạt Pass



1. Các kết quả thử nghiệm ghi trong phần này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
 Test results are valid for the exactly submitted sample(s) only, and this is not a certificate of product.
2. Tên mẫu, số, kích thước được ghi theo yêu cầu của chủ gửi mẫu. / Name of sample(s) and customer are written as customer's request.
3. Độ không đảm bảo do mở rộng được tính từ độ không đảm bảo do chuẩn dẫn với hệ số giãn k = 2, phạm bố chuẩn tương ứng với 95% độ tin cậy.
 The expanded uncertainty of measurement is stated in the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95% confidence level.
4. Không được trích sao một phần phiếu kết quả thử nghiệm này mà không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu <i>Specification</i>	Phương pháp thử <i>Test method</i>	Kết quả thử nghiệm <i>Test result</i>		
THỬ VA ĐẬP Ở - 20 °C IMPACT TEST AT - 20 °C	ASME IX - 2021	Kim loại mối hàn/ Weld metal (Mặt/ Face)		
7.9. Kích thước mẫu thử, <i>Dimension of specimen</i>		10 x 10	10 x 10	10 x 10
7.10. Năng lượng hấp thụ (KV _s), <i>Absorbed energy</i>		100	66	94
		Kim loại mối hàn/ Weld metal (Chân/ Root)		
7.11. Kích thước mẫu thử, <i>Dimension of specimen</i>		10 x 10	10 x 10	10 x 10
7.12. Năng lượng hấp thụ (KV _s), <i>Absorbed energy</i>		82	46	80



1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị dẫn với mẫu đã được kiểm tra và không phải là bằng chứng thực nghiệm.
Test results are valid for the sample submitted/accepted only, and this is not a certificate of product.
 2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của nội dung thử nghiệm. / Sample name and customer name written in customer's request.
 3. Độ không đồng đều do ảnh hưởng được tính từ độ không đồng đều do chuẩn nhân với hệ số phân bố = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.
The reported reported uncertainty of measurement is based on the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
 4. Không được trích sao nội dung phiếu kết quả thử nghiệm này, nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result		
THỬ VA ĐẬP Ở - 20 °C IMPACT TEST AT - 20 °C	ASME IX - 2021	Vùng ảnh hưởng nhiệt/ Heat affected zone (Mặt/ Face) – S355J2		
7.13. Kích thước mẫu thử, Dimension of specimen		10 x 10	10 x 10	10 x 10
7.14. Năng lượng hấp thụ (KV _a), Absorbed energy		168	192	172
		Vùng ảnh hưởng nhiệt/ Heat affected zone (Chân/ Root) – S355J2		
7.15. Kích thước mẫu thử, Dimension of specimen		10 x 10	10 x 10	10 x 10
7.16. Năng lượng hấp thụ (KV _g), Absorbed energy		192	194	192
		Vùng ảnh hưởng nhiệt/ Heat affected zone (Mặt/ Face) – A 572		
7.17. Kích thước mẫu thử, Dimension of specimen		10 x 10	10 x 10	10 x 10
7.18. Năng lượng hấp thụ (KV _g), Absorbed energy		96	96	98
		Vùng ảnh hưởng nhiệt/ Heat affected zone (Chân/ Root) – A572		
7.19. Kích thước mẫu thử, Dimension of specimen		10 x 10	10 x 10	10 x 10
7.20. Năng lượng hấp thụ (KV _g), Absorbed energy		122	94	102

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu đã được kiểm tra và không phải là bằng chứng chấp nhận sản phẩm.
 Test results are valid for the quantity submitted sample only, and this is not a certificate of product.
 2. Tên mẫu, tên khách hàng được ghi trên phiếu của chúng tôi riêng. / Name of sample and customer are written on customer's request.
 3. Độ không đảm bảo do sai lệch được tính là độ không đảm bảo do chuẩn nhân với hệ số phủ k = 2, phần bù chuẩn tương ứng với 95 % độ tin cậy.
 The reported expanded uncertainty of measurement is stated in the reported uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.
 4. Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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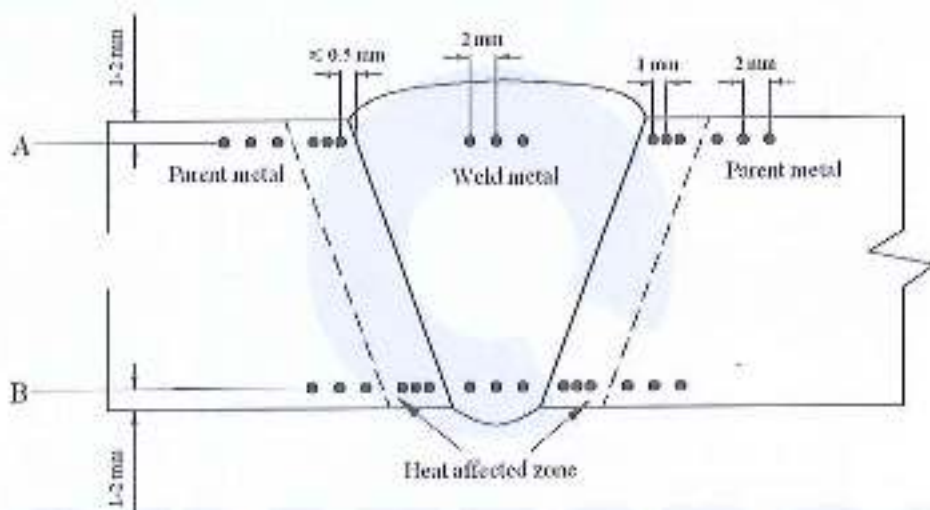
PHIẾU KẾT QUẢ THỬ NGHIỆM
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7. Kết quả thử nghiệm :
Test results

Tên chỉ tiêu Specification	Phương pháp thử Test method	Kết quả thử nghiệm Test result
	ASME IX - 2021	

ĐỘ CỨNG/ HARDNESS HV 10

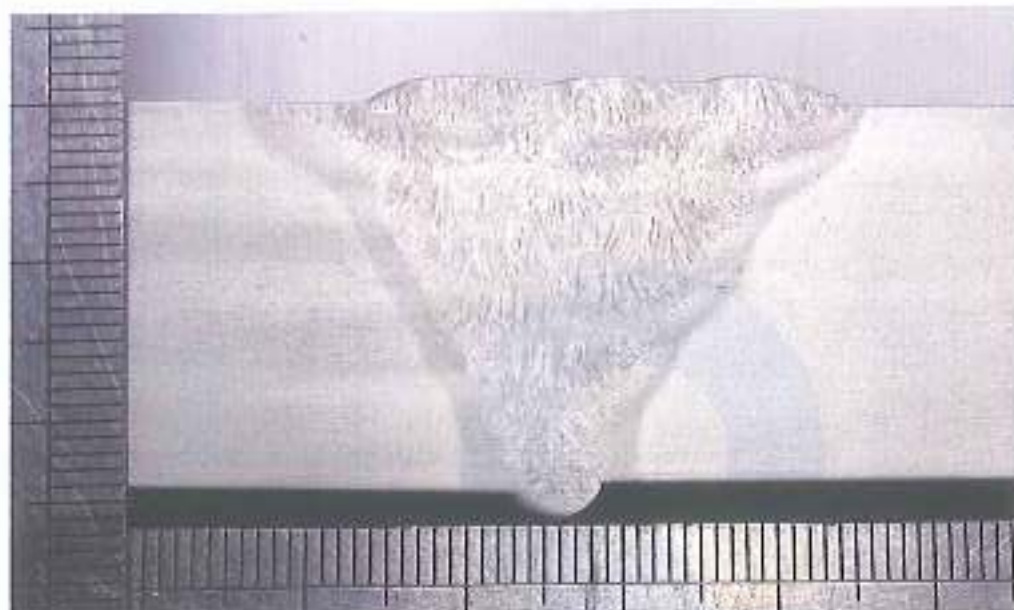


	Kim loại cơ bản Parent metal (S355J2)			Vùng ảnh hưởng nhiệt Heat affected zone (S355J2)			Kim loại mối hàn Weld metal			Vùng ảnh hưởng nhiệt Heat affected zone (A572 Gr.50)			Kim loại cơ bản Parent metal (A572 Gr.50)		
A	191	191	196	253	266	260	205	210	201	245	270	247	191	185	186
B	168	168	159	170	175	183	169	185	173	181	181	177	176	177	178

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the sample submitted and this is not a certificate of product.
2. Từ mẫu, các hình ảnh được ghi theo yêu cầu của tài liệu ghi nhận. + Range of samples and images are taken as customer's request.
3. Độ thẳng của bề mặt mẫu được tính từ độ không thẳng của bề mặt chuẩn so với tỷ lệ số pixel = 2, phần bề mặt chuẩn trong ứng vào 95 % độ tin cậy.
The reported straightness of measurement is based on the straightness of the standard surface with a pixel ratio of 2, the standard surface in the application is 95 % confidence level.
4. Không được trích sao một phần hoặc kết quả thử nghiệm này trừ khi có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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7. Kết quả thử nghiệm : Test results

Tên chỉ tiêu <i>Specification</i>	Phương pháp thử <i>Test method</i>	Kết quả thử nghiệm <i>Test result</i>
TỔ CHỨC THỎ ĐẠI MACRO STRUCTURE	ASME IX - 2021	Mỗi hàn thấu và ngấu hoàn toàn, trong mỗi hàn không có khuyết tật. <i>Full penetration and complete fusion weld, no defect in weld metal</i>

8. Chứng kiến và đã ký trong bản lưu hồ sơ thử nghiệm / *Witnessed and signed in the test records* :

BUREAU VERITAS VIETNAM

Mai Hong Khanh

- [illegible]

IQC	INSPECTOR	3RD PARTY	CLIENT
Evaluated: HOANG CHI THANH IIT LEVEL 2 Signature:   Date: 18/04/2023	Name : Signature: Date:	Name : Signature: Date:	Name : Signature: Date:



	产品名称 PRODUCT
	合同编号 CONTRACT NO.
	交货状态 DELIVERY CONDITION
	证明书编号 CERTIFICATE NO.

✓



検査証明書 INSPECTION CERTIFICATE

需要者 : DAI DUNG CORPORATION
Customer
需要者管理番号 :
Customer's Control No.
注文者 : JFE SHOJI CORPORATION
注文者管理番号 : FB2 -C00560506
Reference No.
品名 : HOT ROLLED STEEL PLATE
Commodity
規格 : ASTM A572 GRADE 50 TYPE 1 2018
Specification

証明書番号 : 201028-
Certificate No. ASHEX-4186-002
CONTINUED

日付 : 2020-10-28
Date

船名 :
Ship No.

工事番号 :
Construction No.

注文番号 : 0A30709
Order No.

5703 H X06

製品番号 Roll No.	行番 Item No.	需要者管理番号 Customer's Control No.	寸法 Size (MM) T X W X L	数量 Quantity	質量 Mass (KG)	化学成分 Chemical Composition (%)										引張試験 Tensile Test			
						C	SI	MN	P	S	CU	Ni	MO	CR	NB	Yield Y.P./Y.S.	引張強 T.S.	引張強 T.S.	引張強 T.S.
SA19903 21056	003		25 X 2050 X 12000	1	4827	15	21	148	19	4	2	1	1	3	20				
SB63601 21065	003		25 X 2050 X 12000	1	4827	15	20	141	17	3	1	1	1	3	19	TC 59.8	76.728		
SB63602 21065	003		25 X 2050 X 12000	1	4827														
SB63603 21065	003		25 X 2050 X 12000	1	4827														
SB63604 21065	003		25 X 2050 X 12000	1	4827														
SB63701 21065	003		25 X 2050 X 12000	1	4827														
SB63702 21065	003		25 X 2050 X 12000	1	4827														
SB63703 21065	003		25 X 2050 X 12000	1	4827														
SB63704 21065	003		25 X 2050 X 12000	1	4827														

(注) 1. 表示記号の付いた数字は、Plate Roll Number
(Notes) 1. The letters & the last 5 figures: mother Plate Number
2. The last 2 figures: divided Plate Number

2. 位置 Position
T: 頂面 Top M: 中面 Middle
B: 底面 Bottom
3. 方向 Direction
L: 縦 Longitudinal
C: 横 Transverse
Z: 厚 Through-thickness

4. 試験片の寸法 Size of Impact Test Specimen
1: 10x10mm 2: 7.5x10mm 3: 2.5x10mm 4: 5x10mm
5: 1/2x10mm 6: 2.5x10mm 7: Thicknessx10mm
5. 試験片の単位 Unit of Impact Test Property
6. 吸収エネルギー Absorbed Energy: KJ/kg, J, C: kgf, m/kgf, J, Joule
7. 試験方法 Test Method: U: Up, F: Ft, L: Lt, B: Bt, C: C
8. 試験結果 Test Result: F: Fracture, S: Shear, B: Brittle
9. 試験結果 Test Result: L: Lateral Expansion: Mmm Limit

5. 試験片の長さ Gauge Length
A: 120mm (A) B: 80mm (B)
C: 150mm (C) D: 100mm (D)
E: 12" (E) F: 10" (F)
G: 12" (G) H: 10" (H)
I: 12" (I) J: 10" (J)
K: 12" (K) L: 10" (L)
M: 12" (M) N: 10" (N)
O: 12" (O) P: 10" (P)
Q: 12" (Q) R: 10" (R)
S: 12" (S) T: 10" (T)

6. 試験方法 Test Method: U: Up, F: Ft, L: Lt, B: Bt, C: C
7. 試験結果 Test Result: F: Fracture, S: Shear, B: Brittle
8. 試験結果 Test Result: L: Lateral Expansion: Mmm Limit

上記商品は検査の結果規定の規格に合格していることを証明いたします。

WE HEREBY CERTIFY THAT THE MATERIAL DESCRIBED HEREIN HAS BEEN TESTED
AND INSPECTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE
REQUIREMENTS OF THE ABOVE SPECIFICATION.

(GB25870021AA)
X A08

D. Yuasa
DALJIRO YUASA

PURCHASER

HAN VINH JOINT STOCK COMPANY

INSPECTION CERTIFICATE

Date of Issue 05/Oct/2026

Certificate No. KG20-002-07-20

TRADE DESIGNATION	SIZE	MFG. No.		APPLICABLE SPECIFICATION & CLASSIFICATION							
TG-S51T	2.4	FKS008B1255		AWS A5.18 ER70S-6							
CHEMICAL COMPOSITION (%) ACCORDING TO EN10204 TYPE 3.1											
Elements	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V	
ROD OR WIRE	0.10	0.86	1.54	0.012	0.013	0.16	0.02	0.04	< 0.01	0.01	
MECHANICAL PROPERTY ACCORDING TO EN10204 TYPE 3.1											
Tensile Test of Weld Metal				Impact Test of Weld Metal							
Yield Strength at 0.2% offset	Tensile Strength		Elongation	Test Temp	Absorbed Energy						
531 MPa	635 MPa		33 %	-30 °C	Avg.	102 J					
Welding Conditions							REMARKS				
Type of Current	DCEN		Shielding Gas		Ar						
Welding Current	180 A										
Welding Voltage	12 V										

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

KOBELCO MIG WIRE (THAILAND) CO., LTD.

CHIEF INSPECTOR



CERTIFIED MATERIAL REPORT

CMR No. : HV - 2022 - 1133

Date of Issue : 2022

 CITY ENERHYUNDAI WELDING VINA
 CÔNG TY TNHH HỖN DAI WELDING VINA
 11-09 3600921745

Customer		G-TECH CORPORATION								
Trade Name		SC-71LH			Size		1.2 mm x 16 kg			
Lot No.		MF22120K71			Purchase's Order No.		B1-2022/GT:HW/PO			
Specification		AWS A5.20 E71T-1C EN ISO 17632-A-T 42 2 P C1 1 H5								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C/hr.)			As Welded			
Mechanical Properties	Tensile Test				Impact Test (CVN - Joule)					
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	527	591	28	-20	87	90	81	85		
AWS Spec.	≥390	450~670	≥22	≥27						
Chemical Composition (%)										
	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
(Weld Metal)	0.04	0.57	1.35	0.016	0.008	0.02	0.02	0.01	0.01	0.02
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.50	≤0.06	≤0.35
Others * AS/NZS ISO 17632-A-T 42 2 P C1 1 H5 * Qty: 6,300 kg										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.										
Approved by  QA Manager: Phạm Đình Anh Vũ										

HYUNDAI WELDING VINA CO., LTD.

Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
ASME VIII, IX:2019

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW+SAW-79.1

WPS-DD-FCAW-79.2

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Jul 19, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

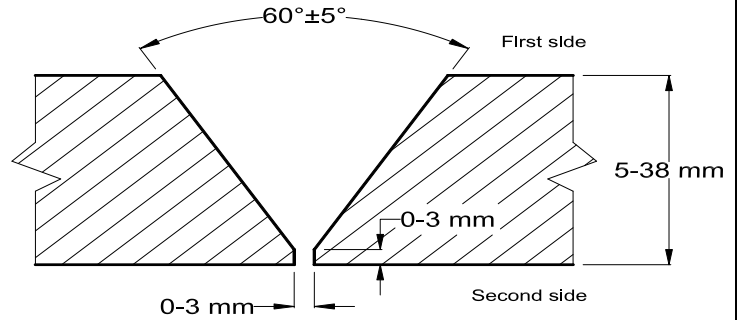
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Applicable Code:	ASME Section VIII, IX - 2019 Edition		
Organization Name:	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION	By:	La Vinh Nhon
Welding Procedure Specification No.:	WPS-DD-FCAW+SAW-79.1	Revision No.	0
Supporting PQR No.(s):	PQR-DD-FCAW+SAW-79	Date	Aug 25, 2023
Welding Process(es)	FCAW and SAW	Date	Dec 14, 2020
	Type(s)(Manual, Automatic, Semi-Automatic): Semi-Automatic, Mechanized, Automatic		

JOINTS (QW-402)

Joint Design	Groove or fillet as per fabrication drawing
Root Spacing	As per fabrication drawing
Backing	With backing or back gouging
Retainer	None
Backing Material	Weld metal for SAW

Details



Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. -	Group No. -	to	P-No. -	Group No. -
OR				
Specification and type/grade or UNS Number	JIS G3101 SS400			
to Specification and type/grade or UNS Number	P-No. 1 Group No. 1 & 2			
OR				
Chem. Analysis and Mech. Prop.	As specification required			
to Chem. Analysis and Mech. Prop.	As specification required			
Thickness Range:				
Base Metal:	5 - 38 (mm) for groove (without CVN test)		All for fillet	
Maximum Pass Thickness ≤ 13 mm:	Yes ☒			
Other	None			

FILLER METALS (QW-404)

Process	FCAW	SAW
Spec. No. (SFA)	SFA-5.20	SFA-5.17
AWS No. (Class)	E71T-1C	EM12K-F7A2
F-No.	6	6
A-No.	1	1
Size of Filler Metals	Φ1.2 mm	Φ3.2 mm
Supplemental Filler Metal	None	None
Filler Metal Product Form	Flux cored wire	Solid wire
Filler Metal Trade Name	KFX-71T or the same spec & class	KW-3 or the same spec & class
Weld Metal		
Deposited Thickness:		
Groove	Max. 24 mm	Max. 26 mm
Fillet	All	All
Electrode-Flux (Class)	None	F7A2-EM12K
Flux Type	None	Neutral flux (Agglomerated)
Flux Trade Name	None	JF-B
Consumable Insert	None	None
Other	None	None





WELDING PROCEDURE SPECIFICATION (WPS) (ASME VIII, IX:2019 Boiler and Pressure Vessel Code)

Page: 2/2

Applicable Code: ASME Section VIII, IX - 2019 Edition

Organization Name: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

By: La Vinh Nhon

Welding Procedure Specification No.: WPS-DD-FCAW+SAW-79.1

Revision No. 0

Date Aug 25, 2023

POSITIONS (QW-405)

Position(s) of Groove	All for FCAW and 1G for SAW	
Position(s) of Fillet	All for FCAW and 1F for SAW	
Welding Progression	FCAW	SAW
Vertical up	Yes	None
Vertical down	None	None

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range	None
Time Range	None
Other	None

GAS (QW-408)

	Percent Composition		
	Gas(es)	Mixture	Flow Rate
Shielding	CO ₂	100%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: only for FCAW Process and minimum purity 99.8% CO₂

PREHEAT (QW-406)

Preheat Temperature, Minimum	10 °C if 5 ≤ T ≤ 32 (mm) or 95 °C if 32 < T ≤ 38 mm (T: Thickness of Base Metal)
Interpass Temperature, Maximum	315 °C
Preheat Maintenance	Yes
Other	Table QW-254/ 255 (QW-406.1) and Table UCS-56-1 (b) (2)

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	FCAW	E71T-1C	1.2	DC EP	140 - 165	None	0.9-1.0	26 - 29	202 - 293	None
Fill	FCAW	E71T-1C	1.2	DC EP	189 - 264	None	1.7-2.8	28 - 33	146 - 251	None
Fill	SAW	F7A2-EM12K	3.2	DC EP	398 - 485	None	4.7-5.3	30 - 37	160 - 205	None
Cap	SAW	F7A2-EM12K	3.2	DC EP	398 - 485	None	5.3-7.1	30 - 37	121 - 182	None

Pulsing Current	None	Heat Input (max.)	None
Tungsten Electrode Size and Type	None		
Mode of Metal Transfer for GMAW (FCAW)	Globular		
Other	None		

TECHNIQUE (QW-410)

Process	FCAW	SAW
String or Weave Bead	Weave (Max. width 16 mm)	String (Max. width 25 mm)
Orifice, Nozzle, or Gas Cup Size	No. 6	None
Initial and Interpass Cleaning (Brushing, Grinding, etc.)	Brushing or Grinding	Brushing or Grinding
Method of Back Gouging	Grinding or Air carbon arc	Grinding or Air carbon arc
Oscillation	None	None
Contact Tube to Work Distance	15 - 25 (mm)	20 - 30 (mm)
Multiple or Single Pass (Per Side)	Multiple or single pass	Multiple or single pass
Multiple or Single Electrode	Single	Single
Electrode Spacing	None	None
Peening	None	None
Other	None	None

Organization DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Certified by Welding Coordinator:

La Vinh Nhon

Approved by: Truong Hoai Phong Vu (QA Manager)

Date: Aug 25, 2023

Date: Aug 25, 2023





WELDING PROCEDURE SPECIFICATION (WPS) (ASME VIII, IX:2019 Boiler and Pressure Vessel Code)

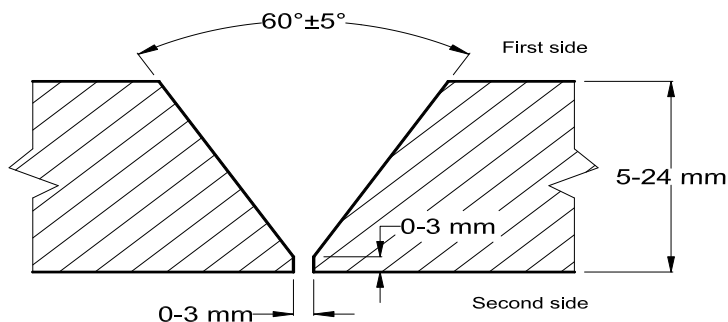
Page: 1/2

Applicable Code: **ASME Section VIII, IX - 2019 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-79.2** Revision No. **0** Date **Aug 25, 2023**
Supporting PQR No.(s): **PQR-DD-FCAW+SAW-79** Date **Dec 14, 2020**
Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Semi-Automatic, Mechanized**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**
Root Spacing **As per fabrication drawing**
Backing **With backing or back gouging**
Retainer **None**
Backing Material **Base metal or weld metal**

Details



Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. - Group No. - to P-No. - Group No. -
OR

Specification and type/grade or UNS Number **JIS G3101 SS400**

to Specification and type/grade or UNS Number **P-No. 1 Group No. 1 & 2**

OR

Chem. Analysis and Mech. Prop. **As specification required**

to Chem. Analysis and Mech. Prop. **As specification required**

Thickness Range:

Base Metal: **5 - 24 (mm) for groove (without CVN test)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other **None**

FILLER METALS (QW-404)

Process **FCAW**

Spec. No. (SFA) **SFA-5.20**

AWS No. (Class) **E71T-1C**

F-No. **6**

A-No. **1**

Size of Filler Metals **Φ1.2 mm**

Supplemental Filler Metal **None**

Filler Metal Product Form **Flux cored wire**

Filler Metal Trade Name **KFX-71T or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 24 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME VIII, IX:2019 Boiler and Pressure Vessel Code)

Page: 2/2

Applicable Code: **ASME Section VIII, IX - 2019 Edition**Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**By: **La Vinh Nhon**Welding Procedure Specification No.: **WPS-DD-FCAW-79.2**Revision No. **0**Date **Aug 25, 2023****POSITIONS (QW-405)**

Position(s) of Groove **All**

Position(s) of Fillet **All**

Welding Progression

Vertical up **Yes**

Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**

Time Range **None**

Other **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**

Interpass Temperature, Maximum **315 °C**

Preheat Maintenance **None**

Other Table QW-255 (QW-406.1)

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	100%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: only for FCAW Process and minimum purity 99.8% CO₂**ELECTRICAL CHARACTERISTICS (QW-409)**

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	FCAW	E71T-1C	1.2	DC EP	140 - 165	None	0.9-1.0	23 - 30	202 - 293	None
Fill	FCAW	E71T-1C	1.2	DC EP	189 - 264	None	1.7-2.8	27 - 34	146 - 251	None
Cap	FCAW	E71T-1C	1.2	DC EP	203 - 264	None	1.8-2.0	27 - 34	202 - 258	None

Pulsing Current **None** Heat Input (max.) **None**

Tungsten Electrode Size and Type **None**

Mode of Metal Transfer for GMAW (FCAW) **Globular**

Other **None**

TECHNIQUE (QW-410)

Process **FCAW**

String or Weave Bead **Weave (Max. width 16 mm)**

Orifice, Nozzle, or Gas Cup Size **No. 6**

Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**

Method of Back Gouging **Grinding or Air carbon arc**

Oscillation **None**

Contact Tube to Work Distance **15 - 25 (mm)**

Multiple or Single Pass (Per Side) **Multiple or single pass**

Multiple or Single Electrode **Single**

Electrode Spacing **None**

Peening **None**

Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

La Vinh Nhon

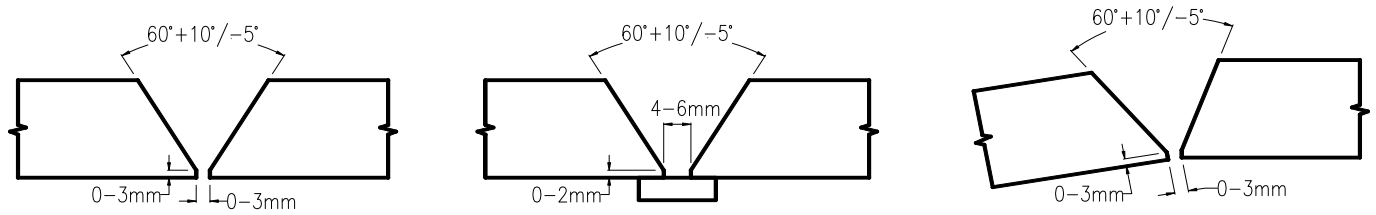
Date: **Aug 25, 2023**Approved by: **Truong Hoai Phong Vu (QA Manager)**Date: **Aug 25, 2023**

Welding Procedure Specification No.: **WPS-FCAW+SAW-79.1 & WPS-FCAW-79.2**

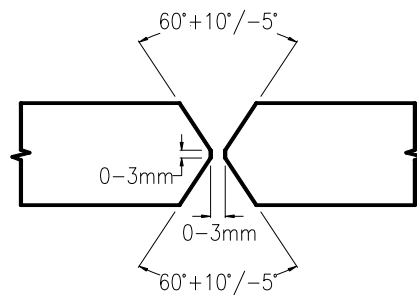
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

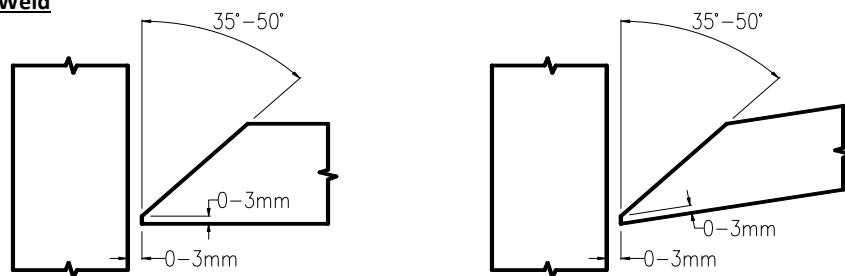
1. Single V Groove Butt Joint



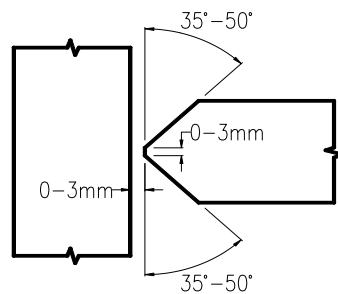
2. Double V Groove Butt Joint



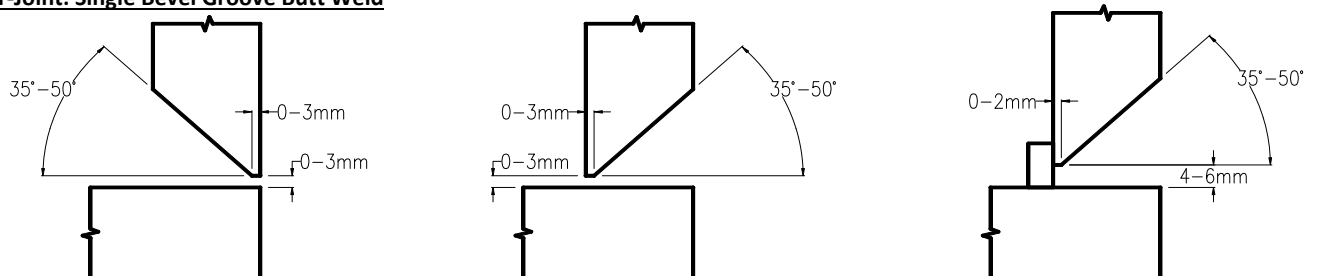
3. T-Joint: Single Bevel Groove Butt Weld



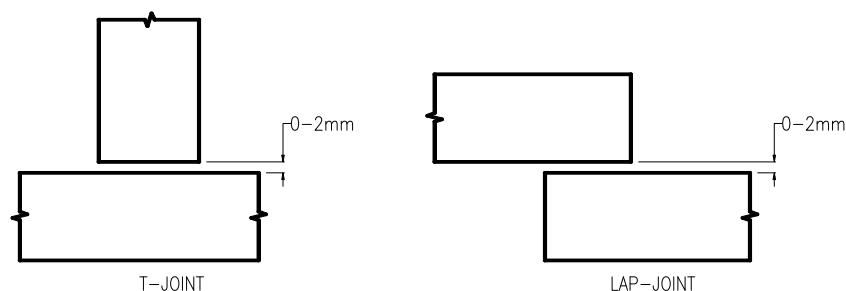
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld



Signature

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW+SAW-79**

Date: **Dec 14, 2020**

Welding Process(es) **FCAW + SAW**

Type(s)(Manual, Automatic, Semi-Automatic) **Semi-Automatic and Automatic**

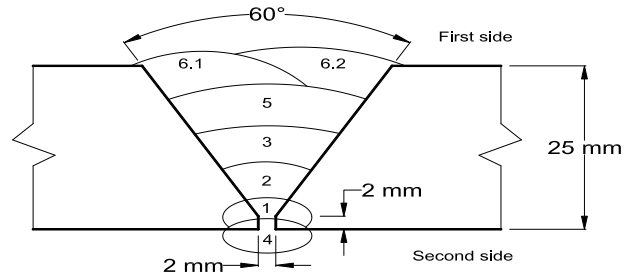
JOINTS (QW-402)

Joint Type **Butt joint, single V groove**

Backing **None**

Retainer **None**

Deposited weld metal thickness:

FCAW 12 mm;
SAW 13 mm


BASE METALS (QW-403)

Material Spec. **JIS G3101 and ASTM**

Type/Grade, or UNS Number **SS400 to A36**

P-No. - Group No. - to P-No. **1** Group No. **1**

Thickness of Test Coupon **25**

Diameter of Test Coupon **Plate**

Maximum Pass Thickness **< 13 mm**

Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**

Time **None**

Other **None**

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	CO2	100%	20 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process	FCAW	SAW
SFA Specification	AWS A5.20	A5.17
AWS Classification	E71T-1C	EM12K
Filler Metal F-No.	6	6
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	1.2 mm	3.2 mm
Filler Metal Product Form	Flux cored wire	Solid wire
Supplemental Filler Metal	None	None
Filler Metal Trade Name	KFX-71T	KW-3
Electrode Flux Classification	-	EM12K-F7A2
Flux Type	-	Agglomerated
Flux Trade Name	-	JF-B
Weld Metal Thickness	12 mm	13 mm
Other	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process	FCAW	SAW
Current and Polarity	DC EP	DC EP
Amps.	See table below	
Volts	See table below	
Mode of Metal Transfer	None	
for GMAW (FCAW)	Globular	None
Other	None	

POSITION (QW-405)

Process	FCAW	SAW
Position(s)	1G	1G
Weld Progression (Uphill, Downhill)	None	None
Other	None	None

PREHEAT (QW-406)

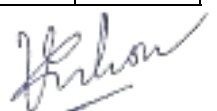
Preheat Temperature	39 °C
Interpass Temperature	Max. 129 °C
Other	None

TECHNIQUE (QW-410)

Process	FCAW	SAW
String or weave Bead	Weave	String
Oscillation	None	None
Method cleaning	Grinding, brushing	
Method back gouge	Air carbon arc, grinding	
Multipass or single pass (per side)	Multipass	Multipass
Multipass or single Electrodes	Single	Single
Peening	None	None
Use of thermal processes	None	None
Other	None	None

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1	FCAW	E71T-1C	1.2	DC EP	140 - 150	26 - 27	255	39	5	20	0.9 - 1.0
2	FCAW	E71T-1C	1.2	DC EP	210 - 230	30 - 31	218	120	3	20	1.7 - 2.0
3	FCAW	E71T-1C	1.2	DC EP	220 - 240	30 - 31	162	125	4	20	2.4 - 2.8
4	FCAW	E71T-1C	1.2	DC EP	225 - 240	30 - 31	224	31	4	20	1.8 - 2.0
5	SAW	EM12K-F7A2	3.2	DC EP	432 - 452	32 - 35	178	70	7	-	4.7 - 5.3
6.1	SAW	EM12K-F7A2	3.2	DC EP	430 - 450	32 - 35	134	115	8	-	6.2 - 7.1
6.2	SAW	EM12K-F7A2	3.2	DC EP	434 - 451	32 - 35	158	129		-	5.3 - 6.0



Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-FCAW+SAW-79** Date: **Dec 14, 2020**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	20.2	24.9	503	243.8	485	Ductile and base metal
Sample 2	20	24.9	498	248.1	498	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1 (side bend)	Pass
Specimen 2 (side bend)	Pass
Specimen 3 (side bend)	Pass
Specimen 4 (side bend)	Pass

Toughness Tests (QW-170) (Not Apply)

Specimen No.	Notch Location	Specimen Size	Test Temperature		Impact Values		Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	

Comments _____

Fillet-Weld Test (QW-180) (Not Apply)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: _____

Other Tests (Not Apply)

Type of Test _____

Deposit Analysis _____

Other _____

Welder's Name **Nguyen Hoang Nhan** Clock No. **None** Stamp (ID) No. **DDC-WD-0042**

Welder's Name **Duong Van Hieu** Clock No. **None** Stamp (ID) No. **DDC-WD-0091**

Tests Conducted by **IQTC** Laboratory Test No. **1311-1/VL-IQTC**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX Edition 2017 of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

LA VINH NHON

Date: **Dec 14, 2020**

Approved by: **TRUONG HOAI PHONG VU**

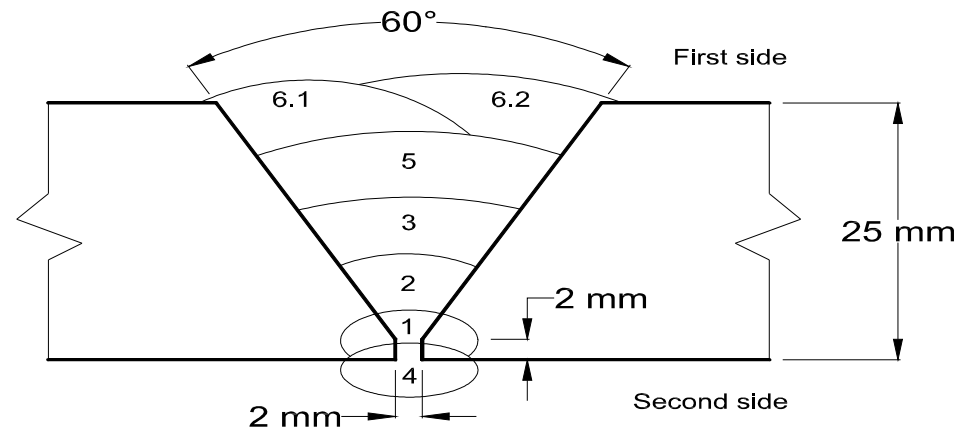
Date: **Dec 14, 2020**

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW+SAW-79)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-FCAW+SAW-79
 Test Coupon: DD-FCAW+SAW-79
 Test position: 1G
 Base metal:
 - Specification: JIS G3101-SS400 to ASTM-A36
 - Dimension: 400x200x25 (mm) x 2
 - Heat/Product No.: SB47909 / PB23726101 & SK53370/PK58967201-5303
 - Certificate No.: 190225-FP01PS-0002A1-0003 & 191008-FPQ8IS-0013A1-0033
 Filler metal:
 - Specification: AWS A5.20 & A5.17
 - Classification: E71T-1C & EM12K
 - Trade name: KFX-71T & KW-3
 - Batch/Lot No.: V20829013 & 190313882
 - Certificates No.: 20090101 & 19032001
 Gas: CO2 (100%)
 Welder Name: Nguyen Hoang Nhan & Duong Van Hieu
 I.D.: DDC-WD-0042 & DDC-WD-0091

Joint Sketch:

Date: Nov 11, 2020



Root face: 2 mm
 Root gap: 2 mm
 Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
FRONT SIDE (First Side):																
1	FCAW	A5.20 E71T-1C	1.2	DC EP	140-150	26-27	1	34	400	255	39	20	0.9-1	5	5	
2	FCAW	A5.20 E71T-1C	1.2	DC EP	210-230	30-31	1	50	400	218	120	20	1.7-2	10	3	
3	FCAW	A5.20 E71T-1C	1.2	DC EP	220-240	30-31	2	28	400	162	125	-	2.4-2.8	14	4	
5	SAW	A5.17 EM12K-F7A2	3.2	DC EP	432-452	32-35	2	15	400	178	31	-	4.7-5.3	23	7	
6.1	SAW	A5.17 EM12K-F7A2	3.2	DC EP	430-450	32-35	2	59	400	134	70	-	6.2-7.1	25	8	Reinforcement: 2 mm
6.2	SAW	A5.17 EM12K-F7A2	3.2	DC EP	434-451	32-35	2	32	400	158	115	-	5.3-6	25		



WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW+SAW-79)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
BACK SIDE (Second Side) with back gouging:																
4	FCAW	A5.20 E71T-1C	1.2	DC EP	225-240	30-31	1	47	400	224	129	20	1.8-2	16	4	Reinforcement: 1 mm
Conducted by Welding Coordinator																
Name		LA VINH NHON														
Signature																
Date		Nov 11, 2020														



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488

- Trụ sở chính: 27/29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- Head office: 27/29 7A Street, Binh Tri Dong Residential Area, Ward Binh Tri Dong B, Binh Tan District, HCM City
- Điện thoại/Tel: 028 38452 303 Fax: 028 38452 397 MST: 0313138782
- Chi nhánh PTN: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- Branch of Laboratory: Binh Tien 2 Hamlet, Duc Hoa Ha Commune, Duc Hoa Dist., Long An Province
- Điện thoại/Tel: 0272 3777 641 Fax: 0272 3777 640 Email: iqtc@iqtc.vn

Số TN/Ref: 1311-1VL-IQTC

Phiếu yêu cầu số/Request No.: 01-121120/YCTN

KẾT QUẢ THÍ NGHIỆM MỐI HÀN THÉP TẤM

TEST RESULTS FOR WELDING STEEL PLATE

- Công trình/Project: THREE RIVERS ENERGY CENTER
- Hạng mục/Item: DUCTING
- Địa điểm/Location: EVAPCO - USA
- Cơ quan yêu cầu/Client: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.
- Tên mẫu/Name of sample: WELDING STEEL PLATE
- FOR WELDING PROCEDURE: DD-FCAW+SAW-79, FCAW+SAW-1G
- Yêu cầu kiểm tra/Testing requirements: + Thử kéo/Tensile test
+ Thử uốn/Bend test
- Mô tả mẫu/Description of sample: 400x400x25mm Mã thợ hàn/Welder code: DDC-WD-0042 & DDC-WD-0091
- Vật liệu/Material: JIS G3101 SS400 to ASTM A36
- Số lượng/Quantity: 01 Mẫu/Sample
- Ngày nhận mẫu/Sampling date: 12/11/2020
- Ngày thí nghiệm/Testing date: 13/11/2020
- Thiết bị thí nghiệm/Testing equipment: Máy điện tử đa năng WAW-1000/ Machine electronic versatile
- Phương pháp thử/Test Method: ASME IX:2017
- Nhóm thí nghiệm/Tested by: Mai Hung Khanh, Nguyen Anh Vu

Tên chỉ tiêu/ Specification	Đơn vị/ Unit	Phương pháp thử nghiệm/ Test method	Yêu cầu/ Requirements	Kết quả/ Results	
				Mẫu 1/ Sample 1	Mẫu 2/ Sample 2
1. Thử kéo/ Tensile test		ASME IX:2017; ASTM A370-17	JIS 3101 SS400; ASTM A36		
- Chiều dày mẫu/Thickness of specimen	mm		-	24.9	24.9
- Chiều rộng mẫu/Width of specimen	mm		-	20.2	20.0
- Tiết diện thực đo/Cross section	mm ²		-	503.0	498.0
- Lực kéo đứt/Tensile load	kN		-	243.8	248.1
- Giới hạn bền kéo/Tensile strength	MPa		400 min	485	498
- Vị trí mẫu đứt/Broken position				Kim loại cơ bản/ Base metal	Kim loại cơ bản/ Base metal

Long An, ngày/date 13 tháng/month 11 năm/year 2020

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

Đ/D NHÓM THÍ NGHIỆM
(LAB REP.)

TRƯỞNG PHÒNG THÍ NGHIỆM
(LAB MANAGER)

KT. GIÁM ĐỐC CHI NHÁNH
REP. BRANCH DIRECTOR

KS/ENG. MAI HUNG KHANH

KS/ENG. NGUYEN ANH DINH

KS/ENG. TRINH DINH TAN

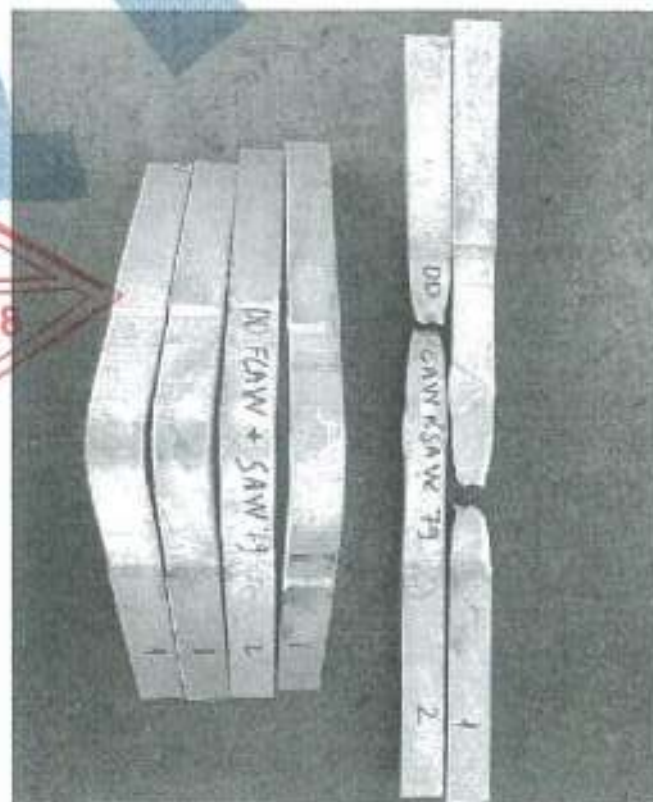
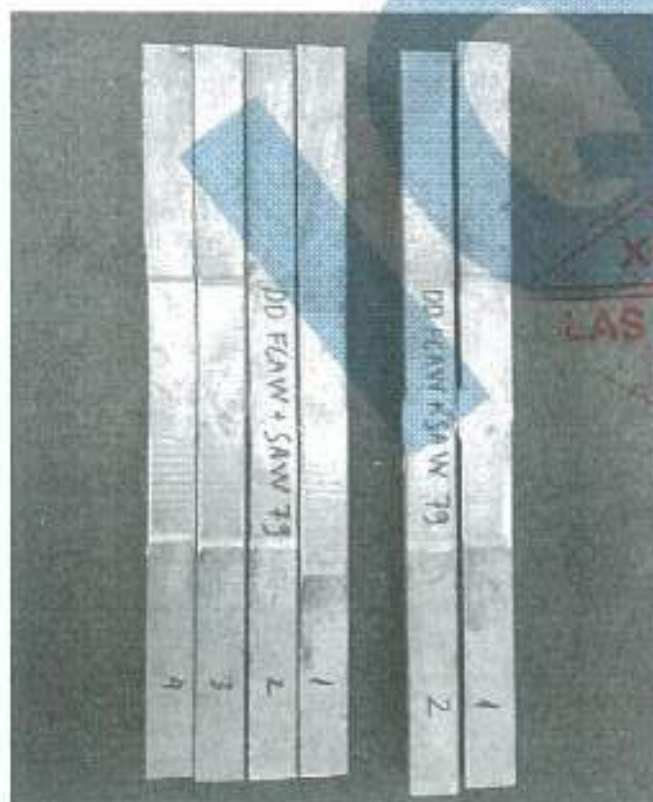
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY (IQTC)



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488

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- Điện thoại/Tel: 028 38452 303 Fax: 028 38452 397 MST: 0313136752
- Chi nhánh PTN: Ấp Bình Tiên 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- Branch of Laboratory: Bình Tiên 2 Hamlet, Đức Hòa Hạ Commune, Đức Hòa Dist., Long An Province
- Điện thoại/Tel: 0272 3777 641 Fax: 0272 3777 640 Email: iqtc@iqtc.vn

Tên chi tiêu/ Specification	Đơn vị/ Unit	Kết quả/ Test Results			
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
2. Thử uốn/ Bend test					
- Vị trí uốn/Location bend		Uốn cạnh/ Side bend	Uốn cạnh/ Side bend	Uốn cạnh/ Side bend	Uốn cạnh/ Side bend
- Chiều dày/Thickness of specimen	mm	10.1	10.1	10.0	10.0
- Chiều rộng/Width of specimen	mm	24.9	24.9	24.9	24.9
- Góc uốn/Bending angle	Độ/°	180	180	180	180
- Đường kính gồ uốn/Plunger diameter	mm	38	38	38	38
- Kết quả/Test results		Đạt/ Pass	Đạt/ Pass	Đạt/ Pass	Đạt/ Pass





CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
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- ❖ Branch of Laboratory: Binh Tien 2 Hamlet, Duc Hoa Ha Commune, Duc Hoa Dist, Long An Province
- ❖ Điện thoại/Tel: 0723 777641 Fax: 0723 777640 Email: qtc@iqtc.vn

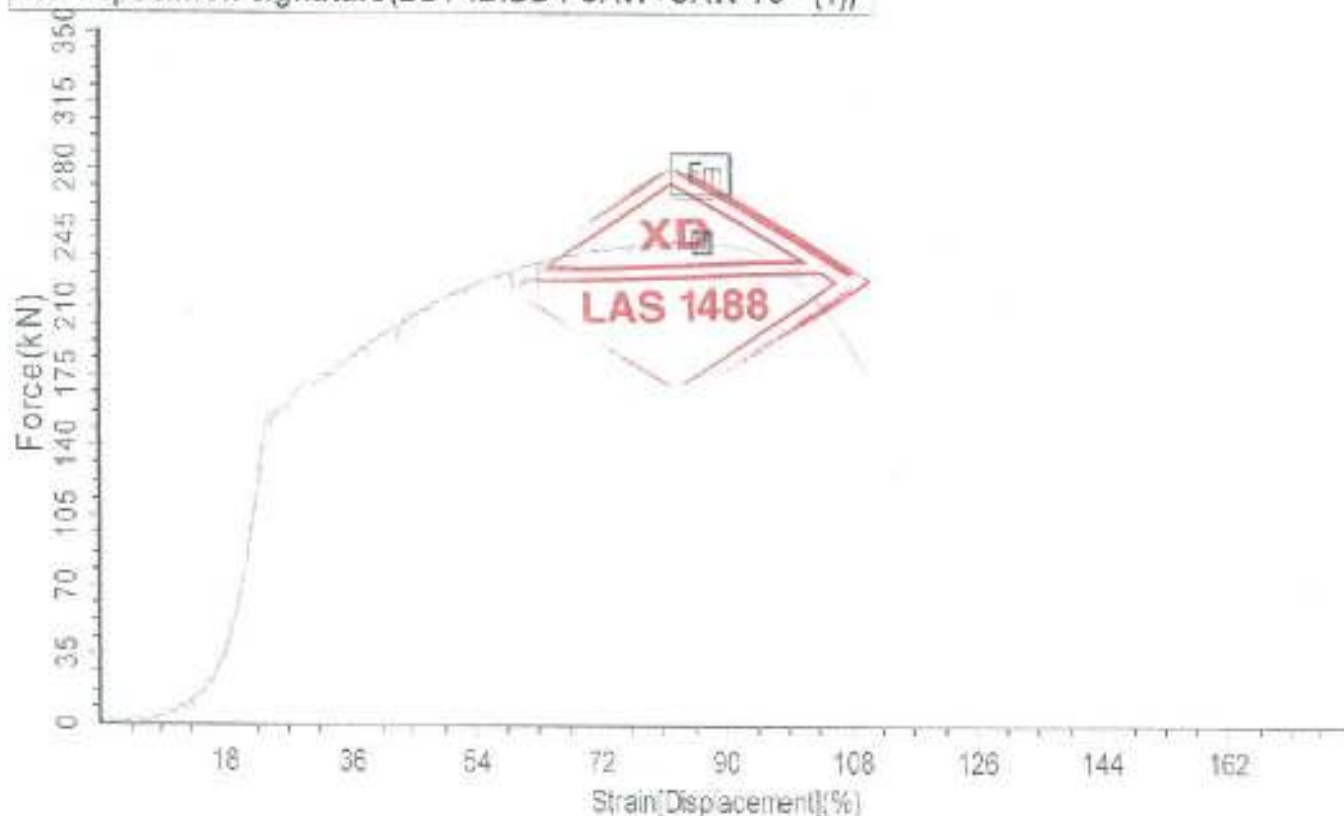
LOT ID: DD-FCAW+SAW-79

Company	DAIDUNG CORPORATION	Test Date	11-13-2020
Standard	ASME IX:2017	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	80%
Machine Model	WAW-1000	Capacity	1000KN

Test Data

No.	Width mm	ao.Ta mm	Fm kN	Rm MPa						
01	20,20	24,90	243,8	485						

No.1 specimen signature(LOT ID:DD-FCAW+SAW-79 - (1))



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

MAI HUNG KHANH



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
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- ✦ Chi nhánh PTM: Ấp Bình Tân 2, Xã Bình Hòa, Huyện Đức Hòa, Tỉnh Long An
- ✦ Branch: Ấp Bình Tân 2, Xã Bình Hòa, Huyện Đức Hòa, Tỉnh Long An
- ✦ Điện thoại/Tel: 0723 777641 Fax: 0723 777640 Email: info@iqtc.vn

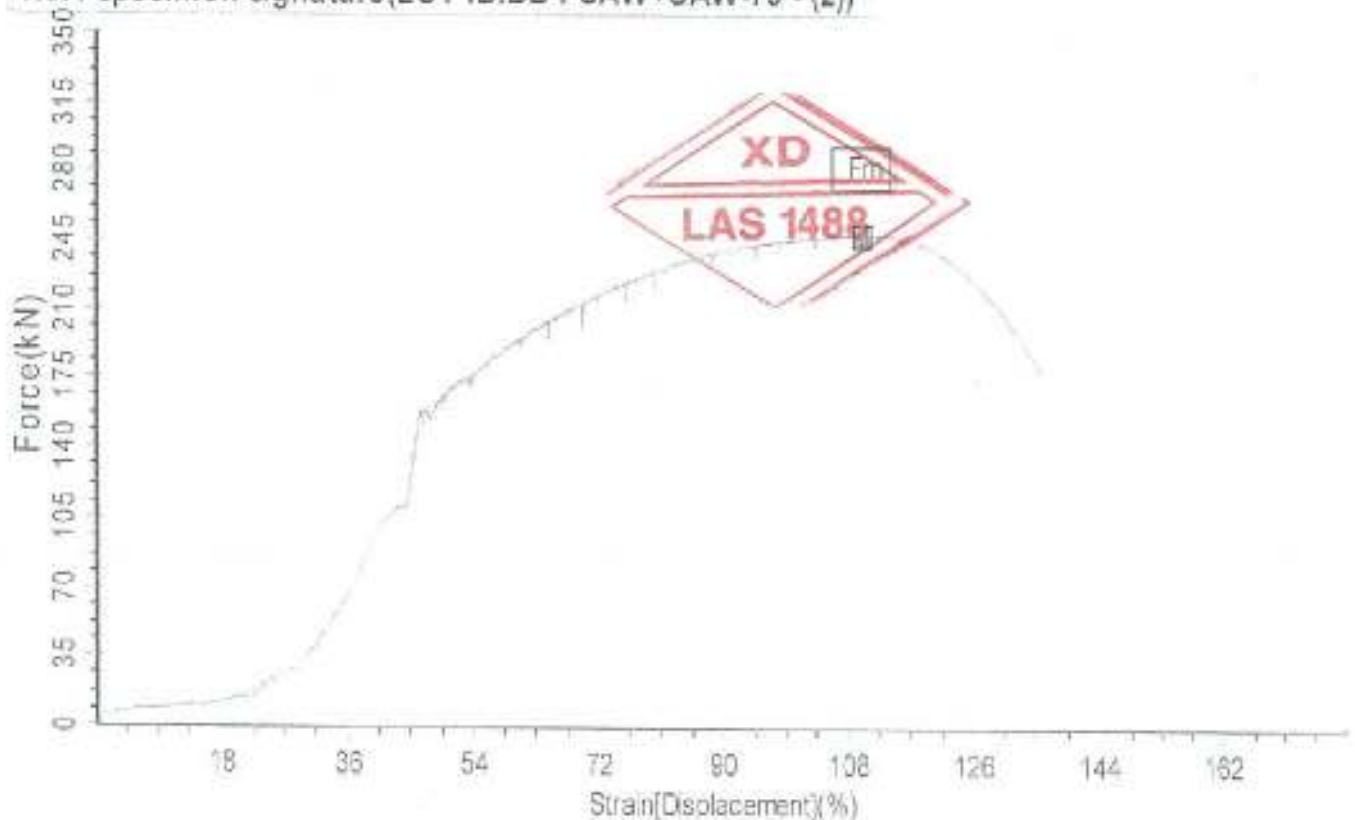
LOT ID: DD-FCAW+SAW-79

Company	DAIDUNG CORPORATION	Test Date	11-13-2020
Standard	ASME IX:2017	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	60%
Machine Model	WAW-1000	Capacity	100CKN

Test Data

No.	Width mm	ao.Ta mm	Fm kN	Rm MPa						
01	20,00	24,90	248,1	498						

No.1 specimen signature(LOT ID:DD-FCAW+SAW-79 - (2))



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

Mai Hùng Khanh



B0915-100534

2020/01-0199



faide's 64ab etc

1. Goods consigned from (Exporter's name, address, country) PT KRAKATAU POSCO JL. APRIKA NO. 02 KAWASAN INDUSTRI KRAKATAU CILEGON 42113, INDONESIA			Reference No. 0094999/JKT/2019 ASEAN TRADE IN GOODS AGREEMENT ASEAN INDUSTRIAL COOPERATION SCHEME CERTIFICATE OF ORIGIN (Combined Declaration and Certificate) FORM D Issued in INDONESIA (Country) See Overleaf Notes		
2. Goods consigned to (Consignee's name, address, country) HOAN PHUC TRADING JSC 14 VU TUNG ST. WARD 2, BINH THANH DIST. HOCHIMINH CITY, VIETNAM			4. For Official Use ☐ Preferential Treatment Given Under ASEAN Trade in Goods Agreement ☐ Preferential Treatment Given Under ASEAN Industrial Cooperation Scheme ☐ Preferential Treatment Not Given (Please state reason(s)) Signature of Authorized Signatory of the Importing Country		
3. Means of transport and route (as far as known) Departure Date: OCTOBER 19, 2019 Vessel's name/Aircraft, etc.: MV ARISTOS II V-19001 Port of Discharge: HOCHIMINH CITY PORT, VIETNAM					
5. Item number	6. Marks and numbers on packages	7. Number and type of packages, description of goods (including quantity where appropriate and HS number of the importing country)	8. Origin criterion (see Overleaf Notes)	9. Gross weight or other Quantity and Value (FOB) where RVC is applied	10. Number and date of invoice
1	MARKING: HEAT NO. PLATE NO. SPECIFICATION DIMENSION HOAN PHUC PC NO PT. KRAKATAU POSCO MADE IN INDONESIA	PRIME HOT ROLLED STEEL PLATES -- OF A THICKNESS EXCEEDING 10 MM TOTAL WEIGHT 3,798,309 MT TOTAL NO. PCS 985 PCS HS: 720851	RVC 21.00 % + CTSH	3,798,309 KGM USD 1,853,371.25	1910084716 OCTOBER 19, 2019
2		PRIME HOT ROLLED STEEL PLATES -- OF A THICKNESS OF 4.75 MM OR MORE BUT NOT EXCEEDING 10 MM TOTAL WEIGHT 1,183,780 MT TOTAL NO. PCS 628 PCS HS: 720852 THIRD PARTY INVOICING: SNP LIMITED SUITE 2103, TRADE TOWER 511, YEONGDONG-DAERO, GANGNAM-GU SEOUL REPUBLIC OF KOREA	RVC 21.00 % + CTSH	1,183,780 KGM USD 521,218.33	1910084716 OCTOBER 19, 2019
11. Declaration by the exporter The undersigned hereby declares that the above dated and signed statement are correct; that all the goods were produced in _____ INDONESIA (Country) and that they comply with the origin requirements specified for these goods in the ASEAN Trade in Goods Agreement for the goods exported to _____ VIET NAM (Importing Country) _____ CILEGON, OCTOBER 23, 2019 Place and date, signature of authorized signatory			12. Certification It is hereby certified, on the basis of control carried out, that the declaration by the exporter is correct ISSUING OFFICE IN PROVINSI DKI JAKARTA  BADIA RAJA H. SIREGAR JAKARTA, OCTOBER 23, 2019 Place and date, signature and stamp of certifying authority		
13. ☒ Third Country Invoicing ☐ Exhibition ☐ Accumulation ☐ De Minimis ☐ Back-to-Back CO ☒ Issued Retroactively ☐ Partial Cumulation					



Order No./계약번호 : Q8S1015812

PO No./주문번호 : SNP-A-19059

Supplier /주문자 : SNP LIMITED

Commodity : PLATE

Customer /고객사 : HOAN PHUC TRADING

Spec & Type : ASTM A36

ORIGINAL

Size/치수	Product No. (제품번호)	Quantity (수량)	Weight (중량) (kg)	Heat No. (재질번호)	Position	Tensile Test (인장 시험)			Division	Chemical Composition/화학 성분											
						YP (MPa)	TS (%)	EL (%)		C (%)	Si (%)	Mn (%)	P (%)	S (%)	Cr (%)	Mo (%)	Cu (%)	Nb (%)	V (%)		
22x2001x12000	PK58956301-6303	3	12,441	SK53370	T	280	438	31	L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002	
22x2001x12000	PK58956601-6603	3	12,441	SK53370	T	280	438	31		P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002
22x2001x12000	PK58956701-6703	3	12,441	SK53370	T	280	438	31		L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002
22x2001x12000	PK58956201-6203	3	12,441	SK53370	T	272	434	34	P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002	
22x2001x12000	PK58956801-6803	3	12,441	SK53370	T	272	434	34		L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002
22x2001x12000	PK58956901-6903	3	12,441	SK53370	T	272	434	34		P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002
22x2001x12000	PK58957001-7003	3	12,441	SK53373	T	289	452	30	L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002	
22x2001x12000	PK58957101-7103	3	12,441	SK53373	T	284	449	32		P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002
22x2001x12000	PK58957201-7203	3	12,441	SK53373	T	284	449	32		L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002
*** Sub Total (070) ***		24	99,528 (kg)						P	0.1558	0.201	0.872	0.0107	0.0049	0.024	0.010	0.016	0.003	0.000	0.002	
25x2001x12000	PK58967201-7202	2	9,422	SK53370	T	280	438	31	L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002	
25x2001x12000	PK58967401-7402	2	9,422	SK53370	T	280	438	31		P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002
25x2001x12000	PK58967501-7502	2	9,422	SK53370	T	280	438	31		L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002
25x2001x12000	PK58967601-7602	2	9,422	SK53370	T	280	438	31	P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002	
25x2001x12000	PK58967701-7702	2	9,422	SK53370	T	280	438	31		L	0.1557	0.183	0.851	0.0115	0.0049	0.025	0.012	0.023	0.003	0.002	0.002
25x2001x12000	PK58967801-7802	2	9,422	SK53370	T	280	438	31		P	0.1526	0.182	0.841	0.0119	0.0057	0.025	0.012	0.021	0.004	0.000	0.002

* Position - T : Top, M : Middle, B : Bottom

* Tensile Test, Direction : Transversal, Gauge Length : 200mm(Reclangular), YP Method : 0.2% off-set

* Division - L : Ladle Analysis, P : Products Analysis

* Supply Condition : As-Rolled unless otherwise Heat Treated

We hereby certify that the material herein has been made in accordance with the order and above specification.

This material has been fully killed and made by basic oxygen process.

This material has been made by vacuum degassing process.

Test Certificate is issued according to EN10204 3.1.

Legal sanction can be imposed on forging. Improper use of product can cause safety issues.

Surveyor To :

Choi Jong Seog

POSCO - 1209 NT

51.1812.409



1. Exporter (Name, address, country) POSCO POSCO CENTER, 440, TEHERAN-RO, GANGNAM-GU, SEOUL, 06194 KOREA	Reference No. K001-19-0151092 Reference Code. 04ae-2b50	ORIGINAL																
CERTIFICATE OF ORIGIN issued by THE KOREA CHAMBER OF COMMERCE & INDUSTRY Seoul, Republic of Korea																		
2. Consignee (Name, address, country) DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION B23/474C, TRAN DAI NGHIA STREET, HAMLET 2, TAN NHUT WARD, BINH CHANH DISTRICT, HOCHIMINH CITY, VIETNAM	3. Country of Origin THE REPUBLIC OF KOREA	5. Remarks 																
4. Transport details FROM : POHANG PORT IN SOUTH KOREA TO : HO CHI MINH PORT, VIETNAM BY : INTERLINK EQUITY ON : 27 FEB., 2019																		
6. Marks & numbers; number and kind of packages; description of goods <table border="0"> <tr> <td>DDC/SPEC/</td> <td colspan="3">HOT ROLLED STEEL PLATE</td> </tr> <tr> <td>SIZE/WEIGHT/</td> <td>522 PCS</td> <td>1,209.253 MT</td> <td>1,209.253 MT</td> </tr> <tr> <td>MANUFACTURER/</td> <td colspan="3">TOTAL</td> </tr> <tr> <td>KOREA</td> <td>522 PCS</td> <td>1,209.253 MT</td> <td>1,209.253 MT</td> </tr> </table>		DDC/SPEC/	HOT ROLLED STEEL PLATE			SIZE/WEIGHT/	522 PCS	1,209.253 MT	1,209.253 MT	MANUFACTURER/	TOTAL			KOREA	522 PCS	1,209.253 MT	1,209.253 MT	7. Quantity
DDC/SPEC/	HOT ROLLED STEEL PLATE																	
SIZE/WEIGHT/	522 PCS	1,209.253 MT	1,209.253 MT															
MANUFACTURER/	TOTAL																	
KOREA	522 PCS	1,209.253 MT	1,209.253 MT															
End Of Page...																		
8. Declaration by the Exporter The undersigned, as an authorized signatory, hereby declares that the above-mentioned goods were produced or manufactured in the country shown in box 3.  HONG-GYUN CHUNG / GROUP LEADER LOGISTICS PLANNING GROUP (Name) GROUP LEADER HONG-GYUN CHUNG		9. Certification The undersigned authority hereby certifies that the goods described above originate in the country shown in box 3 to the best of its knowledge and belief.  Deputy Director IN-SUG KIM 04 MAR 2019  Authorized Signatory																

1/1

*Identify the authenticity of certificate for verification with the website.(http://cert.korcham.net/search)



Order No./계약번호 : 01S3540241

PO No./주문번호 : GHD1812SH-SMI02

Supplier : POSCO Asia Co., Ltd.
/주문자

Commodity : STEEL PLATE
/품명

Customer : DDC
/고객사

Spec & Type : JIS G3101 SS400
/규격

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /재질번호	Position	Tensile Test /인장시험			Division	Chemical Composition/화합성분				
						YP (N/mm ²)	TS (%)	EL (%)		C (%)	Si (%)	Mn (%)	P (%)	S (%)
16x2000x12000	PB23673004-3006	3	9,042	SB48145	T	288	452	30	L	0.1879	0.257	0.498	0.0106	0.0061
16x2000x12000	PB23668103-8105	3	9,042	SB48148	T	316	464	31	L	0.2015	0.244	0.495	0.0109	0.0026
16x2000x12000	PB23668203-8205	3	9,042	SB48148	T	316	464	31	L	0.2015	0.244	0.495	0.0109	0.0026
*** Sub Total (080) ***		10	30,140 (kg)											
18x2000x12000	PB23678501-8504	4	13,554	SB47911	T	284	448	32	L	0.1835	0.252	0.478	0.0128	0.0039
18x2000x12000	PB23679601-9605	5	16,955	SB48138	T	276	448	29	L	0.2073	0.241	0.494	0.0137	0.0045
*** Sub Total (070) ***		9	30,519 (kg)											
20x2000x12000	PB23668801-8804	4	15,072	SB48148	T	280	455	32	L	0.2015	0.244	0.495	0.0108	0.0026
20x2000x12000	PB23668902-8904	3	11,304	SB48148	T	280	455	32	L	0.2015	0.244	0.495	0.0108	0.0026
*** Sub Total (080) ***		7	26,376 (kg)											
25x2000x12000	PB23728101	1	4,709	SB47909	T	278	440	30	L	0.1811	0.241	0.482	0.0123	0.0040
*** Sub Total (090) ***		1	4,709 (kg)											

* Position - T : Top, M : Middle, B : Bottom

* Tensile Test, Direction : Transversal, Gauge Length : 200mm(Rectangular),

YP Method : Upper Point

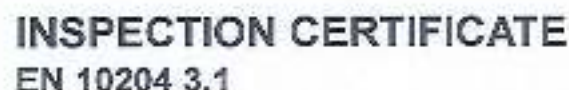
* Division - L : Ladle Analysis

* Supply Condition : As-Rolled unless otherwise Heat Treated.

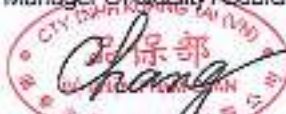
We hereby certify that the material herein has been made in accordance with the order and above specification.
This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
Test Certificate is issued according to EN10204 3.2.

Legal sanction can be imposed on forging. Improper use of product can cause safety issues.

Surveyor To : USER



NHON TRACH IND PARK 3, NHON TRACH, DONG NAI, VIET NAM
TEL: +84 251-3560097 FAX: +84-251-3560096 E-mail: ktvnga@kuanqai.com

Customer	CTY DAI DUNG					Commodity	KFX-71T			Mfg. Date	2020/08		Lot No.	V20829013		
Specification & Classification	AWS A5.20 E71T-1C ASME SECT II PART C- 2017 SFA-5.20 E71T-1C					Dimension	1.2 mm * 15 KG			Issue Date	1-Sep-20		Certificate No.	20090101		
□Chemical Composition for Solid Electrode and Rod (%) / ■Weld Metal Chemical Composition (%)																
Element	C	Mn	Si	S	P	Cr	Ni	Mo	V	Cu						
Requirement	MAX 0.12	MAX 1.75	MAX 0.90	MAX 0.03	MAX 0.03	MAX 0.20	MAX 0.50	MAX 0.30	MAX 0.08	MAX 0.35						
Test Result	0.035	1.35	0.34	0.007	0.011	0.011	0.010	0.012	0.009	0.07						
Category	Tension Test					Impact Test							Welding Condition			
	Yield * Strength N/mm²	Yield ** Strength N/mm²	Tensile Strength N/mm²	Elongation %	Test Temp. °C	Charpy V-Notch Absorbed Energy J (AVG)						Type of Current	Welding Current A	Arc Voltage V	Shielding Gas or Flux	Postweld Heat Treatment
Requirement	MIN 390	NS	490-670	MIN 22	-30	MIN 27						DCEP	NS	NS	C1	NS
Test Result	472	NS	578	28	-30	104	101	107	98	109	(104)	DCEP	NS	NS	C1	NS
Bend Test			Hardness Test	Fillet Weld Test	Radiographic Test	Moisture %	Diffusible Hydrogen mL/100gDM (AVG)		Heat No.		Mn+Ni+Cr+Mo+V					
Face	Rcot	Side														
NR	NR	NR	NR	ACCEPTABLE	ACCEPTABLE	NR	NR		SP54315		NS					
We hereby certify that this report is correct and that all test results are in compliance with the specification described herein. The material described above is free irradiation and free mercury.						Remarks *0.2% offset ** Lower Yield point NS=not specified NR=not required C1=100%CO ₂ M21=Ar+15~25%CO ₂						Certificate: ABS BV DNV .GL NK LR				
ISO 9001:2015 CERTIFICATE NO. NO.:18/2292			Manager Of Quality Assurance 													



INSPECTION CERTIFICATE

EN 10204 3.1

KUANG TAI METAL (VIETNAM) CO., LTD.
 NHON TRACH IND PARK 3, NHON TRACH, DONG NAI, VIET NAM
 TEL: +84 251-3560097 FAX: +84-251-360096 E-mail: ktvnqa@kuangtai.com

Customer	CTY DAI DUNG II <i>FJ 11.10.2019</i>					Commodity	KW-3			Mfg. Date	2019/03		Lot No.	190313882		
Specification & Classification	AWS A5.17 F7A2-EM12K					Dimension	3.2 mm * 25 KG			Issue Date	20-Mar-19		Certificate No.	19032001		
☒ Chemical Composition for Solid Electrode and Rod (%) / ☐ Weld Metal Chemical Composition (%)																
Element	C	Mn	Si	P	S	Cu										
Requirement	0.05~ 0.15	0.80~ 1.25	0.10~ 0.35	MAX 0.030	MAX 0.030	MAX 0.35										
Test Result	0.08	1.00	0.20	0.008	0.006	0.10										
Category	Tension Test				Impact Test						Welding Condition					
	Yield * Strength N/mm ²	Yield ** Strength N/mm ²	Tensile Strength N/mm ²	Elongation %	Test Temp. °C	Charpy V-Notch Absorbed Energy J (AVG)					Type of Current	Welding Current A	Arc Voltage V	Shielding Gas or Flux	Postweld Heat Treatment	
	MIN 400	NS	480-660	MIN 22	-30	MIN 27					DCEP	NS	NS	NS	NS	
Test Result	436	NS	535	28	-30	84	75	78	78	67	(76)	DCEP	NS	NS	JF-B	NS
Bend Test			Hardness	Fillet Weld	Radiographic	Moisture	Diffusible Hydrogen		Heat No.		Mn+Ni+Cr+Mo+V					
Face	Root	Side	Test	Test	Test	%	mL/100gDM (AVG)									
NR	NR	NR	NR	NS	ACCEPTABLE	NR	NR		1818177		NS					
We hereby certify that this report is correct and that all test results are in compliance with the specification described herein. The material described above is free irradiation and free mercury.					Remarks *0.2% offset ** Lower Yield point NS=not specified NR=not required C1=100%CO ₂ M21=Ar+15~25%CO ₂					Certificate: ABS BV						
ISO 9001:2015 CERTIFICATE NO. NO.:18/2292					Manager Of Quality Assurance 											



INSPECTION CERTIFICATE

EN 10204 3.1

KUANG TAI METAL (VIETNAM) CO., LTD.

NHON TRACH IND PARK 3, NHON TRACH, DONG NAI, VIET NAM

TEL:+84 251-3560097 FAX:+84-251-3560096 E-mail:ktvnqa@kuangtai.com

Customer	CTY DAI DUNG				Commodity	JF-B		Mfg. Date	2020/08		Lot No.	200851303		
Specification & Classification	AWS A5.17 F7A2-EM12K				Dimension	25 KG		Issue Date	31-Oct-20		Certificate No.	20103001		
Chemical Composition (%)														
Element	CaO+MgO	Al ₂ O ₃ +MnO	SiO ₂ +TiO ₂	CaF ₂	P	S	H ₂ O							
Requirement	...	...	...	...	...	...	...							
Test Result	...	...	...	...	0.035	0.025	...							
Category	Tension Test				Impact Test						Welding Condition			
	Yield * Strength N/mm ²	Yield ** Strength N/mm ²	Tensile Strength N/mm ²	Elongation %	Test Temp. °C	Charpy V-Notch Absorbed Energy J (AVG)				Type of Current	Welding Current A	Arc Voltage V	Wire	Postweld Heat Treatment
Requirement	MIN 400	NS	480-660	MIN 22	-30	MIN 27				NS	NS	NS	NS	NS
Test Result	420	NS	525	29	-30	72	65	78		(72)	NS	NS	NS	EM12K NS
Bend Test			Hardness Test	Fillet Weld Test	Radiographic Test	Moisture %	Diffusible Hydrogen mL/100gDM (AVG.)		Mesh		Mn+Ni+Cr+Mo+V			
Face	Root	Side												
NR	NR	NR	NR	NR	ACCEPTABLE	0.015	NR		10-60		NS			
We hereby certify that this report is correct and that all test results are in compliance with the specification described herein. The material described above is free irradiation and free mercury.					Remarks *0.2% offset ** Lower Yield point NS=not specified NR=not required C1=100%CO ₂ M21=Ar+15~25%CO ₂					Certificate:				

ISO 9001:2015
CERTIFICATE NO.
NO.:18/2292

Manager Of Quality Assurance

品保部
Chang



Comply:
ISO 9001:2015
ASME IX:2019

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW-133.1

WPS-DD-FCAW-133.2

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Aug 25, 2023	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2019)

Page: 1/2

Applicable Code: **ASME Section IX - 2019 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-GTAW-133.1** Revision No. **0** Date **Aug 25, 2023**

Supporting PQR No.(s): **PQR-DD-GTAW+FCAW-133** Date **Sep 27, 2021**

Welding Process(es) **GTAW** Type(s)(Manual, Automatic, Semi-Automatic): **Manual**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**

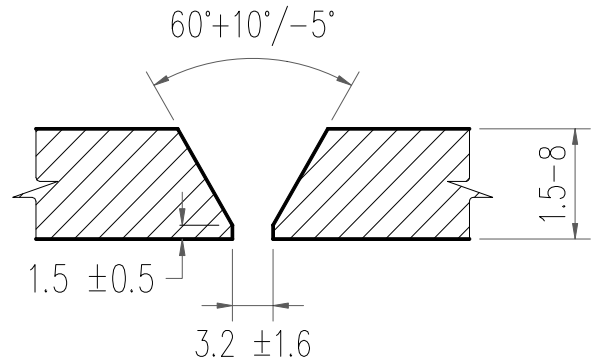
Root Spacing **As per fabrication drawing**

Backing **With or without**

Retainer **None**

Backing Material **Ceramic or P-No. 8 steel or weld metal**

Details



Joint detail as per fabrication drawing. If not specified in drawing, see sketch. All dimension are milimeters.

BASE METALS (QW-403)

P-No. **8** Group No. **1, 2** to P-No. **1** Group No. **1, 2**

OR

Specification and type/grade or UNS Number **Grade 304/304L/316/316L/316Ti for P-No.8 material**

to Specification and type/grade or UNS Number **P-No.1 Group No.1, 2**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **1.5 - 8 (mm) for groove (without CVN test)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other **None**

FILLER METALS (QW-404)

Process **GTAW**

Spec. No. (SFA) **SFA-5.9**

AWS No. (Class) **ER309L**

F-No. **6**

A-No. **8**

Size of Filler Metals **2.4 mm**

Supplemental Filler Metal **None**

Filler Metal Product Form **Solid rod**

Filler Metal Trade Name **Hyundai ST-309L or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 8 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2019)

Page: 2/2

Applicable Code: **ASME Section IX - 2019 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-GTAW-133.1** Revision No. **0** Date **Aug 25, 2023**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	Argon	99.99%	15 - 19 L/min.
Trailing	None	None	None
Backing	Argon	99.99%	15 - 19 L/min.
Other	None	None	None

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**
Interpass Temperature, Maximum **210 °C**
Preheat Maintenance **None**
Other **None**

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	GTAW	ER309L	2.4	DC EN	108 - 132	None	None	16 - 20	51 - 69	None
Fill	GTAW	ER309L	2.4	DC EN	153 - 187	None	None	16 - 20	78 - 106	None
Cap	GTAW	ER309L	2.4	DC EN	153 - 187	None	None	16 - 20	78 - 106	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **Φ2.4 mm, 2% Thoriated Tungsten (EWTh-2)**
Mode of Metal Transfer for GMAW (FCAW) **None**
Other **None**

TECHNIQUE (QW-410)

Process **GTAW**
String or Weave Bead **Weave bead**
Orifice, Nozzle, or Gas Cup Size **All size**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding**
Oscillation **None**
Contact Tube to Work Distance **None**
Multiple or Single Pass (Per Side) **Multiple or single pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

La Vinh Nhon

Date: **Aug 25, 2023**Approved by: **Truong Hoai Phong Vu (QA Manager)**Date: **Aug 25, 2023**



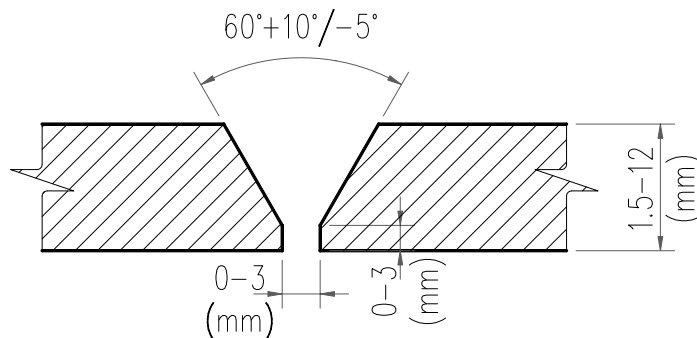
WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2019)

Page: 1/2

Applicable Code: **ASME Section IX - 2019 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-133.2** Revision No. **0** Date **Aug 25, 2022**
Supporting PQR No.(s): **PQR-DD-GTAW+FCAW-133** Date **Sep 27, 2021**
Welding Process(es) **FCAW** Type(s)(Manual, Automatic, Semi-Automatic): **Semi-Automatic, Mechanized**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**
Root Spacing **As per fabrication drawing**
Backing **With backing**
Retainer **None**
Backing Material **Ceramic or P-No. 8 steel or weld metal**

Details

Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **8** Group No. **1, 2** to P-No. **1** Group No. **1, 2**
OR

Specification and type/grade or UNS Number **Grade 304/304L/316/316L/316Ti for P-No.8 material**
to Specification and type/grade or UNS Number **P-No.1 Group No.1, 2**
OR

Chem. Analysis and Mech. Prop. **None**
to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **1.5 - 12 (mm) for groove (without CVN test)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other **None**

FILLER METALS (QW-404)

Process **FCAW**

Spec. No. (SFA) **SFA-5.22**

AWS No. (Class) **E309LT1-1/-4**

F-No. **6**

A-No. **8**

Size of Filler Metals **Ø1.2 mm**

Supplemental Filler Metal **None**

Filler Metal Product Form **Flux cored wire**

Filler Metal Brand Name **Hyundai SW-309L Cored or the same spec & class**

Weld Metal

Deposited Thickness:

Groove **Max. 12 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX - Boiler and Pressure Vessel Code-2019)

Page: 2/2

Applicable Code: **ASME Section IX - 2019 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-FCAW-133.2** Revision No. **0** Date **Aug 25, 2022**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

PREHEAT (QW-406)

Preheat Temperature, Minimum **10 °C**
Interpass Temperature, Maximum **210 °C**
Preheat Maintenance **None**
Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	CO₂	100%	15 - 25 L/min.
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

Note: only for FCAW Process and minimum purity 99.8% CO₂

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	FCAW	E309LT1-1/-4	1.2	DC EP	180 - 220	None	None	28 - 34	207 - 265	None
Fill	FCAW	E309LT1-1/-4	1.2	DC EP	180 - 220	None	None	28 - 34	207 - 265	None
Cap	FCAW	E309LT1-1/-4	1.2	DC EP	153 - 187	None	None	25 - 31	201 - 256	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **None**
Mode of Metal Transfer for GMAW (FCAW) **Globular**
Other **None**

TECHNIQUE (QW-410)

Process **FCAW**
String or Weave Bead **String & Weave (Max. width 15 mm)**
Orifice, Nozzle, or Gas Cup Size **No. 6**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding or Air carbon arc**
Oscillation **None**
Contact Tube to Work Distance **15 - 25 (mm)**
Multiple or Single Pass (Per Side) **Multiple or single pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

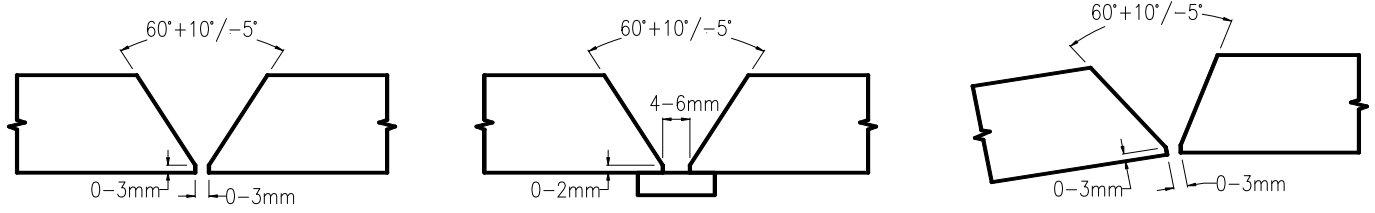
Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Certified by Welding Coordinator:
La Vinh NhonDate: **Aug 25, 2022**Approved by: **Truong Hoai Phong Vu (QA Manager)**Date: **Aug 25, 2022**

Welding Procedure Specification No.: WPS-DD-FCAW-133.2

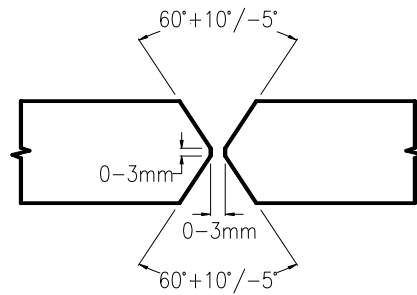
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

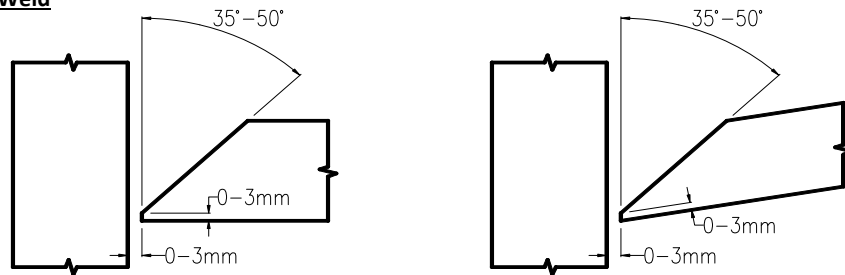
1. Single V Groove Butt Joint



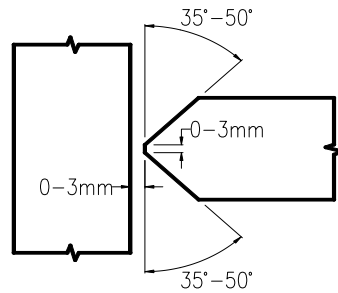
2. Double V Groove Butt Joint



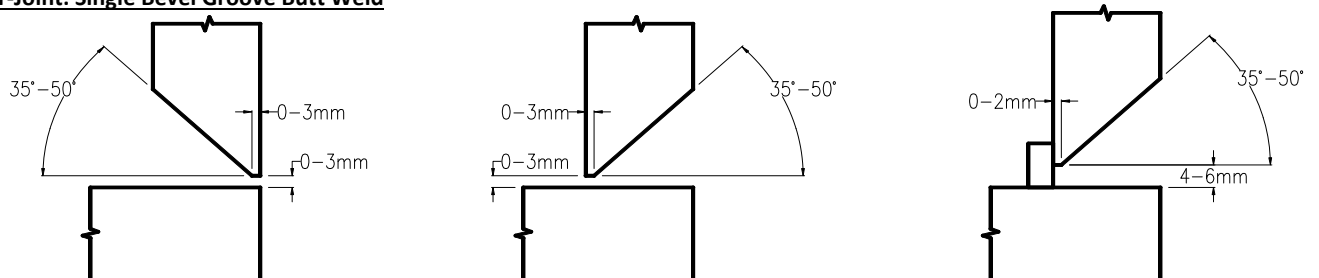
3. T-Joint: Single Bevel Groove Butt Weld



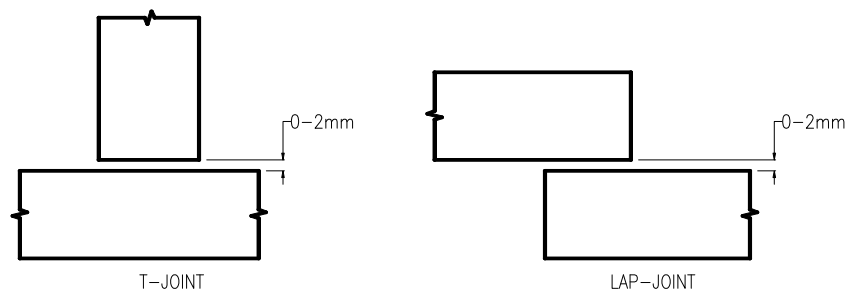
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld



Signature

PROCEDURE QUALIFICATION RECORD (PQR) (ASME IX - Boiler and Pressure Vessel Code-2019)

Page: 1/2

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-GTAW+FCAW-133** Date: **Sep 27, 2021**

Welding Process(es) **GTAW+FCAW**

Type(s)(Manual, Automatic, Semi-Automatic) **Manual + Semi-Automatic**

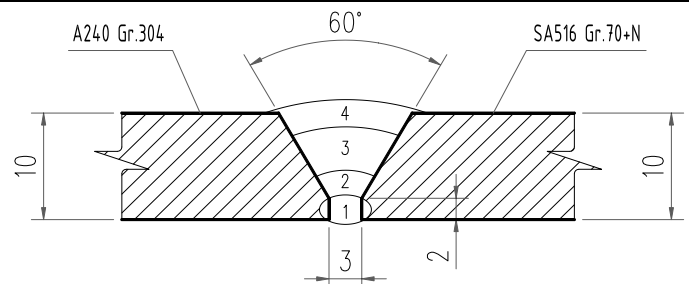
JOINTS (QW-402)

Joint Type **Butt joint, single V groove**

Backing **None**

Retainer **None**

Deposited weld metal thickness:

GTAW 4 mm;
FCAW 6 mm;


BASE METALS (QW-403)

Material Spec. **ASTM A240 to ASME SA516**

Type/Grade, or UNS Number **Gr.304 to Gr.70+N**

P-No. **8** Group No. **1** to P-No. **1** Group No. **2**

Thickness of Test Coupon **10 mm**

Diameter of Test Coupon **Plate**

Maximum Pass Thickness **< 13 mm**

Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**

Time **None**

Other **None**

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	100%	17 L/min
Shielding	CO₂	100%	20 L/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

FILLER METALS (QW-404)

Process	GTAW	FCAW
SFA Specification	ASME SFA-5.9	ASME SFA-5.22
AWS Classification	ER309L	E309LT1-1/-4
Filler Metal F-No.	6	6
Weld Metal Analysis A-No.	8	8
Size of Filler Metal	2.4 mm	1.2 mm
Filler Metal Product Form	Solid rod	Flux cored wire
Supplemental Filler Metal	None	None
Filler Metal Trade Name	ST-309L	SW-309L Cored
Electrode Flux Classification	-	-
Flux Type	-	-
Flux Trade Name	-	-
Weld Metal Thickness	4 mm	6 mm
Other	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process	GTAW	FCAW
Current and Polarity	DC EN	DC EP
Amps.	See table below	
Volts	See table below	
Mode of Metal Transfer		
for GMAW (FCAW)	None	Globular
Other	None	None

TECHNIQUE (QW-410)

Process	GTAW	FCAW
String or weave Bead	Weave	Weave
Oscillation	None	None
Method cleaning	Grinding, brushing	
Method back gouge	None	None
Multipass or single pass (per side)	Multipass	Multipass
Multipass or single Electrodes	Single	Single
Peening	None	None
Use of thermal processes	None	None
Other	None	None

POSITION (QW-405)

Process	GTAW	FCAW
Position(s)	1G	1G
Weld Progression (Uphill, Downhill)	None	None
Other	None	None

PREHEAT (QW-406)

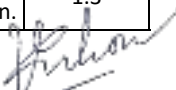
Preheat Temperature **30 °C (Ambient)**

Interpass Temperature **Max. 155 °C**

Other **None**

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1	GTAW	ER309L	2.4	DC EN	120	18	60	30 (Ambient)	2	Ar, 17 L/min.	2.2
2	GTAW	ER309L	2.4	DC EN	170	18	92	85	2	Ar, 17 L/min.	2.0
3	FCAW	E309LT1-1/-4	1.2	DC EP	200	31	230	155	5	CO ₂ , 20 L/min.	1.6
4	FCAW	E309LT1-1/-4	1.2	DC EP	170	28	223	145	2	CO ₂ , 20 L/min.	1.3



Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-GTAW+FCAW-133** Date: **Sep 27, 2021**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	19.4	9.9	192.1	109.8	572	Ductile and base metal
Sample 2	19.4	9.9	192.1	109.7	571	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1 (side bend)	Pass
Specimen 2 (side bend)	Pass
Specimen 3 (side bend)	Pass
Specimen 4 (side bend)	Pass

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	

Comments

Fillet-Weld Test (QW-180)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results:

Other Tests (Not Apply)

Type of Test

Deposit Analysis

Other

Welder's Name **Han Tan Dat** Clock No. **None** Stamp (ID) No. **DDC-WD-0355**

Welder's Name **Le Minh Loi** Clock No. **None** Stamp (ID) No. **DDC-WD-0340**

Tests Conducted by **IQTC** Laboratory Test No. **2309/VL-IQTC**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section VIII, IX - 2019 Edition of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

La Vinh Nhon

Date: **Sep 27, 2021**

Approved by:

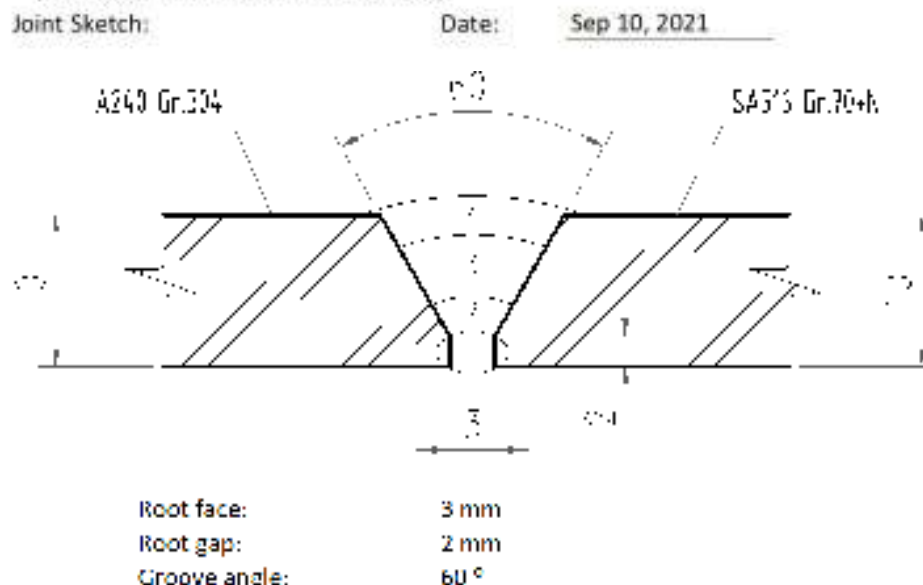


Date:

WELDING PARAMETERS RECORD

(Doc No.: WPR-GTAW+FCAW-133)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GTAW+FCAW-133
 Test Coupon: DD GTAW+FCAW 133
 Test position: 1G
 Base metal:
 - Specification: ASTM A240 Gr.304 - ASME SA516 Gr.70+N
 - Dimension: PI 10 x 180 x 360 PI 10 x 180 x 360 (mm)
 - Heat/Product No.: Y210206050 3 and R71257
 - Certification No.: 20210318026 and 20210103-P5-0001-0003
 Filler metal:
 - Specification: AWS A5.9 and A5.22
 - Classification: ER309L and E309LT1-1/-4
 - Trade name: Hyundai ST 309L and SW 309L Cored
 - Batch/Lot No.: 1219103912 and 20005928CB21EE7
 - Certificates No.: 2107053300002 and 2107053300003
 Gas: Argon (100%) and CO₂ (100%)
 Welder Name: Han Tan Dat and Le Minh Loi
 I.D.: DDC-WD-0355 / U340



Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate: (l/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.						Width	Depth	
1	GTAW	AWS A5.9 / ER309L	2.4	DC EN	120	18	6	2	360	60	30 (Ambient)	Ar, 17 L/min.	2.2	5	2	
2	GTAW	AWS A5.9 / ER309L	2.4	DC EN	170	18	3	56	360	92	85	Ar, 17 l/min.	2.0	6	2	
3	FCAW	AWS A5.22/ E309LT1-1/-4	1.2	DC EP	200	31	1	34	360	230	155	CO ₂ , 20 L/min.	1.6	10	5	
4	FCAW	AWS A5.22/ E309LT1-1/-4	1.2	DC EP	170	28	1	37	360	223	145	CO ₂ , 20 L/min.	1.3	15	2	Reinforcement: 1 mm
		Conducted by Welding Coordinator														
Name		La Vinh Nhon														
Signature																
Date		Sep 10, 2021														



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

✦ Trụ sở chính/ Head office: 27-29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
 ✦ Điện thoại/ Tel: 0272 3777 642 Fax: 02723 777640 Email: iqtc@iqtc.vn MST: 0313138762
 ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



- Số TN/Ref: 2309/VL-IQTC

- Phiếu yêu cầu số/Request No.: 01-220921/YCTN

KẾT QUẢ THỬ NGHIỆM
TEST REPORT

- Công trình/Project: **GREAT PLAINS POWER STATION (MOOSE, JAW, CANADA)**
- Hàng mục/Item: **PROCEDURE QUALIFICATION RECORD (PQR) TEST**
- Địa điểm/Location: **EVAPCO DRY COOLING, INC. - CANADA**
- Cơ quan yêu cầu/Client: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Tên mẫu/Name of sample: **WELDING STEEL PLATE**
FOR WELDING PROCEDURE: DD-GTAW+FCAW-133
GTAW+FCAW-1G. Single V Groove Joint
- Yêu cầu kiểm tra/Testing requirements: **Thử kéo & Thử uốn/Tensile test & Bend test**
- Mô tả mẫu/Description of sample: **Plate 10mm**
- Mã thợ hàn/Welder code: **DDC-WD-0335+0340**
- Vật liệu/Material: **PL10, ASTM A240, Gr.304 weld to PL10, ASTM A516, Gr.70+N**
- Số lượng/Quantity: **01 Mẫu/Sample**
- Ngày nhận mẫu/Sampling date: **22/09/2021**
- Ngày thí nghiệm/Testing date: **22/09/2021**
- Thiết bị thí nghiệm/Testing equipment: **Máy điện tử đa năng WAW-1000/ Machine electronic versatile**
- Phương pháp thử/Test Method: **ASME IX-2019**
- Nhóm thí nghiệm/Tested by: **Mai Hung Khanh, Tran Mong can**

Tên chỉ tiêu/ Specification	Đơn vị/ Unit	Phương pháp thử nghiệm/ Test method	Yêu cầu/ Requirements	Kết quả/ Results	
				Mẫu 1/ Sample 1	Mẫu 2/ Sample 2
1.Thử kéo/ Tensile test		ASME IX-2019	A240 (Gr.304) + A516 (Gr.70+N)		
- Chiều dày mẫu/Thickness of specimen	mm		-	9.9	9.9
- Chiều rộng mẫu/Width of specimen	mm		-	19.4	19.4
- Tiết diện thực đo/Cross section	mm ²		-	192.1	192.1
- Lực kéo đứt/Tensile load	kN		-	109.8	109.7
- Giới hạn bền kéo/Tensile strength	MPa		≥ 485	572	571
- Vị trí mẫu đứt/Broken position				Kim loại cơ bản/ Base metal	Kim loại cơ bản/ Base metal

Long An, ngày/date 23 tháng/month 09 năm/year 2021

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

Đ/D NHÓM THÍ NGHIỆM
(LAB REP.)

PHÓ TRƯỞNG PHÒNG THÍ NGHIỆM
(LAB DEPUTY)

PHÓ GIÁM ĐỐC CHI NHÁNH
(BRANCH VICE DIRECTOR)



TRAN MONG CAN

KS/ENG.MAI HUNG KHANH

PHAM THANH DA



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

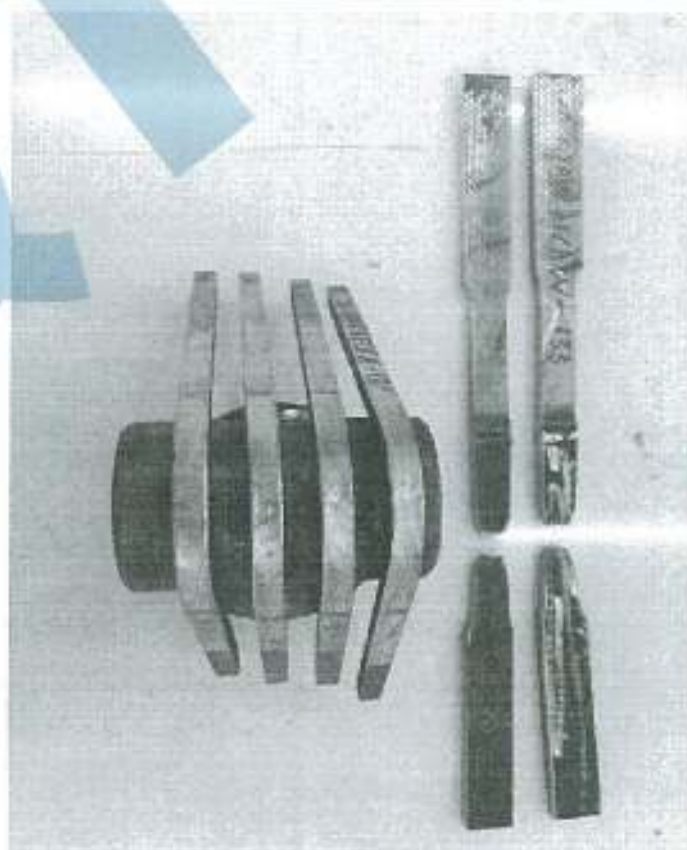
✦ Trụ sở chính/ Head office: 27-29 Đường 7A, KCC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
 ✦ Điện thoại/Tel: 0272 3777 642 Fax: 02723 777640 Email: iqtc@iqtc.vn MST: 0313136782
 ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa N, Huyện Đức Hòa, Tỉnh Long An



- Số TN/Ref: 2309/ML-IQTC

- Phiếu yêu cầu số/Request No.: 01-220921/YCTN

Tên chỉ tiêu/ Specification	Đơn vị/ Unit	Kết quả/ Test Results			
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
2. Thử uốn/ Bend test					
- Vị trí uốn/Location bend		Uốn cạnh/ Side bend	Uốn cạnh/ Side bend	Uốn cạnh/ Side bend	Uốn cạnh/ Side bend
- Chiều dày/Thickness of specimen	mm	10.0	10.1	10.0	10.0
- Chiều rộng/Width of specimen	mm	9.9	9.9	9.9	9.9
- Góc uốn/Bending angle	Độ/°	180	180	180	180
- Đường kính gồ uốn/Plunger diameter	mm	40	40	40	40
- Kết quả/Test results		Đạt/ Pass	Đạt/ Pass	Đạt/ Pass	Đạt/ Pass



Ghi chú/Notice:

- * Phiếu kết quả này chỉ có giá trị cho mẫu thử do khách hàng mang tới.
This test results is value only for sample taken by customer.
- * Tên mẫu và tên khách hàng được ghi theo yêu cầu của khách hàng.
Name of sample and customer are written as customer's request.



Handwritten signature



MILL TEST CERTIFICATE

EN 10204/3.1



Manufacturer: PT. INDONESIA GUANG CHING NICKEL AND STAINLESS STEEL INDUSTRY

ADD: PT IMIF, Desa Fatufin, Kecamatan Bahodopi, Kabupaten Marowali, Provinsi Sulawesi Tengah, Indonesia

Tel : +6282296751870
E-mail: liuy@etsinghan.com

Purchaser		GIA JIH HING XIN CO., LTD		Welding Furnace		RSEF-A30-LF-CCH		Contract No.		ET21/WCE0107-GA-A		Certificate No.		20210318026								
Product Name		HOT ROLLED STAINLESS STEEL COILS, 304 GRADE, NO.1 FINISH		Steel Grade		304		Product Standard		ASTM-A240/A240M-18		Condition of delivery		Hot rolling annealing pickling								
No.	Coil No.	QTY	Heat No.	Size (mm)	Weight (Mt)	Chemical Composition (%)										Tensile Test				Rebar		
						C	Si	Mn	P	S	Cr	Ni	N	Mo	Cu						Rm (MPa)	R _{0.2} (MPa)
1	N2102180441-A	1	Y210207841-2	5.0*1500*C	10.040	0.051	0.37	1.06	0.027	0.002	18.04	8.04	0.039	0.04	0.04			624.4	307.7	/	58.4	86.8
2	N2102180441-B	1	Y210207841-2	5.0*1500*C	10.070	0.051	0.37	1.06	0.027	0.002	18.04	8.04	0.039	0.04	0.04			624.4	307.7	/	58.4	86.8
3	N2102180456-A	1	Y210208043-3	6.0*1500*C	10.540	0.044	0.36	1.04	0.029	0.002	18.13	8.03	0.048	0.04	0.04			640.9	285.2	/	56.7	83.5
4	N2102180456-B	1	Y210208043-3	6.0*1500*C	10.705	0.044	0.36	1.04	0.029	0.002	18.13	8.03	0.048	0.04	0.04			640.9	285.2	/	56.7	83.5
5	N2102180456-C	1	Y210206452-2	8.0*1500*C	10.685	0.041	0.37	1.09	0.030	0.002	18.10	8.05	0.046	0.04	0.03			623.1	286.4	/	57.8	87.0
6	N2102180456-D	1	Y210206452-2	8.0*1500*C	10.525	0.041	0.37	1.09	0.030	0.002	18.10	8.05	0.046	0.04	0.03			623.1	286.4	/	57.8	87.0
7	N2102180483-A	1	Y210206851-5	8.0*1500*C	10.580	0.040	0.38	1.08	0.027	0.003	18.16	8.04	0.046	0.04	0.04			640.9	289.5	/	56.3	83.0
8	N2102180483-B	1	Y210206851-5	8.0*1500*C	10.600	0.040	0.38	1.08	0.027	0.003	18.16	8.04	0.046	0.04	0.04			640.9	289.5	/	56.3	83.0
9	N2102180496-A	1	Y210206850-3	10.0*1500*C	10.535	0.046	0.39	1.05	0.030	0.001	18.12	8.04	0.045	0.04	0.03			621.5	304.6	/	56.0	84.2
10	N2102180496-B	1	Y210206850-3	10.0*1500*C	10.535	0.046	0.39	1.05	0.030	0.001	18.12	8.04	0.045	0.04	0.03			621.5	304.6	/	56.0	84.2
Total		10			106.075																	

Remarks

1. Product marking: steel number, heat No. coil No. size.

2.Surface Quality: ok 3.Shape: ok 4.Dimension: ok 5.Radioactivity Test: ok 6.Non-metallic inclusions: ok 7.intergranular corrosion: ok

- We hereby certify that the material herein described has been manufactured, sampled, tested and inspected in accordance with the requirements of above specifications.
- We hereby certify Product No weld repair; Free of mercury; Free from radiation contamination.
- ISO 9001:2015 Quality management system identified by ZJQC.
- We declare that our testing center has passed ISO/IEC17025, 2017 accreditation review, accreditation body: CNAS, accreditation capability range: Spectral analysis and tensile testing, certificate number: CNAS L12639.
- We certify that the material described below fully conforms to IS 6911:2017 chemical composition and Mechanical properties of product as tested in accordance with the Scheme of Testing and Inspection contained in BIS certification marks license No. CML-4100043562
- Issued in agreement with TÜV SÜD Industrie Service GmbH (April 2018). QMS approved acc. to PED&AD2000, Annex I, Para. 4.3 by Notified Body 0038(Certification no. DGR-0038-QS-W 838/2018/MUC).
- Approved acc. to CPR 305/2011 with certification no. 0036-CPR-M-115-2019.
- Product heat treatment temperature: 1100°C.
- The certificate should be sealed by quality control department dedicate inspection stamp or authorized official stamp of sales.

Typed by	陈妮妮
Rectifying	刘建
Date	2021/3/18

PT. INDONESIA GUANG CHING NICKEL AND STAINLESS STEEL INDUSTRY

Department of Quality Assurance

Special Seal For Inspection

검 사 증 명 서

MILL TEST CERTIFICATE



공진제철소 주식회사 영인길 1400
1400 Butzenup-ro, Songdeup, Daegu, Korea



주문번호
Order No. : F201000027

품 명
Commodity : HOT ROLLED STEEL PLATE

제품규격
Specification : ASME (2019 ED) SA516 Gr.70+N

고객사
Customer : TUNG WANG STEEL

주문자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20210103-PS-0001-003

발행일자
Date of Issue : 2021-01-03

제품규격 Specification		ASME (2019 ED) SA516 Gr.70+N		Contractor		HYUNDAI CORPORATION		Date of Issue																										
제품치수 Dimension	수량 Quantity	중량 Weight (Kg)	재질번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test							충격시험 Impact Test V Notch -45.0 °C			화합성분 Chemical Composition(%)																	단위 Unit	
						YP	TS	EL	YP-EL	RA	YR	YP Mtd	Energy	SF	LE	Division	C	Si	Mn	P	S	Cr	N	B	Cu	Mo	Nb	Ti	V	Se	As	CEQ		
10x2400x12000	1	2,261	R71257	FC2B404701	T	392.3	551.1	27				2	1:236 2:130 3:173 A:180			L	1,542	320	1,170	107	20	18	18	2	138	0	17	2	2	40	407			
10x2400x12000	1	2,261	R71257	FC2B404702	T	392.3	551.1	27				2	1:236 2:130 3:173 A:180			L	1,542	320	1,170	107	20	18	18	2	138	0	17	2	2	40	407			
10x2400x12000	1	2,261	R71257	FC2B404703	T	392.3	551.1	27				2	1:236 2:130 3:173 A:180			L	1,542	320	1,170	107	20	18	18	2	138	0	17	2	2	40	407			
10x2400x12000	1	2,261	R71257	FC2B404801	T	389.2	549.3	25				2	1:242 2:197 3:185 A:203			L	1,542	320	1,170	107	20	18	18	2	138	0	17	2	2	40	407			
NOTE	* Position = T:Top, M:Middle, B:Bottom * Tensile test - Direction:Transverse, Gauge length:200mm (Rectangular), YP Mtd : 2:10, 2% off-set(0.5 (0.5% underload), U (Upper yield point), L (Lower yield point) * Impact test - Direction:Transverse, Sub-size:10mmX7.5mm, SF:Shear fracture, LE:Lateral Expansion * Division = P:Product analysis, L:Ladle analysis * The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.																																	
	* NACE MR0175/MR0103 for hardness test ASME SA516-70 2019 Edition * The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. * This MII Test Certificate is issued in accordance with EN 10204 Type 3.1																																	
	* The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.																																	
상기와 제품검사의 결과 지정된 규격에 합격한 것을 증명합니다. WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS															Signature S. B. Park Chief Of Quality Assurance Team																			



- NOTE
- Position - T: Top, M: Middle, B: Bottom
 - Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mtd: 2 (0.2% of 1-set), 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
 - Impact test - Direction: Transverse, Sub-size: 10mmX7.5mm, SF: Shear fracture, LE: Lateral Expansion
 - Division - P: Product analysis, L: Ladle analysis
 - The energy of sub-size are calculated from a correlation between selected sub-size and standard-size Charpy test.

- NACE MR0175/MR0103 for hardness test ASME SA516-70 2019 Edition
- The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. • This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature
S. B. Park
Chief Of Quality Assurance Team

- * 본 검사증명서는 원본의 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서와 비동기적으로 생성된 사본을 비교하여 확인하실 수 있습니다. (http://sm.hyundai-steel.com/cs/cn/login.jsp)
- * This Mill Test Certificate is a copy that has been printed from original electronic document (with digital signing).
- * You are able to check an original electronic document at hyundai-steel's customer portal (http://sm.hyundai-steel.com/cs/cn/login.jsp) • QRcode scanner App : 'QRcode'
- * [사용상 주의사항] 본 검사증명서와 별다른 규격 외 사용시 제품품질 및 안전상 문제가 발생할 수 있습니다.
- * [Caution for Use] Damage on products and safety problems may arise if used other than the spec stated in the Mill Test Certification.

검 사 증 명 서

MILL TEST CERTIFICATE

(Attached Sheet)



강진제철소 중남 당진시 외안동 국보신로로 1400
1400 Bulgusan-g-ro, Songji-eup, Dangjin-si, Chungnam - Korea



주문번호
Order No. : F201000027

품명
Commodity : HOT ROLLED STEEL PLATE

제품규격
Specification : ASME (2019 ED) SA516 Gr.70+H

고객사
Customer : TUNG WANG STEEL

주최자
Contractor : HYUNDAI CORPORATION

증명서 번호
Certificate No. : 20210103-PS-0001-003

발행일자
Date of Issue : 2021-01-03

제품번호 Product No.	Position	경도시험 Hardness Test			
		HB			
		1	2	3	Avg.
FC28404701	T	156.0	156.0	156.0	156.0
FC28404702	T	156.0	156.0	156.0	156.0
FC28404703	T	156.0	156.0	156.0	156.0
FC28404801	T	156.0	156.0	156.0	156.0



NOTE

• NADE MR0175/MR0103 for hardness test ASME SA516-70 2019 Edition.
• The plate is fully killed and fine grained steel. Basic oxygen process and Vacuum degassing process were applied. • This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS.

Signature

S. B. Park

Chief Of Quality Assurance Team

* 본 검사증명서는 원본인 전자문서(전자서명, 중도, 모방)로부터 출력된 사본입니다. 전자문서의 위조, 변조, 무단복제, 고의도발 등을 확인하실 수 있습니다. (http://sm.hyundai-steel.com/cs/cm/ogn.jsp)
* This Mill Test Certificate is a copy that has been printed from original electronic document with digital signature.
* You are able to check an original electronic document at Hyundai Steel's Customer portal. (http://sm.hyundai-steel.com/cs/cm/ogn.jsp) * QRcode scanner App : 'QRcode'
* [사용상 주의사항] 본 검사증명서에 없거나 규격 외 사용시 제품품질 및 안전성 문제가 발생할 수 있습니다.
* [사용상 주의사항] Products and safety problems may arise if used other than the spec stated in the Mill Test Certificate.

CERTIFIED MATERIAL REPORT

CMR No. : 2107053300002

Date of Issue : 2021-07-05

Customer	GLOBAL TECHNOLOGY COMMERCIAL CORPORA		
Trade Name	ST-309L	Size	2.4mm*1000mm*5kg
Lot No.	1219103912	Purchaser's Order No.	1421_HW_GT
Specification	AWS A5.9 ER309L		

Welding Current		Shielding Gas		Post Weld Heat Treatment (°C × hr.)	
-----------------	--	---------------	--	-------------------------------------	--

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS(MPa)	TS(MPa)	EL(%)	Temp.(°C)	X1	X2	X3	Avg.
	-	-	-		-	-	-	-
AWS Spec.								

Chemical Composition (%)

* According to EN 10204-2004 : 3.1

WIRE	C	Si	Mn	P	S	Cu	Ni	Cr	Mo		
	0.02	0.54	2.26	0.014	0.008	0.01	13.77	23.36	0.02		
AWS Spec.	≤0.03	0.30 -0.65	1.0 -2.5	≤0.03	≤0.03	≤0.75	12.0 -14.0	23.0 -25.0	≤0.75		

Others

* Q'ty : 5kg

ORIGINAL

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

K.W. Park

QA Manager Park K. W.



HYUNDAI
WELDING

HYUNDAI WELDING CO., LTD.

Head Office

507,Teheran-ro,Gangnam-gu,Seoul,Republic of Korea

Pohang Plant

100,Daeseong-ro,Nam-gu,Pohang,Republic of Korea

CERTIFIED MATERIAL REPORT

CMR No. : 2107053300003

Date of issue : 2021-07-05

Customer	GLOBAL TECHNOLOGY COMMERCIAL CORPORA		
Trade Name	SW-309L Cored	Size	1.2mm*15kg
Lot No.	200059280821EE7	Purchaser's Order No.	1421_HW_GT
Specification	AWS A5.22 E309LT1-1/-4		

Welding Current	OCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C × hr.)	AS WELDED
-----------------	------	---------------	-----------------	-------------------------------------	-----------

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS(MPa)	TS(MPa)	EL(%)	Temp.(°C)	X1	X2	X3	Avg.
	456	594	34.8		-	-	-	-
AWS Spec.	-	≥520	≥30	-				

Chemical Composition (%)

* According to EN 10204-2004 : 3.1

Weld Metal	C	Si	Mn	P	S	Cu	Ni	Cr	Mo		
	0.03	0.63	1.17	0.017	0.008	0.01	12.66	22.68	0.04		
AWS Spec.	≤0.04	≤1.0	0.5 ~2.5	≤0.04	≤0.03	≤0.75	12.0 ~14.0	22.0 ~25.0	≤0.75		

Others

* Q'ty : 15kg

ORIGINAL

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

K.W. Park

QA Manager Park K. W.



HYUNDAI
WELDING

HYUNDAI WELDING CO., LTD.

Head Office

507,Teheran-ro,Gangnam-gu,Seoul,Republic of Korea

Pohang Plant

100,Daesong-ro,Nam-gu,Pohang,Republic of Korea

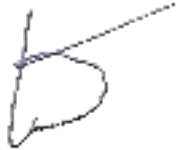


CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DUNG
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
38, Zone C, D1 St, An Ha Industrial Park, Pham Van Hai, Binh Chanh, HCMC
sales@daidung.vn (84-28) 3868 1689 www.daidung.com.vn

Comply:
ISO 9001:2015
ASME IX:2019

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW-121

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Aug 25, 2023	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX-Boiler and Pressure Vessel Code:2021)

Page: 1/2

Applicable Code: **ASME Section IX, 2021 Edition**

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**

Welding Procedure Specification No.: **WPS-DD-GTAW-121** Revision No. **0** Date **Aug 25, 2023**

Supporting PQR No.(s): **PQR-DD-GTAW-121** Date **May 14, 2021**

Welding Process(es) **GTAW** Type(s)(Manual, Automatic, Semi-Automatic): **Manual**

JOINTS (QW-402)

Joint Design **Groove or fillet as per fabrication drawing**

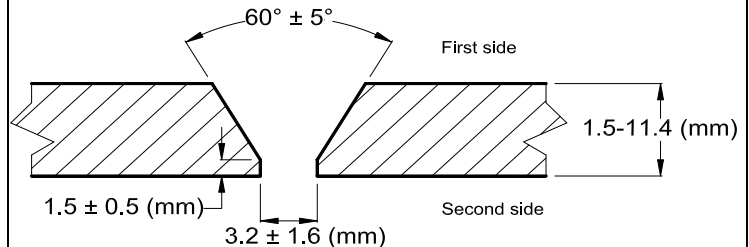
Root Spacing **As per fabrication drawing**

Backing **With or without**

Retainer **None**

Backing Material **Base metal or weld metal or Ceramic**

Details



Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches"

BASE METALS (QW-403)

P-No. **8** Group No. **1** to P-No. **8** Group No. **1**

OR

Specification and type/grade or UNS Number **Grade 304/304L**

to Specification and type/grade or UNS Number **Grade 304/304L**

OR

Chem. Analysis and Mech. Prop. **None**

to Chem. Analysis and Mech. Prop. **None**

Thickness Range:

Base Metal: **1.5 - 11.4 (mm) for groove (without CVN test)** **All for fillet**

Maximum Pass Thickness ≤ 13 mm: **Yes** ☒

Other **None**

FILLER METALS (QW-404)

Process **GTAW**

Spec. No. (SFA) **SFA-5.9**

AWS No. (Class) **ER308L**

F-No. **6**

A-No. **8**

Size of Filler Metals **Ø2.4 mm**

Supplemental Filler Metal **None**

Filler Metal Product Form **Solid rod**

Filler Metal Trade Name **KTS-308L or the same Spec and Class**

Weld Metal

Deposited Thickness:

Groove **Max. 11.4 mm**

Fillet **All**

Electrode-Flux (Class) **None**

Flux Type **None**

Flux Trade Name **None**

Consumable Insert **None**

Other **None**



WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX-Boiler and Pressure Vessel Code:2021)

Page: 2/2

Applicable Code: **ASME Section IX, 2021 Edition**
Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION** By: **La Vinh Nhon**
Welding Procedure Specification No.: **WPS-DD-GTAW-121** Revision No. **0** Date **Aug 25, 2023**

POSITIONS (QW-405)

Position(s) of Groove **All**
Position(s) of Fillet **All**
Welding Progression
Vertical up **Yes**
Vertical down **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **None**
Time Range **None**
Other **None**

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Argon	99.99%	15 - 19 L/min.
Trailing	None	None	None
Backing	Argon	99.99%	15 - 19 L/min.
Other	None	None	None

PREHEAT (QW-406)

Preheat Temperature, Minimum **Sufficient to remove moisture**
Interpass Temperature, Maximum **175 °C (recommendation)**
Preheat Maintenance **None**
Other **None**

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Pass	Process	Filler Metal		Current Type and Polarity	Amps (Range) (A)	Wire Feed Speed (Range) (mm/min)	Energy or Power (Range) (kJ/mm)	Volts (Range) (V)	Travel Speed (Range) (mm/min.)	Other
		Classification	Diameter (mm)							
Root	GTAW	ER308L	2.4	DC EN	68 - 84	None	0.5 - 0.6	9 - 11	66 - 93	None
Fill	GTAW	ER308L	2.4	DC EN	95 - 117	None	0.8 - 0.9	9 - 11	64 - 86	None
Cap	GTAW	ER308L	2.4	DC EN	100 - 123	None	0.9 - 1.1	9 - 11	51 - 78	None

Pulsing Current **None** Heat Input (max.) **None**
Tungsten Electrode Size and Type **Φ2.4 mm, 2% Thoriated Tungsten (EWTh-2)**
Mode of Metal Transfer for GMAW (FCAW) **None**
Other **None**

TECHNIQUE (QW-410)

Process **GTAW**
String or Weave Bead **Weave bead**
Orifice, Nozzle, or Gas Cup Size **All size**
Initial and Interpass Cleaning (Brushing, Grinding, etc.) **Brushing or Grinding**
Method of Back Gouging **Grinding**
Oscillation **None**
Contact Tube to Work Distance **None**
Multiple or Single Pass (Per Side) **Multiple or single pass**
Multiple or Single Electrode **Single**
Electrode Spacing **None**
Peening **None**
Other **None**

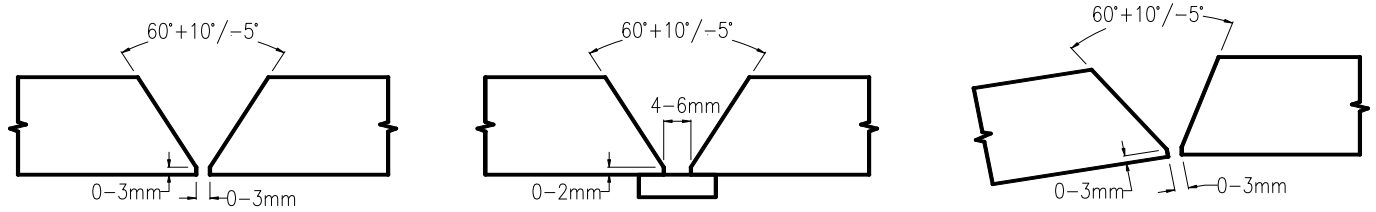
Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**Certified by Welding Coordinator:
La Vinh NhonApproved by: **Truong Hoai Phong Vu (QA Manager)**Date: **Aug 25, 2023**Date: **Aug 25, 2023**

Welding Procedure Specification No.: **WPS-GTAW-121**

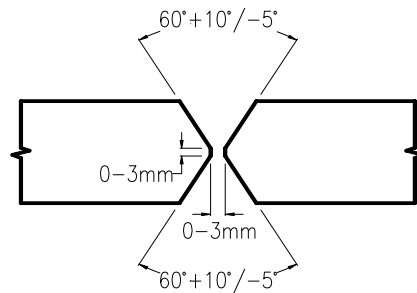
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld map. Below is typical only

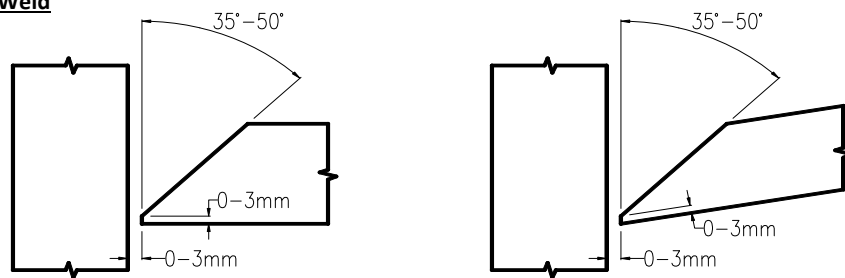
1. Single V Groove Butt Joint



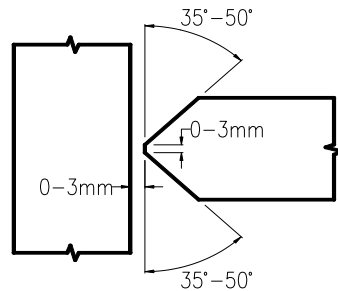
2. Double V Groove Butt Joint



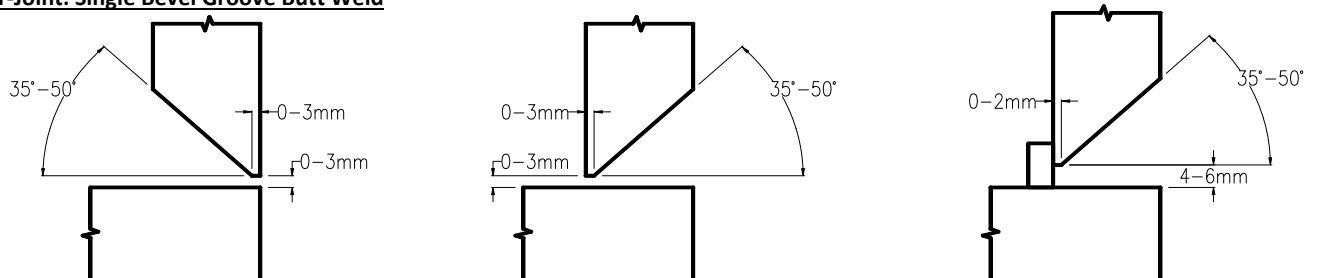
3. T-Joint: Single Bevel Groove Butt Weld



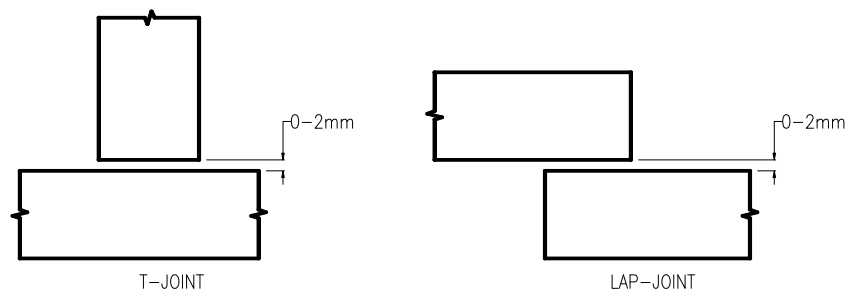
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-Joint: Single Bevel Groove Butt Weld



6. Fillet Weld

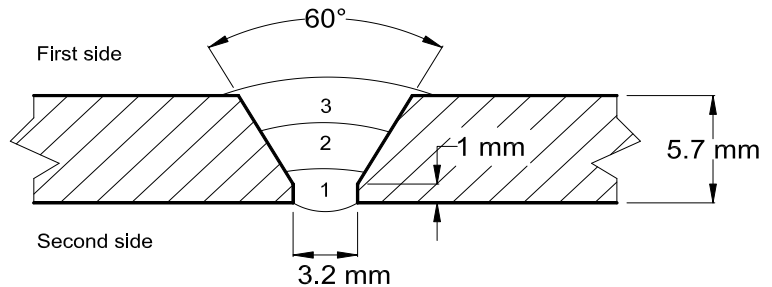


Signature

Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
 Procedure Qualification Record No. **PQR-DD-GTAW-121** Date: **May 14, 2021**
 Welding Process(es) **GTAW**
 Type(s)(Manual, Automatic, Semi-Automatic) **Manual**

JOINTS (QW-402)

Joint Type **Butt joint, single V groove**
 Backing **None**
 Retainer **None**



BASE METALS (QW-403)

Material Spec. **ASTM A312**
 Type/Grade, or UNS Number **TP304/304L**
 P-No. **8** Group No. **1** to P-No. **8** Group No. **1**
 Thickness of Test Coupon **5.7 mm**
 Diameter of Test Coupon **Plate**
 Maximum Pass Thickness **< 13 mm**
 Other **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature **None**
 Time **None**
 Other **None**

GAS (QW-408)

	Percent Composition		Flow Rate
	Gas(es)	(Mixture)	
Shielding	Argon	100%	17 L/min
Trailing	None	None	None
Backing	Argon	100%	17 L/min
Other	None	None	None

FILLER METALS (QW-404)

Process **GTAW**
 SFA Specification **ASME SFA-5.9**
 AWS Classification **ER308L**
 Filler Metal F-No. **6**
 Weld Metal Analysis A-No. **8**
 Size of Filler Metal **2.4 mm**
 Filler Metal Product Form **Solid rod**
 Supplemental Filler Metal **None**
 Filler Metal Trade Name **KTS-308L**
 Electrode Flux Classification **-**
 Flux Type **-**
 Flux Trade Name **-**
 Weld Metal Thickness **5.7 mm**
 Other **None**

ELECTRICAL CHARACTERISTICS (QW-409)

Process **GTAW**
 Current and Polarity **DC EN**
 Amps. **See table below**
 Volts **See table below**
 Mode of Metal Transfer
 for GMAW (FCAW) **None** **None**
 Other

TECHNIQUE (QW-410)

Process **GTAW**
 String or weave Bead **Weave**
 Oscillation **None**
 Method cleaning **Grinding, brushing**
 Method back gouge **None**
 Multipass or single pass (per side) **Multipass**
 Multipass or single Electrodes **Single**
 Peening **None**
 Use of thermal processes **None**
 Other **None**

POSITION (QW-405)

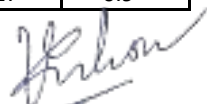
Process **GTAW**
 Position(s) **2G + 5G**
 Weld Progression (Uphill, Downhill) **Uphill**
 Other **None**

PREHEAT (QW-406)

Preheat Temperature **36 °C (Ambient)**
 Interpass Temperature **Max. 153 °C**
 Other **None**

WELDING PARAMETER

Weld Pass	Process	Electrode/Filler		Current Type and Polarity	Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/ Interpass Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat input (kJ/mm)
		AWS Class	Dia (mm)								
1.1	GTAW	ER308L	2.4	DC EN	72	11	78	36	1.7	17	0.6
1.2	GTAW	ER308L	2.4	DC EN	80	9	81	100	1.7	17	0.5
1.3	GTAW	ER308L	2.4	DC EN	80	9	79	99	1.7	17	0.5
2	GTAW	ER308L	2.4	DC EN	106	10	75	112	2.5	17	0.8
3.1	GTAW	ER308L	2.4	DC EN	121	10	68	125	2.5	17	1.1
3.2	GTAW	ER308L	2.4	DC EN	102	9	60	153	2.5	17	0.9



Organization Name **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Procedure Qualification Record No. **PQR-DD-GTAW-121** Date: **May 14, 2021**

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress, (MPa)	Type of Failure and Location
Sample 1	19.03	5.7	108.5	65.3	602	Ductile and base metal
Sample 2	19.06	5.7	108.6	66.1	608	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 1 (face bend)	Pass
Specimen 2 (face bend)	Pass
Specimen 3 (root bend)	Pass
Specimen 4 (root bend)	Pass

Toughness Tests (QW-170) (Not Apply)

Specimen No.	Notch Location	Specimen Size	Test Temperature		Impact Values		Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	

Comments _____

Fillet-Weld Test (QW-180) (Not Apply)

Result - Satisfactory: Yes _____ No _____ Yes _____ No _____

Macro - Results: _____

Other Tests (Not Apply)

Type of Test _____

Deposit Analysis _____

Other _____

Welder's Name **Nguyen Dinh Chinh** Clock No. **None** Stamp (ID) No. **DDC-WD-0363**

Welder's Name _____ Clock No. _____ Stamp (ID) No. _____

Tests Conducted by **IQTC** Laboratory Test No. **1405/VL-IQTC**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX Edition 2019 of the ASME Boiler and Pressure Vessel Code.

Organization **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by Welding Coordinator:

LA VINH NHON

Date: **May 14, 2021**

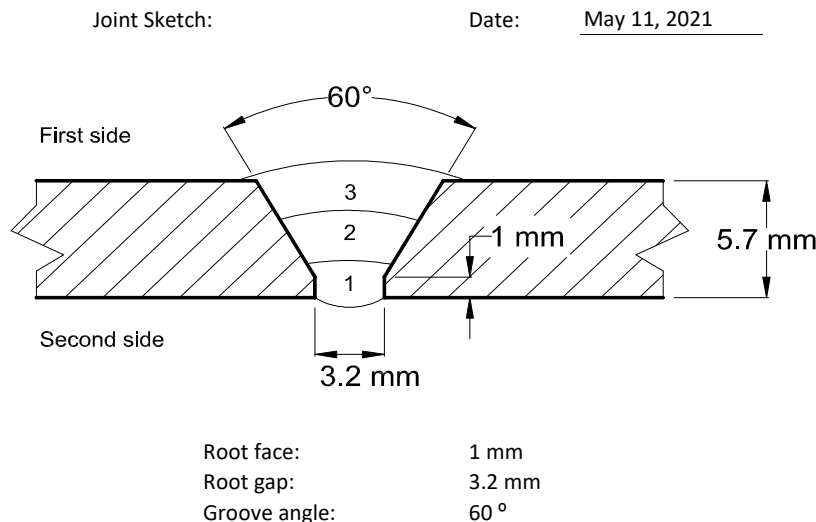
Approved by:



Date: _____

WELDING PARAMETERS RECORD (Doc No.: WPR-GTAW-121)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GTAW-121
 Test Coupon: DD-GTAW-121
 Test position: 2G+5G
 Base metal:
 - Specification: ASTM A312 TP304/304L
 - Dimension: O.D 101.6 x 5.7 (mm) x 2 pcs
 - Heat/Product No.: 161218V08
 - Certificate No.: ST170505-08
 Filler metal:
 - Specification: ASME SFA-5.9/ AWS A5.9
 - Classification: ER308L
 - Trade name: KTS-308L
 - Batch/Lot No.: V19706026
 - Certificates No.: 16012005
 Gas: Argon (100%)
 Welder Name: Nguyen Dinh Chinh
 I.D.: DDC-WD-0363



Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Backing Gas Flow Rate (L/min.)	Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth		
FRONT SIDE (First Side):																	
1	GTAW	SFA-5.9 ER308L	2.4	DC EN	72	11	1	25	110	78	36 (Ambient)	17	0.6	5	1.7	17	Position: 2G
	GTAW	SFA-5.9 ER308L	2.4	DC EN	80	9	1	10	95	81	100	17	0.5	5		17	
	GTAW	SFA-5.9 ER308L	2.4	DC EN	80	9	1	20	105	79	99	17	0.5	5		17	
2	GTAW	SFA-5.9 ER308L	2.4	DC EN	106	10	4	7	310	75	112	17	0.8	8	2.5	17	
3	GTAW	SFA-5.9 ER308L	2.4	DC EN	121	10	2	21	160	68	125	17	1.1	11	2.5	17	Position: 5G
	GTAW	SFA-5.9 ER308L	2.4	DC EN	102	9	2	41	160	60	153	17	0.9	11		17	
		Prepared by															
Name		LA VINH NHON															
Signature																	
Date		May 11, 2021															



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488



- Trụ sở chính: 27/29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- Head office: 27/29 7A Street, Binh Tri Dong Residential Area, Ward Binh Tri Dong B, Binh Tan District, HCM City
- Điện thoại/Tel: 028 38452 303 Fax: 028 38452 397 MST: 0313138782
- Chi nhánh PTN: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- Branch of Laboratory: Binh Tien 2 Hamlet, Duc Hoa Ha Commune, Duc Hoa Dist., Long An Province
- Điện thoại/Tel: 0272 3777 641 Fax: 0272 3777 640 Email: iqtc@iqtc.vn

- Số TN/Ref: 1405/VL-IQTC

- Phiếu yêu cầu số/Request No.: 02-120521/YCTN

KẾT QUẢ THỬ NGHIỆM
TEST REPORT

- Công trình/Project: **THREE RIVERS ENERGY CENTER**
- Hạng mục/Item: **CONDENSATE TANK**
- Địa điểm/Location: **EVAPCO - USA**
- Cơ quan yêu cầu/Client: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Tên mẫu/Name of sample: **WELDING STEEL PIPE**
FOR WELDING PROCEDURE: DD-GTAW-121, GTAW-(2G+5G)-SINGLE V
- Yêu cầu kiểm tra/Testing requirements: Thử kéo & Thử uốn/Tensile test & Bend test
- Mô tả mẫu/Description of sample: **O.D 101.6x5.7**
- Mã thợ hàn/Welder code: **DDC-WD-0363**
- Vật liệu/Material: **ASTM A312 TP304/304L**
- Số lượng/Quantity: **01 Mẫu/Sample**
- Ngày nhận mẫu/Sampling date: 12/05/2021
- Ngày thí nghiệm/Testing date: 13/05/2021
- Thiết bị thí nghiệm/Testing equipment: Máy điện tử đa năng WAW-1000I Machine electronic versatile
- Phương pháp thử/Test Method: **ASME IX:2019**
- Nhóm thí nghiệm/Tested by: Mai Hung Khanh, Nguyen Anh Vu

Tên chỉ tiêu/ Specification	Đơn vị/ Unit	Phương pháp thử nghiệm/ Test method	Yêu cầu/ Requirements	Kết quả/ Results	
				Mẫu 1/ Sample 1	Mẫu 2/ Sample 2
1.Thử kéo/ Tensile test		ASME IX:2019	ASTM A312 TP304/304L		
- Chiều dày mẫu/Thickness of specimen	mm		-	5.7	5.7
- Chiều rộng mẫu/Width of specimen	mm		-	19.03	19.06
- Tiết diện thực đo/Cross section	mm ²		-	108.5	108.6
- Lực kéo đứt/Tensile load	kN		-	65.3	66.1
- Giới hạn bền kéo/Tensile strength	MPa		515 min	602	608
- Vị trí mẫu đứt/Broken position				Kim loại cơ bản/ Base metal	Kim loại cơ bản/ Base metal

Long An, ngày/date 14 tháng/month 05 năm/year 2021

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

Đ/D NHÓM THÍ NGHIỆM
(LAB REP.)

TRƯỞNG PHÒNG THÍ NGHIỆM
(LAB MANAGER)

KT. GIÁM ĐỐC CHI NHÁNH
REP. BRANCH DIRECTOR

KS/ENG. MAI HUNG KHANH

KS/ENG. NGUYEN ANH DINH

KS/ENG. TRINH DINH TAN

INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY (IQTC)



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488

- ❖ Trụ sở chính: 27/29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- ❖ Head office: 27/29 7A Street, Binh Tri Dong Residential Area, Ward Binh Tri Dong B, Binh Tan District, HCM City
- ❖ Điện thoại/Tel: 028 38452 303 Fax: 028 38452 397 MST: 0313138782
- ❖ Chi nhánh PTN: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- ❖ Branch of Laboratory: Binh Tien 2 Hamlet, Duc Hoa Ha Commune, Duc Hoa Dist., Long An Province
- ❖ Điện thoại/Tel: 0272 3777 641 Fax: 0272 3777 640 Email: iqtc@iqtc.vn

Tên chỉ tiêu/ Specification	Đơn vị/ Unit	Kết quả/ Test Results			
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
2. Thử uốn/ Bend test					
- Vị trí uốn/Location bend		Uốn mặt/ Face bend	Uốn mặt/ Face bend	Uốn chân/ Root bend	Uốn chân/ Root bend
- Chiều dày/Thickness of specimen	mm	5.7	5.7	5.7	5.7
- Chiều rộng/Width of specimen	mm	38.0	38.1	38.1	38.0
- Góc uốn/Bending angle	Độ	180	180	180	180
- Đường kính gồ uốn/Plunger diameter	mm	40	40	40	40
- Kết quả/Test results		Đạt/ Pass	Đạt/ Pass	Đạt/ Pass	Đạt/ Pass

Ghi chú/Notice:

- * Phiếu kết quả này chỉ có giá trị cho mẫu thử do khách hàng mang tới.
This test results is value only for sample taken by customer.
- * Tên mẫu và tên khách hàng được ghi theo yêu cầu của khách hàng.
Name of sample and customer are written as customer's request.



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488



- ✦ Trụ sở chính: 27/29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- ✦ Head office: 27/29 7A Street, Binh Tri Dong Residential Area, Ward Binh Tri Dong B, Binh Tan District, HCM City
- ✦ Điện thoại/Tel: 08 38452 303 Fax: 08 38452 397 Email: igtc@igtc.vn
- ✦ Chi nhánh FTN: Ấp Bình Tân 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- ✦ Branch of Laboratory: Binh Tan 2 Hamlet, Duc Hoa Ha Commune, Duc Hoa Dist., Long An Province
- ✦ Điện thoại/Tel: 0723 777641 Fax: 0723 777640 Email: igtc@igtc.vn

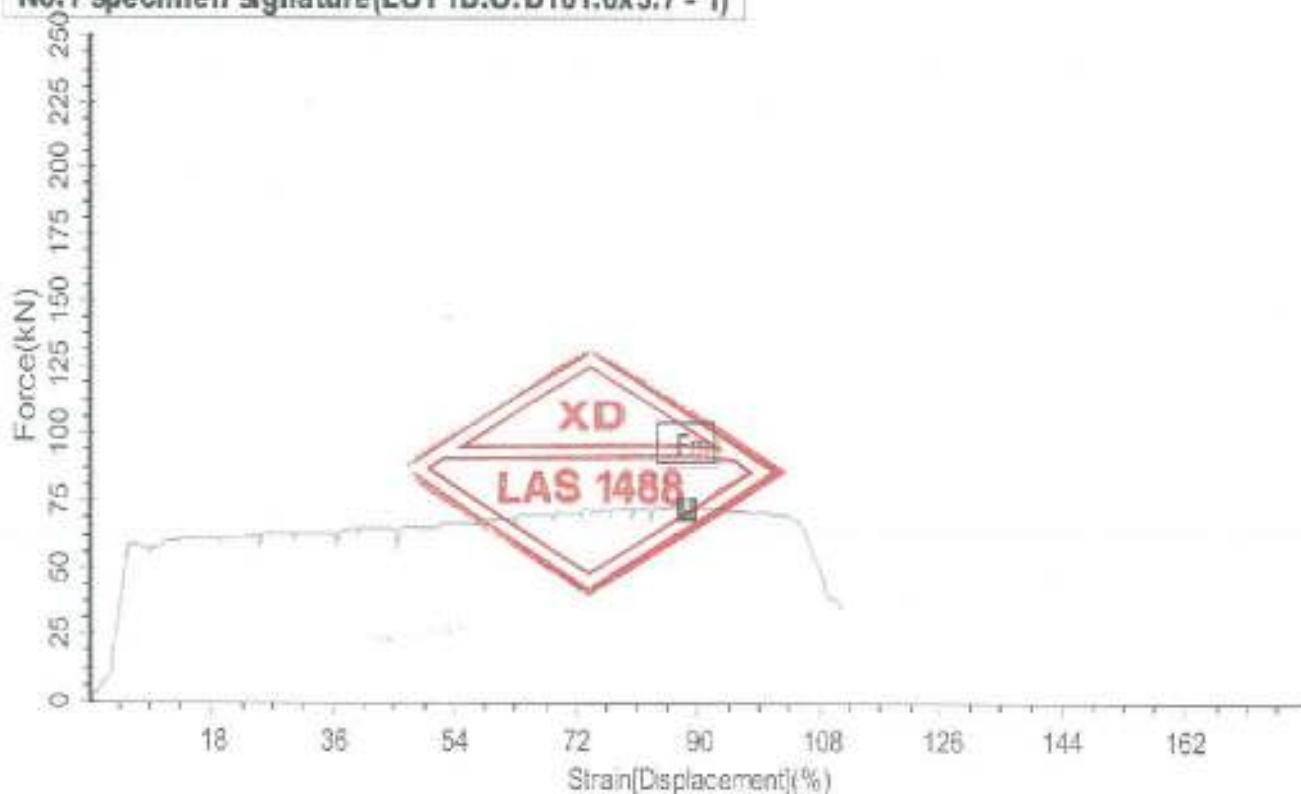
LOT ID: O.D101.6x5.7 - 1

Company	DAIDUNG CORPORATION	Test Date	13-05-2021
Standard	ASME IX:2019	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	80%
Machine Model	WAW-1000	Capacity	1000KN

Test Data

No.	Width mm	ao.Ta mm	Fm kN	Rm MPa						
01	19,03	5,70	65,3	602						

No.1 specimen signature(LOT ID:O.D101.6x5.7 - 1)



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

Mai Hùng Khanh



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY
PHÒNG THÍ NGHIỆM VLXD VÀ KIỂM ĐỊNH CHẤT LƯỢNG CÔNG TRÌNH LAS-XD 1488
BUILDING MATERIAL TESTING AND CIVIL ENGINEERING CONSTRUCTION QUALITY
CONTROL LABORATORY LAS-XD 1488



- ✦ Trụ sở chính: 27/29 Đường T4, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- ✦ Head office: 27/29 T4 Street, Bình Trị Đông Residential Area, Ward Bình Trị Đông B, Bình Tân District, HCM City
- ✦ Điện thoại/Tel: 08 38452 303 Fax: 08 38452 397 Email: info@iqtc.vn
- ✦ Chi nhánh PTN: Ấp Bình Tiên 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An
- ✦ Branch of Laboratory: Bình Tiên 2 Hamlet, Đức Hòa Hạ Commune, Đức Hòa Dist., Long An Province
- ✦ Điện thoại/Tel: 0723 777541 Fax: 0723 777540 Email: info@iqtc.vn

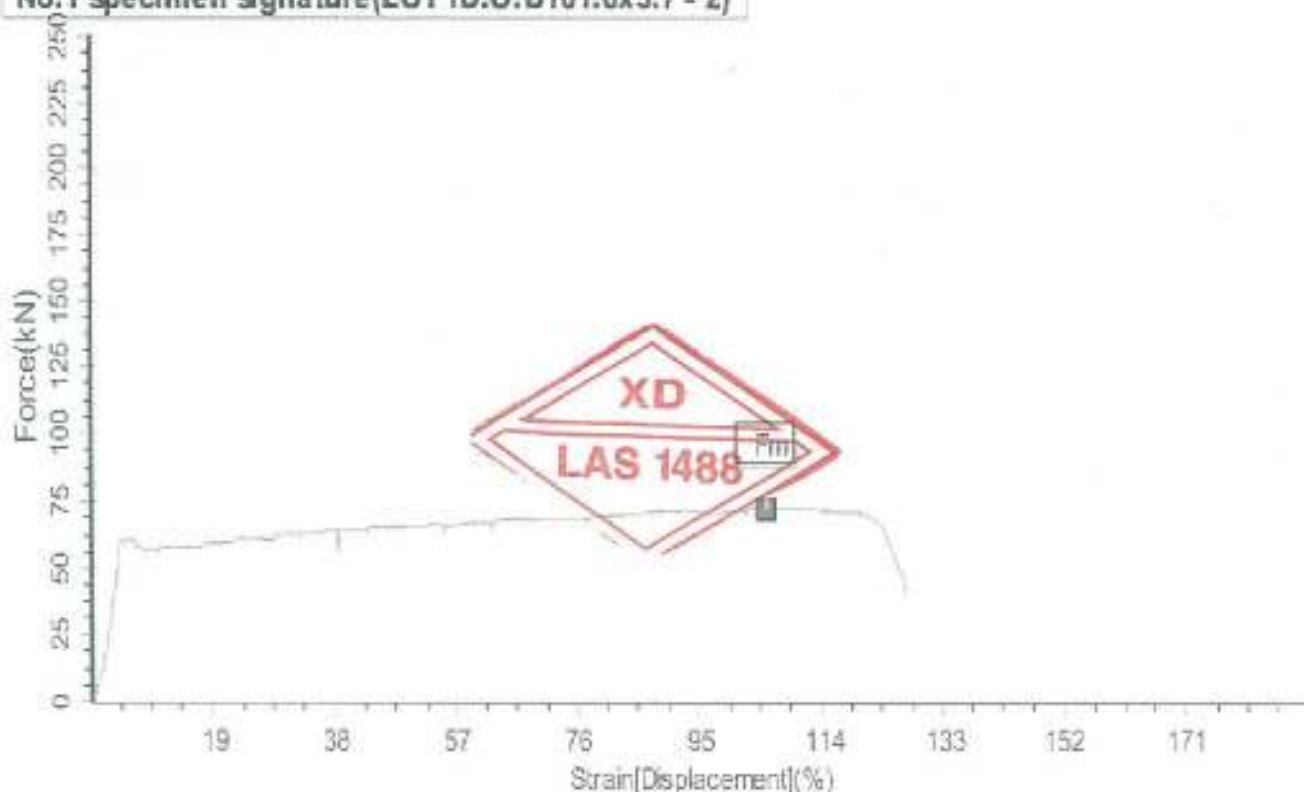
LOT ID: O.D101.6x5.7- 2

Company	DAIDUNG CORPORATION	Test Date	13-05-2021
Standard	ASME IX:2019	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	80%
Machine Model	WAW-1000	Capacity	1000KN

Test Data

No.	Width mm	ad,Ta mm	Fm kN	Rm MPa						
01	19,06	5,70	66,1	608						

No.1 specimen signature(LOT ID:O.D101.6x5.7 - 2)



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

Mai Hùng Khanh



ZHEJIANG STELLAR PIPE INDUSTRY CO.,LTD

XIAOZHI INDUSTRIAL PARK, WENXI TOWN, QINGTIAN COUNTY, LISHUI CITY, ZHEJIANG, CHINA, 323900

MILL TEST CERTIFICATES to EN 10204/3.1 PED 97/23

PED Certificate no.:331/2007/MUC

Customer : SUZHOU YULI PIPELINE INDUSTRY CO.,LTD

Contract No.: T172WL0410-12

Specification : ASTM A312-2015

Steel Grade : TP304/304L

Goods : STAINLESS STEEL SEAMLESS PIPE

Order No.

Cert No. : ST170505-08

Delivery Condition : Solution Treated

Appearance : Pickling

Heat No.		Chemical Composition (%)									
		C	Si	Mn	P	S	Cr	Ni	Mo	Ti	N
Spec.	Min.						18.00	8.00			
	Max.	0.035	1.00	2.00	0.045	0.030	20.00	13.00			
Ladle Analysis	161218V08	0.025	0.35	1.35	0.024	0.004	18.25	8.13			
Product Analysis		0.027	0.35	1.33	0.022	0.003	18.22	8.15			
Batch No.		Size		Quantity						Visual Examination	
2T170420-17		101.8*5.7*6000		5 PCS		407 KGS				OK	
Mechanical Properties (Acc to ASTM A370)											
Test No.		Tensile Strength	Proof Strength	Elongation	Flattening Test	Ring Tensile Test	Drift Expanding Test	Ring Expanding Test	Hardness Test	Grain Size No.	
		MPa	MPa	%					HRE	E112	
Spec.	Min. Max.	515	205	35					90		
L1		585	280	55	GOOD	/	/	/	77	/	
Leak Tightness Test			Non-destructive Testing			Intergranular Corrosion Test					
Hydrostatic Test		Eddy Current Test	Ultrasonic Test		Acc to ASTM A262-E						
Acc to ASTM A999		Acc to ASTM E426	Acc to ASTM E213								
GOOD		GOOD	GOOD		/						
Additional Remarks:											
1. Country of Origin: China 2. 100% PMI 3. No weld repair 4. Solution Temperature: 1050°C. Hold 30 minutes. Rapid Water Cooling. 5. Hardness Acc. to NACE MR 0103 / NACE MR 0175											
We hereby certify that the material described above has been tested and complies with the terms of the Contract & the specification and we confirm that P.M.I has been done.											
MAY 05/2017 Date											





KUANG TAI METAL (VIETNAM) CO., LTD.

MILL CERTIFICATE

Customer:

Commodity	Size	Lot No.	Specification	Certificate No.	Date of Issue						
KTS-308L	2.4mm	V19706026	ASME SFA-5.9 AWS A5.9 ER308L	16012005	13-07-2019						
CHEMICAL COMPOSITION %											
Elements	C	Mn	Si	P	S	Cu	Ni	Cr	Mo	V	
Requirement	0.03	1.0-2.5	0.3-0.65	0.03	0.03	0.75	9.0-11.0	19.0-22.0	0.75		
Weld Metal	0.019	1.90	0.64	0.014	0.010	0.00	9.90	19.95	0.00		
	Tensile Test of Deposited Metal			Impact Test of Deposited Metal		Soundness Test	Welding Condition				
	Yield Point MPa	Tensile Strength MPa	Elongation %	Test Temp. °C	Impact Value J		Type of Current	Amperage	Arc Voltage	Shielding Gas or Flux	
Requirement		MIN 520	MIN 35								
Actual Result		595	40					---	---		Ar
Bent Test		Fillet Weld Test	Hardness HRC	Moisture %	Core Wire Heat No.	Remarks:					
Face	Root										
			---	---	B3500972	WE HEREBY CERTIFY THAT THIS REPORT CORRECT AND THAT ALL TEST RESULTS ARE IN COMPLIANCE WITH THE SPECIFICATION DESCRIBED HEREIN					
Certificate		Quality System Certificate: ISO 9001:2008									





CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

123 Bạch Đằng, Phường Tân Sơn Hòa, TP. Hồ Chí Minh, Việt Nam

sale@daidung.vn

(84) 28 3868 1689

www.daidung.com

Comply:
ISO 9001:2015
ASME VIII, IX

WELDING PROCEDURE

Doc. No.: DDC-WPS-GTAW-240

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Oct 31, 2023	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)
(ASME IX-Boiler and Pressure Vessel Code:2021
ASME B31.3-Process Piping:2022)

Page: 1/2

Applicable Code: ASME Section IX, 2021 Edition & ASME B31.3-Process Piping, 2022 Edition & UT-MID-60-WPR2-210010
Organization Name: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION By: Truong Hoa Phang Vu
Welding Procedure Specification No.: WPS-DD-GTAW-240 Revision No.: 00 Date: 31 October, 2023
Supporting PQR No.(s): PQR-DD-GTAW-240 Date: 31 October, 2023
Welding Process(es): GTAW Type(s)(Manual, Automatic, Semi-Automatic): Manual

JOINTS (QW-402)		Details
Joint Design	Groove or fillet as per fabrication drawing	Joint detail as per fabrication drawing, if not specified in drawing, see "Page for Sketches."
Root Spacing	As per fabrication drawing	
Backing	With or without	
Retainer	None	
Backing Material	Base metal or weld metal or Ceramic	

BASE METALS (QW-403)

P-No. 1 Group No. 1, 2 to P No. 1 Group No. 1, 2

OR

Specification and type/grade or UNS Number None

to Specification and type/grade or UNS Number None

OR

Chem. Analysis and Mech. Prop. None

to Chem. Analysis and Mech. Prop. None

Thickness Range

Base Metal: 1.5 - 16 (mm) for groove (without CVN test) All for fillet

Maximum Pass Thickness ≤ 13 mm: Yes ☒

Other None

FILLER METALS (QW-404)

Process GTAW

Spec. No. (SFA) SFA-5.18

AWS No. (Class) ER70S-6

F No. 6

A No. 1

Size of Filler Metals ø2.4 mm

Supplemental Filler Metal None

Filler Metal Product Form Solid rod

Filler Metal Trade Name ST-50.6 (Hyundai)

Weld Metal

Deposited Thickness:

Groove Max. 16 mm

Fillet All

Electrode Flux (Class) None

Flux Type None

Flux Trade Name None

Consumable Insert None

Other None

The
SG

[Signature]



WELDING PROCEDURE SPECIFICATION (WPS) (ASME IX-Boiler and Pressure Vessel Code:2021 ASME B31.3-Process Piping:2022)

Page: 2/2

Applicable Code: ASME Section IX, 2021 Edition & ASME B31.3-Process Piping, 2022 Edition & UT-MID-60-WPR2-210010

Organization Name: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

By: Trương Huân Phong Vũ

Welding Procedure Specification No.: WPS-DD-GTAW-240

Revision No.: 00

Date: 31 October, 2023

POSITIONS (QW-405)

Position(s) of Groove: All
Position(s) of Fillet: All
Welding Progression:
Vertical up: Yes
Vertical down: None

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: None
Time Range: None
Other: None

PREHEAT (QW-406)

Preheat Temperature, Minimum: 10°C
Interpass Temperature, Maximum: 152 °C
Preheat Maintenance: None
Other: None

GAS (QW-408)

	Percent Composition		
	Gas(es)	Mixture	Flow Rate
Shielding	Argon	99.99%	14 - 18 L/min.
Trailing	None	None	None
Jacking	None	None	None
Other	None	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Weld Layer	Process	Filler Metal		Current Type and Polarity	Amps [Range] [A]	Wire Feed Speed [Range] [mm/min]	Energy or Power [Range] [kJ/mm]	Volts [Range] [V]	Travel Speed [Range] [mm/min]	Other
		Classification	Diameter [mm]							
Root	GTAW	ER70S-6	2.4	DC EN	108 - 117	None	0.70 - 1.11	10 - 13	82 - 92	None
Fill	GTAW	ER70S-6	2.4	DC LN	128 - 163	None	0.70 - 1.90	10 - 14	72 - 109	None
Cap	GTAW	ER70S-6	2.4	DC EN	122 - 138	None	0.75 - 1.27	10 - 14	91 - 97	None

Pulsing Current: None
Tungsten Electrode Size and Type: Ø2.4 mm, 2% Thoriated Tungsten (EWTh-2)
Mode of Metal Transfer for GMAW (FCM): None
Other: None

TECHNIQUE (QW-410)

Process: GTAW
Stitch or Weave bead: Weave bead
Orifice, Nozzle, or Gas Cup Size: All size
Initial and Interpass Cleaning (Brushing, Grinding, etc): Brushing or Grinding
Method of Back Chipping: None or Grinding or Air carbon arc
Oscillation: None
Contact Tip to Work Distance: 15-25 mm
Multiple or Single Pass (Per Side): Multiple pass
Multiple or Single Electrode: Single
Electrode Spacing: None
Peening: None
Other: None

Organization: DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION

Certified by:



Approved by: Trương Huân Phong Vũ (CA Manager)

Vu Nhu Minh

Date: 31 October, 2023 *Hồ Anh Dũng*

Date: 31 October, 2023

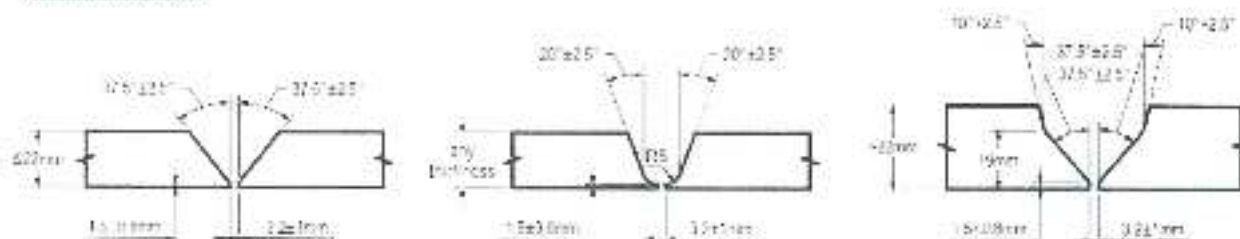


Welding Procedure Specification No.: WPS-DD-GTAW-240

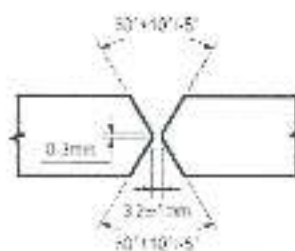
JOINT DETAILS (QW-402)

Joint designs refer to fabrication drawing or weld meta. Below is typical only

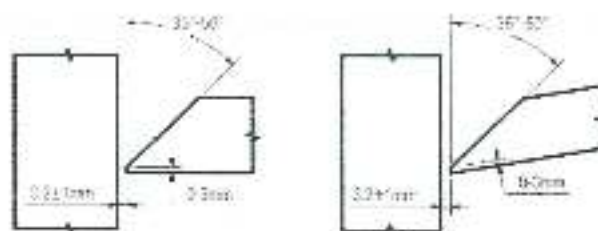
1. Typical Butt Weld



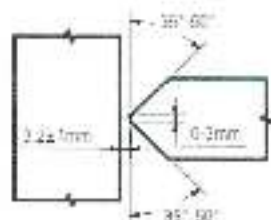
2. Double V Groove Butt Joint



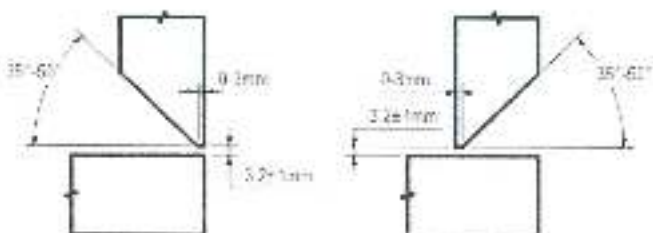
3. T-Joint: Single Bevel Groove Butt Weld



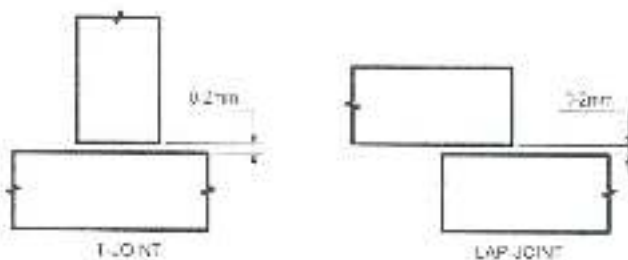
4. T-Joint: Double Bevel Groove Butt Weld



5. Corner-joint: Single Bevel Groove Butt Weld



6. Fillet Weld





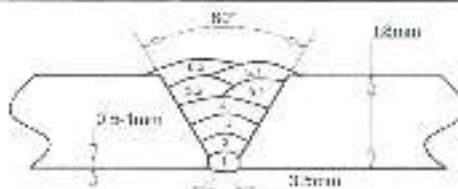
PROCEDURE QUALIFICATION RECORD (PQR)
(ASME IX-Boiler and Pressure Vessel Code 2021
ASME B31.3-Process Piping:2022)

Page: 01/02

Organization Name: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**
 PQR No.: **PQR-DD-GTAW-240** Date: **October 31, 2023**
 WPS No.: **WPS-DD-GTAW-240**
 Welding Process(es): **GTAW**
 Type(s) (Manual, Automatic, Semi-Automatic): **Manual**

JOINTS (QW-402)

Joint Design: **Single V groove butt joint**
 Root Spacing: **As per details**
 Backing: **Without**
 Backing Material: **None**



BASE METALS (QW-403)

Material Spec: **ASTM A572 to EN 10025-2**
 Type/Grade, or UNS Number: **Gr.50 to S35512**
 Thickness of Test Coupon: **8mm**
 Diameter of Test Coupon: **Plate**
 Maximum Pass Thickness: **≤ 8 mm**
 Other: **None**

POSTWELD HEAT TREATMENT (QW-407)

Temperature: **None**
 Time: **None**
 Other: **None**

FILLER METALS (QW-404)

Process: **GTAW**
 SFA Specification: **SFA-5.18**
 AWS Classification: **ER70S-6**
 Filler Metal P-No.: **6**
 Weld Metal Analysis A No.: **1**
 Size of Filler Metal: **2.4 mm**
 Trade Name: **ST-S0.6 (Hyundai)**
 Filler Metal Product Form: **Solid Rod**
 Weld Metal Thickness: **8 mm**
 Other: **None**

GAS (QW-408)

	Gases	Percent Composition (Mixture)	Flow Rate
Shielding	Argon	99.99%	15-16l/min
Trailing	None	None	None
Backing	None	None	None
Other	None	None	None

ELECTRICAL CHARACTERISTICS (QW-409)

Process: **GTAW**
 Polarity: **DCEN**
 Amperage: **See table below**
 Volts: **See table below**
 Other: **None**

POSITION (QW-405)

Process: **GTAW**
 Position(s): **1G**
 Weld Progression (Direction of Travel): **None**
 Other: **None**

TECHNIQUE (QW-410)

Process: **GTAW**
 String or weave: **Both**
 Orbits, Nozzle, or Gas Cup Size: **No. 6**
 Method cleaning: **Brushing, Grindling**
 Method back gouge: **None**
 Multipass or single pass (per side): **Multipass**
 Multipass or single Electrodes: **Single**
 Closed to out chamber: **None**
 Manual or automatic: **Manual**
 Peening: **None**
 Use of thermal processes: **None**
 Other: **None**

PREHEAT (QW-406)

Preheat Temperature: **Ambient (11°C)**
 Interpass Temperature: **Max. 107°C**
 Other: **None**

WELDING PARAMETER

No	Layer	Process	Electrode/Filler		Amperage (A)	Voltage (V)	Travel Speed (mm/min)	Preheat/Preheat Temp.(°C)	Deposited Thickness (mm)	Flow rate (L/min)	Heat Input (KJ/mm)
			AWS Class	Dia(mm)							
1	1	GTAW	ER70S-6	2.4	111-114	11-12	88	51	2	15-15	0.83-0.93
2	2	GTAW	ER70S-6	2.4	137-139	11-12	100	48	2	15-15	0.83-0.92
3	3	GTAW	ER70S-6	2.4	156-158	12-13	75	53	1.5	15-15	1.42-1.56
4	4	GTAW	ER70S-6	2.4	156-158	12-13	72	45	1.5	15-15	1.55-1.71
5	5.1	GTAW	ER70S-6	2.4	132-134	11-12	83	37	2	15-15	1.32-1.14
6	5.2	GTAW	ER70S-6	2.4	147-149	12-13	88	91	2	15-15	1.2-1.32
7	6.1	GTAW	ER70S-6	2.4	126-128	12-13	97	107	1	15-15	0.96-1.03
8	6.2	GTAW	ER70S-6	2.4	132-134	11-12	91	95	1	15-15	0.90-1.06

Thay m
SGS St

PQR No.: PQR-00-GTAW-240

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile (MPa)	Ultimate Tensile Stress (MPa)	Type of Failure and Location
S1-1	19.07	7.79	148.6	83.6	564	Ductile and base metal
S1-2	19.03	7.9	150.1	83.6	578	Ductile and base metal

Guided-Bend Tests (QW-160)

Type and Figure No.	Result
Specimen 3f Face bend	No Crack - Pass
Specimen 3f Edge bend	No Crack - Pass
Specimen 3f Back bend	No Crack - Pass
Specimen 4f Back bend	No Crack - Pass

Toughness Tests (QW-170) (Not Apply)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values		Drop Weight Weight (J/s)
				Full or J	% Absorbed	

Comments:

Filler-Weld Test (QW-180) (Not Apply)

Result - Satisfactory: ☒ Yes ☐ No ☐ Yes ☐ No
 Macro - Result: ☐ Yes ☐ No

Other Tests

Type of Test: **Macroetch**
 Deposit Analysis: **Full penetration and complete fusion weld**
 Other: **Magnetic Particle Testing: TZ 93 BB MT-A0009(PQ) - Acceptable**

Welder's Name: **Nguyen Van An** Check No.: **None** Stamp (IC) No.:
 Tests Conducted by: **SGS** Laboratory Test No.: **CN123/0348-1**

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX, Edition: 2021 of the ASME Bailer and Pressure Vessel Code.

Organization: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORPORATION**

Certified by:
SGS APPROVAL
 Date: **October 31, 2023**
 Sign: **Hồ Anh Dũng**

Approved by:
Vu Nhu Minh
 Date: **October 31, 2023**

Trương Hoài Phong Vũ

TUVNORD
 TUV-100 Vietnam
 On behalf of
 Date: **14/10/23**
 WITD REV 2 VER 02

công ty
Vietnam Ltd

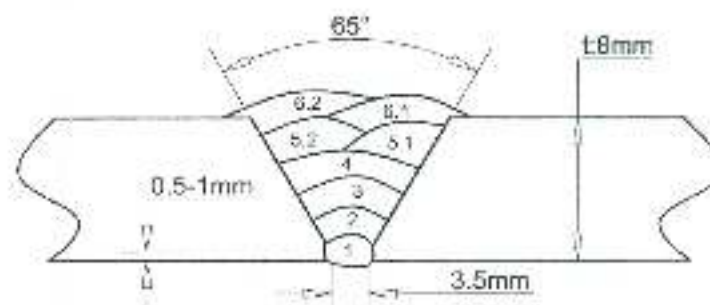
WELDING PARAMETERS RECORD

Company name: Daidung Metallic Manufacture Construction & Trade Corporation
 PQR No.: WPR-DD-GTAW-240
 Test Coupon: DD-GTAW-240
 Test position: 1G
 Base metal: T1 T2
 Specification: ASTM A572 Gr 50 (P-1 Gr 1) EN 10025-2 S355 J2 (P-1 Gr 2)
 Dimension: P: 8x200x650mm PL: 8x200x650mm
 Heat/Product No.: 130263 2204137A
 Certification No.: Q24022401409 220321200214
 Filler metal: AWS A-5.18
 Specification: ER70S-6
 Classification: LR70S-6
 Trade name: Hyundai SI 50.6
 Batch/Lot No.: 22616508
 Certificates No.: 2308073300002
 Gas: Ar (89.9%)
 Welder Name: Nguyễn Văn An

Doc No.: WPR-DD-GTAW-240

Date: 13-Oct-2023

Joint Sketch:



Root face: 0.5-1 mm

Root gap: 3.5 mm

Groove angle: 65°

Layer/ pass	Process	Filler metal		Electric Current		Times		Length (mm)	Travel Speed (mm/min)	Preheat/ Interpass Temp. (°C)	Shielding Gas Flow Rate (L/min.)	Heat Input (kJ/mm)	Weld bead		Backing Gas Flow Rate (L/min.)	Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.					Width	Depth		
1	GTAW	ER70S-6	2.4	DCEN	115-114	11-12	137	200	68	31	15-16	0.88-0.93	8	2	N/A	
2	GTAW	ER70S-6	2.4	DCEN	137-139	11-12	110	200	109	46	15-16	0.83-0.92	9	2	N/A	
3	GTAW	ER70S-6	2.4	DCEN	156-158	12-13	152	200	79	55	15-16	1.42-1.56	10	1.5	N/A	
4	GTAW	ER70S-6	2.4	DCEN	150-158	12-13	167	200	77	45	15-16	1.36-1.71	10	1.5	N/A	
5.1	GTAW	ER70S-6	2.4	DCEN	132-134	11-12	141	200	65	37	15-16	1.02-1.14	11	2	N/A	
5.2	GTAW	ER70S-6	2.4	DCEN	147-149	12-13	137	200	88	91	15-16	1.2-1.32	10	2	N/A	

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min)	Preheat/ Interpass Temp. (°C)	Shielding Gas		Heat input (kJ/mm)	Weld bead		Backing Gas Flow Rate (L/min.)	Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)	Volts (V)	min.	sec.				Flow Rate (L/min.)			Width	Depth		
6.1	GTAW	ER70S-6	2.4	DCEN	125-128	12-13		124	200	87	107	15-16		0.94-1.03	10	1	N/a	
6.2	GTAW	ER70S-6	2.4	DCEN	137-134	11-12		137	200	91	96	15-16		0.95-1.06	10	1	N/a	
Name		Manufacturer				Third party (SGS)				Examiner or examining body								
Signature		<i>[Signature]</i>				<i>[Signature]</i>				SGS Reviewed Date: 31/10/2023 Sign: <i>[Signature]</i>								
Date		10-Oct-2023				20/Oct/23												

Vu Nhu Minh



BRN

[Signature]

Pham Trung Dung

Hồ Anh Dũng



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

✦ **Tổng chi nhánh head office:** 77/21 Đường T4, Khu Dân Cư Đông, Phường Bình Tân Đông B, Quận Bình Tân, TP HCM
✦ **Điện thoại:** 0279.3777.640 Fax: 0279.3777.640 Email: info@iqtc.vn Website: www.iqtc.vn
✦ **Phòng Nghiệm Laboratory:** Phòng Dân Cư, 35 Đường Hòa Hạ, Huyện Bình Hòa, Tỉnh Long An



Report No.: TZ-90-BB-MT-ACC000(PQR)
Request No: DDC-WPQR-RQ-002, Rev2
Procedure No: TZ-TAN-84-55J1-030026 Rev01
Date: 13/Oct/2023
Page No: 1 of 1

MAGNETIC PARTICLE REPORT

Client:	ĐẠI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION				
Project:	EAST AFRICAN CRUDE OIL PIPELINE PROJECT				
Item:	WPQR - WELDING PROCEDURE QUALIFICATION RECORD				
Location:	ĐẠI DUNG FACTORY				
Inspection date:	13/Oct/2023				
Drawing No	NA	Quantity	1 Pcs	Description	WPQR-DD-GTAW-240

MAGNETIC PARTICLES INFORMATION

Magnetic particles material	Brand-Batch No.	Color	Temperature
Contrast paint	NABAKEM - MP35	White	26°C - 37°C
Magnetic particles	NABAKEM - MP35	Black	26°C - 37°C

METHOD OF EXAMINATION

Magnetizing material status	☒ Wet ☐ Dry	Equipment type: MP-A2 Serial No: MP-5552
Magnetizing Method:	☒ AC ☐ DC	Demagnetization: ☐ Yes ☒ No

MATERIAL INFORMATION

Material Base: ASTM A572 Gr.50 (P-1 Gr 1), EN 10025 2 S355J2 (P-1 Gr 2)	Product Type: WPQR	Standard: Clause S4.3.2 ASME B31.3	Welding process: GTAW
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EVALUATION

No.	Weld identification	Welder No.	Length & Thickness (mm)	Indication/Defect			Evaluation		Remarks
				Location	Length (mm)	Type	Accept	Reject	
1.	WPQR-DD-GTAW-240								
	DD-GTAW-240	NGUYEN VAN AN	950 ± (6x8)	-	-	-	Accept		Discard 50mm of 2 ends
-End of Documents-									

Attach:



WPQR-DD-GTAW-240

BBN
Pham Trung Dung

Abbreviation:	CP - Check Passably	SD - Surface Defect	CP - Check	LD - Undercut
DD - Accurately	CP - Checkably	CP - Overlap	SI - Signification	IP - Inside inspection

IQTC	DDC QC	CONTRACTOR QC (BBNTUV)	COMPANY QC (CPY)
Evaluated: VO VAN TAN M-LEVEL 3 TESTING JOINT STOCK Signature:	Name: Signature:	Name: Vu Nhu Minh Signature: TUVNORD TNV-130 Vietnam On behalf of Date: 14.10.23	Name: Signature: Date: REV 2
Date: 14/Oct/2023	Date: 14.10.2023	Date: REV 2	Date: REV 2

Client's name : DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
Tên khách hàng
CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Client's address : LOT NO. 38, AREA C, D1 ROAD, AN HA INDUSTRIAL PARK, PHAM VAN HAI COMMUNE, BINH CHANH DISTRICT, HO CHI MINH CITY, VIETNAM
Địa chỉ
LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : WELDING / MÔI HÀN
Name / Tên : DD-GTAW-240
1G – Single V butt joint
Size / Kích thước : PL8x200x650mm welded to PL8x200x650mm
Material / Vật liệu : ASTM A572 Gr.50 (P-1 Gr.1) + EN 10025-2 S355J2 (P-1 Gr.2)
Filler / Que hàn : Hyundai ST-50.6
Welder / thợ hàn : NGUYEN VAN AN

Project / Dự án : EACOP PROJECT

Item / Hạng mục : Procedure Qualification Record (PQR) Test for Piping

Quantity / Số lượng mẫu : 01 sample / mẫu

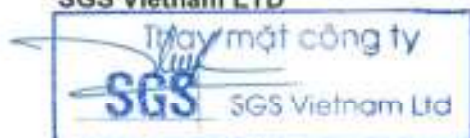
Date of receiving / Ngày nhận mẫu : 19-Oct-2023

Date of testing / Ngày thử nghiệm : 23-Oct-2023

Test method / Phương pháp thử : ASME BPVC.IX-2021 Welding, Brazing, and Fusing Qualifications
ASME B31.3-2022 Process Piping
ASTM E340 - 15 Standard Practice for Macroetching Metals and Alloys

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

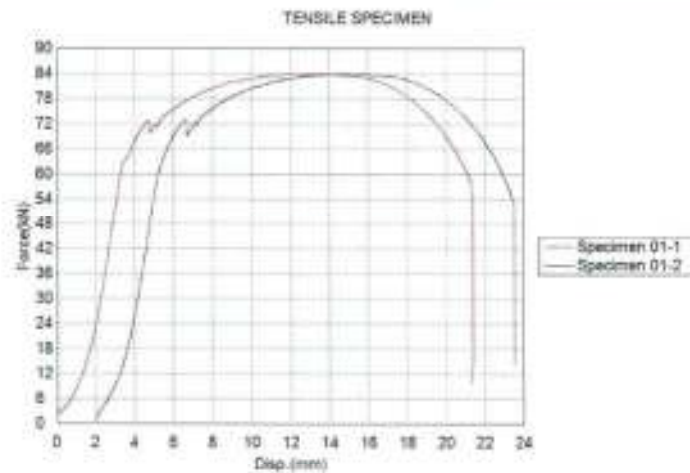


Pham Quang Hien
NR OGC and I&E Laboratory Manager
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test / Thử kéo theo phương ngang	1	2
• Width of specimen / Chiều rộng mẫu mm	19.07	19.03
• Thickness of specimen / Chiều dày mẫu mm	7.79	7.90
• Cross section area / Tiết diện mm ²	148.6	150.3
• Tensile load / Lực kéo đứt kN	83.8	83.8
• Tensile strength / Giới hạn bền kéo MPa	564	558
• Broken position / Vị trí mẫu đứt	In base metal	In base metal



The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	8.03	8.04	8.02	8.02
• Width of specimen / Chiều rộng mm	38.01	38.01	38.01	38.01
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	32	32	32	32
• Test result / Kết quả	No Crack/ Không nứt	No Crack/ Không nứt	No Crack/ Không nứt	No Crack/ Không nứt

*** TO BE CONTINUED ***

(Handwritten signatures)

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C. Characteristic / Tên chỉ tiêu : Macro etch (*) / Tổ chức thô đại

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

Note / Ghi chú :

(*) The test was accredited ISO 17025:2017 by VILAS / Chỉ tiêu được công nhận ISO 17025:2017 bởi VILAS.
ILAC: International Laboratory Accreditation / Tổ chức hợp tác công nhận Phòng thử nghiệm quốc tế.

MRA: Mutual Recognition Arrangement / Thỏa ước thừa nhận lẫn nhau.

ILAC-MRA -VILAS: VILAS is the full member and signatory of Mutual Recognition Arrangement with International Laboratory Accreditation / VILAS là thành viên đầy đủ và ký thỏa ước thừa nhận lẫn nhau với tổ chức công nhận phòng thử nghiệm quốc tế.

(**) The sample was tested at CONSTRUCTION MATERIAL LABORATORY SGS VIETNAM Co., LTD on 23-Oct-2023, witnessed by the client and related parties / Mẫu thử được thử nghiệm tại PHÒNG THỬ NGHIỆM VẬT LIỆU XÂY DỰNG CÔNG TY SGS VIỆT NAM vào ngày 23/10/2023, có sự chứng kiến của khách hàng và các bên có liên quan:

- Mr. Hoang Van Dung (SGS)
- Mr. Nguyen Dang Tri (TUV)

*** END OF THE REPORT ***

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产品质量证明书
INSPECTION CERTIFICATE

[illegible]



MILL CERTIFICATE

台塑河靜鋼鐵興業責任有限公司

FORMOSA HA TINH STEEL CORPORATION

ADDRESS: Trung An Economic Zone, Ky Luong Ward, Ky Anh Town,
Ha Tinh Province, Viet Nam

TEL: +84 239 3722 123 Ext.2066 Fax: +84 239 3722 112

PURCHASER:	CITICOM COMMERCIAL JOINT STOCK COMPANY	COMMODITY:	HOT ROLLED COIL, NON-SKINPASSED
SPECIFICATION and GRADE:	TCC'S 40:2019/11S ASTM A572/A572M GRADE 50	FHS CONTRACT NO:	2020110023
INSPECTION:	FHS MILL INSPECTION	CERTIFICATE NO:	025A073401469
END-USER NAME:	CITICOM COMMERCIAL JOINT STOCK COMPANY	MC ISSUE DATE:	2023/04/26
		PURCHASER ORDER NO:	
		INVOICE NO:	
		CUSTOMER NO:	



COIL NO.	PRODUCT INFORMATION				HEAT NO.	SAMPLE ID	DBI	TENSILE TEST				BENDING	CHEMICAL ANALYSIS RESULT %																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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SGS Reviewed
Date: 31/10/2023
Sign: *[Signature]*
Ho Anh Dung

Sub Total	3 pcs	85.715 kg	TOTAL	3 pcs	85.715 kg
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NOTES	COMMODITY on LABEL: HOT ROLLED COIL, NON-SKINPASSED SPECIFICATION on LABEL: TCC'S 40:2019/11S GRADE on LABEL: ASTM A572/A572M GRADE 50 THIS MILL CERTIFICATE IS ISSUED IN ACCORDANCE WITH EN 10204 TYPE 3.1	WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.	SGS: YIELD POINT TYPE T: TRACE. THE ANALYSIS RESULT COMPLIES WITH SPECIFICATION REQUIREMENT
SURVEYOR TO			chou chonghua MANAGER, TECHNOLOGY DEPT.



Item 1

CERTIFIED MATERIAL REPORT

CMR No. : 2305073300002

Date of Issue : 2023-08-07

Customer	GLOBAL TECHNOLOGY COMMERCIAL CORPORA		
Trade Name	S-50.6	Size	2.4mm*1000mm*5kg
Lot No.	22595103	Purchaser's Order No.	0422_HW_GT
Specification	AWS A5.18 E 470B C EN ISO 638-A - W 391		
Welding Current	DC IN	Shielding Gas	Ar
		Post Weld Heat Treatment (°C x hr.)	AS WEL (DL)

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)			
	YS(MPa)	TS(MPa)	EL(%)	Temp.(°C)	X1	X2	X3
	450	560	29.0	-30	110	115	120
AWS Spec.	≥ 400	≥ 480	≥ 22	≥ 27			

Chemical Composition (%)

* According to EN 10204-2004 : 3.1

W.R.F	C	Si	Mn	P	S	Cu	N	Cr	Mo	V	
	0.08	0.97	1.47	0.016	0.018	0.17	0.01	0.04	0.01	0.01	
AWS Spec.	0.06 ~ 0.15	0.20 ~ 1.15	1.40 ~ 1.85	≤ 0.025	≤ 0.035	≤ 0.60	≤ 0.15	≤ 0.15	≤ 0.15	≤ 0.03	

*Radiographic Test : ACCEPT

Others

The product supplied will meet the requirements of the applicable AWS standard, when tested in accordance with that standard.

* Qty : 1,00kg



Hồ Anh Dũng

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

K.W. Park

QA Manager Park K. W.



HYUNDAI WELDING CO., LTD.

Head Office

507 Teheran-ro, Gangnam-gu, Seoul, Republic of Korea

Pohang Plant

100 Daesong-ro, Nam-gu, Pohang, Republic of Korea